

Principles of Education

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T. Gale



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TABLE OF CONTENTS

Preface

xi

Chapter 1	Universal Basic Education	1
	General Administration in Education	2
	Gender Equality in Basic Education	5
	Etymological Meaning of Education	5
	Meaning and Importance of Education	6
	Meaning	6
	The importance of education	6
	Education in Developing Countries: Evidence and Implications	8
	A Changing World	9
	A Look at the Evidence	10
	Getting From Here to There	14
	The Future of Education	14
	Lessons from the Ancient World	15
	Eight Driving Forces	17
	The Age of Hyper-Individuality	22
	Transition from Consumers to Producers	23
	Archiving Knowledge	26
	Defining the New, New Era in Learning	27
	Twelve Dimensions of the Future Courseware Architecture	27
	Standard Courseware Unit	27
	Understanding the System Entry-Point	30
	The Social Environments of Learning	31
	Thinking Through the Transition for Existing School Systems	31
	Changing Revenue Streams	32

Chapter 2	Education and the Practice Environment	34
	Introduction	34
	Background and Rationale of the Study	35
	Structure of Services and Practice Environment	36
	Education and Training.....	39
	Outcomes and Performance Measures	40
	Best Practices in Community-Based Social Work.....	41
	Social Work Education and Practice in the United States	42
	Historical Development	42
	Social Work Today	43
	Regulation of Practice: Registration, Certification and Licensure	44
	Social Work Professional Associations	44
	Reforms of Higher Education in E&E: Impact on Social Work.....	45
	Best Practice Model Applied to the Profession of Social Work	46
	Environmental Education	46
	Definition	46
	Scope	46
	Need and Importance of Environmental Studies	47
	Concept of Environmental Education	48
	Historical Background of Environmental Education	50
	Celebration of Various Days in Relation with Environment	52
	Plastic Recycling and Prohibition of Plastic Bag/Cover	54
	Role of School in Environmental Conservation and Sustainable Development.....	60
	Pollution Free Eco-System	62
Chapter 3	Educational Philosophy and Thoughts	72
	Subject Matter	73
	Meaning of Philosophy	73
	Meaning of Philosophy and Education	73
	Concept of Philosophy of Education	74
	John Locke.....	77
	Educational Implications of Locke's Philosophy	77
	Jean-Jacques Rousseau	78
	On Education	78
	Definition of Education by Redden	80
	John Dewey	80
	On Education	81
	Gandhi	83
	On Education	83
	Zakir Hussain.....	86
	Dr. Hussain's Views On Education.....	86

Chapter 4	National Education Systems: Theory and Practice	89
	National Education Systems	89
	Education and Social Cohesion	90
	Education and Social Conflict	90
	Education and Personal Growth	90
	Education and Civil Society	91
	Education and Economic Development	92
	Global Enrollment Trends Since the Mid-20th Century	92
	Primary-level School Enrollments	93
	Secondary-level School Enrollments	93
	Definition	94
	Related Terms	94
	History	96
	Theory	99
	Behaviourism	99
	Practice	101
	Synchronous and Asynchronous	101
	Training Management System	109
	Learning Objects	109
	Settings	110
	Corporate and Professional	113
	Benefits	115
	Disadvantages	116
	Sociocultural Criticism	118
	Teacher Training	120
	Assessment	121
	Analytics	122
	Expenditure	122
	Careers	122
Chapter 5	Theories of Organisation and Management in Education	123
	Management of School Building	123
	Contingency Theory	126
	Systems Theory	126
	Management in India	128
	Management Functions and Change	130
	The Management of Education	131
	National Level	131
Chapter 6	Theory of Intelligence	134
	Intelligence as a Process	135
	Nature of Intelligence	135

Social Intelligence	137
Scientific Studies	138
Measuring Social Intelligence	139
Emotional Intelligence	140
History	140
Definitions	140
Criticisms of the Theoretical Foundation of EI	145
Criticisms of Measurement Issues	146
Distribution of Intelligence	147
Intelligence Tests as Clinical Instruments	148
Intelligence	148
Motivation	154
Integrity Inventory	157
International Test Commission	158
Risk Intelligence	159
Mayer-Salovey-Caruso Emotional Intelligence Test	160
Ammons Quick Test	161
Army Alpha	161
Block Design Test	170
Bracken School Readiness Assessment	171
Cattell Culture Fair III	172
Educational Quotient	173
Figure Reasoning Test	173
G Factor (Psychometrics)	173
Theory of Intelligences: Single, Two Factor and Multifactor Theories	190
Development of the Theory	190
Chapter 7 Teacher Education: Quality Assurance and its Indicators	192
Definitions	192
Teacher	196
Professional Educators	197
Pedagogy and Teaching	197
Rights to Enforce School Discipline	198
Obligation to Honour Students Rights	200
Teacher Enthusias	200
Misconduct	202
University	202
A Methodological Approach to Integrating Web-Based Learning into Traditional Teaching	203
The Method of Case Studies- Dealing with Proto-Scientific Ideas in Education	206
A Case Study Concerning the Learning and Teaching of Ecology: Context, Impediments and Aspirations	207

Web Based Learning: New Technologies with Different Pedagogies ...	208
Qualities of Effective Teachers	209
Impact of Teacher Effectiveness on Student Achievement	211
Highly Qualified Versus Highly Effective	212
Chapter 8 Role of Indian Society	219
Society and Gender Roles	220
Gender Stereotypes for Males and Females	221
Gender Socialisation	221
The Family as Gendered Relationships: Influences on Gender Socialization Process	222
A Relational Approach to Gender Roles Socialization	223
Feminist Effects on Society	224
Women's Rights	224
International Law	225
Middle Ages Women in Islamic Society	228
Cultural and Intellectual Pursuits of Muslim Women	232
Religion in Tribal Societies	234
Chief Feature of Tribal Societies: Easy Form of Religion	234
Role of Caste in Indian Society	243
Society Family	245
Position of Women in Herding Societies	248
Education of the Young in Herding Societies	249
Agrarian Society	249
Development of Horticulture and Agriculture	250
Agriculture Today	251
Industrial	251
Post-Industrial	252
Structure of Indian Society	252
Untouchability	255
Slavery	256
Purushartha, Ashrama and Samskara	256
Chapter 9 Modern Trends and New Dimensions in Education	258
Value Based Education	258
New Approaches and Topics in Higher Education	261
Advancement of Knowledge by Research	263
The Potential and the Challenge of Technology	265
From 'Brain Drain' to 'Brain Gain'	267
Financing of Higher Education as a Public Service	268
Technical Description of Computer Education	269
Applications of Computer	269

Advantages of Computer	270
Scopes of Computer	270
Limitations of Computer	270
Teleconferencing	271
Technical Description of Teleconferencing	271
Effective Content Environment	272
Advantages of Teleconferencing	272
Experiments in Teleconferencing	273
Videodisc as Non-Print Media	273
Advantages of Videodisc	274
Application of Videodisc	274
Videotex as Non-Print Media	275
Teletext—A System Related to Videotex	277
Teletext Application in India	277
Use of Computer in India	278
Demographics of Online Instructors	278
Emerging Technology	278
Enormous Learner Demands	279
Enhanced Pedagogy	280
Group Work in Online Education	280
Technological Determinism and the Network of Learning Society	283
Online Educational Counseling	287
Ongoing Programme Advising	287
The Digital Library	287
Access for Students with Disabilities	288
Student Rights and Access to Ombuds Services	288
The Online Learner's Role in Governance	289
Learner Satisfaction Monitoring and Environmental Scanning	290
Current Online Teaching Situation	290
Online Experience	290
Usefulness of Web-based Tools for Teaching and Learning	294
Useful Online Class Tools	294
Future Online Teaching Situation	301
Predicted Instructional Time Online	301
Institutional Motives and Decision Making	302
<i>Bibliography</i>	305
<i>Index</i>	309

Preface

The theory of education and learning's objectives, applications, and interpretations is known as education theory. It is more of an umbrella concept than a singular explanation of how humans learn and how teachers should approach their work. Instead, it is influenced by a number of things, such as epistemological viewpoint and theoretical perspective.

Education has frequently been strongly associated with being a fundamentally hopeful human undertaking marked by hopes for advancement and improvement. Many people believe it to be a strategy for getting beyond obstacles, promoting greater equality, and obtaining money and social position. Children are seen as having the opportunity to develop at school in accordance with their own requirements and capabilities. It is also regarded as one of the most effective ways to promote social equality.

The success in life of an individual is greatly influenced by their education. Students receive education that teaches them life skills that will better prepare them for the workforce in the future on a physical, mental, and social level. Most people agree that education is the cornerstone of a society that leads to economic growth, social prosperity, and political stability. The preparation of healthcare professionals, informed healthcare consumers, and population maintenance are all made possible by higher education, which supports the maintenance of a healthy society.

We can be sure that the advancement of technology won't harm learning and education, so there's no need to presume that it would. The creation and design of modern education to face the problems of the 21st century are covered in a wide range of topics in this book. It also offers a thorough review of the initiatives taken to revive education.

Author

Universal Basic Education

Basic education featured heavily in the 1997 ISCED document, but the term was not included in the glossary. Each country interpreted the term in different ways, and leading up to the 2011 revision, a discussion paper was issued to seek clarification.

In most countries, ISCED 1 corresponds to the nationally designated primary education, and basic education includes that and also ISCED 2 lower secondary education (the lower level of secondary school). In other countries, where there is no break between primary and lower secondary education “basic education” covers the entire compulsory school period. For statistical reasons, ISCED 1 is then considered to be the first six years of schooling.

Universal basic education is regarded as a priority for developing countries and is the focus of the Education For All movement led by UNESCO. It is also included in the Millennium Development Goals as goal number 2: achieve universal primary education by 2015.

An extensive number of studies have proven its benefits for public health (*e.g.*, lower spread of HIV/AIDS; better vaccination; prevention and medication of disease; better nutrition; lower maternal, infant, and child mortality), demography (*e.g.*, longer life expectancy, accelerated demographic transition through better birth control) and the economy (*e.g.*, increased purchase power, increased productivity in traditional sectors, increased demand on service sectors). Other benefits, although more difficult to measure, include a beneficial impact on democracy, human rights, governance, and political stability through increased understanding of non-violent ways to solve problems and mutual understanding between groups in conflict.

The Convention on the Rights of the Child (CRC), established by UNICEF in 1989, protects children's inalienable rights by setting standards for multiple issues, one of which is education.

GENERAL ADMINISTRATION IN EDUCATION

It is being argued that “all administration is the same. Whether one looks at administration from the standpoint of its purpose, its task, its situational milieu or its process, he will find much that is common in the management of Business, Government, Army, Education and other organisations.”

Those who see no significant difference between the two say that human behaviour, cult of efficiency and bureaucracy are common to all administration.

Educational administration includes most of the general administrative functions such as appraisal of the organisational purposes and coordination of all the forces—staffing, training, public relations, community service, raising funds for institutions, maintenance of accounts and a score of other things to achieve the objectives.

On this account they say that educational administration very well fits into the pattern of administration in general:

- Despite the common elements, there are differences at least in degree between the administrative processes related to various social organisations and educational institutions. Educational administration is a “non-profit making enterprise”. It is chiefly welfare or service oriented. The purpose of an industrial administration is to produce goods, of course, to meet the needs of the people but their motive is to make profits. No worthwhile business administration can run even on ‘no profit, no loss’ basis whereas educational administration worth the name aims at the development of human personality without caring for the monetary losses. In fact, educational enterprise is a “welfare enterprise” first and last and more often than not runs in financial losses which—are met by donations, public contribution or by levies or taxes etc.
- Peculiarities of educational administration arise from the objectives of education in our society. Education is closely linked to life. It needs to be transferred into a powerful instrument of social change. National development can be brought about only through universal education. Even the survival of a democratic society is possible through education. “Destiny of India is now being shaped in her classrooms.” This, we believe is no mere rhetoric. In a world based on science and technology it is education that determines the level of prosperity, welfare and security of the people. On the quality and number of persons coming out of our schools and colleges will depend our success in the great enterprise of our national reconstruction; whose principal objective is to raise the standard of living of our people.—Indian Education Commission (1964-66).

- Good educational institutions free men to rise to the level of their natural abilities. Hope for personal advancement and the advancement of one's children is, of course, one of great wellsprings of human energy. The schools and colleges only sustain this hope. Public education has an inbuilt corrective of our society. Only through general public education can we succeed as the largest democracy of the world. The educational administrators have, therefore, a special responsibility.
- Education deals with people directly and intimately. "The most intimate relationship the school/college has, is with children and youth. The pupils represent the dearest possessions of their parents and these parents, by and large, are greatly concerned with how the school/college is treating their offspring. In other words, the school/college is confronted with the emotional attachment of parents to their children and these attachments are much stronger than those which most parents have for other public or private services." The educational administrator is thus concerned with a group of clients who may not be his by choice. But without whose cooperation the institution, he is responsible for, cannot function properly.
- The main concern of educational administration being, people and their welfare, the techniques of educational administration are mostly persuasive while those in other kinds of administration are mostly coercive, dictatorial and authoritative. An educational administrator has to look to the interaction of the persons involved, whereas in other types of administration one can afford to ignore the reactions of the persons affected by his decisions—though sometimes he too has to pay through the nose for the neglect.
- Another special feature of the educational enterprise is the nature of the learning process. Learning is a complicated affair. It comes more from informal situations than from the class room lessons. Much informal education goes on without any body's intending to learn—simply as a process of social interaction. Much of what a pupil learns is learnt as a consequence of social interaction among pupils. Thus learning is preceded by a number of factors. It does not merely mean acquiring some knowledge or understanding but it may provoke many emotional commitments. Shaheed-i-Azam Bhagat Singh who shook the foundations of the British Empire, Subhash Chandra Bose who alienated the heavenly born Indian Civil Service and Lala Lajpat Rai, the "Lion of the Punjab", achieved more than learning in their school and college days. The channels of contact and communication have to be always kept open and free with the teachers, the students and their parents. The educational administrator has to be highly visible and accessible to the public in general too as the educational institutions are the public affair.

- Evaluation of results in education is another area which poses special problems to educational administrators. In business, one can use sales of production or the profits as evaluative measures. In the military, objectives are definite and it is not difficult to assess whether these goals have been achieved or not. Likewise the targets of the town architects and dam builders can be measured to a large extent as these are more or less concrete. But it is not so in the case of educational objectives and targets. The “purpose of all education at all levels varies from literacy to critical thinking, intellectual leisure and facilitation of social mobility.” Literacy could of course be taken as Something tangible, *i.e.* reading, writing and numeration but critical thinking, intellectual leisure or facilitation of social mobility may almost mean anything to anybody. To say, therefore, about any institution as to how far these abstract goals have been achieved is difficult if not impossible. People may disagree even on their interpretations.
- Educational administration is different from other types of administrations in one more way—that is staffing. The teaching staff of a college is more or less equally qualified. All of them hold post-graduate degrees if not Ph.D. or M.Phil degrees and generally with peer merit. Many members of the staff may even be better trained and experienced than the principal. They may have even more of research work and teaching merit to their credit. The educational administrator is thus working with professionals who feel, and often rightly that they know more about teaching and learning than he does.
Gatzels and Cuba have pointed out that professional trained people, in contrast to day labourers or skilled people or production workers, are more inclined towards ‘idiographic’ than nomothetic’ behaviour. A college organisation as contrasted to any business enterprise is more persuasive and idiographic. In other words, the training of teachers and the institutional settings in which the teachers work produce behaviour which stresses personality-factors more than role-factors.
- Authority, of course, serves as fuel on which all formal organisations are run. But authority is far more sophisticated than simple “command and obey” situation or relationship and the role of authority in the academic organisation is even more discreet than it is in other enterprises. The zone of acceptance in the case of an academic faculty is much narrower than in other enterprises.
- We often find that the governing bodies or the boards of control of educational institutes know very little about the complexities of learning and teaching. The unfortunate part of the whole situation is that they sit in judgement on the educational programmes they do not fully understand. But this thing seldom happens in other organisations. The manager of a firm or a factory works for a board most of whom are thoroughly familiar with the factory affairs, its organisation and objectives.

- Lastly the educational administrator in a college has to perform two major functions. Firstly, he is principal of the college, and secondly, he needs to continuously reshape his organisation. The principal of a college has not only to be an administrator but an educational statesman too. He has to achieve the immediate objectives of the organisation and also to plan the long-range development and growth of the institution. He is not only to implement the educational policies framed by the powers that be, but is expected to suggest the changes in the policies. A general administrator is bound to plan and operate a programme designed to implement certain policies. He has to work within the given steel framework. If he ever trespasses the framework, he has to pay the price for it; whereas an educational administrator who is worth his salt spearheads new tracks. He may be, even ploughing a lonely furrow.

GENDER EQUALITY IN BASIC EDUCATION

Gender equality in education has traditionally been narrowly equated with gender parity at different levels of formal education. Gender has been a traditional factor of inequality and disparity in education, most often to the disadvantage of girls and women. Yet there has been significant progress in narrowing the gap around the world since 2000, with a larger proportion of girls and women accessing different levels of formal education. Indeed, gender parity in primary education has been achieved in Central and Eastern Europe, Central Asia, East Asia and the Pacific, Latin America and the Caribbean, North America and Western Europe. In addition, significant progress has been made since 2000 in narrowing the gender gap, particularly in South and West Asia and to a lesser degree in sub-Saharan Africa and the Arab States. However, despite the significant progress made, the majority of out-of-school children are girls, while two-thirds of youth and adults with low levels of Literacy in the world are women. To help ensure women's empowerment, boys and men must also be engaged in the fight against gender inequality. This must begin with basic education.

ETYMOLOGICAL MEANING OF EDUCATION

The first thing we must do in order to approach our subject is to define its true nature and meaning in clear terms; *i.e.*, to make a terminological specification. It is particularly necessary to do this in the case of education so that we may become aware of the existence of a basic misunderstanding, or confusion that must be eliminated. Education has often been considered as synonymous to "instruction," *i.e.*, the imparting of knowledge and information. Up until a short time ago and even now) the majority of schools of all levels has aimed, more or less openly, at such a goal, and has intended to carry out such a function. That education, in the true meaning of the word, is something quite different, something that is much more inclusive, and that in a certain sense, even

has the opposite meaning! Instruction means to infuse, to put something in that is lacking, to fill a vacuum. The etymological meaning of the word “education”, however, means to “draw out” (from the Latin “educere”)...to lead, to draw out that which is within; *i.e.* to bring to light what is hidden, to render actual what is only potential, to develop. It also means to draw out of conditions that limit; in other words, it is the favouring of a process of growth. Of course, education also includes the imparting of ideas, but this must be seen only as a first step or stage, as an instrument or necessary means, and not as an end in itself. Both aspects and concepts are included in the common usage of the word “education”, and this easily creates confusion and misunderstandings. It would, therefore, be useful to distinguish between them and to always specify, for example, by using the terms “informative education” and “formative education”.

MEANING AND IMPORTANCE OF EDUCATION

MEANING

To know the meaning of education we discuss about its etymological, narrow and broader meaning of education.

- *Etymological Meaning of Education:* The term education is derived from Latin word educere, educare, and educatum which means ‘to learn’, ‘to know’, and to ‘lead out’. That is education means to lead out internal hidden talent of a child or person.
- *Narrow Meaning of Education:* The education provided under the premises of schools, colleges and universities is the narrow concept of education. The narrow meaning of education limited under the premises of educational institutions. It doesn’t include the education outside of four walls of institution. Narrow meaning of education emphasize on bookish knowledge.
- *Broader meaning of education:* Opposite of narrow meaning of education is broader meaning of education. That is the education provided under the educational institution’s premises is only education is wrong. According to its concept education is universal, we can gain education from anywhere, anytime. There is no bound of place and time. Education is the long life process. It starts from cradle and ends to the grave.

THE IMPORTANCE OF EDUCATION

Education is very important for an individual’s success in life. Education provides pupils teaching skills that prepare them physically, mentally and socially for the world of work in later life. Education is generally seen as the foundation of society which brings economic wealth, social prosperity and political stability. Higher education helps in maintaining a healthy society which prepares health care professionals, educated

health care consumers and maintaining healthy population. Education is major aspect of development of any modern society since if there is a deficit of educated people then society will stop its further progress. Government should pay serious attention to education and support it economically and morally all over the country. Education is the best investment for the people because well educated people have more opportunities to get a job which gives them satisfaction. Educated individuals enjoy respect among their colleagues and they can effectively contribute to the development of their country and society by inventing new devices and discoveries. Today's ever growing numbers of people mostly are not satisfied with their basic education and try to get secondary or tertiary education in order to meet the demands of contemporary society. Some of them enter higher educational institutions and some search additional information on the internet. Good People sacrifices their time and money and sometimes even their health to raise educational level because they realize that education is their passport to the future and for tomorrow. Main purpose of education is to educate individuals within society, to prepare and qualify them for work in economy as well as to integrate people into society and teach them values and morals of society.

Role of education is means of socializing individuals and to keep society smoothing and remain stable. Education in society prepares youngsters for adulthood so that they may form the next generation of leaders. It will yield strong families and strong communities. Indeed, parents taking an active role in their child education produce a willingness in children to learn. Education and society provides a forum where teachers and scholars all over the world are able to evaluate problems in education and society from a balanced and comparative social and economic perspective. Education is an important aspect of the work of society and it will raise the countryside issues and promote knowledge and understanding of rural communities. One of the education essential tasks is to enable people to understand themselves. Students must be equipped with knowledge and skills which are needed to participate effectively as member of society and contribute towards the development of shared values and common identity. Education has a vital role to play in assisting students to understand their cultural identity. Education acts as the distribution mechanism of the cultural values such as it more layered the society and participate in society that carries the culture. Education and society provides a forum where teachers and scholars all over the world are able to evaluate problems in education and society from a balanced and comparative social and economic perspective. Education is an important aspect of the work of society and it will raise the countryside issues and promote knowledge and understanding of rural communities.

One of the education essential tasks is to enable people to understand themselves. Students must be equipped with knowledge and skills which are needed to participate effectively as member of society and contribute towards the development of shared values and common identity. Education has a vital role to play in assisting students to

understand their cultural identity. Education acts as the distribution mechanism of the cultural values such as it more layered the society and participate in society that carries the culture. In our culture today, there is a great emphasis on higher education. In a society, more educated you are, better off you are. Every society has specialized individuals that require extended education to fulfill certain main positions. These persons are normally known as professors, priests, doctors, mechanics or artists. Education has been a higher part of every culture on earth and education is a systemic project. Whole society should care for and support the education patriotism, cause and socialism among the young people. Everyone must do work hard to cultivate moral conduct. Education mainly begins at home; one does not acquire knowledge from a teacher, one can learn and get knowledge from a parent or a family member. In almost all societies, receiving education and attending school is very necessary is one wants to achieve success. Education is the key to move in the world, seek better jobs and ultimately succeed in life. Schools play a vital role in preparing our children and young people for effective participation and responsible citizenship in society. The development of education and educational opportunities is built on creativity tempered by knowledge and wisdom gain through the experience of learning. Investment in human capital, life long learning and quality education help in the development of society. Teachers are the most important factors for an innovative society because teachers' knowledge and skills not only enhance the quality and efficiency of education, but also improve the prerequisites of research and innovation. Many members of our society are not provided with a safe and secure environment in which children can develop, child abuse, violence against women and interpersonal violence cause a cancer on our society. Society play a key role in the realization of life long learning. The improvement of social education facilities such as libraries and the learning opportunities are implemented by the local governments. Students today are exposed to loads of technology and information at everywhere.

EDUCATION IN DEVELOPING COUNTRIES: EVIDENCE AND IMPLICATIONS

The world in which young people grow up today is very different from that of their parents or grandparents. This may be especially true in developing countries. Compared with the youth of past generations, young people today have more opportunities and challenges. They are likely to have more independence from their parents and spend more time in school. They are likely to have widespread access to the radio and television and, increasingly, to the Internet and mobile phones. They are also entering adolescence earlier and healthier, postponing marriage and childbearing until later, and are more likely to have sex before marriage. In response to these major societal changes, educators, researchers, policymakers and parents alike have become increasingly interested in the

potential for sexuality education to help meet the needs of young people. The quality and quantity of evaluation research in this field has improved dramatically over the last decade, and there is now clear evidence that sexuality education programmes can help young people to delay sexual activity and improve their contraceptive use when they begin to have sex. Moreover, studies to date provide an evidence base for programmes that go beyond just reducing the risks of sexual activity—namely, unintended pregnancy and sexually transmitted infections (STIs)—to instead address young people’s sexual health and well-being more holistically.

Yet, the fact remains that, too often, young people do not get even the most basic sexuality education and that misinformation about sex and its consequences remains common. Indeed, implementing comprehensive sexuality education programmes remains a challenge in many parts of the world. To address these challenges, experts say that stronger responses are needed to engage governments, communities, families and young people themselves in sexuality education policies and programmes. Specifically, they assert that sexuality education policies and programmes must be based in human rights and respond to the interests, needs and experiences of young people themselves.

A CHANGING WORLD

Developing countries are now confronting what industrialized countries have faced over the last century: the emergence of “adolescence” and the social changes around sexuality that came with it. In the past, young men and women tended to move directly from childhood into adult roles. In many ways, marriage marked the turning point, especially for young women. The transition into marriage and the assumption of other adult responsibilities, such as establishing one’s own household and having children, commonly took place soon thereafter.

Today, however, the transition from childhood to adulthood takes longer—and as a result, adolescence as a stage in life has gained in significance. Compared even with 20 years ago, young people are completing their education and beginning full-time employment later, and as schooling and work has become more a part of women’s lives, marriage and parenthood have tended to occur at older ages, especially for women.

Most young people today begin to have sex at about the same age as in the past: in their middle to late teens. By their 18th birthday, more than 40% of women in Latin America and the Caribbean report having had sex, as do close to 60% in Sub-Saharan Africa. (The age at which young women in the United States typically initiate sex is similar: By age 18, about 52% of U.S. women have had sex.) For the majority of young men, sex occurs prior to marriage; however, premarital sex has also become more common among females, at least in part because of delays in the age of marriage.

Parents and other family members, of course, have always played a critical role in the physical, emotional and sexual development of young people. At the same time,

there is increasing acceptance of the notion that, in today's world, these sources of education are insufficient and that more organized, formal approaches are called for. There is also a growing advocacy movement—including at the global level within the United Nations—for the recognition of comprehensive sexuality information and education as a basic human right. Still, sexuality education of any kind is not available in many regions of the world, adolescents' knowledge of sexual and reproductive health is not detailed, and myths are common. For example, many adolescents think that a young woman cannot get pregnant the first time she has sexual intercourse or if she has sex standing up. Some adolescents still report a belief that HIV can be transmitted through a mosquito bite or that a man who is HIV-positive can be cured by having sex with a virgin.

In regions and communities throughout the developing world, therefore, policymakers and youth-serving professionals are grappling with how best to address the wide-ranging needs of young people. They are weighing what is possible, considering both the political realities and context, and are taking a close look at the evidence for different approaches.

A LOOK AT THE EVIDENCE

The last few decades have seen a proliferation of curriculum-based interventions, both in and out of school. For a long time, these curricula emphasized the medical aspects of sex and reproduction, human anatomy and development. Today, although having a basic understanding of human biology and the reproductive system is still considered crucial, programmes have evolved to include a broader range of topics. Two basic approaches have emerged, each supported by different perspectives on what is best for children and young people. The abstinence-only approach focuses primarily, if not exclusively, on promoting abstinence outside of marriage, on moral as well as public health grounds. The comprehensive approach, on the other hand, supports young people's ability to decide whether and when to have sex, but also recognizes that sexual debut in adolescence is normative behaviour and thus seeks to prepare youth with the knowledge and skills they need for healthy sexual lives.

Abstinence-only Approaches

Over the last three decades, especially in the United States but also in parts of the developing world, much of the focus of sexuality education—at least among politicians, if not programme planners—has been on trying to convince young people to delay the initiation of sex, generally until after marriage. This approach is based on the premises that sex before marriage itself is a problem because it is morally wrong and that young people can be convinced to wait, even well into their 20s. These “abstinence-only-until-marriage” programmes focus primarily or exclusively on the putative benefits of abstaining from sex. They may also distort and actively denigrate the effectiveness of contraceptives and safer-sex behaviours.

This kind of education has become increasingly marginalized, as several well-designed studies conducted over the last 15 years have shown just how futile the focus on stopping young people from having sex is. For example, in 2007, investigators at the Centre for Evidence-Based Intervention at the University of Oxford conducted an international literature search for randomized or quasirandomized trials of abstinence-only programmes in high-income countries. (The researchers assumed that high-income settings may present optimal conditions for showing the effectiveness of abstinence-only programmes.) Despite its international focus, the search found only 13 studies that met the standards for inclusion—all conducted in the United States, and with a total sample of nearly 16,000 youth. The researchers concluded that programmes that exclusively encourage abstinence are ineffective, saying “when compared with a variety of control groups, the participants in these 13 abstinence-only programme trials did not report differences in risk behaviours or biological outcomes.”

These findings are similar to those of another comprehensive review of sex and HIV education programmes published in 2008. The analysis, conducted by Douglas Kirby, reviewed 56 studies with a strong experimental or quasiexperimental design, including eight studies of abstinence programmes based in the United States.

(Notable among these was an evaluation—conducted by Mathematica Policy Research at a cost of nearly \$8 million—of four model abstinence-only programmes that were carefully selected as having the most promise.) Study results indicate that abstinence-only programmes are not effective at stopping or even delaying sex. “Taken as a whole, this evidence certainly does not justify the widespread replication of abstinence sexuality education programmes,” says Kirby. “[N]o abstinence programmes evaluated with rigorous experimental designs show evidence that they delayed adolescents’ initiation of sex.”

Moreover, research suggests that strategies that promote abstinence while withholding information about contraceptives can actually place young people at increased risk of pregnancy and STIs. For example, young people who take “virginity pledges” are just as likely as those who do not to have sex, but they are less likely to use condoms or other forms of contraception when they become sexually active. These virginity pledges, which are a centerpiece of many abstinence-only programmes, originated in the United States in the early 1990s, but have since been implemented in developing countries as well. For example, leaders of True Love Waits, probably the best known of these programmes, report that nearly one million young people across Africa have signed their pledge.

More Comprehensive Approaches

Further promotion of abstinence-only approaches would not only run counter to the evidence, but also to the desires of many educators, parents and adolescents. But there is a diversity of opinions about the primary purpose of this education. Some adults,

parents and politicians—who may be conservative but pragmatic—believe that although sex among adolescents is troubling, it is inevitable and unavoidable, and society must accept this reality and concentrate on helping adolescents avoid the negative consequences of sex. Others go further and assert that the formation and testing of romantic attachments and the physical expression of sexual feelings are a natural and developmentally appropriate part of the transition to adulthood. Therefore, they say, young people should be approached with respect and equipped with the knowledge and skills they need to feel comfortable and confident about their sexuality.

These different motivations for comprehensive sexuality education have gradually made their way into policies and programmes, which vary between countries. For example, in the United States, the focus on keeping young people safe has been translated into prevention-oriented programmes. Although these programmes may cover a wide range of topics—from fertility and reproduction to STIs, from relationships and communication to gender norms, culture and society—they are primarily aimed at helping adolescents minimize their risk of adverse outcomes. Northern European countries such as Sweden and the Netherlands, by contrast, embrace a more positive attitude towards adolescent sexuality, based on the premise that young people are “rights-holders,” and therefore are entitled to information and education, as well as the right to express and enjoy their sexuality. These rights-based or “holistic” programmes are concerned, of course, with equipping young people to avoid unintended pregnancy and STIs, but they are focused less on behaviour and outcomes per se, and more on reflection and choice. The underlying assumption is that empowering young people to make considered, informed decisions about their own lives and helping them to develop the critical thinking skills and sense of self necessary to do so will result in better sexual and reproductive health in the broadest sense—including pleasure, love and sexual well-being.

It has long been recognized that those countries that have a more open and positive attitude towards sexuality have better sexual health outcomes. Cross-national comparisons show that, despite similar levels of sexual activity, adolescent pregnancy rates are consistently lower in many Western European countries than in other regions of the world. Experts say this is because, in Western Europe, sex among adolescents is generally accepted, with little to no societal pressure to remain abstinent. But with that acceptance comes strong cultural norms that emphasize that young people who are having sex should take actions to protect themselves and their partners from pregnancy and STIs. In keeping with this view, government-supported schools in many Western European countries provide—and even require—comprehensive sexuality education and offer easy access to reproductive health services.

In fact, the evidence for a positive impact on behaviour from evaluations of comprehensive sexuality education programmes throughout the world is strong. According to a rigorous 2008 review of the evidence of comprehensive sexuality

education's impact on sexual behaviour, effective programmes can not only reduce misinformation, but also increase young people's skills to make informed decisions about their health. Commissioned by the United Nations Educational, Scientific and Cultural Organization (UNESCO) as part of the development of the *International Technical Guidance on Sexuality Education*, the review included 87 studies from around the world with experimental or quasiexperimental designs: 29 from developing countries, 47 from the United States and 11 from other developed countries. Nearly all of the programmes increased knowledge, and two-thirds had a positive impact on behaviour: Many delayed sexual debut, reduced the frequency of sex and number of sexual partners, increased condom or contraceptive use, or reduced sexual risk-taking. More than one-quarter of programmes improved two or more of these behaviours. And most tended to lower risky sexual behaviour by, very roughly, one-fourth to one-third.

In addition, at least one study has demonstrated that comprehensive sexuality education programmes are potentially cost-effective as well. In 2010, UNESCO commissioned a study of the health impact and cost-effectiveness of school-based sexuality education in Estonia. Sexuality education in that country is included as a component of compulsory human studies courses for grades 5–7 and, importantly, is strongly linked to youth-friendly sexual health services in the community. According to the study, between 2001 and 2009, after the introduction of sexuality education in Estonia, there were significant improvements in adolescent sexual and reproductive health: Nearly 4,300 unintended pregnancies, 7,200 STIs and 2,000 HIV infections among adolescents aged 15–19 were averted. If even 4% of the reduction in HIV infections were attributed to sexuality education, the researchers estimate that the programme would result in a net savings.

Finally, no study of comprehensive programmes to date has found evidence that providing young people with sexual and reproductive health information and education results in increased sexual risk-taking. These studies also demonstrate that it is possible, within the same programmes, to delay sexual intercourse and to increase the use of condoms or other forms of contraception. According to UNESCO, “[A] dual emphasis on abstinence together with use of protection for those who are sexually active is not confusing to young people. Rather, it can be both realistic and effective.”

These findings can be extremely useful in gauging the impact of various comprehensive programmes on those sexual behaviours that directly affect pregnancy and sexual transmission of HIV and other STIs. But because they are so focused on behaviours, they provide little insight into how well these strategies work to achieve other desired outcomes—such as greater gender equality, critical thinking skills, a sense of confidence and belief in the future, and sexual pleasure. Measureable indicators of these kinds of outcomes have yet to be developed. This is understandable, as much of the focus of programmes to date has been on pregnancy and disease prevention. But rigorous evaluations of new approaches with a broader range of outcome measures are

needed to understand how programmes can be most effectively taught in different settings. Sexuality education—like other aspects of education—should expose future citizens to material that not only reduces their risk of unplanned pregnancy and disease, but also enlightens and empowers them.

GETTING FROM HERE TO THERE

Notwithstanding the wealth of evidence in support of comprehensive sexuality education, implementing programmes that actually provide young people the sexuality education they need is easier said than done. Each country and region is different, and there is wide variation in policies, government structures, traditions, resources and limitations. But even as implementers strive to be context-specific, certain themes and considerations emerge.

Meanwhile, concerted actions must be taken to build and sustain political will for these programmes.

In many communities worldwide, government officials, school principals, teachers and parents may not all be convinced of the need for sexuality education, or else may be reluctant to provide it because they do not want to be perceived as promoting sexual activity. Vocal groups that oppose sexuality education may be in the minority, but they can be extremely effective at pressuring policymakers and school districts to not allow sexuality education. One starting point for building and sustaining more widespread support for sexuality education is to demonstrate what the evidence has to say about the benefits of these programmes. Scientific evidence can keep public attention focused on the rights and needs of young people, help frame the public policy debate and help mobilize public support—all of which can make a difference in policies, programmes and practice.

THE FUTURE OF EDUCATION

While many people are making predictions about the direction that education systems are headed, we have found the best predictors to be hidden in the participative viral systems springing to life in the online world, such as iTunes and Amazon. These bottom-up approaches are quick to develop, participant-driven systems that are closely aligned to the demands of the marketplace.

In this paper we will focus on the key missing elements that will cause the disruptive next generation education systems to emerge. These missing pieces will likely be created within the next two years through private funding and will cause a dramatic educational shift in less than five years.

The primary missing pieces are a standard architecture for an organic courseware module and the software necessary to build this courseware. The solution to these missing pieces will be a participative courseware-builder that allows the general public to create

courses on any conceivable topic. We expect many companies will attempt to solve this problem, but the market will quickly gravitate towards the one it likes best.

Once the market begins to gravitate towards a favourite courseware-builder, a number of new systems will be developed to grow the courseware library, build integrity, make it universally distributed, archive results, and add functionality.

LESSONS FROM THE ANCIENT WORLD

During the time of the ancient Greek civilization, several mathematicians became famous for their work. People like Archimedes, Pythagoras, Euclid, Hipparchus, Posidonius and Ptolemy all brought new elements of thinking to society, furthering the field of math, building on the earlier work of Babylonian and Egyptian mathematicians.

A few generations later the Romans became the dominant society on earth, and the one aspect of Roman society that was remarkably absent was the lack of Roman mathematicians. Rest assured, the scholarly members of Roman society came from a good gene pool and they were every bit as gifted and talented as the Greeks. But Roman society was being held hostage by its own systems. One of the primary culprits for the lack of Roman mathematicians was their numbering system – Roman numerals and its lack of numeric positioning.

While it's easy for us today to look at Roman numerals and say that it was a pretty stupid numbering system, it was just one of many inferior numbering systems in ancient times. But the feature that made Roman numerals so bad was the fact that each number lacked specific numeric positioning and was in fact an equation, and this extra layer of complexity prevented people from doing higher math.

Roman numerals were a system problem, and a huge one at that. They prevented an entire civilization from furthering the field of math and science. Romans were so immersed in their numbering system that they had no clue that it was preventing them from doing even rudimentary math such as adding a column of numbers or simple multiplication or division, a feat still handled by abacus. It also prevented them from creating some of the more sophisticated banking and accounting systems and restricted academia from moving forward in areas of science, astronomy, and medicine.

Ratchet forward to today. We live in a society where virtually everything is different from the days of the Roman Empire. But what seems so counterintuitive to most is that we are even more dependent today on our systems than the Romans ever were. Most of these systems we take for granted – systems for weights and measurement, accounting, banking, procurement, traffic management, and food labeling. With each of these systems we are much like the Romans, immersed in the use of these systems to a point where we seldom step back and question the reasoning and logic behind them.

Our systems govern virtually every aspect of our lives. They determine how we live and where we live, what we eat and where we work, where and when we travel, how much money we will make, the job we do, the friends we have, who we marry, and even

how long we will live. But much like fish not understanding what water is, we seldom step back to fully understand the context of our existence.

As a starting point, one question we should be asking is, “What systems do we employ today that are the equivalent of Roman numerals, preventing us from doing great things?”

This simple question is very revealing. It has a way of opening a Pandora’s box full of friction points, inefficiencies, and flow restrictors that we contend with every day. Our systems are what control the flow of commerce, govern our effectiveness as members of society, and create much of the stress we face on a daily basis.

After studying American systems and applying this “equivalency to Roman numerals” test, it is easy to conclude that we, as a society, are operating at somewhere just between 5-10 percent efficiency, maybe less. The upside is huge. So what are some examples of restrictive systems that are preventing us from doing great things?

Here are just a few examples:

- *Income Tax System:* The income tax system is currently the mother of all boat anchors, slowing commerce and the pace of business to a crawl. Currently somewhere in the neighbourhood of 64, 000 pages in length, the United States tax code in use today will stand as a shining example throughout history as one of the world’s most incomprehensible systems.
- *Half-Implemented Metric System:* We are using a half-implemented metric system where we are purchasing cars with 3.2 litre engines and filling them with quarts of oil.
- *Keyboards:* We use keyboards that were designed to slow the speed of typing by placing the most frequently used keys randomly across the face of the keyboard. Keyboards in any configuration are an extremely inefficient way to transfer knowledge from one person to another.
- *Laws:* We now have more laws on the books in the United States than any country at any time in history. There aren’t even any good estimates as to the number of laws on the books in the United States. With each city, county, state, federal agency, and taxing district able to issue their own regulations, mandates, ordinances, rules, and law, we have created a legal snake pit of intertwined and overlapping rules that we are expected to live by.

Lest you think the United States is the only country with system problems, consider some of the major issues plaguing other countries:

- *Chinese Alphabet:* The number of Chinese characters contained in the Kangxi dictionary is approximately 47, 035, although a large number of these are rarely-used variants accumulated throughout history. Studies carried out in China have shown that full literacy requires knowledge of between three and four thousand characters.

So as you can see, we are a long way from optimizing the systems that govern our lives. The freedom that we value so highly in the United States is only a fraction of what it can be if we begin to seriously reinvent society one system at a time. And the system that we see as the highest leverage point for improving society is our education system.

EIGHT DRIVING FORCES

The following are eight key trends that are driving change in the world of education. These trends will eventually define the size, scope, and speed of the emerging new system along with the characteristics needed for a global-scale adoption.

As you read through the following trends, it is our hope that you will begin to feel the forces at play, gain a sense of the undercurrent of influencers, and begin to understand the dramatic changes that will be happening only a few short years ahead.

Transition from Teaching to Learning

Education has traditionally consisted of the two fundamental elements of teaching and learning, with a heavy emphasis on teaching. Throughout history, the transfer of information from the teacher to the learner has been done on a person-to-person basis. A teacher stands in front of a room and imparts the information for a student to learn. Because this approach requires the teacher to be an expert on every topic that they teach, this is referred to as the “sage on stage” form of education.

While lecture-style teaching has been used for centuries to build today’s literate and competent society, it ends up being a highly inefficient system, in many respects, the “equivalent of using Roman numerals.” For any new topic to be taught, a new expert needs to be created, and this universal need for more and more experts has become a serious chokepoint for learning. To illustrate this point, let’s look at the example of a new topic that cannot be taught because the expert on this topic lives on the other side of the world. A teacher-dependent education system is also time-dependent, location-dependent, and situation-dependent. The teachers act as a control valve, turning on or off the flow of information. The education system of the future will undergo a transition from a heavy emphasis on teaching to a heavy emphasis on learning. Experts will create the courseware and the students will learn anytime or anywhere at a pace that is comfortable for them, learning about topics that they are interested in. In the future, teachers will transition from topic experts to a role in which they act more as guides and coaches.

Exponential Growth of Information

During the time of Gutenberg, people tended to live and die within 20 miles of where they were born, not because they were afraid to travel, but because they had no reliable maps. People during this era had a very limited understanding of the world

around them. The flow of information was controlled by just a few elite members of society, and they understood well the concept of knowledge equaling power. We have gone from that time, just 500 years ago, where information was precious and few, to today, a time where information is so plentiful that we feel like we are drowning in it – information overload.

However, we still see many of the same “information control” issues permeating society today. Elite members of society still control the flow of information, perpetuating the notion that only doctors can understand medicine, only physicists can understand how the universe works, and only teachers know how to prepare us for the world to come.

There are many ways to talk about the rapid growth of information that we have experienced over the past few years. But it is important to pay attention to the changing dimensions of information as well as the sheer volume of it. Information is no longer just text-based, but graphical, musical, audio and visual.

Consider the following statistics:

- The number of songs available on iTunes – over 3.5 million.
- The number of books on Amazon – over 4 million.
- The number of blogs available online – over 60 million.
- The number of entries on Wikipedia – over 4 million.
- The number of user accounts on MySpace – over 100 million
- The number of videos on YouTube – over 6.1 million.

Courseware Vacuum

After viewing the data above and thinking about the size and shape of information around the world, now consider the number of courses available, either online or in a classroom. Information is exploding around us in every possible form. Yet, we do not have an easy way to translate these blocks of information into courseware. While some attempts are currently being made to unleash the public on this problem, we remain a long ways from solving the problem.

Open Education Movement – The open-education movement was inspired by the open-source software movement (*i.e.*, Linux). It mixes in the powerful communication abilities of the Internet and applies the result to teaching and learning materials, such as course notes and textbooks. Open educational materials include text, images, audio, video, interactive simulations, and games that are free to be used and also re-used in new ways by anyone around the world.

It is estimated that more than 150 well-intentioned initiatives have been launched in this area. Over time, the increasing levels of attention and activity will cause one initiative to stand out and become the “industry leader.” This leader will, by default, set the standards for everyone that follows. Some open-education projects are already attracting a large number of users per month. Some, like the MIT OpenCourseWare project and

its OCW Consortium, are top-down organized institutional repositories that showcase their institutions' courses. Others, like Rice University's Connexions, Wikiversity, and Moodle are grassroot efforts that encourage contributions from all comers.

- MIT Open Course Ware makes the course materials that are used in the teaching of almost all MIT's 1, 400 undergraduate and graduate subjects available on the Web, free of charge, to any user anywhere in the world. MIT now claims 1.4 million visits per month from learners "in every single country on the planet."
- The Open Course Ware Consortium is an extension of what MIT began. Students don't have to register for classes but need only to log on to more than 1, 800 potential courses at 12 universities that provide the course materials such as syllabi, video or audio lectures, notes, homework assignments, and illustrations.
- Connexions claims more than one million people from 194 countries are tapping into its 3, 768 modules and 199 courses developed by a worldwide community of authors.
- Wikiversity is a division of Wikipedia serving as a community for the creation and use of free learning materials and activities. Wikiversity is a multidimensional social organization dedicated to learning, teaching, research and service. Its primary goals are to create and host free content, multimedia learning materials, resources, and curricula for all age groups in all languages.
- Moodle is a course management system using a free, Open Source software package designed to help educators create effective online learning communities. Moodle claims over 20, 000 participating sites listing over 820, 000 courses.
- Curriki.org is an education development resource with over 3, 000 members and 450 courses in development.

While we applaud these efforts, there are some critical elements missing. The learning system of the future will have a single access point for all of its courses. Moodle is claiming over 820, 000 courses but they are spread over 20, 000 sites and many courses are duplicates. We estimate the number of unique and different courses to be less than 50, 000, not in the millions like the number of available books and songs.

Using books as a close analogy, it can be argued that every available book has the potential of being translated into courseware and, most often, multiple courses. There are currently far more topics discussed in books than there are courses to teach the material. This leaves an obvious courseware vacuum waiting to be filled, and the key to unlocking this vacuum is the participative courseware-builder described below.

Expanding Gulf Between Literates and Super-Literates

According to the New York Times, there are an additional 20, 000 new words added to the English language every year. The primary driver behind this ever-expanding dimension of vocabulary is the ongoing development of science and technology. Along with the creation of new science and technology comes the need to explain its attributes,

its function in technical terms, and its overall purpose. New words and their associated colloquialisms help create meaning and structure around the emerging new concepts as they attract more research and come into focus.

Young students can learn new words quickly: on average, 3, 000 new words each year, which works out to 8 words a day. This, of course varies significantly from one student to the next. In the English language, the 2, 000 most frequently used words account for 80-85 percent of the words used in non-specialized written texts and about 90-95 percent in conversational speech.

However, the total number of words in the English language tops out around one million words, and the vocabulary of some of our most gifted scientist and engineers tops out around 200, 000 words. The distance between the functionally literate and the super literate is growing. Some people who have become expert on a specific topic have pushed the envelope of understanding far beyond the comprehension of the rest of the world. And in doing so, have created whole new vocabularies to describe the concepts and phenomenon they encountered. These super experts often live in a research community where they are often the only living person who truly understands the topic of their research.

Until now the primary tool for these super literates to pass along their understanding of research to future generations has been through papers that are published in technical journals. Because of the rigid requirements for publication, these papers often take months to compose, and are written in a vocabulary few can comprehend.

An alternative to publishing papers will soon be the creation of courseware. While developing courseware in the past has been laborious and poorly utilized, the new courseware builder described below has the potential to change all that. Courseware will become an alternative to publishing papers or writing books, and will serve as an additional channel for the super literates to disseminate their understanding of the world.

Our “Touch Points” for Interfacing with Society are Changing

“Touch points” are the places where we come in contact with the rest of the world. As an example, the average person comes in contact with the physical world through three primary physical touch points or interfaces – the shoes that we walk in, the bed that we sleep in, and the chairs that we sit in. These are the primary touch points for our physical body. While it is important to study the touch points for our physical body, it is even more important to understand the touch points for our mind. How does our mind interface with the rest of the world, and how can we improve the touch points to improve our abilities and capabilities?

The Classroom Touch Point: There has long been the pervasive notion that learning can take place only in a classroom. Even though schools use field trips and outdoor experiences to enhance education, the classroom remains the dominant central fixture of today’s educational systems.

Classrooms are designed to focus attention, close off the rest of the world, and create a controllable environment where learning can take place. Architects refer to schools as a “place, ” and over the years place-makers have attempted to create the ultimate classroom – a place where learning can be optimized and students can excel.

Most educators will argue that the real learning takes place inside the classroom. Even though external activities such as doing homework, reading assignments, or writing papers happen outside the bounds of the school, the primary education interface remains the classroom. Using classrooms as the primary “touch point” for learning creates many problems. The person or education system that controls the classroom also controls the time when learning can take place, the students who will participate, the lighting, the sounds, the media used, the tools, the pace, the subject matter, and in many cases, the results.

However, classroom-centric education is not necessary for learning.

Learning takes place from the moment a person wakes up in the morning until they fall asleep at night. In fact, learning continues even while a person is sleeping. We may not be learning about math and science while we watch a movie, but we learn about the characters in the movie, the plot, the setting, the drama, the resolution of the problem, the kind of popcorn a theater serves, and how comfortable the seats are.

Indeed some topics like math and science require a more structured form of learning for most students to grasp the information being imparted, but learning is not dependent upon the classroom. In some cases the classroom may be the optimal environment for learning to take place, but most often it is not. Important new touch points for our mind include our computers, electronic newspapers, video magazines, handheld televisions, cellphones, MP3 players, video games, artwork, and much more.

Learning Drivers

Why do people need to learn? Why do people want to learn? What are their motivations? What are the drivers that control a person’s desire to fill their minds with knowledge and information? Maslow’s Hierarchy of Human Needs is a theory in psychology that Abraham Maslow proposed in his 1943 paper A Theory of Human Motivation. His theory contends that as humans meet ‘basic needs’, they seek to satisfy successively ‘higher needs’ that occupy a set hierarchy. Maslow’s initial hierarchy was based on two groupings: deficiency needs and growth needs. Within the deficiency needs, each lower need must be met before moving to the next higher level. Once each of these needs has been satisfied, if at some future time a deficiency is detected, the individual will act to remove the deficiency. The first four levels were:

- Physiological: hunger, thirst, bodily comforts, etc.;
- Safety/security: out of danger;
- Belonginess and Love: affiliate with others, be accepted; and
- Esteem: to achieve, be competent, gain approval and recognition.

According to Maslow, an individual is ready to act upon the growth needs if and only if the deficiency needs are met. Maslow's early thinking included only one growth need – self-actualization.

Self-actualized people were characterized by:

- Being problem-focused;
- Incorporating an ongoing freshness of appreciation of life;
- A concern about personal growth; and
- The ability to have peak experiences. Maslow later add a new dimension to the growth need of self-actualization, defining two lower-level growth needs below self-actualization and one above that level. They were:
 - Cognitive: to know, to understand, and explore;
 - Aesthetic: patterns, symmetry, order, and beauty;
 - Self-actualization: to find self-fulfillment and realize one's potential; and
 - Self-transcendence: to connect to something beyond the ego or to help others find self-fulfillment and realize their potential.

Our motivations for learning form similar patterns. Maslow's basic concept is that the higher needs in the hierarchy come into focus only once all the needs that are lower down in the pyramid are mainly or entirely satisfied.

For this reason, our desires to learn, and the topics we want to learn about, transition depending on the situation we find ourselves in. As an example, we will have very little desire to learn math and science if we are worried about survival. However, we will have a great desire to learn about survival topics.

The problem sets that surround us, and our ability to solve those problems, are a constantly refocusing lens into our learning motivators.

Maslow's basic position is that as people become more self-actualized and self-transcendent, they develop wisdom and automatically know what to do in a wide variety of situations. His ultimate conclusion that the highest levels of self-actualization are transcendent in their nature may be one of his most important contributions in this area of study.

THE AGE OF HYPER-INDIVIDUALITY

As a society we are less and less interested in the status competition involved in "keeping up with the Joneses." We are not all that concerned about what kind of car our neighbour drives, what kind of TV they are watching, or what kind of cell phone they are using. Instead, we are much more concerned about finding products that will satisfy our own particular needs.

We live in an era where we are approaching 100 million products in the marketplace, and depending on how you define a product, some would argue that we have already far exceeded that number. Suffice it to say that we now have products that are much more aligned with the needs of a very wide range of consumers, and consumers are voting

with their debit cards for uniqueness and individuality. So much so that we have dubbed this the age of hyper-individuality.

When the cable TV companies started offering 500 different channels they found that all of the channels had an audience. When Amazon started offering over 2 million different books for sale on its web site it found that all of the books had a market. This phenomenon is best described in Chris Anderson's book "The Long Tail" which explains how the Internet has driven the cost of shelf space in the online world to a number approaching zero, and in doing so has enabled online merchants like iTunes, Amazon, Buy.com, and YouTube to carry millions of different products.

Our need for hyper-individualized solutions is driven by several factors including our time, our personality, and an overwhelming need to feel special in a world of over 6 billion other people wanting many of the same things.

Today the average person sleeps two hours less than a person in the 1920s. We have gone from 8.9 hours per night to 6.9 hours per night, and many people today, if they could do without, would skip sleeping altogether.

With time being one of our major constraints, we are continually searching for products that will save us time, and if we can find that left-handed, counter-balanced, pocket-sized device that we can operate efficiently on moon-lit nights when the stars are aligned, we will make the purchase.

TRANSITION FROM CONSUMERS TO PRODUCERS

As we transition from a predominantly passive society to a more active one, people no longer want to just sit on the sidelines and watch. They want to participate. And a whole new generation of tools and equipment are allowing people to shift their role from consumer to producer. This transition began with the introduction of comment sections at the end of online news posts. People began to voice their thoughts on whether or not a piece of news was accurate, timely, or in any way news-worthy. Many commenters added additional information.

When Evan Williams and Meg Hourihan's company, Pyra Labs, launched Blogger (later purchased by Google) in 1999, a major shift began in the world of user-generated content for the Web. Suddenly it became easy for anyone to create a blog site, and millions of people began to experiment. In July of 2003, MySpace was founded by Tom Anderson, Chris DeWolfe, and a small team of programmers. As a site that allowed users to generate their own web site and connection to friends, MySpace quickly became the dominant player in the emerging category of social networking with the 100 millionth user account created in August 2006.

Similarly, when Chad Hurley, Steve Chen, and Jawed Karim launched YouTube in February of 2005, it became very easy for people to produce and post videos online. As an enormously popular and free video sharing web site, YouTube lets users upload, view, share, and rate video clips. As a result, millions of people have transitioned from

video consumers to video producers with an average of 65,000 new video clips uploaded onto YouTube every day. While each of these are examples of runaway success stories, the world of user-generated content is not without its own set of problems. Each has managed to handle the challenges in their own unique way. But what these examples best illustrate is the public's driving need to participate and lend their own thoughts and ideas to the world around them.

Setting the Stage

It is easy to fall into the trap of thinking in terms of past top-down approaches. Instead, we need to focus on the key elements, the seeds of innovation, which will allow this new organic form of education to spring to life. New forms of education are not achieved by putting an umbrella over our existing education systems and networking them with hopes that they will get better. And they're not achieved by simply recording the lectures for later broadcast. Education in its current state is the equivalent of Roman numerals, a system that is preventing us from achieving great things.

Unlimited Shelf Space

Since most people still believe that education must take place in the classroom, and only educators can create new courses, we have placed a very constrictive valve on the inflow of new courseware. The retail world had very similar restrictions just a few years ago, with the cost of shelf space being one of the primary constraints to the introduction of new products. But the online marketplace has given us unlimited shelf space at a near-zero cost. This combination of super cheap and unlimited shelf space has caused an exponential growth curve in the introduction of new products.

The notion that education can take place only in a classroom is similar to the notion that purchasing a product can only take place when you see it on a store shelf. Removing the classroom constraints to learning is similar to removing the shelf space constraints to the marketplace.

At the same time that we have been experiencing the exponential growth of information, the amount of available courseware has remained rather static and consequently, our education systems have not kept pace.

Only a small percentage of the information being developed today is being passed on to future generations in the form of classes or courseware. This growing gap between information and courseware is what we call a "courseware vacuum," where supply has clearly not kept up with demand.

Certifying Accuracy – Truth vs. Untruth

As we contemplated the education system of the future, one of our biggest concerns was finding a way to ensure the accuracy of the information presented. Initially our thinking centered around the idea of selecting a central authority, some sort of governing

truth authority, to authenticate the accuracy of information in each of the courseware modules. But this approach became unworkable as we considered the implications.

To begin with, a high percentage of what is taught in classes today is theoretical, ranging from theories of gravity, to theories of evolution, to music theory. None of these topics end up being 100 percent provable, and so from the standpoint of passing muster with a governing truth authority, none of these topics could be included.

Further, we realized that virtually every aspect of society has its own version of truth – religious truths, scientific truths, legal truths, etc.

For this reason we concluded that any governing truth authority would quickly deteriorate into a highly politicized authority, and the politicalization of any aspect of this future learning system will quickly compromise its usefulness.

As an alternative, we are proposing a checks-and-balances system where individual groups can create their own central truth authority and place their tags of approval or disapproval on courses. These tags will be a central feature of the search criteria used by the smart profiler and the recommendation engine.

For example, organizations like the American Chemical Society, People for the Ethical Treatment of Animals, Focus on the Family, American Civil Liberties Union, National Rifle Association, National Education Association, or the Catholic Church can all review the new courseware that is being introduced and make a determination as to whether or not it meets their criteria.

We think that such organizations will quickly gear up to develop their own line of courseware so they can have better control of the content.

This type of tagging system holds value on many levels. First, it creates ways for virtually everyone to participate, and in fact, demands participation. Participation is an essential ingredient in a truly pervasive education system. It allows the learning system to develop organically without any central gatekeepers telling people what they can and can't learn.

Confidence-Based Learning

While many different learning methodologies will be experimented with, one that holds considerable promise is confidence-based learning. Some experiments in this area have demonstrated a significant reduction in learning time.

The first Confidence-Based Assessment study appeared in the *Journal of Social Psychology* in 1932 by Kate Hevner. Her goal was to improve the validity and reliability of standard musical assessments at the time. She did so by adding confidence assessment to knowledge. From 1932-1967, Confidence-Based Assessment focused on statistical validity and reliability. In 1967 it was discovered that Confidence-Based Assessment also improved memory retention. This is the discovery where people began to realize that the process of taking a test can and will make you smarter. What started as a breakthrough approach for measuring knowledge and confidence is now moving front-

and-center into corporate training centers in the form of a fast and accurate learning methodology. Confidence-based learning is on the rise among organizations that are transitioning their companies from training organizations to learning organizations.

Confidence-based learning is designed to ensure that learning actually takes place and mastery of a topic is achieved. It is much more than simply delivering information to students. It ensures learning by assessing precisely what people know and what they don't know without guesswork and doubt skewing the results. It then works to rapidly remediate a learner's gaps in knowledge and confidence. Here's a summary of how it works. Determining what people need to learn starts by understanding what ignorance, doubts and misinformation presently exists.

Unlike traditional learning methodologies that measure only how many questions someone answers correctly, confidence-based learning assesses:

- Correct answers that are answered with confidence, indicating competency;
- Correct answers that are answered with doubt;
- Correct answers that are total guesses, equivalent to no knowledge;
- Incorrect answers that are answered with confidence, indicating misinformation.

Distinguishing between a person guessing correctly and one who answers correctly with confidence can have a major impact. An assessment process that is built into the system can capture and validate knowledge confidence because of the unique structure of its multiple-choice questions and detailed analysis.

With the completion of an assessment, confidence-based learning is designed to close knowledge gaps at the moment users are most inclined to learn – right after being evaluated and their own misinformation and doubt has been exposed.

We have speculated that if confidence-based learning is somehow designed into the courseware-builder that the speed of learning can be increased dramatically, by as much as 50% or more.

ARCHIVING KNOWLEDGE

While rarely viewed as such, education is a system for archiving a culture by passing down the knowledge of one generation to the next. Museums, written documents, books, photos, videos and audio recordings typically come to mind when considering a cultural archive. But certain aspects of a culture need to be experienced in order to be preserved, and that's where the notion of education as an archival medium comes in. Craftsman guilds such as tilers and bricklayers are a good example of trades that require hands-on experience. The intricacies and nuances of piecing together building materials into artistic patterns cannot be adequately conveyed through books or even video. The tactile feel of textures, tapping, testing for hollow spots, and cleaning off excess material are all part of the experiential learning that cannot be conveyed through some other medium.

The concept of archiving knowledge is just one of many theories that will emerge as strategies around the new system begin to develop.

DEFINING THE NEW, NEW ERA IN LEARNING

As we begin to peel back the layers of the system we are envisioning, we will use a number of different descriptors and definitions to describe the nature of the new, new era of learning. The key to this whole system is an easy-to-use courseware builder that catches the imagination of the general public and inspires participation. Several features will be necessary to give this system both the range and functionality of a truly rich learning environment.

TWELVE DIMENSIONS OF THE FUTURE COURSEWARE ARCHITECTURE

We have identified 12 critical dimensions of the future learning system. However, only the first two need to be in place for the revolution to begin.

The two critical components that will define education for centuries to come will be a standardized architecture for developing a courseware unit, and an organic distribution system that allows anyone around the world access to it.

STANDARD COURSEWARE UNIT

In the past, creating a standard was often a long and tedious process where smart people gathered around tables and argued about where to place the comma in a sentence. But times have changed, and so has the development process for standards. Very likely the development of a standard will happen concurrently with the early testing of courseware modules, with the drafting of the standard document happening in parallel to the testing of the architecture. The final standard may not be finished for several years.

The reason it's important to create a standard is simply to focus energy. Competing standards can be divisive, creating temporary chaos in the industry, and greatly delaying market acceptance. Also, the path to market acceptance would end up being far more expensive. So, what will a standard courseware unit look like? The short answer is that it is too early to know, but we have to have a starting point. For this reason we have put together a list of attributes and features that we think will be necessary components:

- *The Courseware Builder*: Envisioned as a smooth, fill-in-the-blanks templated process, the courseware builder will carefully step courseware producers through the design, build, and launch phases of each course.
- *60 Minute (Approx) Learning Experience*: An hour is an international unit of measurement recognized around the globe. We schedule our time in one-hour units, we plan our days in one-hour units, so building educational modules around one hour units makes sense. Some learning experiences may involve a grouping of 2, 5, or even 10 units, but the majority will be centered around the basic one-hour unit.
- *Modality Agnostic, Language Agnostic*: Learning comes in many forms ranging from reading text, to listening to audio, to watching video, to hands-on

experiences, and more. The Standard Courseware Unit needs to accommodate all modes of sensory input and learning experiences.

- *Confidence-Based Learning*: All units will use some form of testing to validate competency or fluency in the topic, as well as the student's confidence in their answers. Test scores that are lower than minimum required proficiency levels will force students to repeat portions of the learning experience until students have achieved mastery.
- *Smart Profiler*: In addition to the basic name and address type of information found in most profiles, students will be asked to participate in regularly scheduled assessment surveys to determine primary and secondary areas of interest. The smart profiler will continually expand the profile of the student throughout their life, recalibrating topical interest levels, building a comprehensive understanding of the individual student as they evolve over time. The Smart Profiler will feed information directly to the Personal Recommendation Engine for prioritizing course selections.
- *Multi-Dimensional Tagging Engine*: Much of the system usability will be driven by the multi-dimensional nature of the tagging engines. Attributes include:
 - *Personal rating tags*: Upon completion, each student will be asked to rate the courseware. Courses will be graded on accuracy, quality of the learning experience, ease of use, and overall effectiveness.
 - *Approval and Disapproval Tags*: Each courseware unit will be set up to allow various organizations to either put their stamp of approval or disapproval on it. Since every political interest group will want to influence the direction of courseware development, it will be impossible to build courses around the likes and dislikes of all of the various interest groups. For this reason, each course will allow groups to place their stamp of approval or disapproval on it.
 - *Taxonomy Tags–Folksonomy*: Folksonomy is the emerging science of user generated taxonomies. Since each courseware topic will be understood differently by each student, it will be necessary to allow students to place descriptor tags on all completed courses. This self-tagging approach will create the necessary taxonomy for the Personal Recommendation Engine as well as other search engines.
 - *Prerequisite and Post-Requisite Tags*: Knowledge builds on knowledge. As an example, students cannot study literature until they know how to read, and they cannot study computer programming until they know math and algebra. Much of today's learning needs to happen sequentially. Therefore it becomes imperative to create some system for sequencing courses based on the order of which learning must take place.

- *Comment Tags*: Comment sections will allow students to voice their thoughts on each course
- *Personal Recommendation Engine*: Each time the student completes a course, the Personal Recommendation Engine will present a number of possible future courses based on personal interests and past courses. This engine should offer an expanded view of possible directions the student can take, listing a variety of learning options as well as the certification endpoints.
- *Certification Inputs*: Every profession, personal skill, or area of learning has logical points where experts in that field would consider the necessary learning to be sufficiently complete. But every profession or skill is different. As an example, the courses necessary to become a master cigar maker are vastly different than the courses need to become a C+ programmer or airline pilot or registered nurse. Most professions and skills will use a combination of courses completed and a certification exam to validate student proficiency.
- *New Achievement Standards*: Descriptors like grade-level, graduate-level, and undergraduate-level will begin to disappear from our vocabulary. Initially, a set of equivalency units will be used to describe achievements (equivalent to sixth grade, or equivalent to a BA degree). Over time, the systems for illustrating achievement will change with the use of charts and graphs to explain the breadth and depth of a person's understanding.
- *Official Record Keeping System*: Building a system with impeccable integrity means that the system for archiving the accomplishments of every participant must be secure, private, and managed by an organization with impeccable credentials. While many people will think that a government-run archive is the best solution, the best possible record-keeping system will be one that transcends governmental boundary lines.
- *Participative Wealth Pricing*: The revenue stream generated by each courseware unit will be divided between the courseware producer, distribution company, transaction company, system operations company, and the official record keeping system. Maybe more. Courseware prices need to be kept low to make courseware accessible to anyone interested in learning.
- *Global Distribution System*: Think of the nature and functionality of iTunes with the following features:
 - A single online access point
 - Content aggregator
 - Search/sort capabilities
 - Accepting user generated content
 - Recommendation engine
 - Uniform pricing.

UNDERSTANDING THE SYSTEM ENTRY-POINT

Before a child can tap into a system with courseware as described above, a certain number of skills must first be in place – ability to read, follow directions, and respond to questions. However, once basic motor skills have been mastered, it is conceivable that very young children can begin this type of learning with the aid of some future design of the early childhood workstation. Since age is not a good way to determine a child's capabilities, new systems will need to be created to assess the overall readiness of a young student to participate.

Priming the Pump

Creating the initial Standard Courseware Units will be time-intensive, poorly understood, and topically spotty. People creating the first units will most likely be educators working on a tiny budget. Budget limitations will cause courseware development to happen slowly, and this will leave gaps in many topical areas. Gaps will include areas such as craftsman trades that use more of an apprentice-style approach to learning, personal interest topics such as gardening and home repair, and social system education on topics like water law and import-export laws.

One way to “prime the pump” will be through a system of grants. Perhaps the creation of the first 1, 000-3, 000 Standard Courseware Units could be funded through a system offering \$5, 000-\$10, 000 individual development grants. Once the initial courseware is developed and tested, smaller amounts of grant money could be made available in the range of \$2, 000-\$5, 000 depending on whether it's a new topic or degrees of improvement over past courseware. For example, a Courseware Unit using a combination of video animation, expert interviews, and advance mathematical modeling will naturally provide a superior, content-rich learning experience over a standard text-based course and could therefore be eligible for a larger grant. With this approach there should be a sufficient critical mass of courseware to inspire other people to begin creating their own courses without grants.

The After Effects

We are on the verge of radical shifts in our education systems, and not everyone will be happy to see them develop. Teacher unions and other people dependent upon the existing education system will provide much of the early resistance. Because of the many facets of the architecture outlined above, the system will not be created all at once. It will be phased in, starting with the courseware builder and distribution system, and later followed by the official record keeping system and various groups providing inputs. Adoption will be spotty at first. We see home schoolers and foreign students as being some of the earliest adopters, followed by private schools and charter schools, and later public schools. Initially these courses will be used to supplement traditional classroom-based courses, but will later develop into a complete learning curriculum.

Learning Camps

Many kinds of learning camps are already in existence, but we will go through an explosive growth in this area. Many kinds of learning are best achieved through hands-on touch and feeling experiences. Marine biology is best learned through working with marine life in all its many forms. The best way to learn history is to travel to the battlefields, take tours of the castles, walk through the ancient ruins, dress up in the ancient clothing, and sleep overnight in a wigwam or cliff dwelling. The best way to become a plumber is to work with a skilled plumber and perform hands-on work-related tasks to fix real world plumbing problems. Learning camps, ranging from one-day camps to multi-week camps, will begin to proliferate around specific topics. Some camps will be more academic-related areas of study such as math and science, while others will deal with more skill-related topics like woodworking or auto repair. Each camp will have its own identity, use its own in-house experts, and will focus on a specific learning experience that is tied to courseware with a built-in testing system to validate competency.

THE SOCIAL ENVIRONMENTS OF LEARNING

Much of today's learning happens inside a social context. When a classmate asks a question, the whole class learns. When one student laughs at a teacher's joke, all the rest of the students perk up. These are pieces of a learning environment that may disappear if the learning process becomes too hyper-individualized and too much of a solemn, one-person experience. Technology has a way of isolating people. As an example, many young girls today grab their cell phone and start talking as soon as the final classroom bell rings. This makes them unapproachable to boys who would like to find a good time to strike up a conversation with them. Sitting at a classroom computer or watching a video are other forms of technology isolation created by placing social barriers around individuals. However, lasting relationships are based on common, shared experiences, and any learning system that does not address the need for building social relationships will be missing a critical dimension of the learning environment. Learning camps are only a partial answer. Courseware designed around strengthening relationships may address this problem, but the power of socialization and the need for building relationships cannot be underestimated.

THINKING THROUGH THE TRANSITION FOR EXISTING SCHOOL SYSTEMS

We see the existing school systems going through a complicated transition which may not always be smooth. Below is a description of some of the anticipated changes that will happen to students, teachers, buildings, and school districts or systems. Students – Perhaps the people who will be quickest to adapt to the new system will be the students. Instead of being forced to learn specific courses that are often of little interest to them, students will be free to select the topics that they are most interested in.

Most students will have the opportunity to travel to various learning camps around the country. As more and more students begin using the system, the demand for new courses will cause more and more people to develop courseware.

School Buildings – Some school buildings will transition into learning centers that are open 24 hours a day, accommodating both child and adult learners, providing support staff to assist people who struggle with the system or on a specific topic.

Other school buildings, or portions of buildings, will transition into production centers filled with the tools and equipment for people to produce new courseware. Staff people will also be on hand to assist in courseware design and creation.

Teachers – Teachers will have many good options to consider as the changes begin to happen. Some teachers will remain with the school buildings and work more as guides, coaches, and tutors for students needing help. Others will move into event planners and experience designers as each facility experiments with re-engineering the social side of learning.

Other teachers will choose to develop their own learning camp or series of camps. Learning camps will specialize in a specific experiential topic that is tied to specific courseware. These teachers will effectively operate their own enterprise with revenues driven by the number of students opting to go to their camp.

Some of the more entrepreneurial-minded teachers may choose to become full-time courseware producers. The techniques for creating good and effective new courseware will be an iterative process going through multiple evolutionary stages as new and better tools become available.

CHANGING REVENUE STREAMS

So how does all this get funded, and how do the existing revenue streams change to accommodate the new era in learning? Because of the sheer volume of students that will be involved, it is recommended that the price point for courseware modules be kept very low – as low as 99 cents per module. Some of the more specialized areas of learning will likely charge a higher amount, but general courseware should be pushed to the lowest possible price point.

Initially people will pay for their own courses, or the courses completed by their children. Later, systems for grants and loans will allow a broader range of students to participate. Eventually government money will begin to shift and cover student expenses.

Because the learning camps will demand a more aggressive involvement of people and technology, the pricing of camps will be done on a camp-by-camp basis according to market demand or according to some district or state pricing schedule, possibly with matching funds available to cover costs.

Speculating on government-sponsored funding mechanisms, students could each be granted an account with sufficient credits to complete the equivalent of a college bachelor's degree, roughly on the order of 20,000 credits. Once those credits have been

exhausted, the students would automatically enter into a loan arrangement where any additionally used credits would have to be paid back over time. A second dimension of funding may be an annual amount of credits that could be spent on learning camps. Since prices of the camps would vary, student would be allowed to roll over funding from one year to the next. Existing streams of funding for school district will not go away, but will be scaled to appropriate levels for staffing and maintaining buildings as each phase of the transition takes place.

Education and the Practice Environment

INTRODUCTION

According to an agreement between the International Association of Schools of Social Work and the International Federation of Social Workers, the social work profession “promotes social change, problem solving in human relationships and the empowerment and liberation of people to enhance well-being. Utilizing theories of human behaviour and social systems, social work intervenes at the points where people interact with their environments. Principles of human rights and social justice are fundamental to social work.” As such, social workers are a key component of modern social service provision. Among other things, they provide counseling, assist individuals in accessing social services and other benefits, lobby for the disenfranchised, and engage in actions designed to influence social policies.

During the Soviet era, social problems were either unrecognized or minimized in the Europe and Eurasia (E&E) region, which limited the availability of social and other support services. Former Communist regimes in the region relied heavily on financial allocations, many of them universal allowances, to solve social problems while services were severely underdeveloped. Social issues, including child abandonment, disabilities, domestic violence, or unemployment were hidden in homes or large institutions, which provided medical care and accommodation. Social workers (as well as many other types of service providers familiar to those who live in the West) did not exist. Thus, in attempting to create modern systems of community-based social services, E&E countries

have had to build the profession of social work from the ground up in many cases. While some progress is being made in this regard across the region, it is uneven, with some countries being further along than others. Building the social work profession is a complex undertaking that involves creating education programmes, developing and strengthening curricula, helping to nurture professional associations for social workers, developing licensing and practice standards, and raising awareness about the need for social workers, among other things. Moreover, building the profession must be closely coordinated with other programme activities aimed at creating and strengthening social services.

Even in those E&E countries where progress has been the strongest, many obstacles remain. Government officials and the general population often do not understand what social workers do and how their work differs from other professionals (*e.g.*, teachers or social pedagogues).

In some countries, schools of social work are now producing cohorts of trained social workers, but these individuals are struggling to secure employment because those who provide social services may not understand the role of social workers or may not have the funds to pay them. Social workers in the region earn notoriously low wages and are often forced to seek higher paying employment in other fields.

BACKGROUND AND RATIONALE OF THE STUDY

Social work as a profession is relatively new to the E&E region, yet a well-trained social work workforce is a key component in the creation of an effective system of social services that can provide a continuum of community care to vulnerable individuals and groups.

USAID Missions with an interest in social services programming (related to child welfare, disabilities, trafficking in persons (TIP) and domestic violence, social assistance social benefits and cash transfers, and related issues), have identified an active and viable social work profession as critical in such programming.

As USAID Missions tackle programming related to a wide array of social services and social assistance programmes, there is a strong need for examples of best practices and creative thinking about how to facilitate recruitment and retention, training, and advocating for social workers in host countries.

The target countries for this study are: Albania, Armenia, Azerbaijan, Belarus, Bosnia, Bulgaria, Croatia, Georgia, Kazakhstan, Kosovo, Kyrgyzstan, Macedonia, Moldova, Romania, Russia, Serbia, Montenegro, Tajikistan, Turkmenistan, Ukraine, and Uzbekistan.

Purpose of Study: The purpose of this study is to collect, organize and report on information about model programmes, best practices, and lessons learned in addressing key issues related to social workers in the E&E Region.

Objectives: The study aims to:

- Describe the current status of social work practice and education in the region;
- Describe the gaps between the reality of social work practice and education (*i.e.*, what is desired, and strategies for improving outcomes for social work);
- Provide examples of model programmes and best practices from the region for key issues;
- Discuss lessons learned and challenges to implementation of best practices;
- Provide sources of information and tools to enhance programming related to key issues through web sites, titles of documents and locations, professional guidelines, toolkits, and regional contacts.

Target Audience: The primary target audience for this activity is USAID Missions and USAID/Washington personnel who are involved in Investing in People (IIP) programming related to social services and vulnerable groups, or social assistance. Additional audiences include anyone interested in detailed information about the role of social workers in providing social services in the E&E region, such as the U.S., Department of State, other U.S., government agencies, other donors, and NGOs that work on this issue in the region.

Data collection methodology: This desk-top study was conducted April-July, 2008 and utilized the following sources:

- *Literature review:* A review of the social work literature on the professionalization of social work, outcomes studies on models of practice, development and application of standards of practice, and studies on social work and social work practice specific to the Region was completed.
- *Documents review:* Copies of relevant policies, laws and strategic plans; descriptions of programmes and services available for different groups (public and private); USAID publications and web sites; available data that included at-risk populations, workforce, financial resources, etc.; evaluation and research on programmes and services; curricula from education and training programmes and related documents were obtained through an internet search and personal contacts.
- *Field research:* Dr. Sergei Lukoshov, Technical Expert, EveryChild, worked directly in the field to provide information from selected countries of the Former Soviet Union. Dr. Rebecca Davis conducted field research in Romania during the Rutgers Study Abroad Programme. Interviews were held with university faculty, students, professionals, ministry and local level authorities, professional associations, and NGOs.

STRUCTURE OF SERVICES AND PRACTICE ENVIRONMENT

Definition of Social Work and Social Work Practice: Across the region, the meaning of “social worker,” and related terms, as defined in the legislation and practice, is

confusing. The Former Yugoslav Republics have a more consistent definition and understanding of social work and social work practice, in part, due to the well-established social work schools and Centers for Social Work. For the most part, in other countries, there is confusion between social services, social assistance and social work. Social work as a profession and practice means persons that are working in social services providing basic services such as food, clothing and financial benefits (social assistance). In Ukraine, Russia and Armenia, the term “specialist of social work” is a degree that is somewhere between the BA and MA level degrees. It refers to someone that has higher education and credentials than “social workers.” In Romania, the “case manager” is at a specialist and supervisor, with an MA in Social Work at least 3-5 years, whereas in other countries the term usually refers to someone that is less experienced with lesser professional status (usually at the BA level).

- *Job Functions:* Most social workers are employed in public social services. Social work tasks primarily focus on completing administrative tasks related to eligibility for various social and financial benefits and providing concrete services such as housing, clothing, food, and meeting health-related needs such as medications. These tasks to require a lot of paperwork. Across all countries, there is concern that social workers are not performing the work that they are trained to do and they feel is needed such as family support and psycho-social counseling. There have been recommendations to separate the tasks related to eligibility for benefits (make that a clerical job) and family support and psychosocial services.
- *Salaries and Low Professional Status:* One of the most consistent and pervasive issues discussed was low salaries for social workers and difficult working conditions (large caseloads, excessive paper work, and limited resources for clients). Combined with low status when compared to other human services professionals (*i.e.*, doctors, nurses, teachers, psychologists), recruitment and retention of qualified social workers is affected across all countries. Social workers often leave traditional social service positions to take jobs in other sectors and ministries. Therefore, social service positions are also frequently filled by unqualified individuals, even if the position is for a qualified social worker. This is even more prevalent in rural towns and villages where the salary differentials may be even greater. Some caution against comparing social work salaries to those of health care professionals because of “under the table” payments given to doctors, nurses, etc., which makes the disparity even greater.
- *High Status as a “Helper:”* In spite of the negatives of salary and professional status, there is status and value in “being a helper” of those who have special needs. In Georgia, it was noted that there is a certain “prestige” to being a social worker as a new profession. A study on perceptions of social work in Croatia revealed that, although there was a negative attitude towards the

profession of social work, there was a more positive attitude towards the work that they do.

- *Social Work as a Career:* The recognized trajectory for professional social workers is: social work as a job/occupation Æ social work as a profession Æ social work as a career. Experienced social workers expressed concern about their future with a career in social work. There are few opportunities to advance, or to make a career in social work, as there are few management and supervisory positions available and limited opportunities for diversified practice, and especially specialized services.
- *Role of Independent, Private Practice:* Seen as the hallmark of a legitimate profession and a career option, only Romania has a law that allows for private, independent practice although, in reality, there is not yet a market for private practice. Some emerging tension was observed related to the prestige associated with more psycho-social (clinical) work versus traditional case management within the public sector. This is an ethical and professional dilemma for an increasingly professional workforce as private practice is seen as serving those with middle and upper incomes. The authority to have an independent (clinical) practice is also viewed as another criterion of legitimate
- *Professional status:* But, the conflict remains concerning social work's mission to serve the poor and disenfranchised.
- *External Labour Migration:* For the countries in Central and Eastern Europe in which social work is more developed and the borders have opened for easier travel, trained and experienced social workers are leaving their countries to find social work jobs that have higher pay, better benefits, and more professional jobs.

Social workers are actively recruited by Western Europe, and especially the UK. Social workers find better working conditions and salaries and hope for a social work career that can provide financial and professional security. One of the aims of the higher education reforms of the Bologna Process is to make it easier for cross-border employment, at least between and among those countries of the EU.

- *Internal Labour Migration and Access to Services:* Internal migration, rural to urban, as experienced in Belarus, also contributes to the uneven distribution of a qualified workforce. Social protection strategies call for increased outreach and case-finding. This is especially a challenge in rural areas, not only due to transportation issues, availability of services and specialists, and overall access to services, but also due to increased isolation and workloads. Creative solutions need to be explored such as training paraprofessionals and integrating social workers into health and education services.
- *Risk Management and Safety of Social Workers:* An emerging issue noted within

this framework relates to safety issues for social workers in particular regions. This has been highlighted in specialized fields of practice: human trafficking, domestic violence, and infectious diseases such as HIV/AIDS.

- *Transfer of Knowledge and Specialization:* Social work in most countries has grown out of child protection, directly in response to deinstitutionalization. Transferring knowledge and experience across fields of practice is beginning. In Romania, experience in community care for children was transferred to the fields of disability and aging. In Croatia, the experience in dealing with trauma recovery for war victims is now being applied to the field of domestic violence and child abuse. The emerging cross-discipline and cross-border models in work with victims of trafficking and domestic violence have direct application to child welfare, developmental disabilities and mental health and addictions, all growing fields of practice.
- *The Social Worker as Manager and Supervisor:* There is an overwhelming recognition about the significant contribution that social work supervision has on quality of services. There is also the reality that supervision, as practiced in the region, is primarily administrative in nature and does not address individual casework/clinical practice and professional coaching and mentoring. For those countries that have developed a Masters Degree in Social Work Supervision (*i.e.*, Romania, Bosnia and Herzegovina, and Croatia), the challenge has been implementation of the supervisory job, with the title and function of a social work supervisor that reflects what is being taught. The model that has been developed in St. Petersburg through EveryChild is a promising model. In addition, managers and administrators do not receive formal training in strategic planning, programme development, and programme evaluation, all which are considered to be part of a supervisor and manager's professional job functions.

EDUCATION AND TRAINING

University Education: Social work education is alive and well across the Region. Although some social work education programmes are in early developmental stages, there are others that are well- established. The Former Yugoslav Republics have had established training programmes since the 1950s with Bulgaria, Romania, Russia and the Ukraine establishing programmes in the early 1990s. Other countries are just establishing both the BA and MA programmes, while some are just now establishing a Ph.D. programme, although primarily in related disciplines. The aspect of university education that causes the most concern is the heavy reliance on theory. With the emphasis on theory in universities, students expressed concern about learning how theory is integrated into practice. Field education in the majority of countries has not been established as formalized internships with trained field supervisors and clearly defined learning contracts. Rather the students are viewed as "passive learners."

- *Teacher Training:* In selected countries of the region, there are faculty fellowship programmes through OSI and the U.S., Department of State that serve as a valuable resource for mutual exchange, curriculum development and teaching methodologies. Assisting faculty in accessing scholarships and grants that currently exist and expanding the use of internet-based learning and exchanges need to be explored.
- *Social Work Literature:* There is a wealth of literature being published in English language journals and books by scholars and practitioners in the region. Publishing in English provides an avenue for communication and exchanges across borders and continents. The primary limitation is that the literature in the students' and practitioners' first language, reflective of the local reality, is more limited. Since there is some evidence that English language skills are more prevalent among urban city dwellers, this further increases the rural/urban divide in terms of access to social work practice literature.
- *Training and Continuing Education:* Opportunities for training and continuing education are limited for social workers due to resources and geographic constraints. Those opportunities that do exist tend to be project-specific, time-limited, and have limited expansion funding. These types of project-specific trainings do not help with career development/job mobility, whereas more generic training would be relevant across practice settings.
- *Professional Associations:* While the countries studied have established legal frameworks for the existence of professional associations and they exist in almost every country, they are not well-established as a "voice of the profession of social work." Most are involved in providing continuing education and conferences, with little emphasis on professional advocacy and standard setting for the profession. Also, as membership organizations, there is a general feeling that the organizations do not understand how to service the membership. This is a lost opportunity to utilize the professional network as a way to reduce personal and professional isolation.

OUTCOMES AND PERFORMANCE MEASURES

- *Social Work Practice Standards:* Although there is consensus in many countries about the need for standards, most of this work has been done through social protection legislation and the development of standards of care for services. There are some good models that have been developed but dissemination and adoption by public services entities has been difficult.
- *Outcomes in Deinstitutionalization:* Decreased deinstitutionalization results from the social worker gatekeepers being better trained and more professional and having a standardized case recording format and assessment forms. University and field partnerships need to be formed in order to

- Develop evidence-based research to scientifically test interventions and evaluate outcomes on specific populations.
- *Performance Measure (Indicators) and Client Outcomes:* The difficulty in developing standards is linked to the absence of performance measures (indicators) and client outcomes. In terms of evaluation and impact, the focus has traditionally been on project outcomes as opposed to client outcomes and well-being. One example is the positive results seen in Georgia in reduction of child institutionalization in social services units that had a trained social worker. Yet the indicator was not directly related to child outcomes in terms of health, education, and quality of parenting. The problem of outcome tracking is exacerbated by the absence of functional information management systems available to public services and NGOs in the region.

BEST PRACTICES IN COMMUNITY-BASED SOCIAL WORK

The best practices framework consists of four pillars, deemed a comprehensive model of community-based social services for vulnerable groups. The framework incorporates the legal framework of the profession including laws related to practice within the social welfare system, the structure of services and the work environment for social work, the human capacity dimension focused primarily on education and training, and outcomes and performance measured through research and evaluation of professional practice.

The specific elements for each pillar as they apply to social work include:

- *Policy and Legal Framework:* This pillar includes policies and laws that reflect internationally recognized standards for the social work profession, the legal mandate and policy mandates for social work, regulation of practice through legislated Code of Ethics and certification or licensure, and policies relating to standards of practice that reflect best practices in family and community care.
- *Structure of Services and the Practice Environment:* This pillar includes the programmes and services in which social workers practice, level of qualifications, job functions, salaries and status of social workers, relationships with clients, other professionals, and the public authorities. This pillar also includes development and implementation of standards of practice that are consistent with professional social work practice inclusive of community outreach and accessibility of services, citizen and user involvement through initiatives such as in-home services and volunteerism, and public education about social services.
- *Human Capacity:* This pillar focuses on the acquisition of knowledge, values and skills for social workers providing direct services and those managing and supervising services. This includes academic and professional education and

training, curriculum development activities, field education, and conferences and workshops provided by a range of providers. In addition, this pillar includes role and activities of professional associations and quality improvement approaches to standard setting and professional accountability.

- *Performance Outcomes and Measures:* This refers to indicators used to measure outcomes for social work interventions, systems for monitoring social work inputs, cost-benefit analyses, development of evidence-based practices, research on the professionalization of social work such as salaries, standards, opinions and attitudes, client satisfaction, and client outcomes through research and evaluation.

The table below illustrates good practices in each of the four pillars. It describes how each pillar can function effectively to provide the best outcomes for society, from vulnerable populations to taxpayers.

SOCIAL WORK EDUCATION AND PRACTICE IN THE UNITED STATES

HISTORICAL DEVELOPMENT

The roots of the social work profession in North America and England are found in the Charitable Organization Societies (COSs) serving the poor and a desire by those leading that movement to provide consistent learning and education to “friendly visitors”. From its origins, social work recognized the value of real life/practical experience along with theoretical knowledge. So much so that when Mary Richmond first published *Social Diagnosis* in 1917, with a goal of establishing a professional base for casework through training, she was doubtful of university based education, which historically did not utilize field education as a model. In a paper published in 1897, when discussing the creation of the Charitable Organization Societies, Richmond spoke of the “right” of those doing the work to demand the right to further education and professional development, and a living wage. This article included a discussion on what should come first: professional standards of practice or a training school to establish them.

Essentially, in the United States the creation of formal social work education came from two paths, the COS’s and the Settlement Houses. Mary Richmond’s COS partnered with the New York School of Philanthropy to create technical training for case work and agency practice. Ultimately, this work came under the auspices of Columbia University in the 1940s. At the same time, Jane Addams’ Settlement House movement began working with the University of Chicago, having identified the need for an educational foundation based in social policy and social philosophy so that students would be prepared “for their role in constructing a better society”. The profession first emphasized attempting to determine the causes of individual problems through a focus

on internal causes, and interventions were focused on rehabilitation of those internal problems. As time went on, the profession began to recognize the role societal causes played in individual problems, and began to focus its attention on social change.

SOCIAL WORK TODAY

Some might say that social work is a well developed field of practice in the US today. This perception is based on the profession meeting criteria used to measure the legitimacy of all professions: a systematic body of knowledge, a service mission, professional authority, self-regulation such as licensing or certification, and a regulative code of ethics. A quick review of the social work profession today indicates all those are in fact present. The higher education opportunities include two degrees, Bachelor in Social Work (BA) and Masters in Social Work (MSW) regulated by a national body, the Council on Social Work Education (CSWE) – an oversight body that sets standards for curriculum and field education. There is also a doctoral degree, which prepares social workers to carry out research in social work rather than relying on research in other disciplines.

In addition to the status gained by achieving one of these degrees, most social workers also enjoy the benefits of being identified as a social worker. In many ways, this professional identity was solidified with the emergence of social work licensure. And while it was originally developed as a consumer or public protection mechanism, it has also served to carve out a scope of practice for social workers particularly in direct or clinical practice. Both the education system established for social workers and the licensing system that has emerged support the premise that theory and practice go hand in hand. Every social work education programme accredited by the Council on Social Work Education (CSWE) includes both classroom study and field experience. Similarly, licensing requirements generally establish the need for continued classroom learning as well as peer or clinical supervision.

The value placed on practical experience (through field education) is one of the defining attributes of social work in the US. However, like other professions, social work's status is determined to some degree by those they serve. The status of the profession is low because clients are typically held in low regard. Historically, this has impacted funding opportunities, work conditions, compensation, and prestige. As a result, recruitment and retention present unique challenges at any given time in the history of the profession. The research that underpins this report revealed the presence and/or emergence of similar elements as the profession of social work continues to develop in the former Soviet Bloc: specialization in educational programmes with a prescribed curriculum, recognition of particular degrees and training to assure certain knowledge and skill development; professional regulation through licensure or certification as a means of consumer protection; development and implementation of best practice standards in response to problem or crises identification. Also notable is

the relationship of an identified crisis or social need and the development of social work practice and standards of care.

REGULATION OF PRACTICE: REGISTRATION, CERTIFICATION AND LICENSURE

In the U.S., The first social work regulatory bill was introduced in California in 1929, but failed to pass. In 1945, legislation was enacted in California for voluntary registration of social workers. In the meantime, Puerto Rico had enacted legislation in 1934, making it the first state to regulate social work practice. Since 1960, every state, Puerto Rico, the U. S. Virgin Islands and the District of Columbia have some form of social work regulation, either certification or licensure², which are used almost interchangeably. Voluntary registration is no longer used as a form of practice regulation in US jurisdictions. A major issue today is the lack of uniformity in the terminology and levels of licensure and certification across the US. The Association of Social Work Boards (ASWB) is the national association of social work boards from each state that regulates social work practice. ASWB develops and maintains the social work licensing examination used across the country and in several Canadian provinces. It is also a central resource for information on the legal regulation of social work. ASWB also serves as a coordinating body for individual state boards to share information and work together on common issues of concern. ASWB has proposed a model social work practice act that has three levels of licensure: Licensed Baccalaureate Social Worker (LBSW), Licensed Master Social Worker (LMSW), and Licensed Clinical Social Worker (LCSW).

SOCIAL WORK PROFESSIONAL ASSOCIATIONS

The National Association of Social Workers is the largest membership organization for professional social workers involved in all areas of social work practice. In the US prior to 1955, there were numerous professional associations existing at any given time, all with independent missions, focused on different specialty practice areas, and with varying numbers of members, offering varying levels of services to members. Then in the late 1940s, the association representing schools of social work initiated discussions with the membership organizations regarding possible unification. Over a period of several years, work was undertaken to develop one central organization, representing the needs of all professional social workers. The National Association of Social Workers (NASW) began operations on October 1, 1955. There is a chapter in every state as well as in a number of territories, each with a separate governing body, mission, and staff. They are all responsible for public education, professional development, advocacy, and public policy work. These functions are important to note as they are essential to the development of both policy and practice in the United States. It has been the collective voice of the professional associations and the educational institutions that helped raise the status and visibility of the profession.

The practice of social work is guided by the NASW Code of Ethics, which defines the facets of ethical social work practice at all levels. Professional social workers also receive guidance from their professional association through the Standards for Social Work Practice which have been developed for specific fields of practice (*i.e.*, child welfare, adolescence, substance use) as well as areas of social work knowledge (*i.e.*, cultural competence, technology, continuing education). Despite the intentions of the founders of NASW, social workers in the US have continued to form and join multiple organizations as they become concerned that NASW cannot or does not represent their individual needs. Some examples of the professional organizations that exist side by side with NASW are: The National Association of Black Social Workers, The Society for Clinical Social Workers, The Society for Transplant Social Workers, The National Network for Social Work Managers, The Association for the Advancement of Social Work with Groups, and perhaps the largest other entity – the Council on Social Work Education.

REFORMS OF HIGHER EDUCATION IN E&E: IMPACT ON SOCIAL WORK

The Bologna Process, initiated in 1999, is aimed at modernizing European Higher Education and creating a European Higher Education Area (EHEA) by 2010. The long term objective is to facilitate speedy entrance of educated professionals into the job market. Other objectives of the programme are to enhance cross-border mobility of students and job seekers, and to increase the competitiveness of European higher education internationally. The Bologna Declaration calls for:

- Adoption of a system of easily readable and comparable degrees;
- Adoption of a degree structure based on two primary degrees (undergraduate and graduate); and
- Adoption of a system of credits such as European College Transfer System (ECTS) (for ease of transfer and accumulation of credits).

Most of the social work programmes in universities in the former Soviet Bloc countries, including the Former Soviet Union, have joined the Bologna Process. This means incorporating the required changes in programme structure and curriculum that aims to reform the nature of education – to make it more practical and applied. One major change is that the bachelor's degree in social work takes three years rather than four to complete, and the MA takes two years rather than one. By 2010, the plan is to have standard recognized European degrees at the bachelor's and master's or doctoral level in much the same way as exists in the United States. The implementation of this has proven to be difficult as there is resistance among some of the established universities in Western Europe as well as in the countries of the former Soviet Bloc. It is beyond the scope of this project to assess the impact of the Bologna Process. However, the benefits of a standardized curriculum and credit system may be outweighed by increased out-migration of social workers to countries where social work is more established in search of better working conditions and higher salaries.³

BEST PRACTICE MODEL APPLIED TO THE PROFESSION OF SOCIAL WORK

The current report builds on two previous USAID-funded studies on the evolution of community-based services in E&E countries that highlight good practice models that have been established as part of professional social work. This report explores the social work profession as a key dimension of social sector reforms and follows the same four-pillar framework and best practices matrix as the earlier Davis publications with an emphasis on the profession of social work as it is changing within the practice environment. The key principles of the framework and best practices model are consistent with international standards for social work education and social work practice.

ENVIRONMENTAL EDUCATION

DEFINITION

The word environment is derived from the french word ‘environ’ meaning surroundings. Hence, everything surrounding us is called “Environment”. Every organism is surrounded by materials and forces that constitute its environment. It is the environment from where every organism must derive its requirement. The environment creates favourable conditions for the existence and development of living organisms. The survival of any organism requires a steady supply of materials and removal of waste products from its environment.

The degradation of the environment has become a serious problem for the existence of human beings. Pollution of soil, water and air causes harm to living organisms as well as loss to valuable natural resources. Environmental studies involves educating the people for preserving the quality of environment.

The scope of environmental studies include:

- Developing an awareness and sensitivity to the total environment and its related problems
- Motivating people for active participation in environmental protection and improvement
- Developing skills for active identification and development of solutions to environmental problems
- Imbibe and inculcate the necessity for conservation of natural resources
- Evaluation of environmental programmes in terms of social, economic, ecological and aesthetic factors.

SCOPE

Scope of environmental studies is so wide that it has got relation with every science and scientific aspects in general and biology in particular.

The scope of environmental studies in numerous fields is given below:

- Conservation and management of natural resources like forest, water, air, mineral, etc.
- Conservation of biodiversities like genetic diversity, species diversity, ecosystem diversity, landscape diversity, etc.
- Control of environmental pollutions like air pollution, water pollution, soil pollution, solid waste pollution, noise pollution, electronic waste pollution, e-pollution, etc.
- Replacement of development (like green revolution, urbanization,) economic growth, industrialization, etc.) With sustainable development
- Control of human population.
- Ecology

Main Objectives of Environmental Studies are given below:

- Acquire an awareness of the environment as a whole and its related problems.
- Gain a variety of experiences and acquire a basic understanding and knowledge about the environment and its allied problems.
- Acquire an attitude of concern for the environment.
- Acquire the skills for identifying and solving environmental problems.
- Participate in improvement and protection of environment.
- Develop the ability to evaluate measures for the improvement and protection of environment.

To sum up, the objectives of environmental studies are to develop a world in which persons are aware of and concerned about environment and the problems associated with it, and committed to work individually as well as collectively towards solutions of current problems and prevention of future problems.

NEED AND IMPORTANCE OF ENVIRONMENTAL STUDIES

Need

Increasing population, Urbanization and poverty have generated pressure on the natural resources and lead to a degradation of the environment. To prevent the environment from further degradation, the supreme court has ordered and initiated environmental protection awareness through government and non-government agencies to take part in protecting our environment. Environmental pollution cannot prevented by laws alone. Public participation is equally important with regard to environmental protection.

Environmental Education (EE) is a process of learning by giving an overall perspective of knowledge and awareness of the environment. It sensitizes the society about environmental issues and challenges interested individuals to develop skills and expertise thereby providing appropriate solutions. Climate change, loss of biodiversity, declining fisheries, ozone layer depletion, illegal trade of endangered species, destruction of

habitats, land degradation, depleting ground water supplies, introduction of alien species, environmental pollution, solid waste disposal, storm water and sewage disposal pose a serious threat to ecosystems in forest, rural, urban and marine ecosystems.

Both formal and informal education on the environment will give the interested individual the knowledge, values, skills and tools needed to face the environmental challenges on a local and global level.

Importance of Environmental Studies

In the industrialized era that we live today, every component that we intake - be it, air, water or food are contaminated by industrial activities. There is no zero pollution. To minimize this problem, knowledge of environmental studies is essential.

An in-dept study of environmental studies will help us in the following ways:

- We will begin to appreciate and adopt the idea of “Development without Destruction of the Environment”
- Knowledge about “Various Types of Environments and Different Environmental Hazards”
- Playing an effective role in protecting the environment by “Demanding Changes in Law and Enforcement Systems”.
- Having a “Positive Impact” on “Quality of Life”.
- Creating a “Concern and Respect for the Environment”.

CONCEPT OF ENVIRONMENTAL EDUCATION

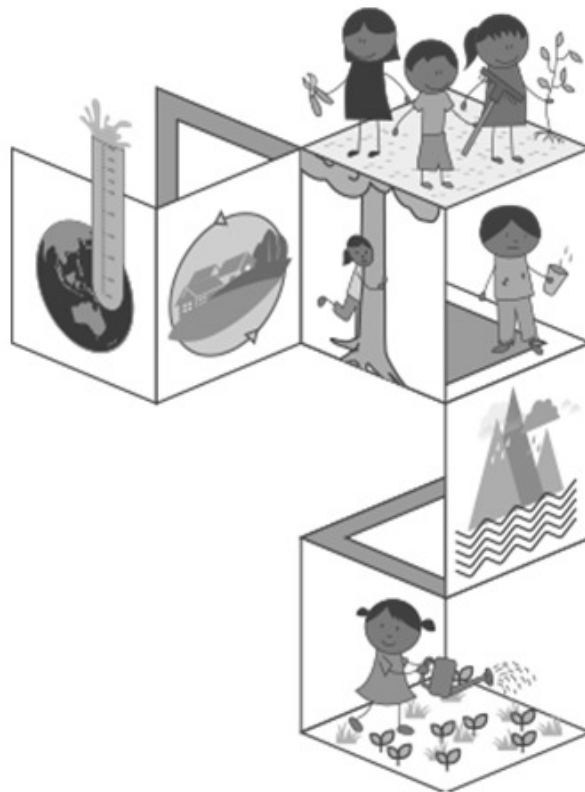
The term environment refers to our surrounding which includes physical, biological and socio-cultural aspects. The non-living things like air, water, land, etc. Come under physical aspects and plants and trees comes under biological aspects and the man-made aspect like culture, religion, customs, etc. come under socio cultural aspects. There is a harmonious relationship among the components of the environment. The status of soil, water and other elements determines the biodiversity because different animals and plants have special adaptability cyclic relationship. Environmental condition determines our health also.

Environment is the source of all our needs. It provides all the things for our livelihood. However, the activities of human beings are not environment friendly. Environment Education is the educational process that teaches everything about the environment. It teaches us the way to keep our environment clean and preserve the environmental elements.

According to UNESCO in 1975, environment education can be defined as, Education that concerns about the total environment and its associated problems and commitment to work individually and collectively towards the solution of current problems and prevention of new one.

Importance of Environment Education

Environment helps to fulfill all our demands. Human activities have resulted into natural calamities, environmental degradation, environmental pollution and many other problems. Environment education plays a significant role to control these problems and maintain the balance of the environment.



The importance of education environment are:

- *Develops the concept of wise use of natural resources:* Many people do not have the concept that their activities interfere the status of the environment. They use the natural resources randomly. They do not even know the hazards of their malpractices. They do not try to replace the lack by the means of afforestation and other regeneration programme mes. Environment education teaches people about wise use of natural resources.
- *Encourage for the creation of healthy environment:* Healthy environment refers to the environment which is free from the dust, germ and other elements that cause many hazardous diseases. Environment education provides knowledge about various things such as personal hygiene, environmental sanitation and other topic related to healthy environment which prompts us to create a healthy environment.

- *Cultivates awareness about environmental:* Environmental degradation has been a great threat to the existence of all flora, fauna and human beings. Environment education gives knowledge about environmental pollution, create awareness.
- *Helps to provide integrated knowledge:* Environment education incorporates the elements that are related with health and population education. It helps to control the rapidly growing population, to keep the surrounding clean and to give emphasis on personal hygiene.

HISTORICAL BACKGROUND OF ENVIRONMENTAL EDUCATION

The roots of environmental education can be traced back as early as the 18th century when Jean-Jacques Rousseau stressed the importance of an education that focuses on the environment in *Emile: or, On Education*. Several decades later, Louis Agassiz, a Swiss-born naturalist, echoed Rousseau's philosophy as he encouraged students to "Study nature, not books." These two influential scholars helped lay the foundation for a concrete environmental education programme, known as nature study, which took place in the late 19th century and early 20th century.

The nature study movement used fables and moral lessons to help students develop an appreciation of nature and embrace the natural world. Anna Botsford Comstock, the head of the Department of Nature Study at Cornell University, was a prominent figure in the nature study movement and wrote the *Handbook for Nature Study* in 1911, which used nature to educate children on cultural values.

Comstock and the other leaders of the movement, such as Liberty Hyde Bailey, helped Nature Study garner tremendous amounts of support from community leaders, teachers, and scientists and change the science curriculum for children across the United States.

A new type of environmental education, Conservation Education, emerged as a result of the Great Depression and Dust Bowl during the 1920s and 1930s. Conservation Education dealt with the natural world in a drastically different way from Nature Study because it focused on rigorous scientific training rather than natural history. Conservation Education was a major scientific management and planning tool that helped solve social, economic, and environmental problems during this time period.

The modern environmental education movement, which gained significant momentum in the late 1960s and early 1970s, stems from Nature Study and Conservation Education. During this time period, many events – such as Civil Rights, the Vietnam War, and the Cold War – placed Americans at odds with one another and the U.S., government.

However, as more people began to fear the fallout from radiation, the chemical pesticides mentioned in Rachel Carson's *Silent Spring*, and the significant amounts of air pollution and waste, the public's concern for their health and the health of their natural environment led to a unifying phenomenon known as environmentalism.

Environmental education was born of the realization that solving complex local and global problems cannot be accomplished by politicians and experts alone, but requires “the support and active participation of an informed public in their various roles as consumers, voters, employers, and business and community leaders” One of the first articles about environmental education as a new movement appeared in the *Phi Delta Kappan* in 1969, authored by James A. Swan. A definition of “Environmental Education” first appeared in *The Journal of Environmental Education* in 1969, authored by William B. Stapp. Stapp later went on to become the first Director of Environmental Education for UNESCO, and then the Global Rivers International Network.

Ultimately, the first Earth Day on April 22, 1970 – a national teach-in about environmental problems – paved the way for the modern environmental education movement. Later that same year, President Nixon passed the National Environmental Education Act, which was intended to incorporate environmental education into K-12 schools. Then, in 1971, the National Association for Environmental Education (now known as the North American Association for Environmental Education) was created to improve environmental literacy by providing resources to teachers and promoting environmental education programmes.

Internationally, environmental education gained recognition when the UN Conference on the Human Environment held in Stockholm, Sweden, in 1972, declared environmental education must be used as a tool to address global environmental problems. The United Nations Education Scientific and Cultural Organization (UNESCO) and United Nations Environment Programme (UNEP) created three major declarations that have guided the course of environmental education.

Stockholm Declaration

June 5–16, 1972 - The Declaration of the United Nations Conference on the Human Environment. The document was made up of 7 proclamations and 26 principles “to inspire and guide the peoples of the world in the preservation and enhancement of the human environment.”

Belgrade Charter

October 13–22, 1975 - The Belgrade Charter was the outcome of the International Workshop on Environmental Education held in Belgrade, Yugoslavia (now Serbia). The Belgrade Charter was built upon the Stockholm Declaration and adds goals, objectives, and guiding principles of environmental education programmes. It defines an audience for environmental education, which includes the general public.

Tbilisi Declaration

October 14–26, 1977 - The Tbilisi Declaration “noted the unanimous accord in the important role of environmental education in the preservation and improvement

of the world's environment, as well as in the sound and balanced development of the world's communities." The Tbilisi Declaration updated and clarified The Stockholm Declaration and The Belgrade Charter by including new goals, objectives, characteristics, and guiding principles of environmental education. Later that decade, in 1977, the Intergovernmental Conference on Environmental Education in Tbilisi, Georgia emphasized the role of Environmental Education in preserving and improving the global environment and sought to provide the framework and guidelines for environmental education. The Conference laid out the role, objectives, and characteristics of environmental education, and provided several goals and principles for environmental education.

CELEBRATION OF VARIOUS DAYS IN RELATION WITH ENVIRONMENT

- World Wetlands Day: 02-02: February 2
- International Polar Bear Day: 02-27: February 27
- World Wildlife Day: 03-03: March 3
- Solar Appreciation Day: second Friday of March
- International Day of Action for Rivers: 03-14: March 14
- World Consumer Rights Day: 03-15: March 15
- World Sparrow Day: 03-20: March 20
- International Day of Forests: 03-21: March 21
- World Planting Day: 03-21: March 21
- World Wood Day: 03-21: March 21
- World Water Day: 03-22: March 22
- Earth Day: 04-22: April 22
- World Fish Migration Day: April 21, 2018
- Day of Remembrance for all Victims of Chemical Warfare: 04-29: April 29
- Green Up Day: 05: first Saturday of May in Vermont
- Greenery Day: 05-04: May 4 in Japan (previously April 29)
- World Migratory Bird Day: second Saturday in May
- International Migratory Bird Day: second Saturday in May (in the U.S. and Canada) and second Saturday in October (in Mexico, Central and South America, and the Caribbean)
- International Recycling Day: May 17th, 2017
- Endangered Species Day: 05-19: May 19
- International Day for Biological Diversity (World Biodiversity Day): 05-22: May 22
- World Turtle Day: 05-23: May 23
- European day of parks (Europarc): 05-24: May 24
- Bike-to-Work Day: 05: third Friday in May
- World Environment Day: 06-05: June 5

- World Oceans Day: 06-08: June 8
- Coral Triangle Day: 06-09: June 9
- Global Wind Day: 06-15: June 15
- World Sea Turtle Day: 06-16: June 16
- World Day to Combat Desertification and Drought: 06-17: June 17
- World Giraffe Day: 06-21: June 21
- World Population Day: 07-11: July 11
- International Tiger Day: 07-29: July 29
- World Lion Day: 08-10: August 10
- World Elephant Day: 08-12: August 12
- World Orangutan Day: 08-19: August 19
- National Honey Bee Day: 08-22: August 22
- Amazon Day: 09-05: September 5
- World Cleanup Day 2018: 09-08: September 8, 2018
- International Day for the Preservation of the Ozone Layer: 09-16: September 16
- Clean Up the World: 09-17 to 09-19: weekend - September 16-18
- World Water Monitoring Day: 09-18: September 18
- Zero Emissions Day: 09-21: September 21
- Car Free Day: 09-22: September 22
- World Rhino Day: 09-22: September 22
- Ecological Debt Day (Earth Overshoot Day) 09-23: September 23 in 2008, but receding
- World Environmental Health Day: 09-26: September 26 since 2011 (IFEH)
- World Rivers Day: 09: last Sunday in September
- World Habitat Day: 10: first Monday in October
- World Animal Day: 10-04: October 4
- International Day for Natural Disaster Reduction: 10: second Wednesday in October, October 13.
- Sustainability Day: 10: fourth Wednesday of October
- International Snow Leopard Day: October 23
- International Day of Climate Action: 10-24: October 24
- World Vegan Day: 11-01: November 1
- International Day for Preventing the Exploitation of the Environment in War and Armed Conflict: 11-06: November 6
- America Recycles Day: 11-15: November 15
- World Soil Day: 12-05: December 5
- International Mountain Day: 12-11: December 11
- Ozone Action Day: at certain times during the summer months
- eDay: variable date

PLASTIC RECYCLING AND PROHIBITION OF PLASTIC BAG/COVER

Plastic Recycling

Plastic recycling is the process of recovering scrap or waste plastic and reprocessing the material into useful products. Since the vast majority of plastic is non-biodegradable, recycling is a part of global efforts to reduce plastic in the waste stream, especially the approximately eight million metric tonnes of waste plastic that enter the Earth's ocean every year. This helps to reduce the high rates of plastic pollution.

Plastic recycling includes taking any type of plastic, sorting it into different polymers and then chipping it and then melting it down into pellets. After this stage, it can then be used to make items of any sort such as plastic chairs and tables. Soft Plastics are also recycled such as polyethylene film and bags. This closed-loop operation has taken place since the 1970s and has made the production of some plastic products amongst the most efficient operations today. Compared with lucrative recycling of metal, and similar to the low value of glass, plastic polymers recycling is often more challenging because of low density and low value. There are also numerous technical hurdles to overcome when recycling plastic. A macro molecule interacts with its environment along its entire length, so total energy involved in mixing it is largely due to the product side stoichiometry. Heating alone is not enough to dissolve such a large molecule, so plastics must often be of nearly identical composition to mix efficiently.

When different types of plastics are melted together, they tend to phase-separate, like oil and water, and set in these layers. The phase boundaries cause structural weakness in the resulting material, meaning that polymer blends are useful in only limited applications. The two most widely manufactured plastics, polypropylene and polyethylene behave this way, which limits their utility for recycling. Recently, the use of block copolymers as “molecular stitches” or “macromolecular welding flux” has been proposed to overcome the difficulties associated with phase separation during recycling. Another barrier to recycling is the widespread use of dyes, fillers, and other additives in plastics. The polymer is generally too viscous to economically remove fillers, and would be damaged by many of the processes that could cheaply remove the added dyes. Additives are less widely used in beverage containers and plastic bags, allowing them to be recycled more often. Yet another barrier to removing large quantities of plastic from the waste stream and landfills is that many common but small plastic items lack the universal triangle recycling symbol and accompanying number. An example is the billions of plastic utensils commonly distributed at fast-food restaurants or sold for use at picnics. The percentage of plastic that can be fully recycled, rather than downcycled or go to waste, can be increased when manufacturers of packaged goods minimize mixing of packaging materials and eliminate contaminants. The Association of Plastics Recyclers have issued a “Design Guide for Recyclability”. The use of biodegradable plastics is increasing.

Processes

Before recycling, most plastics are sorted according to their resin type. In the past, plastic reclaimers used the resin identification code (RIC), a method of categorization of polymer types, which was developed by the Society of the Plastics Industry in 1988. Polyethylene terephthalate, commonly referred to as PET, for instance, has a resin code of 1. Most plastic reclaimers do not rely on the RIC now; they use automatic sort systems to identify the resin, ranging from manual sorting and picking of plastic materials to mechanized automation processes that involve shredding, sieving, separation by rates of density *i.e.* air, liquid, or magnetic, and complex spectrophotometric distribution technologies *e.g.* UV/VIS, NIR, Laser, etc. Some plastic products are also separated by colour before they are recycled. The plastic recyclables are then shredded. These shredded fragments then undergo processes to eliminate impurities like paper labels. This material is melted and often extruded into the form of pellets which are then used to manufacture other products.

Thermal Depolymerization

Another process involves the conversion of assorted polymers into petroleum by a much less precise thermal depolymerization process. Such a process is able to accept almost any polymer or mix of polymers, including thermoset materials such as vulcanized rubber tire separation of wastes and the biopolymers in feathers and other agricultural waste. Like natural petroleum, the chemicals produced can be made into fuels as well as polymers. A pilot plant of this type exists in Carthage, Missouri, United States, using turkey waste as input material. Gasification is a similar process, but is not technically recycling, since polymers are not likely to become the result.

Waste Plastic Pyrolysis to Fuel Oil

Plastic pyrolysis can convert petroleum-based waste streams such as plastics into quality fuels and carbons. Given below is the list of suitable plastic raw materials for pyrolysis:

- Mixed plastic (HDPE, LDPE, PE, PP, Nylon, Teflon, PS, ABS, FRP etc.)
- Mixed-waste plastic from waste paper mill
- Multi-layered plastic.

Heat Compression

Yet another process that is gaining ground with startup companies (especially in Australia, United States and Japan) is heat compression, which takes all unsorted, cleaned plastic in all forms, from soft plastic bags to hard industrial waste, and mixes the load in tumblers (large rotating drums resembling giant clothes dryers). The most obvious benefit to this method is that all plastic is recyclable, not just matching forms. However, criticism rises from the energy costs of rotating the drums, and heating the post-melt pipes.

Distributed Recycling

For some waste plastics, technical devices called recyclebots enable a form of distributed recycling. Preliminary life-cycle analysis (LCA) indicates that such distributed recycling of HDPE to make filament of 3-D printers in rural regions is energetically favourable to either using virgin resin or conventional recycling processes because of reductions in transportation energy

Other Processes

A process has also been developed in which many kinds of plastic can be used as a carbon source in the recycling of scrap steel. There is also a possibility of mixed recycling of different plastics, which does not require their separation. It is called compatibilization and requires use of special chemical bridging agents compatibilizers. It can help to keep the quality of recycled material and to skip often expensive and inefficient preliminary scanning of waste plastics streams and their separation/purification.

Applications

PET

Post-consumer polyethylene terephthalate (PET or PETE) containers are sorted into different colour fractions, and baled for onward sale. PET recyclers further sort the baled bottles and they are washed and flaked (or flaked and then washed). Non-PET fractions such as caps and labels are removed during this process. The clean flake is dried. Further treatment can take place *e.g.* melt filtering and pelletising or various treatments to produce food-contact-approved recycled PET (RPET).

RPET has been widely used to produce polyester fibres. This sorted post-consumer PET waste is crushed, chopped into flakes, pressed into bales, and offered for sale.

One use for this recycled PET is to create fabrics to be used in the clothing industry. The fabrics are created by spinning the PET flakes into thread and yarn. This is done just as easily as creating polyester from brand new PET. The recycled PET thread or yarn can be used either alone or together with other fibres to create a very wide variety of fabrics. Traditionally these fabrics are used to create strong, durable, rough, products, such as jackets, coat, shoes, bags, hats, and accessories since they are usually too rough for direct skin contact and can cause irritation. However, these types of fabrics have become more popular as a result of the public's growing awareness of environmental issues. Numerous fabric and clothing manufacturers have capitalized on this trend.

Other major outlets for RPET are new containers (food-contact or non-food-contact) produced either by (injection stretch blow) moulding into bottles and jars or by thermoforming APET sheet to produce clam shells, blister packs and collation trays. These applications used 46% of all RPET produced in Europe in 2010. Other applications, such as strapping tape, injection-moulded engineering components and

building materials, account for 13% of the 2010 RPET production. In the United States the recycling rate for PET packaging was 31.2% in 2013, according to a report from The National Association for PET Container Resources (NAPCOR) and The Association of Postconsumer Plastic Recyclers (APR). A total of 1,798 million pounds was collected and 475 million pounds of recycled PET used out of a total of 5,764 million pounds of PET bottles.

HDPE

Plastic #2, high-density polyethylene (HDPE) is a commonly recycled plastic. It is typically downcycled into plastic lumber, tables, roadside curbs, benches, truck cargo liners, trash receptacles, stationery (*e.g.* rulers) and other durable plastic products and is usually in demand.

PS

Most polystyrene products are not recycled due to the lack of incentive to invest in the compactors and logistical systems required. As a result, manufacturers cannot obtain sufficient scrap. Expanded polystyrene (EPS) scrap can easily be added to products such as EPS insulation sheets and other EPS materials for construction applications. When it is not used to make more EPS, foam scrap can be turned into clothes hangers, park benches, flower pots, toys, rulers, stapler bodies, seedling containers, picture frames, and architectural molding from recycled PS.

Recycled EPS is also used in many metal casting operations. Rastra is made from EPS that is combined with cement to be used as an insulating amendment in the making of concrete foundations and walls. Since 1993, American manufacturers have produced insulating concrete forms made with approximately 80% recycled EPS.

Other Plastics

The white plastic polystyrene foam peanuts used as packing material are often accepted by shipping stores for reuse.

Successful trials in Israel have shown that plastic films recovered from mixed municipal waste streams can be recycled into useful household products such as buckets.

Similarly, agricultural plastics such as mulch film, drip tape and silage bags are being diverted from the waste stream and successfully recycled into much larger products for industrial applications such as plastic composite railroad ties. Historically, these agricultural plastics have primarily been either landfilled or burned on-site in the fields of individual farms.

CNN reports that Dr. S. Madhu of the Kerala Highway Research Institute, India, has formulated a road surface that includes recycled plastic: aggregate, bitumen (asphalt) with plastic that has been shredded and melted at a temperature below 220 degrees C (428 °F) to avoid pollution. This road surface is claimed to be very durable and monsoon

rain resistant. The plastic is sorted by hand, which is economical in India. The test road used 60 kg of plastic for an approximately 500-metre-long, 8-metre-wide, two-lane road.

The process chops thin-film road-waste into a light fluff of tiny flakes that hot-mix plants can uniformly introduce into viscous bitumen with a customized dosing machine. Tests at both Bangalore and the Indian Road Research Centre indicate that roads built using this ‘KK process’ will have longer useful lives and better resistance to cold, heat, cracking, and rutting, by a factor of three.

Recycling Rates

The quantity of post-consumer plastics recycled has increased every year since at least 1990, but rates lag far behind those of other items, such as newspaper (about 80%) and corrugated fibreboard (about 70%). Overall, U.S., post-consumer plastic waste for 2008 was estimated at 33.6 million tons; 2.2 million tons (6.5%) were recycled and 2.6 million tons (7.7%) were burned for energy; 28.9 million tons, or 85.5%, were discarded in landfills.

Economic and Energy Potential

In 2008, the price of PET dropped from \$370/ton in the US to \$20 in November. PET prices had returned to their long-term averages by May 2009. Recycling one ton of plastic can save 5,774 kWh of energy, 98,000,000 btu of energy, 1,000–2,000 gallons of gasoline, 685 gallons of oil, 30 cubic yards of landfill space, 48,000 gallons of water.

Consumer Education

In the UK, the amount of post-consumer plastic being recycled is relatively low, due in part to a lack of recycling facilities. The Plastics 2020 Challenge was founded in 2009 by the plastics industry with the aim of engaging the British public in a nationwide debate about the use, reuse and disposal of plastics, and hosts a series of online debates on its web site framed around the waste hierarchy.

There is a facility in Worksop capable of recycling 60–80 thousand metric tonnes a year. In Northern Ireland, the rate of recycling is 37.4%. However, emerging technologies are helping to increase the recycling rates of items previously landfilled *e.g.* mixed hard plastics.

Plastic Identification Code

Seven groups of plastic polymers, each with specific properties, are used worldwide for packaging applications. Each group of plastic polymer can be identified by its plastic identification code (PIC), usually a number or a letter abbreviation. For instance, low-density polyethylene can be identified by the number “4” or the letters “LDPE”. The PIC appears inside a three-chasing-arrow recycling symbol. The symbol is used to

indicate whether the plastic can be recycled into new products. The PIC was introduced by the Society of the Plastics Industry, Inc., to provide a uniform system for the identification of various polymer types and to help recycling companies separate various plastics for reprocessing. Manufacturers of plastic products are required to use PIC labels in some countries/regions and can voluntarily mark their products with the PIC where there are no requirements. Consumers can identify the plastic types based on the codes usually found at the base or at the side of the plastic products, including food/chemical packaging and containers. The PIC is usually not present on packaging films, since it is not practical to collect and recycle most of this type of waste.

United States

Low national plastic recycling rates have been due to the complexity of sorting and processing, unfavourable economics, and consumer confusion about which plastics can actually be recycled. Part of the confusion has been due to the use of the resin identification code which is not on all plastic parts but just a subset that includes the recycling symbol as part of its design. The resin identification code is stamped or printed on the bottom of containers and surrounded by a triangle of arrows. The intent of these symbols was to make it easier to identify the type of plastics used to make a particular container and to indicate that the plastic is potentially recyclable. The question that remains is which types of plastics can be recycled by local recycling centers. In many communities, not all types of plastics are accepted for sidewalk recycling collection programmes due to the high processing costs and complexity of the equipment required to recycle certain materials. There is also sometimes a seemingly low demand for the recycled product depending on a recycling center's proximity to entities seeking recycled materials. Another major barrier is that the cost to recycle certain materials and the corresponding market price for those materials sometimes does not present any opportunity for profit. The best example of this is polystyrene (commonly called styrofoam), although some communities, like Brookline, Massachusetts, are moving towards banning the distribution of polystyrene containers by local food and coffee businesses.

Prohibition of Plastic Bag/Cover

The ACT banned plastic bags on 1 November 2011. Single-use plastic shopping bags litter our environment, harm wildlife and require valuable resources to manufacture. The Northern Territory, South Australia and Tasmania have also banned single-use plastic shopping bags. More states and territories will likely follow.

The ban has reduced plastic bag use, with shoppers encouraged to bring their own bags. Overall the ban has reduced the volume of plastic bag waste going to landfill by around one-third. The ban applies to all retailers in the ACT for single-use, lightweight polyethylene polymer plastic bags that are less than 35 microns in thickness (these are the thin plastic bags with handles that were typically supplied at supermarkets check-

outs). The ban does not apply to other bags such as barrier bags for fruit and vegetables. The ACT's plastic bag ban was reviewed in 2012 and 2014, including through community surveys. The 2014 survey reported that:

- More than 70% of people surveyed did not want the ban overturned
- 65% of Canberra grocery shoppers supported the ban for environmental reasons and agreed it had a positive effect on the environment.

ROLE OF SCHOOL IN ENVIRONMENTAL CONSERVATION AND SUSTAINABLE DEVELOPMENT

As we close in on Earth Day 2012, it seems fitting to reflect on the school's dual role in environmental protection. Like all entities, schools have an environmental footprint. Those in the school generate trash. They use energy for heating, lighting, photocopying and so on. Schools are cleaned using chemicals that have environmental impacts. The list continues. But schools also have a very unique role in environmental protection: They help students become conscientious stewards of (and advocates for) the environment, protecting it for themselves and generations to come. As we look to schools in the effort to preserve our environment, we should consider both roles to maximize their impact. Of course, simply by minimizing their own environmental footprint, schools can do a great deal to help the environment. As Redmond (WA) environmental science teacher Mike Town has pointed out, "in Redmond and in most of the small cities in the United States — suburban cities, rural cities — the single biggest greenhouse point source is the local high school." In addition to raising thermostats in the summer and turning out lights when classrooms and auditoriums are not in use, one relatively easy step that schools and districts can take to lessen their environmental impact is "cleaning green." In a recent edition of *The State Education Standard*, the National Association of State Boards of Education (NASBE) offered "A Roadmap to Implementing Green Cleaning in Districts and Schools," suggesting a few simple steps to help schools (and districts) start a green cleaning programme:

- Develop a comprehensive programme focusing on low-hanging fruit or a significant challenge. It should include prevention (in terms of both dirt entering the building and exposure of students and staff to chemicals), conservation (of energy, water and chemicals), education (of the community and staff) and evaluation (of both cost and quality of cleaning and products)
- Switch to green cleaning products as identified by standards set by third-party organizations like Green Seal, EcoLogo and the EPA's Design for the Environment. Such a switch is often cost-neutral
- Introduce green equipment and supplies such as auto scrubbers that reduce water consumption and chemical use and high-quality entryway mats that reduce dirt contaminations. While the upfront costs of such materials may be higher than traditional options, they can save money as well as the environment in the long-term

- Adopt green cleaning procedures adjusting the frequency, technique or time of a cleaning. Such changes can have environmental and cost impacts as large as switching products and equipment
- Share the responsibility to ensure that teachers, administrators, students, outside contractors and others join the custodial staff in efforts to maintain a green cleaning programme.

Inspire

In addition to acting themselves, schools can also educate and inspire students on environmental issues, equipping them to make informed choices about the impact of their (and others) actions.

Some might argue that this task does not fall under the purview of the school. But, as Town has said, there are a number of other reasons to teach environmental science. It is a great hook for getting students interested in science and math. And there is an occupational component — green jobs, some claim, are the way of the future. Schools need to ensure students are prepared to fill them. The Partnership for 21st century skills considers environmental literacy, a key requirement for success in the global community.

Sometimes opportunities for environmental education occur by chance. For example, at Maryland's Crellin Elementary School, science camp students noticed orange water seeping from a playground area into a stream. Investigation revealed that the seepage was acid mine drainage from water flowing under coal and mine waste that had been spilled back when Crellin was a coal mining town. The school and community joined forces, working together to reclaim the land and develop an environmental education lab for use by the school.

But environmental issues should also be deliberately addressed, either in science, math, social studies or other courses or in dedicated environmental science courses, like Town's. His students have designed green buildings and developed new, greener transportation plans for their community. He also engaged his students in creating the Cool School Challenge. They measured the carbon footprint for everything that happened in their school (from transportation to electricity to heating and waste). With partners, they developed a replicable model for reducing the carbon footprint of a school (available for free on the web) and got buy-in from their school community to implement it.

As is evidenced by the Cool School Challenge (and energy conservation programmes in St. Tammany Parish, Louisiana, York County, Virginia, and Scarsdale, New York, to name just a few examples), the school's two roles in environmental protection are not mutually exclusive. Most schools and districts that have successfully decreased their environmental footprint have done so in a comprehensive manner that includes both administrator actions and educational experiences for students. As we look forward to Earth Day, hopefully more schools and districts will follow their lead.

POLLUTION FREE ECO-SYSTEM

Pollution is the introduction of contaminants into the natural environment that cause adverse change. Pollution can take the form of chemical substances or energy, such as noise, heat or light. Pollutants, the components of pollution, can be either foreign substances/energies or naturally occurring contaminants. Pollution is often classed as point source or non-point source pollution.

Air pollution has always accompanied civilizations. Pollution started from prehistoric times when man created the first fires. According to a 1983 article in the journal *Science*, “soot” found on ceilings of prehistoric caves provides ample evidence of the high levels of pollution that was associated with inadequate ventilation of open fires.” Metal forging appears to be a key turning point in the creation of significant air pollution levels outside the home. Core samples of glaciers in Greenland indicate increases in pollution associated with Greek, Roman and Chinese metal production, but at that time the pollution was comparatively small and could be handled by nature. The burning of coal and wood, and the presence of many horses in concentrated areas made the cities the cesspools of pollution. The Industrial Revolution brought an infusion of untreated chemicals and wastes into local streams that served as the water supply. King Edward I of England banned the burning of sea-coal by proclamation in London in 1272, after its smoke became a problem. But the fuel was so common in England that this earliest of names for it was acquired because it could be carted away from some shores by the wheelbarrow.

It was the industrial revolution that gave birth to environmental pollution as we know it today. London also recorded one of the earlier extreme cases of water quality problems with the Great Stink on the Thames of 1858, which led to construction of the London sewerage system soon afterwards. Pollution issues escalated as population growth far exceeded view ability of neighbourhoods to handle their waste problem. Reformers began to demand sewer systems, and clean water.

In 1870, the sanitary conditions in Berlin were among the worst in Europe. August Bebel recalled conditions before a modern sewer system was built in the late 1870s:

- “Waste-water from the houses collected in the gutters running alongside the curbs and emitted a truly fearsome smell. There were no public toilets in the streets or squares. Visitors, especially women, often became desperate when nature called. In the public buildings the sanitary facilities were unbelievably primitive....As a metropolis, Berlin did not emerge from a state of barbarism into civilization until after 1870.”

The primitive conditions were intolerable for a world national capital, and the Imperial German government brought in its scientists, engineers and urban planners to not only solve the deficiencies but to forge Berlin as the world’s model city. A British expert in 1906 concluded that Berlin represented “the most complete application of science, order and method of public life,” adding “it is a marvel of civic administration, the most modern and most perfectly organized city that there is.”

The emergence of great factories and consumption of immense quantities of coal gave rise to unprecedented air pollution and the large volume of industrial chemical discharges added to the growing load of untreated human waste. Chicago and Cincinnati were the first two American cities to enact laws ensuring cleaner air in 1881. Pollution became a major issue in the United States in the early twentieth century, as progressive reformers took issue with air pollution caused by coal burning, water pollution caused by bad sanitation, and street pollution caused by the 3 million horses who worked in American cities in 1900, generating large quantities of urine and manure. As historian Martin Melosi notes, The generation that first saw automobiles replacing the horses saw cars as “miracles of cleanliness.” By the 1940s, however, automobile-caused smog was a major issue in Los Angeles. Other cities followed around the country until early in the 20th century, when the short lived Office of Air Pollution was created under the Department of the Interior. Extreme smog events were experienced by the cities of Los Angeles and Donora, Pennsylvania in the late 1940s, serving as another public reminder. Air pollution would continue to be a problem in England, especially later during the industrial revolution, and extending into the recent past with the Great Smog of 1952.

Awareness of atmospheric pollution spread widely after World War II, with fears triggered by reports of radioactive fallout from atomic warfare and testing. Then a non-nuclear event, The Great Smog of 1952 in London, killed at least 4000 people. This prompted some of the first major modern environmental legislation, The Clean Air Act of 1956.

Pollution began to draw major public attention in the United States between the mid-1950s and early 1970s, when Congress passed the Noise Control Act, the Clean Air Act, the Clean Water Act and the National Environmental Policy Act.

Severe incidents of pollution helped increase consciousness. PCB dumping in the Hudson River resulted in a ban by the EPA on consumption of its fish in 1974. Long-term dioxin contamination at Love Canal starting in 1947 became a national news story in 1978 and led to the Superfund legislation of 1980. The pollution of industrial land gave rise to the name brownfield, a term now common in city planning.

The development of nuclear science introduced radioactive contamination, which can remain lethally radioactive for hundreds of thousands of years. Lake Karachay, named by the Worldwatch Institute as the “most polluted spot” on earth, served as a disposal site for the Soviet Union throughout the 1950s and 1960s. Chelyabinsk, Russia, is considered the “Most polluted place on the planet”.

Nuclear weapons continued to be tested in the Cold War, especially in the earlier stages of their development. The toll on the worst-affected populations and the growth since then in understanding about the critical threat to human health posed by radioactivity has also been a prohibitive complication associated with nuclear power. Though extreme care is practiced in that industry, the potential for disaster suggested by incidents such as those at Three Mile Island and Chernobyl pose a lingering specter

of public mistrust. Worldwide publicity has been intense on those disasters. Widespread support for test ban treaties has ended almost all nuclear testing in the atmosphere.

International catastrophes such as the wreck of the Amoco Cadiz oil tanker off the coast of Brittany in 1978 and the Bhopal disaster in 1984 have demonstrated the universality of such events and the scale on which efforts to address them needed to engage. The borderless nature of atmosphere and oceans inevitably resulted in the implication of pollution on a planetary level with the issue of global warming. Most recently the term persistent organic pollutant (POP) has come to describe a group of chemicals such as PBDEs and PFCs among others. Though their effects remain somewhat less well understood owing to a lack of experimental data, they have been detected in various ecological habitats far removed from industrial activity such as the Arctic, demonstrating diffusion and bioaccumulation after only a relatively brief period of widespread use.

A much more recently discovered problem is the Great Pacific Garbage Patch, a huge concentration of plastics, chemical sludge and other debris which has been collected into a large area of the Pacific Ocean by the North Pacific Gyre. This is a less well known pollution problem than the others described above, but nonetheless has multiple and serious consequences such as increasing wildlife mortality, the spread of invasive species and human ingestion of toxic chemicals. Organizations such as 5 Gyres have researched the pollution and, along with artists like Marina DeBris, are working towards publicizing the issue.

Pollution introduced by light at night is becoming a global problem, more severe in urban centres, but nonetheless contaminating also large territories, far away from towns.

Growing evidence of local and global pollution and an increasingly informed public over time have given rise to environmentalism and the environmental movement, which generally seek to limit human impact on the environment.

Forms of Pollution

The major forms of pollution are listed below along with the particular contaminant relevant to each of them:

- Air pollution: the release of chemicals and particulates into the atmosphere. Common gaseous pollutants include carbon monoxide, sulfur dioxide, chlorofluorocarbons (CFCs) and nitrogen oxides produced by industry and motor vehicles. Photochemical ozone and smog are created as nitrogen oxides and hydrocarbons react to sunlight. Particulate matter, or fine dust is characterized by their micrometre size PM_{10} to $PM_{2.5}$.
- Light pollution: includes light trespass, over-illumination and astronomical interference.
- Littering: the criminal throwing of inappropriate man-made objects, unremoved, onto public and private properties.

- Noise pollution: which encompasses roadway noise, aircraft noise, industrial noise as well as high-intensity sonar.
- Soil contamination occurs when chemicals are released by spill or underground leakage. Among the most significant soil contaminants are hydrocarbons, heavy metals, MTBE, herbicides, pesticides and chlorinated hydrocarbons.
- Radioactive contamination, resulting from 20th century activities in atomic physics, such as nuclear power generation and nuclear weapons research, manufacture and deployment.
- Thermal pollution, is a temperature change in natural water bodies caused by human influence, such as use of water as coolant in a power plant.
- Visual pollution, which can refer to the presence of overhead power lines, motorway billboards, scarred landforms (as from strip mining), open storage of trash, municipal solid waste or space debris.
- Water pollution, by the discharge of wastewater from commercial and industrial waste (intentionally or through spills) into surface waters; discharges of untreated domestic sewage, and chemical contaminants, such as chlorine, from treated sewage; release of waste and contaminants into surface run-off flowing to surface waters (including urban run-off and agricultural run-off, which may contain chemical fertilizers and pesticides); waste disposal and leaching into groundwater; eutrophication and littering.
- Plastic pollution: involves the accumulation of plastic products in the environment that adversely affects wildlife, wildlife habitat, or humans.

Pollutants

A pollutant is a waste material that pollutes air, water or soil. Three factors determine the severity of a pollutant: its chemical nature, the concentration and the persistence.

Cost of Pollution

Pollution has cost. Manufacturing activities that cause air pollution impose health and clean-up costs on the whole society, whereas the neighbours of an individual who chooses to fire-proof his home may benefit from a reduced risk of a fire spreading to their own houses. If external costs exist, such as pollution, the producer may choose to produce more of the product than would be produced if the producer were required to pay all associated environmental costs. Because responsibility or consequence for self-directed action lies partly outside the self, an element of externalization is involved. If there are external benefits, such as in public safety, less of the good may be produced than would be the case if the producer were to receive payment for the external benefits to others.

Sources and Causes

Air pollution comes from both natural and human-made (anthropogenic) sources. However, globally human-made pollutants from combustion, construction, mining,

agriculture and warfare are increasingly significant in the air pollution equation. Motor vehicle emissions are one of the leading causes of air pollution. China, United States, Russia, India Mexico, and Japan are the world leaders in air pollution emissions. Principal stationary pollution sources include chemical plants, coal-fired power plants, oil refineries, petrochemical plants, nuclear waste disposal activity, incinerators, large livestock farms (dairy cows, pigs, poultry, etc.), PVC factories, metals production factories, plastics factories, and other heavy industry. Agricultural air pollution comes from contemporary practices which include clear felling and burning of natural vegetation as well as spraying of pesticides and herbicides

About 400 million metric tons of hazardous wastes are generated each year. The United States alone produces about 250 million metric tons. Americans constitute less than 5% of the world's population, but produce roughly 25% of the world's CO₂, and generate approximately 30% of world's waste. In 2007, China has overtaken the United States as the world's biggest producer of CO₂, while still far behind based on per capita pollution - ranked 78th among the world's nations.

In February 2007, a report by the Intergovernmental Panel on Climate Change (IPCC), representing the work of 2,500 scientists, economists, and policymakers from more than 120 countries, said that humans have been the primary cause of global warming since 1950. Humans have ways to cut greenhouse gas emissions and avoid the consequences of global warming, a major climate report concluded. But to change the climate, the transition from fossil fuels like coal and oil needs to occur within decades, according to the final report this year from the UN's Intergovernmental Panel on Climate Change (IPCC).

Some of the more common soil contaminants are chlorinated hydrocarbons (CFH), heavy metals (such as chromium, cadmium—found in rechargeable batteries, and lead—found in lead paint, aviation fuel and still in some countries, gasoline), MTBE, zinc, arsenic and benzene. In 2001 a series of press reports culminating in a book called *Fateful Harvest* unveiled a widespread practice of recycling industrial byproducts into fertilizer, resulting in the contamination of the soil with various metals. Ordinary municipal landfills are the source of many chemical substances entering the soil environment (and often groundwater), emanating from the wide variety of refuse accepted, especially substances illegally discarded there, or from pre-1970 landfills that may have been subject to little control in the U.S., or EU. There have also been some unusual releases of polychlorinated dibenzodioxins, commonly called *dioxins* for simplicity, such as TCDD.

Pollution can also be the consequence of a natural disaster. For example, hurricanes often involve water contamination from sewage, and petrochemical spills from ruptured boats or automobiles. Larger scale and environmental damage is not uncommon when coastal oil rigs or refineries are involved. Some sources of pollution, such as nuclear power plants or oil tankers, can produce widespread and potentially hazardous releases

when accidents occur. In the case of noise pollution the dominant source class is the motor vehicle, producing about ninety percent of all unwanted noise worldwide.

Effects

Human Health

Adverse air quality can kill many organisms including humans. Ozone pollution can cause respiratory disease, cardiovascular disease, throat inflammation, chest pain, and congestion. Water pollution causes approximately 14,000 deaths per day, mostly due to contamination of drinking water by untreated sewage in developing countries. An estimated 500 million Indians have no access to a proper toilet, Over ten million people in India fell ill with waterborne illnesses in 2013, and 1,535 people died, most of them children. Nearly 500 million Chinese lack access to safe drinking water. A 2010 analysis estimated that 1.2 million people died prematurely each year in China because of air pollution. The WHO estimated in 2007 that air pollution causes half a million deaths per year in India. Studies have estimated that the number of people killed annually in the United States could be over 50,000. Oil spills can cause skin irritations and rashes. Noise pollution induces hearing loss, high blood pressure, stress, and sleep disturbance. Mercury has been linked to developmental deficits in children and neurologic symptoms. Older people are majorly exposed to diseases induced by air pollution. Those with heart or lung disorders are at additional risk. Children and infants are also at serious risk. Lead and other heavy metals have been shown to cause neurological problems. Chemical and radioactive substances can cause cancer and as well as birth defects.

Environment

Pollution has been found to be present widely in the environment. There are a number of effects of this:

- Biomagnification describes situations where toxins (such as heavy metals) may pass through trophic levels, becoming exponentially more concentrated in the process.
- Carbon dioxide emissions cause ocean acidification, the ongoing decrease in the pH of the Earth's oceans as CO₂ becomes dissolved.
- The emission of greenhouse gases leads to global warming which affects ecosystems in many ways.
- Invasive species can out compete native species and reduce biodiversity. Invasive plants can contribute debris and biomolecules (allelopathy) that can alter soil and chemical compositions of an environment, often reducing native species competitiveness.
- Nitrogen oxides are removed from the air by rain and fertilise land which can change the species composition of ecosystems.

- Smog and haze can reduce the amount of sunlight received by plants to carry out photosynthesis and leads to the production of tropospheric ozone which damages plants.
- Soil can become infertile and unsuitable for plants. This will affect other organisms in the food web.
- Sulfur dioxide and nitrogen oxides can cause acid rain which lowers the pH value of soil.

Environmental Health Information

The Toxicology and Environmental Health Information Programme (TEHIP) at the United States National Library of Medicine (NLM) maintains a comprehensive toxicology and environmental health web site that includes access to resources produced by TEHIP and by other government agencies and organizations. This web site includes links to databases, bibliographies, tutorials, and other scientific and consumer-oriented resources. TEHIP also is responsible for the Toxicology Data Network (TOXNET) an integrated system of toxicology and environmental health databases that are available free of charge on the web.

TOXMAP is a Geographic Information System (GIS) that is part of TOXNET. TOXMAP uses maps of the United States to help users visually explore data from the United States Environmental Protection Agency's (EPA) Toxics Release Inventory and Superfund Basic Research Programmes.

Worker Productivity

A number of studies show that pollution has an adverse effect on the productivity of both indoor and outdoor workers.

Regulation and Monitoring

To protect the environment from the adverse effects of pollution, many nations worldwide have enacted legislation to regulate various types of pollution as well as to mitigate the adverse effects of pollution.

Pollution Control

Pollution control is a term used in environmental management. It means the control of emissions and effluents into air, water or soil. Without pollution control, the waste products from overconsumption, heating, agriculture, mining, manufacturing, transportation and other human activities, whether they accumulate or disperse, will degrade the environment. In the hierarchy of controls, pollution prevention and waste minimization are more desirable than pollution control. In the field of land development, low impact development is a similar technique for the prevention of urban run-off.

Practices:

- Recycling
- Reusing
- Waste minimisation
- Mitigating
- Preventing
- Compost

Pollution control devices:

- Air pollution control
 - Thermal oxidizer
- Dust collection systems
 - Baghouses
 - Cyclones
 - Electrostatic precipitators
- Scrubbers
 - Baffle spray scrubber
 - Cyclonic spray scrubber
 - Ejector venturi scrubber
 - Mechanically aided scrubber
 - Spray tower
 - Wet scrubber
- Sewage treatment
 - Sedimentation (Primary treatment)
 - Activated sludge biotreaters (Secondary treatment; also used for industrial wastewater)
 - Aerated lagoons
 - Constructed wetlands (also used for urban run-off)
- Industrial wastewater treatment
 - API oil-water separators
 - Biofilters
 - Dissolved air flotation (DAF)
 - Powdered activated carbon treatment
 - Ultrafiltration
- Vapor recovery systems
- Phytoremediation.

Perspectives

The earliest precursor of pollution generated by life forms would have been a natural function of their existence. The attendant consequences on viability and population levels fell within the sphere of natural selection. These would have included the demise

of a population locally or ultimately, species extinction. Processes that were untenable would have resulted in a new balance brought about by changes and adaptations. At the extremes, for any form of life, consideration of pollution is superseded by that of survival.

For humankind, the factor of technology is a distinguishing and critical consideration, both as an enabler and an additional source of byproducts. Short of survival, human concerns include the range from quality of life to health hazards. Since science holds experimental demonstration to be definitive, modern treatment of toxicity or environmental harm involves defining a level at which an effect is observable. Common examples of fields where practical measurement is crucial include automobile emissions control, industrial exposure (*e.g.* Occupational Safety and Health Administration (OSHA) PELs), toxicology (*e.g.* LD₅₀), and medicine (*e.g.* medication and radiation doses).

“The solution to pollution is dilution”, is a dictum which summarizes a traditional approach to pollution management whereby sufficiently diluted pollution is not harmful. It is well-suited to some other modern, locally scoped applications such as laboratory safety procedure and hazardous material release emergency management. But it assumes that the dilutant is in virtually unlimited supply for the application or that resulting dilutions are acceptable in all cases.

Such simple treatment for environmental pollution on a wider scale might have had greater merit in earlier centuries when physical survival was often the highest imperative, human population and densities were lower, technologies were simpler and their byproducts more benign. But these are often no longer the case. Furthermore, advances have enabled measurement of concentrations not possible before. The use of statistical methods in evaluating outcomes has given currency to the principle of probable harm in cases where assessment is warranted but resorting to deterministic models is impractical or infeasible. In addition, consideration of the environment beyond direct impact on human beings has gained prominence.

Yet in the absence of a superseding principle, this older approach predominates practices throughout the world. It is the basis by which to gauge concentrations of effluent for legal release, exceeding which penalties are assessed or restrictions applied. One such superseding principle is contained in modern hazardous waste laws in developed countries, as the process of diluting hazardous waste to make it non-hazardous is usually a regulated treatment process. Migration from pollution dilution to elimination in many cases can be confronted by challenging economical and technological barriers.

Greenhouse Gases and Global Warming

Carbon dioxide, while vital for photosynthesis, is sometimes referred to as pollution, because raised levels of the gas in the atmosphere are affecting the Earth's climate. Disruption of the environment can also highlight the connection between areas of pollution that would normally be classified separately, such as those of water and air. Recent studies have investigated the potential for long-term rising levels of atmospheric

carbon dioxide to cause slight but critical increases in the acidity of ocean waters, and the possible effects of this on marine ecosystems.

Most Polluting Industries

The Pure Earth, an international non-for-profit organization dedicated to eliminating life-threatening pollution in the developing world, issues an annual list of some of the world's most polluting industries.

- Lead-Acid Battery Recycling
- Industrial Mining and Ore Processing
- Lead Smelting
- Tannery Operations
- Artisanal Small-Scale Gold Mining
- Industrial/Municipal Dumpsites
- Industrial Estates
- Chemical Manufacturing
- Product Manufacturing
- Dye Industry

World's Worst Polluted Places

The Pure Earth issues an annual list of some of the world's worst polluted places.

- Agbogbloshie, Ghana
- Chernobyl*, Ukraine
- Citarum River, Indonesia
- Dzershinsk*, Russia
- Hazaribagh, Bangladesh
- Kabwe*, Zambia
- Kalimantan, Indonesia
- Matanza Riachuelo, Argentina
- Niger River Delta, Nigeria
- Norilsk*, Russia.

Educational Philosophy and Thoughts

Philosophy is a search for a general understanding of values and reality by chiefly speculative rather than observational means. It signifies a natural and necessary urge in human beings to know themselves and the world in which they live and move and have their being.

Western philosophy remained more or less true to the etymological meaning of philosophy in being essentially an intellectual quest for truth. Hindu philosophy is intensely spiritual and has always emphasized the need for practical realisation of Truth. Philosophy is a comprehensive system of ideas about human nature and the nature of the reality we live in.

It is a guide for living, because the issues it addresses are basic and pervasive, determining the course we take in life and how we treat other people. Hence we can say that all the aspects of human life are influenced and governed by the philosophical consideration. As a field of study philosophy is one of the oldest disciplines. It is considered as a mother of all the sciences. In fact it is at the root of all knowledge. Education has also drawn its material from different philosophical bases.

Education, like philosophy is also closely related to human life. Therefore, being an important life activity education is also greatly influenced by philosophy. Various fields of philosophy like the political philosophy, social philosophy and economic philosophy have great influence on the various aspects of education like educational procedures, processes, policies, planning and its implementation, from both the theoretical and practical aspects.

SUBJECT MATTER

MEANING OF PHILOSOPHY

The word philosophy literally means love of wisdom; It is derived from two Greek words *i.e.* 'phileo' (love) and 'Sophia' (wisdom). This tells us something about the nature of philosophy, but not much, because many disciplines seek wisdom. Since times immemorial there have been various pursuits for unfolding the mystery of the universe, birth and death, sorrow and joy. Various ages have produced different thoughts throwing light upon the mystic region. The ultimate truth is yet to be found out. This eternal quest for truth 'lends the origin of philosophy. A love of wisdom is the essence for any philosophy investigation.

On the standard way of telling the story, humanity's first systematic enquiries took place within a mythological or religious framework: wisdom ultimately was to be derived from sacred traditions and from individuals thought to possess privileged access to a supernatural realm, whose own access to wisdom, in turn, generally was not questioned. However, starting in the sixth century BCE, there appeared in ancient Greece a series of thinkers whose enquiries were comparatively secular. Presumably, these thinkers conducted their enquiries through reason and observation, rather than through tradition or revelation. These thinkers were the first philosophers. Although this picture is admittedly simplistic, the basic distinction has stuck: philosophy in its most primeval form is considered nothing less than secular enquiry itself.

The subject of philosophical enquiry is the reality itself. There are different schools of philosophy depending on the answers they seek to the question of reality. It is the search for understanding of man, nature and the universe. There are different branches of philosophy-Epistemology, Metaphysics, etc. There are different fields of philosophy such as educational philosophy, social philosophy, political philosophy, economic philosophy etc. There are also different philosophical approaches such as idealism, naturalism, pragmatism, materialism, and so on.

MEANING OF PHILOSOPHY AND EDUCATION

Meaning of Education

Etymologically, the word education is derived from educare (Latin) "bring up", which is related to educere "bring out", "bring forth what is within", "bring out potential" and ducere, "to lead". Education in the largest sense is any act or experience that has a formative effect on the mind, character or physical ability of an individual. In its technical sense, education is the process by which society deliberately transmits its accumulated knowledge, skills and values from one generation to another.

Webster defines education as the process of educating or teaching (now that's really useful, isn't it?) Educate is further defined as "to develop the knowledge, skill, or character

of..." Thus, from these definitions, we might assume that the purpose of education is to develop the knowledge, skill, or character of students. In ancient Greece, Socrates argued that education was about drawing out what was already within the student. (As many of you know, the word education comes from the Latin *e-ducere* meaning "to lead out.") At the same time, the Sophists, a group of itinerant teachers, promised to give students the necessary knowledge and skills to gain positions with the city-state. Thus we see that there are different views and understandings of the meaning of the term education. In the modern times it has acquired two different shades of meaning namely:

1. An institutional instruction, given to students in school colleges formally; and
2. A pedagogical science, studied by the student of education.

The words of Adam education is the dynamic side of philosophy. Philosophy takes into its orbit, all the dimensions of human life. Similarly education also reflects the multifaceted nature of human life. Therefore, education is closely related to various aspects of human life and environment. Hence, the term education has a wide connotation. It is difficult to define education by single definition. Philosophers and thinkers from Socrates to Dewey in west and a host of Indian philosophers have attempted to define education. However education can be understood as the deliberate and systematic influence exerted by a mature through instruction, and discipline. It means the harmonious development of all the powers of the human being; physical social, intellectual, aesthetic and spiritual. The essential elements in the educative process are a creative mind, a well integrated self, socially useful purposes and experience related to the interests of the individual, needs and abilities of the individual as a of a social group.

In the historical development of man, education has been the right of a privileged few. It is only in recent centuries that education has come to be recognised as a human right. All have equal right to be educated as education has become *sine qua non* of civilization. In order to understand the concept of Philosophy of education it is necessary to first understand the meaning of the two terms; Philosophy and Education.

CONCEPT OF PHILOSOPHY OF EDUCATION

All human societies, past and present, have had a vested interest in education; and some wits have claimed that teaching (at its best an educational activity) is the second oldest profession. While not all societies channel sufficient resources into support for educational activities and institutions, all at the very least acknowledge their centrality- and for good reasons.

For one thing, it is obvious that children are born illiterate and innumerate, and ignorant of the norms and cultural achievements of the community or society into which they have been thrust; but with the help of professional teachers and the dedicated amateurs in their families and immediate environs (and with the aid, too, of educational resources made available through the media and nowadays the internet), within a few

years they can read, write, calculate, and act (at least often) in culturally-appropriate ways. Some learn these skills with more facility than others, and so education also serves as a social-sorting mechanism and undoubtedly has enormous impact on the economic fate of the individual. Put more abstractly, at its best education equips individuals with the skills and substantive knowledge that allows them to define and to pursue their own goals, and also allows them to participate in the life of their community as full-fledged, autonomous citizens.

Equips individuals with the skills and substantive knowledge that allows them to define and to pursue their own goals, and also allows them to participate in the life of their community as full-fledged, autonomous citizens. But this is to cast matters in very individualistic terms, and it is fruitful also to take a societal perspective, where the picture changes somewhat. It emerges that in pluralistic societies such as the Western democracies there are some groups that do not wholeheartedly support the development of autonomous individuals, for such folk can weaken a group from within by thinking for themselves and challenging communal norms and beliefs; from the point of view of groups whose survival is thus threatened, formal, state-provided education is not necessarily a good thing. But in other ways even these groups depend for their continuing survival on educational processes, as do the larger societies and nation-states of which they are part; for as John Dewey put it in the opening chapter of his classic work *Democracy and Education* (1916), in its broadest sense education is the means of the "social continuity of life".

Dewey pointed out that the "primary ineluctable facts of the birth and death of each one of the constituent members in a social group" make education a necessity, for despite this biological inevitability "the life of the group goes on" (Dewey, 3). The great social importance of education is underscored, too, by the fact that when a society is shaken by a crisis, this often is taken as a sign of educational breakdown; education, and educators, become scapegoats. It is not surprising that such an important social domain has attracted the attention of philosophers for thousands of years, especially as there are complex issues aplenty that have great philosophical interest. Abstractly, at its best education equips individuals with the skills and substantive knowledge that allows them to define and to pursue their own goals, and also allows them to participate in the life of their community as full-fledged, autonomous citizens.

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It is not surprising that such an important social domain has attracted the attention of philosophers for thousands of years, especially as there are complex issues aplenty that have great philosophical interest. The following are some issues that philosophers have deeply thought about and philosophy is still in the process of answering these questions.

Is Education as transmission of knowledge versus education as the fostering of enquiry and reasoning skills that are conducive to the development of autonomy (which, roughly, is the tension between education as conservative and education as progressive, and also is closely related to differing views about human "perfectibility"-issues that historically have been raised in the debate over the aims of education); the question of what this knowledge, and what these skills, ought to be-part of the domain of philosophy of the curriculum; the questions of how learning is possible, and what is it to have learned something-two sets of issues that relate to the question of the capacities and potentialities that are present at birth, and also to the process (and stages) of human development and to what degree this process is flexible and hence can be influenced or manipulated; the tension between liberal education and vocational education, and the overlapping issue of which should be given priority-education for personal development or education for citizenship (and the issue of whether or not this is a false dichotomy); the differences (if any) between education and enculturation; the distinction between educating versus teaching versus training versus indoctrination; the relation between education and maintenance of the class structure of society, and the issue of whether different classes or cultural groups can-justly-be given educational programmes that differ in content or in aims; the issue of whether the rights of children, parents, and socio-cultural or ethnic groups, conflict-and if they do, the question of whose rights should be dominant; the question as to whether or not all children have a right to state-provided education, and if so, should this education respect the beliefs and customs of all groups and how on earth would this be accomplished; and a set of complex issues about the relation between education and social reform, centring upon whether education is essentially conservative, or whether it can be an (or, the) agent of social change.

It is here that philosophy of education plays an important role in providing direction to education on the following issues as well as providing a theory of knowledge for education to work upon. Philosophy of education is essentially a method of approaching

educational experience rather than a body of conclusions. It is the specific method which makes it philosophical. Philosophical method is critical, comprehensive and synthetic.

JOHN LOCKE

John Locke (August 1632 – 28 October 1704), widely known as the *Father of Liberalism*, was an English philosopher and physician regarded as one of the most influential of Enlightenment thinkers. Considered one of the first of the British empiricists, following the tradition of Francis Bacon, he is equally important to social contract theory. His work had a great impact upon the development of epistemology and political philosophy. His writings influenced Voltaire and Rousseau, many Scottish Enlightenment thinkers, as well as the American revolutionaries. His contributions to classical republicanism and liberal theory are reflected in the American Declaration of Independence. Locke's theory of mind is often cited as the origin of modern conceptions of identity and the self, figuring prominently in the work of later philosophers such as Hume, Rousseau and Kant. Locke was the first to define the self through a continuity of *consciousness*. He postulated that the mind was a blank slate or *tabula rasa*. Contrary to pre-existing Cartesian philosophy, he maintained that we are born without innate ideas, and that knowledge is instead determined only by experience derived from sense perception.

EDUCATIONAL IMPLICATIONS OF LOCKE'S PHILOSOPHY

The following are some of the central points in Locke's educational theory:

- *Purpose Of Education:* The purpose of education is to produce an individual with a sound mind in a sound body, so as to better serve his country.
- *Different Curriculum:* Locke thought that the content of education ought to depend upon one's station in life. The common man only required moral, social and vocational knowledge. He could do quite well with the Bible and a highly developed vocational skill that would serve to support him in life and offer social service to others. However, the education of gentlemen ought to be of the very highest quality. The gentleman must serve his country in a position of leadership.
- *The Education Of The Gentleman:* Locke believed that the gentleman must have a thorough knowledge of his own language. The schools of the Puritans in England broke with tradition completely. They sought to educate one for the society in which he would live. The schools were called, therefore, schools of social realism. Locke, in keeping with Milton and other Puritans, held that the content of the curriculum must serve some practical end. He recommended the introduction of contemporary foreign languages, history, geography, economics,

math and science. Locke proposed the following for the education of the gentleman:

- a. *Moral Training*: All Christians must learn to live virtuously.
- b. *Good Breeding*: The gentleman must develop the poise, control and outward behaviour of excellent manners. Education must aim, therefore, at developing correct social skills.
- c. *Wisdom*: The gentleman ought to be able to apply intellectual and moral knowledge in governing his practical affairs.
- d. *Useful Knowledge*: The gentleman must receive education which will lead to a successful life in the practical affairs of the society, as well as that which leads to the satisfaction derived from scholarship and good books.

JEAN-JACQUES ROUSSEAU

Jean-Jacques Rousseau (28 June 1712 – 2 July 1778) was a major Genevan philosopher, writer, and composer of 18th-century Romanticism. His political philosophy heavily influenced the French Revolution, as well as the overall development of modern political, sociological and educational thought. His novel, *Émile: or, On Education* is a seminal treatise on the education of the whole person for citizenship. His sentimental novel, *Julie, ou la nouvelle Héloïse*, was of great importance to the development of pre-romanticism and romanticism in fiction. Rousseau's autobiographical writings: his *Confessions*, which initiated the modern autobiography, and his *Reveries of a Solitary Walker* were among the pre-eminent examples of the late 18th-century movement known as the *Age of Sensibility*, featuring an increasing focus on subjectivity and introspection that has characterized the modern age.

His *Discourse on the Origin of Inequality* and his *On the Social Contract* are cornerstones in modern political and social thought and make a strong case for democratic government and social empowerment. Rousseau was also a successful composer and made important contributions to music as a theorist. During the period of the French Revolution, Rousseau was the most popular of the *philosophes* among members of the Jacobin Club.

He was interred as a national hero in the Panthéon in Paris, in 1794, 16 years after his death.

ON EDUCATION

The focus of *Émile* is upon the individual tuition of a boy/young man in line with the principles of 'natural education'. This focus tends to be what is taken up by later commentators, yet Rousseau's concern with the individual is balanced in some of his other writing with the need for public or national education. In *A Discourse on Political Economy* and *Considerations for the Government of Poland*

we get a picture of public education undertaken in the interests of the community as a whole.

- From the first moment of life, men ought to begin learning to deserve to live; and, as at the instant of birth we partake of the rights of citizenship, that instant ought to be the beginning of the exercise of our duty. If there are laws for the age of maturity, there ought to be laws for infancy, teaching obedience to others: and as the reason of each man is not left to be the sole arbiter of his duties, government ought the less indiscriminately to abandon to the intelligence and prejudices of fathers the education of their children, as that education is of still greater importance to the State than to the fathers: for, according to the course of nature, the death of the father often deprives him of the final fruits of education; but his country sooner or later perceives its effects. Families dissolve but the State remains. (Rousseau 1755: 148-9).

‘Make the citizen good by training’, Jean-Jacques Rousseau writes, ‘and everything else will follow’. In *Émile* Rousseau drew on thinkers that had preceded him - for example, John Locke on teaching - but he was able to pull together strands into a coherent and comprehensive system - and by using the medium of the novel he was able to dramatize his ideas and reach a very wide audience. He made, it can be argued, the first comprehensive attempt to describe a system of education according to what he saw as ‘nature’. It certainly stresses wholeness and harmony, and a concern for the person of the learner. Central to this was the idea that it was possible to preserve the ‘original perfect nature’ of the child, ‘by means of the careful control of his education and environment, based on an analysis of the different physical and psychological stages through which he passed from birth to maturity’. This was a fundamental point. Rousseau argued that the momentum for learning was provided by the growth of the person —and that what the educator needed to do was to facilitate opportunities for learning. The focus on the environment, on the need to develop opportunities for new experiences and reflection, and on the dynamic provided by each person’s development remain very powerful ideas. We’ll quickly list some of the key elements that we still see in his writing:

- A view of children as very different to adults - as innocent, vulnerable, slow to mature - and entitled to freedom and happiness. In other words, children are naturally good.
- The idea that people develop through various stages - and that different forms of education may be appropriate to each.
- A guiding principle that what is to be learned should be determined by an understanding of the person’s nature at each stage of their development.
- An appreciation that individuals vary within stages - and that education must as a result be individualized. ‘Every mind has its own form’

- Each and every child has some fundamental impulse to activity. Restlessness in time being replaced by curiosity; mental activity being a direct development of bodily activity.
- The power of the environment in determining the success of educational encounters. It was crucial - as Dewey also recognized - that educators attend to the environment. The more they were able to control it - the more effective would be the education.
- The controlling function of the educator - The child, Rousseau argues, should remain in complete ignorance of those ideas which are beyond his/her grasp.
- The importance of developing ideas for ourselves, to make sense of the world in our own way. People must be encouraged to reason their way through to their own conclusions - they should not rely on the authority of the teacher. Thus, instead of being taught other people's ideas, Émile is encouraged to draw his own conclusions from his own experience. What we know today as 'discovery learning' One example, Rousseau gives is of Émile breaking a window - only to find he gets cold because it is left unrepaired.
- A concern for both public and individual education.

We could go on - all we want to do is to establish what a far reaching gift Rousseau gave. We may well disagree with various aspects of his scheme- but there can be no denying his impact then – and now. It may well be, as Darling has argued, that the history of child-centred educational theory is a series of footnotes to Rousseau.

DEFINITION OF EDUCATION BY REDDEN

Redden define education as the deliberate and systematic influence exerted by the adult person upon the not-yet adult, through instruction and discipline for the harmonious becoming of all the potentialities of the human being, namely:

- Physical,
- Social,
- Intellectual,
- Moral,
- Aesthetic, and
- Spiritual.

JOHN DEWEY

John Dewey (October 20, 1859 – June 1, 1952) was an American philosopher, psychologist and educational reformer whose ideas have been influential in education and social reform. Dewey was an important early developer of the philosophy of pragmatism and one of the founders of functional psychology. He was a major representative of the progressive and progressive populist philosophies of schooling during the first half of the 20th century in the USA. Although Dewey is known best for

his publications concerning education, he also wrote about many other topics, including experience, nature, art, logic, enquiry, democracy, and ethics. In his advocacy of democracy, Dewey considered two fundamental elements—schools and civil society—as being major topics needing attention and reconstruction to encourage experimental intelligence and plurality. Dewey asserted that complete democracy was to be obtained not just by extending voting rights but also by ensuring that there exists a fully formed public opinion, accomplished by effective communication among citizens, experts, and politicians, with the latter being accountable for the policies they adopt.

ON EDUCATION

Dewey's educational theories were presented in *My Pedagogic Creed* (1897), *The School and Society* (1900), *The Child and the Curriculum* (1902), *Democracy and Education* (1916) and *Experience and Education* (1938). Throughout these writings, several recurrent themes ring true; Dewey continually argues that education and learning are social and interactive processes, and thus the school itself is a social institution through which social reform can and should take place. In addition, he believed that students thrive in an environment where they are allowed to experience and interact with the curriculum, and all students should have the opportunity to take part in their own learning. The ideas of democracy and social reform are continually discussed in Dewey's writings on education.

Dewey makes a strong case for the importance of education not only as a place to gain content knowledge, but also as a place to learn how to live. In his eyes, the purpose of education should not revolve around the acquisition of a pre-determined set of skills, but rather the realization of one's full potential and the ability to use those skills for the greater good. He notes that "to prepare him for the future life means to give him command of himself; it means so to train him that he will have the full and ready use of all his capacities". In addition to helping students realize their full potential, Dewey goes on to acknowledge that education and schooling are instrumental in creating social change and reform. He notes that "education is a regulation of the process of coming to share in the social consciousness; and that the adjustment of individual activity on the basis of this social consciousness is the only sure method of social reconstruction". In addition to his ideas regarding what education is and what effect it should have on society, Dewey also had specific notions regarding how education should take place within the classroom. In *The Child and the Curriculum*, Dewey discusses two major conflicting schools of thought regarding educational pedagogy. The first is centered on the curriculum and focuses almost solely on the subject matter to be taught. Dewey argues that the major flaw in this methodology is the inactivity of the student; within this particular framework, "the child is simply the immature being who is to be matured; he is the superficial being who is to be deepened". He argues that in order for education to be most effective, content must be presented in a way that allows the student to relate

the information to prior experiences, thus deepening the connection with this new knowledge. At the same time, Dewey was alarmed by many of the “child-centered” excesses of educational-school pedagogues who claimed to be his followers, and he argued that too much reliance on the child could be equally detrimental to the learning process. In this second school of thought, “we must take our stand with the child and our departure from him. It is he and not the subject-matter which determines both quality and quantity of learning”. According to Dewey, the potential flaw in this line of thinking is that it minimizes the importance of the content as well as the role of the teacher. In order to rectify this dilemma, Dewey advocated for an educational structure that strikes a balance between delivering knowledge while also taking into account the interests and experiences of the student. He notes that “the child and the curriculum are simply two limits which define a single process. Just as two points define a straight line, so the present standpoint of the child and the facts and truths of studies define instruction”. It is through this reasoning that Dewey became one of the most famous proponents of hands-on learning or experiential education, which is related to, but not synonymous with experiential learning. He argued that “if knowledge comes from the impressions made upon us by natural objects, it is impossible to procure knowledge without the use of objects which impress the mind”. Dewey’s ideas went on to influence many other influential experiential models and advocates. Many researchers even credit him with the influence of Project Based Learning which places students in the active role of researchers. Dewey not only re-imagined the way that the learning process should take place, but also the role that the teacher should play within that process. According to Dewey, the teacher should not be one to stand at the front of the room doling out bits of information to be absorbed by passive students. Instead, the teacher’s role should be that of facilitator and guide. As Dewey explains it:

- The teacher is not in the school to impose certain ideas or to form certain habits in the child, but is there as a member of the community to select the influences which shall affect the child and to assist him in properly responding to these influences.

Thus the teacher becomes a partner in the learning process, guiding students to independently discover meaning within the subject area. This philosophy has become an increasingly popular idea within present-day teacher preparatory programmes. As well as his very active and direct involvement in setting up educational institutions such as the University of Chicago Laboratory Schools and The New School for Social Research, many of Dewey’s ideas influenced the founding of Bennington College in Vermont, where he served on the Board of Trustees. Dewey’s works and philosophy also held great influence in the creation of the short-lived Black Mountain College in North Carolina, an experimental college focused on interdisciplinary study, and whose faculty included Buckminster Fuller, Willem de Kooning, Charles Olson, Franz Kline, Robert Duncan, and Robert Creeley, among others. Black Mountain College was the

locus of the “Black Mountain Poets” a group of avant-garde poets closely linked with the Beat Generation and the San Francisco Renaissance.

GANDHI

The real difficulty is that people have no idea of what education truly is. We assess the value of education in the same manner as we assess the value of land or of shares in the stock-exchange market. We want to provide only such education as would enable the student to earn more. We hardly give any thought to the improvement of the character of the educated. The girls, we say, do not have to earn; so why should they be educated? As long as such ideas persist there is no hope of our ever knowing the true value of education. In a piece published some years ago, Krishna Kumar, Professor of Education at Delhi University, wrote that ‘no one rejected colonial education as sharply and as completely as Gandhi did, nor did anyone else put forward an alternative as radical as the one he proposed’. Gandhi’s critique of Western, particularly English, education was part of his critique of Western civilization as a whole. There is a story that, on arriving in Britain after he had become famous, someone asked him the question: ‘Mr Gandhi, what do you think of civilization in England?’ to which he replied ‘I think that it would be something worth trying!’

ON EDUCATION

Given Gandhi’s values and his vision of what constituted a truly civilized and free India, it was not surprising that he developed firm views on education. Education not only moulds the new generation, but reflects a society’s fundamental assumptions about itself and the individuals which compose it. His experience in South Africa not only changed his outlook on politics but also helped him to see the role education played in that struggle. He was aware that he had been a beneficiary of Western education and for a number of years while he was in South Africa he still tried to persuade Indians to take advantage of it. However, it was not until the early years of this century, when he was in his middle thirties, that he became so opposed to English education that he could write about ‘the rottenness of this education’ and that ‘to give millions a knowledge of English is to enslave them.. that, by receiving English education, we have enslaved the nation’. He was enraged that he had to speak of Home Rule or Independence in what was clearly a foreign tongue, that he could not practice in court in his mother tongue, that all official documents were in English as were all the best newspapers and that education was carried out in English for the chosen few. He did not blame the colonial powers for this. He saw that it was quite logical that they would want an elite of native Indians to become like their rulers in both manners and values. In this way, the Empire could be consolidated. Gandhi blamed his fellow Indians for accepting the situation. Later in his life he was to declare that ‘real freedom will come only when we free ourselves of the

domination of Western education, Western culture and Western way of living which have been ingrained in us... Emancipation from this culture would mean real freedom for us'. As we have seen, Gandhi had not only rejected colonial education but also put forward a radical alternative. So what was this alternative? What was so radical about it? First of all, I need to say a word about Gandhi's attitude to industrialization. He was, in fact, absolutely opposed to modern machinery.

In his collected works, he refers to machinery as having impoverished India, that it was difficult to measure the harm that Manchester had done to them by producing machine-made cloth which, in turn, ruined the internal market for locally produced handwoven goods. Typically of Gandhi, however, he does not blame Manchester or the mill owners. 'How can Manchester be blamed?' he writes. 'We wore Manchester cloth and this is why Manchester wove it'. However, he notes that where cloth mills were not introduced in India, in places such as Bengal, the original hand-weaving occupation was thriving. Where they did have mills *e.g.* in Bombay, he felt that the workers there had become slaves. He was shocked by the conditions of the women working in the mills of Bombay and made the point that before they were introduced these women were not starving.

He maintained that 'if the machinery craze grows in our country, it will become an unhappy land'. What he wanted was for Indians to boycott *all* machine-made goods not just cloth. He was quite clear when he asked the question 'What did India do before these articles were introduced?' and then answered his own question by stating 'Precisely the same should be done today. As long as we cannot make pins without machinery, so long will we do without them. The tinsel splendour of glassware we will have nothing to do with, and we will make wicks, as of old, with home-grown cotton and use hand-made earthen saucers or lamps. So doing, we shall save our eyes and money and support *swadeshi* and so shall we attain Home Rule'. Within this context of the need for a machine-less society, Gandhi developed his ideas on education. The core of his proposal was the introduction of productive handicrafts in the school curriculum. The idea was not simply to introduce handicrafts as a compulsory school subject, but to make the learning of a craft the centrepiece of the entire teaching programme. It implied a radical restructuring of the sociology of school knowledge in India, where productive handicrafts had been associated with the lowest groups in the hierarchy of the caste system. Knowledge of the production processes involved in crafts, such as spinning, weaving, leather-work, pottery, metal-work, basket-making and bookbinding, had been the monopoly of specific caste groups in the lowest stratum of the traditional social hierarchy.

Many of them belonged to the category of 'untouchables'. India's own tradition of education as well as the colonial education system had emphasized skills such as literacy and acquisition of knowledge of which the upper castes had a monopoly. Gandhi's proposal intended to stand the education system on its head. The social philosophy and the curriculum of what he called 'basic education' thus favoured the child belonging to

the lowest stratum of society. in such a way it implied a programme of social transformation. It sought to alter the symbolic meaning of 'education' and to change the established structure of opportunities for education. Why Gandhi proposed the introduction of productive handicrafts into the school system was not really as outrageous as may appear. What he really wanted was for the schools to be self-supporting, as far as possible.

There were two reasons for this. Firstly, a poor society such as India simply could not afford to provide education for all children unless the schools could generate resources from within. Secondly, the more financially independent the schools were, the more politically independent they could be. What Gandhi wanted to avoid was dependence on the state which he felt would mean interference from the centre. Above all else, Gandhi valued self-sufficiency and autonomy. These were vital for his vision of an independent India made up of autonomous village communities to survive. It was the combination of *swaraj* and *swadeshi* related to the education system. A state system of education within an independent India would have been a complete contradiction as far as Gandhi was concerned.

He was also of the opinion that manual work should not be seen as something inferior to mental work. He felt that the work of the craftsman or labourer should be the ideal model for the 'good life'. Schools which were based around productive work where that work was for the benefit of all were, therefore, carrying out education of the whole person - mind, body and spirit. The right to autonomy that Gandhi's educational plan assigns to the teacher in the context of the school's daily curriculum is consistent with the libertarian principles that he shared with Tolstoy. Gandhi wanted to free the Indian teacher from interference from outside, particularly government or state bureaucracy. Under colonial rule, the teacher had a prescribed job to do that was based on what the authorities wanted the children to learn. Textbooks were mandatory so that Gandhi found that 'the living word of the teacher has very little value. A teacher who teaches from textbooks does not impart originality to his pupils'.

Gandhi's plan, on the other hand, implied the end of the teacher's subservience to the prescribed textbook and the curriculum. It presented a concept of learning that simply could not be fully implemented with the help of textbooks. Of equal, if not more importance, was the freedom it gave the teacher in matters of curriculum. It denied the state the power to decide what teachers taught and what they did in the classroom. It gave autonomy to the teacher but it was, above all, a libertarian approach to schooling that transferred power from the state to the village. Gandhi's basic education was, therefore, an embodiment of his perception of an ideal society consisting of small, self-reliant communities with his ideal citizen being an industrious, self-respecting and generous individual living in a small co-operative community. For informal educators, we can draw out a number of useful pointers. First, Gandhi's insistence on autonomy and self-regulation is reflected in the ethos of informal education. Gandhi's conception

of basic education was concerned with learning that was generated within everyday life which is the basis on which informal educators work. It was also an education focused on the individual but reliant on co-operation between individuals. There is also a familiar picture of the relationships between educators and students/learners:

- A teacher who establishes rapport with the taught, becomes one with them, learns more from them than he teaches them. He who learns nothing from his disciples is, in my opinion, worthless. Whenever I talk with someone I learn from him. I take from him more than I give him. In this way, a true teacher regards himself as a student of his students. If you will teach your pupils with this attitude, you will benefit much from them.

Lastly, it was an education that aimed at educating the whole person, rather than concentrating on one aspect. It was a highly moral activity.

ZAKIR HUSSAIN

A graceful stylist, a sensitive soul vibrant with deep emotions, practicing educationist of long standing, Dr. Hussain is by far one of the best representatives of a secular, free and civilized India. Known for his devotion and sacrifice to the educational cause he is one of the chief exponents of Basic Education. It is a great privilege for the people of our country that two of her most eminent educationists, Dr. Radhakrishnan and Dr. Zakir Hussain, have functioned as presidents of our country. By virtue of their office they have dignified the teaching profession and given a new leadership and direction in national practice.

DR. HUSSAIN'S VIEWS ON EDUCATION

According to Dr. Zakir Hussain education in this country is sadly handicapped. He considers education as the very life sustaining sap of a cultured society. The problems for this state of affairs are as follows:

1. Sitting for long hours silently in the classroom receiving lessons in complete passivity in a manner, which is most un-child like conformity. This system kills the natural ardor the child once had and transforms him into a listless non-reaching entity.
2. Scarcity of good teachers. What is available in the system are teachers with cruelty, harshness and strict regimentation.
3. Another source of indiscriminate hardship is the heavy and meaningless curriculum.
4. The existing educational system enables the democratic educator to dispose of the mistaken notion that education must be the shaping of the education according to a given generic type, according to a ready-made educational ideal with a sharp delineated content. He believes that education is not the

process of pressing into shape but a letting loose and setting free which respects the unique and specific individuality of the education. The success of Indian education depends on the ideas and principles and how its evolution helps in the growth and development of the democratic way of life, on how it provides for the full growth and development of individuality, on how it harnesses harmoniously developed individuality to social ends, and how it masters the mysteries of selflessness.

To him the basic principle of education in a democracy should be reverence to the individuality of the child, the child who is to grow into the citizen through his intelligent and willing participation in education. For democracy is nothing but the full discharge of duties to himself and to the society by every citizen. This is possible only if the competencies are discovered and developed to their fullest by education. Regarding the process of education, he compares it with the growing development of human body, from its embryonic beginnings grows and develops to its full stature by means of agreeable, assimilable food, movement and exercise, in accordance with physical and chemical laws, so does the mind grow and develop from its original disposition to its full evolutionary cultivation by means of mental food and mental exercise according to the laws of mental growth. This development of individual human mind must start from infancy. He firmly believed that the process of education is a continuing one in which the journey is as important as the destination. Dr. Hussain expresses his deep concern for the fact that more often than not the means become the ends in due course. It appears to him that schools are the creation of an evil mind; otherwise they could not have been what they actually are. He says, "Schooling is not equated with learning the three R's but it is on the other hand, a gradual initiation to life". He would like schools to be familiar with the society, family and individuals. He believed in having and acquiring only that knowledge which is functional and in support of this he cites an English educator:- "knowledge is idle in a community if it becomes the private possession of an esoteric coterie".

He wanted education to provide effective leadership in a world of change. Dr. Hussain's primary objective is to provide leadership to community, which gives sustenance to its academic organ. His views on university education are very clear. He considers 'work' as the main instrument of education. He wants education to be life long. It should be relevant and provide effective leadership to the community to which they belong to. This is possible only by learning to respect 'work' and treat it as an instrument of educating mind so that universities could repay their debt to the society. In this connection Ramaswamy Iyer observes that the principal purpose of university education is to enable one to serve the country in its variegated scientific and technological necessities. He dislikes the peculiar reaction to a problem by an Indian, which he says, is 'evasion'. No matter a university has to face it must never evade an issue but face it boldly and squarely. A true university should possess a passion for excellence. He has reservations on the people having patience especially with poverty,

dirt, disease and incompetence. He says, unless patience is combined with diligence, it is a crime. If a university discharges its obligations well, there is no earthly reason why a new world of which Dr. Hussain so fervently and sincerely dreams of may not come to pass. Dr. Hussain has his clear view on the long-standing controversy between 'science' and 'humanities'. He believes that the advance modern world has made in the recent past is entirely due to the application of scientific knowledge and its application to social and political problems that man is faced with. He laments that the specialization in each field of technology has reached to such an extent that the jargon of is hardly understood by those in the other field of specialization. This phenomenon is rightly described by C. P. Snow, "never was the exclusion of one branch of knowledge from another so pronounced, and neither was one faculty so developed at the expense of the other as we find today". Dr. Hussain concludes that humanities and science are not something mutually contradictory but complimentary. One should realize the fact that science is devoid of values especially moral and ethical values. He points out that science is a system of philosophy without ethics. Science devoid of ethical judgement becomes an ally of everyone – of the good as well as the bad and is of service in changing the world into a paradise or reducing it to a veritable hell. Regarding the importance of education, he said in his speech after being sworn in as the President of India in the following words. "I maintain that education is a prime instrument of national purpose and that the quality of its education is inseparably involved in the quality of the nation".

Dr. Hussain Laid Stress on the Following aims of Education:

- Education should develop a sense of common national ethos.
- Education should develop higher values of life.
- Education should develop qualities of citizenship.
- Education should also derive inspiration from traditional knowledge and real work experience.
- Education should develop positive attitude.
- Education should develop a sense of social responsibility.
- Education should develop vocational efficiency.

His views on education and politics are that education is the master and politics is its servant. About education and morality he says it is necessary to combine power with morality as well as with science and technology. Regarding education, science and technology his views are very distinct. The scientists and technologists must keep in mind social welfare. Education thus should develop the totality of the child. He laments that Indian education is lacking in the following major drawbacks.

- Indian education has been like stagnant water for quite some time.
- Indian education ignores new ideas and fresh thinking in educational matters.

National Education Systems: Theory and Practice

NATIONAL EDUCATION SYSTEMS

One of the most significant phenomena of the 20th century was the dramatic expansion and extension of public (*i.e.*, government-sponsored) education systems around the world—the number of schools grew, as did the number of children attending them. Similarly, the subjects taught in schools broadened from the basics of mathematics and language to include sciences and the arts.



Fig. Learn about India's efforts to improve its education system, 2009 film.

Various explanations have been given for the substantial increase in numbers of youths as well as adults attending government-sponsored schools; social scientists tend to categorize the reasons for these enrollment increases as products of either conflict or consensus in the process of social change. In most cases these perspectives are rooted in theories of social science that were formulated in the late 19th and early 20th centuries.

EDUCATION AND SOCIAL COHESION

One major school of thought is represented in the work of French sociologist Émile Durkheim, who explained social phenomena from a consensus perspective. According to him, the achievement of social cohesion—exemplified in Europe’s large-scale national societies as they experienced industrialization, urbanization, and the secularization of governing bodies—required a universalistic agency capable of transmitting core values to the populace. These values included a common history that contributed to cultural continuity, social rules that instilled moral discipline and a sense of responsibility for all members of the society, and occupational skills that would meet the society’s complex and dynamic needs. Durkheim recognized that public schooling and teachers—as agents of a larger, moral society—served these necessary functions. As he observed in *The Rules of Sociological Method* (1895), “Education sets out precisely with the object of creating a social being.”

Durkheim’s thoughts, expressed near the turn of the 20th century, were reflected in the policies of newly sovereign states in the post-World War II period. Upon achieving their independence, governments throughout Africa and Asia quickly established systems of public instruction that sought to help achieve a sense of national identity in societies historically divided by tribal, ethnic, linguistic, religious, and geographic differences.

EDUCATION AND SOCIAL CONFLICT

The German political theorist and revolutionary Karl Marx viewed public schooling as a form of ideological control imposed by dominant groups. This perspective saw education not as building social cohesion but as reproducing a division of labour or enabling various status groups to gain control of organizations and to influence the distribution of valued resources. The German sociologist Max Weber regarded educational credentials as one such resource, in that credentials function as a form of “cultural capital” that can generally preserve the status quo while granting social mobility to select members of society.

EDUCATION AND PERSONAL GROWTH

The American philosopher John Dewey believed that education should mean the total development of the child. On the basis of the observations he made at the University of Chicago Laboratory Schools—the experimental elementary schools that he founded

in 1896—Dewey developed revolutionary educational theories that sparked the progressive education movement in the United States. As he propounded in *The School and Society* (1899) and *The Child and the Curriculum* (1902), education must be tied to experience, not abstract thought, and must be built upon the interests and developmental needs of the child. He argued for a student-centred, not subject-centred, curriculum and stressed the teaching of critical thought over rote memorization.

Later, in *Experience and Education* (1938), he criticized those of his followers who took his theories too far by disregarding organized subject matter in favour of vocational training or mere activity for their students. If prudently applied, progressive education could, Dewey believed, “shape the experiences of the young so that instead of reproducing current habits, better habits shall be formed, and thus the future adult society be an improvement on their own.” Concurrent pedagogies appeared in European institutions such as Ovide Decroly’s *École de l’Ermitage* (the Hermitage School), which envisioned students utilizing the classroom as a workshop, and Maria Montessori’s *Casa dei Bambini* (“Children’s House”), which incorporated experiential and tactile learning methods through students’ use of “didactic materials.”

EDUCATION AND CIVIL SOCIETY

Towards the end of the 20th century, comprehensive theories—such as those represented by the consensus and conflict models—were increasingly viewed as oversimplifications of social processes and, in many quarters, gave way to more particularized interpretations. One such perspective viewed educational expansion and extension less as a function of national interest and more as a by-product of religious, economic, political, and cultural changes that had occurred across most of Europe. Especially in the wake of the Enlightenment, an emphasis on the glorification of God was joined by the growing celebration of human progress (ultimately defined as economic growth), while concerns for the salvation of the soul were augmented by the cultivation of individual potential. As nation-states with centralized governments extended citizenship rights in the 18th century, state sponsorship of schools began to supersede the church-supported instruction that had become the norm in the 16th and 17th centuries. According to such scholars as John Meyer and Michael Hannan in *National Development and the World System: Educational, Economic, and Political Change, 1950–1970* (1979), formal systems of education not only represent the means by which nation-states have modernized and prospered economically but are also the surest route to enhancing the talents of individuals. As a requirement for all children and youths between certain ages and as an institution regulated by the state, schooling also became the primary agency for creating citizens with equal responsibilities and rights.

These values emerged in education systems throughout the world, especially in the late 20th century as education professionals promoted them in developed and less-developed countries alike. As such, schools effectively carried modernity into many

parts of the world, where it was met with varying degrees of resistance and acceptance. Teachers, non-governmental organizations (NGOs), and government agencies contributed, for example, to standardization in the shape and style of the classroom, types of curricula, and goals for school enrollments. In the first half of the 20th century, schools in most industrialized countries came to exhibit similar characteristics—that is, schools could be identified as schools. By the second half of the 20th century, these traits had become prominent in most schools around the world.

EDUCATION AND ECONOMIC DEVELOPMENT

One explanation for the changes evidenced in this “institutionalist” view of education can be found in the human-capital theory first popularized by American economist Theodore Schultz in “Investment in Human Capital,” his presidential address to the American Economic Association in 1960. According to this theory, education is not a form of consumption that represents a costly expenditure for government but instead serves as an investment that improves the economic worth of individuals (*e.g.*, human capital) and thereby raises a country’s overall productivity and economic competitiveness. In other words, governments support education because it ultimately strengthens their countries.

GLOBAL ENROLLMENT TRENDS SINCE THE MID-20TH CENTURY

Each of these theories partially explains the widespread increase in enrollments, as reported by UNESCO (the United Nations Educational, Scientific and Cultural Organization), in all levels of education during the last half of the 20th century. Broadly speaking, enrollments increased substantially for school-age children and youths, while adult illiteracy rates decreased significantly. In the second half of the 20th century, the proportion of children worldwide enrolled at all levels (from primary through tertiary) increased from less than half to approximately two-thirds of the relevant age-groups.

Much of this enrollment growth was a product of political change. Most countries in a postcolonial phase expand their education systems, largely because it is something governments can do at a reasonable cost with significant effect. With the opening of schools to many who were once denied education under semifeudal, colonial, or totalitarian systems, it has not been uncommon to find large numbers of overage students enrolled. First-grade classes might have an age range from 6 to 11. Overall, primary-school enrollments more than tripled in the last half of the 20th century, from slightly more than 200 million to some 670 million; secondary education increased more than ninefold, from more than 40 million to nearly 400 million; and tertiary education increased more than 12-fold, from about 7 million to nearly 90 million. Higher levels of enrollment are usually sustained, in part, because “credentialing”—the attainment of degrees or certificates of achievement—has become a social necessity. Employers tend to seek highly schooled individuals while depending on the education system to

prepare and distinguish job candidates. In addition, enrollments have been known to gain momentum through the “queuing” effect; that is, when people line up to participate in something, others soon join the crowd in the belief that something of value will be obtained.

PRIMARY-LEVEL SCHOOL ENROLLMENTS

In not only the industrially developed world but also in other regions (*e.g.*, Latin America and East and South Asia), gross primary-school enrollment rates had reached 95 to 100 percent by the beginning of the 21st century, while in Africa they had achieved an average of about 80 percent. Some of the world’s least-developed countries took the most dramatic steps towards offering universal primary education in the final decades of the 20th century. As late as 1970 less than half of the relevant school-age population attended primary schools in such countries, but by 1997 primary-school enrollments in the least-developed countries had grown to include more than 70 percent of school-age children. Between 1999 and 2005, the overall number of children entering primary education worldwide increased by 4 percent, from 130 million to 135 million. Worldwide total enrollment for primary education increased 6 percent, to 688 million. The biggest gains for entering students took place in sub-Saharan Africa, with an increase of 40 percent. Some countries, however, continued to lag behind this trend. Some of the world’s lowest primary-school enrollment rates persisted in countries such as Niger and Djibouti (both less than 40 percent). Although primary education, as compared with higher levels of schooling, is the least costly to maintain and the easiest to expand, a 2008 UNESCO Global Monitoring Report, “Education for All, ” concluded that a number of sub-Saharan African and Arab countries were not likely to achieve universal primary education by 2015. Another significant challenge is to provide continuing education opportunities for those who complete basic schooling.

SECONDARY-LEVEL SCHOOL ENROLLMENTS

Educational technology is “the study and ethical practice of facilitating learning and improving performance by creating, using, and managing appropriate technological processes and resources”. Educational technology is the use of both physical hardware and educational theoretics. It encompasses several domains, including learning theory, computer-based training, online learning, and, where mobile technologies are used, m-learning. Accordingly, there are several discrete aspects to describing the intellectual and technical development of educational technology:

- Educational technology as the theory and practice of educational approaches to learning
- Educational technology as technological tools and media that assist in the communication of knowledge, and its development and exchange
- Educational technology for learning management systems (LMS), such as tools

for student and curriculum management, and education management information systems (EMIS)

- Educational technology as back-office management, such as training management systems for logistics and budget management, and Learning Record Store (LRS) for learning data storage and analysis.
- Educational technology itself as an educational subject; such courses may be called “Computer Studies” or “Information and communications technology (ICT)”.

An educational technologist is someone who is trained in the field of educational technology. Educational technologists try to analyze, design, develop, implement and evaluate process and tools to enhance learning. While the term *educational technologist* is used primarily in the United States, *learning technologist* is synonymous and used in the UK as well as Canada.

DEFINITION

Richey defined educational technology as “the study and ethical practice of facilitating learning and improving performance by creating, using and managing appropriate technological processes and resources”. The Association for Educational Communications and Technology (AECT) denoted instructional technology as “the theory and practice of design, development, utilization, management, and evaluation of processes and resources for learning”. As such, educational technology refers to all valid and reliable applied education sciences, such as equipment, as well as processes and procedures that are derived from scientific research, and in a given context may refer to theoretical, algorithmic or heuristic processes: it does not necessarily imply physical technology. Educational technology is the process of integrating technology into education in a positive manner that promotes a more diverse learning environment and a way for students to learn how to use technology as well as their common assignments

RELATED TERMS

Given this definition, educational technology is an inclusive term for both the material tools *and* the theoretical foundations for supporting learning and teaching. Educational technology is not restricted to high technology. Education technology is anything that enhances classroom learning in the utilization of blended or online learning.

However, modern electronic educational technology is an important part of society today. Educational technology encompasses e-learning, instructional technology, information and communication technology (ICT) in education, EdTech, learning technology, multimedia learning, technology-enhanced learning (TEL), computer-based instruction (CBI), computer managed instruction, computer-based training (CBT), computer-assisted instruction or computer-aided instruction (CAI), internet-based

training (IBT), flexible learning, web-based training (WBT), online education, digital educational collaboration, distributed learning, computer-mediated communication, cyber-learning, and multi-modal instruction, virtual education, personal learning environments, networked learning, virtual learning environments (VLE) (which are also called learning platforms), m-learning, ubiquitous learning and digital education.

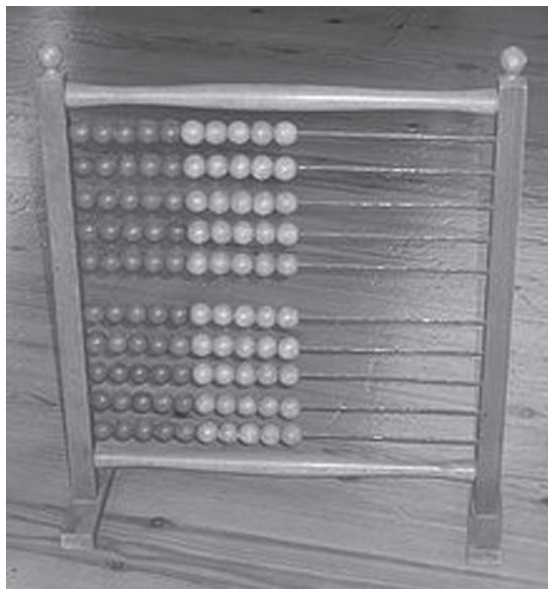


Fig. Early 20th century abacus used in a Danish elementary school.

Each of these numerous terms has had its advocates, who point up potential distinctive features. However, many terms and concepts in educational technology have been defined nebulously; for example, Fiedler's review of the literature found a complete lack agreement of the components of a personal learning environment. Moreover, Moore saw these terminologies as emphasizing particular features such as digitization approaches, components or delivery methods rather than being fundamentally dissimilar in concept or principle. For example, m-learning *emphasizes* mobility, which allows for altered timing, location, accessibility and context of learning; nevertheless, its purpose and conceptual *principles* are those of educational technology.

In practice, as technology has advanced, the particular “narrowly defined” terminological aspect that was initially emphasized by name has blended into the general field of educational technology. Initially, “virtual learning” as narrowly defined in a semantic sense implied entering an environmental simulation within a virtual world, for example in treating posttraumatic stress disorder (PTSD). In practice, a “virtual education course” refers to any instructional course in which all, or at least a significant portion, is delivered by the Internet. “Virtual” is used in that broader way to describe a course that is not taught in a classroom face-to-face but through a substitute mode that can conceptually be associated “virtually” with classroom teaching, which means that

people do not have to go to the physical classroom to learn. Accordingly, virtual education refers to a form of distance learning in which course content is delivered by various methods such as course management applications, multimedia resources, and videoconferencing.

Educational content, pervasively embedded in objects, is all around the learner, who may not even be conscious of the learning process: students may not have to do anything in order to learn, they just have to be there. The combination of adaptive learning, using an individualized interface and materials, which accommodate to an individual, who thus receives personally differentiated instruction, with ubiquitous access to digital resources and learning opportunities in a range of places and at various times, has been termed smart learning. Smart learning is a component of the smart city concept.

HISTORY

Helping people learn in ways that are easier, faster, surer, or less expensive can be traced back to the emergence of very early tools, such as paintings on cave walls. Various types of abacus have been used. Writing slates and blackboards have been used for at least a millennium. From their introduction, books and pamphlets have held a prominent role in education.



Fig. 19th century classroom, Auckland

From the early twentieth century, duplicating machines such as the mimeograph and Gestetner stencil devices were used to produce short copy runs (typically 10–50 copies) for classroom or home use. The use of media for instructional purposes is generally traced back to the first decade of the 20th century with the introduction of educational films (1900s) and Sidney Pressey's mechanical teaching machines (1920s). The first all multiple choice, large-scale assessment was the Army Alpha, used to assess the intelligence and more specifically the aptitudes of World War I military recruits. Further large-scale use of technologies was employed in training soldiers during and after WWII using films and other mediated materials, such as overhead projectors. The concept of hypertext is traced to the description of memex by Vannevar Bush in 1945.

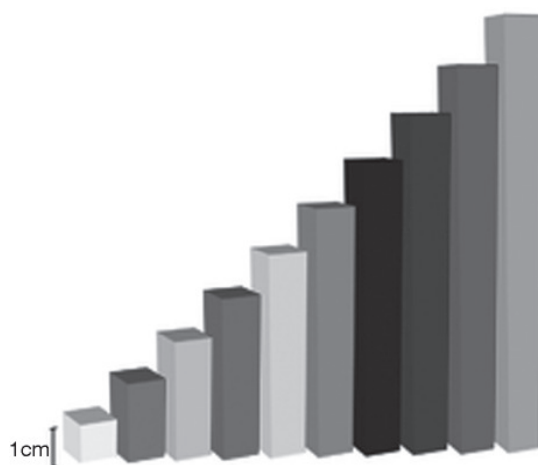


Fig. Cuisenaire rods

Slide projectors were widely used during the 1950s in educational institutional settings. Cuisenaire rods were devised in the 1920s and saw widespread use from the late 1950s. In the mid 1960s Stanford University psychology professors Patrick Suppes and Richard C. Atkinson experimented with using computers to teach arithmetic and spelling via Teletypes to elementary school students in the Palo Alto Unified School District in California. Stanford's Education Programme for Gifted Youth is descended from those early experiments. In 1971, Ivan Illich published a hugely influential book called, *Deschooling Society*, in which he envisioned "learning webs" as a model for people to network the learning they needed. The 1970s and 1980s saw notable contributions in computer-based learning by Murray Turoff and Starr Roxanne Hiltz at the New Jersey Institute of Technology as well as developments at the University of Guelph in Canada. In the UK, the Council for Educational Technology supported the use of educational technology, in particular administering the government's National Development Programme in Computer Aided Learning (1973–77) and the Microelectronics Education Programme (1980–86).

By the mid-1980s, accessing course content became possible at many college libraries. In computer-based training (CBT) or computer-based learning (CBL), the learning interaction was between the student and computer drills or micro-world simulations. Digitized communication and networking in education started in the mid-1980s. Educational institutions began to take advantage of the new medium by offering distance learning courses using computer networking for information. Early e-learning systems, based on computer-based learning/training often replicated autocratic teaching styles whereby the role of the e-learning system was assumed to be for transferring knowledge, as opposed to systems developed later based on computer supported collaborative learning (CSCL), which encouraged the shared development of knowledge. Videoconferencing was an important forerunner to the educational technologies known

today. This work was especially popular with museum education. Even in recent years, videoconferencing has risen in popularity to reach over 20, 000 students across the United States and Canada in 2008–2009. Disadvantages of this form of educational technology are readily apparent: image and sound quality is often grainy or pixelated; videoconferencing requires setting up a type of mini-television studio within the museum for broadcast, space becomes an issue; and specialised equipment is required for both the provider and the participant.



Fig. Multimedia space Moldova Alliance Française.

The Open University in Britain and the University of British Columbia (where Web CT, now incorporated into Blackboard Inc., was first developed) began a revolution of using the Internet to deliver learning, making heavy use of web-based training, online distance learning and online discussion between students. Practitioners such as Harasim (1995) put heavy emphasis on the use of learning networks. With the advent of World Wide Web in the 1990s, teachers embarked on the method using emerging technologies to employ multi-object oriented sites, which are text-based online virtual reality systems, to create course web sites along with simple sets of instructions for its students. By 1994, the first online high school had been founded. In 1997, Graziadei described criteria for evaluating products and developing technology-based courses that include being portable, replicable, scalable, affordable, and having a high probability of long-term cost-effectiveness. Improved Internet functionality enabled new schemes of communication with multimedia or webcams. The National Center for Education Statistics estimate the number of K-12 students enrolled in online distance learning programmes increased by 65 percent from 2002 to 2005, with greater flexibility, ease of communication between teacher and student, and quick lecture and assignment feedback. According to a 2008 study conducted by the U.S Department of Education, during the 2006–2007 academic year about 66% of postsecondary public and private schools participating in student financial aid programmes offered some distance learning courses; records show 77% of enrollment in for-credit courses with an online component. In 2008, the Council of Europe passed a statement endorsing e-learning's potential to

drive equality and education improvements across the EU. Computer-mediated communication (CMC) is between learners and instructors, mediated by the computer. In contrast, CBT/CBL usually means individualized (self-study) learning, while CMC involves educator/tutor facilitation and requires scenarization of flexible learning activities. In addition, modern ICT provides education with tools for sustaining learning communities and associated knowledge management tasks.

Students growing up in this digital age have extensive exposure to a variety of media. Major high-tech companies such as Google, Verizon and Microsoft have funded schools to provide them the ability to teach their students through technology, in the hope that this would lead to improved student performance. 2015 was the first year that private non-profit organizations enrolled more online students than for-profits, although public universities still enrolled the highest number of online students. In the fall of 2015, more than 6 million students enrolled in at least one online course.

THEORY

Various pedagogical perspectives or learning theories may be considered in designing and interacting with educational technology. E-learning theory examines these approaches. These theoretical perspectives are grouped into three main theoretical schools or philosophical frameworks: behaviourism, cognitivism and constructivism.

BEHAVIOURISM

This theoretical framework was developed in the early 20th century based on animal learning experiments by Ivan Pavlov, Edward Thorndike, Edward C. Tolman, Clark L. Hull, and B.F. Skinner. Many psychologists used these results to develop theories of human learning, but modern educators generally see behaviourism as one aspect of a holistic synthesis. Teaching in behaviourism has been linked to training, emphasizing the animal learning experiments. Since behaviourism consists of the view of teaching people how to do something with rewards and punishments, it is related to training people. B.F. Skinner wrote extensively on improvements of teaching based on his functional analysis of verbal behaviour and wrote “The Technology of Teaching”, an attempt to dispel the myths underlying contemporary education as well as promote his system he called programmed instruction. Ogden Lindsley developed a learning system, named Celeration, that was based on behaviour analysis but that substantially differed from Keller’s and Skinner’s models.

Cognitivism

Cognitive science underwent significant change in the 1960s and 1970s. While retaining the empirical framework of behaviourism, cognitive psychology theories look beyond behaviour to explain brain-based learning by considering how human memory

works to promote learning. The Atkinson-Shiffrin memory model and Baddeley's working memory model were established as theoretical frameworks. Computer Science and Information Technology have had a major influence on Cognitive Science theory. The Cognitive concepts of working memory (formerly known as short term memory) and long term memory have been facilitated by research and technology from the field of Computer Science. Another major influence on the field of Cognitive Science is Noam Chomsky. Today researchers are concentrating on topics like cognitive load, information processing and media psychology. These theoretical perspectives influence instructional design.

Constructivism

Educational psychologists distinguish between several types of constructivism: individual (or psychological) constructivism, such as Piaget's theory of cognitive development, and social constructivism. This form of constructivism has a primary focus on how learners construct their own meaning from new information, as they interact with reality and with other learners who bring different perspectives. Constructivist learning environments require students to use their prior knowledge and experiences to formulate new, related, and/or adaptive concepts in learning (Termos, 2012).

Under this framework the role of the teacher becomes that of a facilitator, providing guidance so that learners can construct their own knowledge. Constructivist educators must make sure that the prior learning experiences are appropriate and related to the concepts being taught. Jonassen (1997) suggests "well-structured" learning environments are useful for novice learners and that "ill-structured" environments are only useful for more advanced learners.

Educators utilizing a constructivist perspective may emphasize an active learning environment that may incorporate learner centered problem-based learning, project-based learning, and enquiry-based learning, ideally involving real-world scenarios, in which students are actively engaged in critical thinking activities. An illustrative discussion and example can be found in the 1980s deployment of constructivist cognitive learning in computer literacy, which involved programming as an instrument of learning. *LOGO*, a programming language, embodied an attempt to integrate Piagetian ideas with computers and technology. Initially there were broad, hopeful claims, including "perhaps the most controversial claim" that it would "improve general problem-solving skills" across disciplines. However, *LOGO* programming skills did not consistently yield cognitive benefits. It was "not as concrete" as advocates claimed, it privileged "one form of reasoning over all others," and it was difficult to apply the thinking activity to non-*LOGO*-based activities. By the late 1980s, *LOGO* and other similar programming languages had lost their novelty and dominance and were gradually de-emphasized amid criticisms.

PRACTICE

The extent to which e-learning assists or replaces other learning and teaching approaches is variable, ranging on a continuum from none to fully online distance learning. A variety of descriptive terms have been employed (somewhat inconsistently) to categorize the extent to which technology is used. For example, ‘hybrid learning’ or ‘blended learning’ may refer to classroom aids and laptops, or may refer to approaches in which traditional classroom time is reduced but not eliminated, and is replaced with some online learning. ‘Distributed learning’ may describe either the e-learning component of a hybrid approach, or fully online distance learning environments.

SYNCHRONOUS AND ASYNCHRONOUS

E-learning may either be synchronous or asynchronous. Synchronous learning occurs in real-time, with all participants interacting at the same time, while asynchronous learning is self-paced and allows participants to engage in the exchange of ideas or information without the dependency of other participants’ involvement at the same time. Synchronous learning refers to the exchange of ideas and information with one or more participants during the same period. Examples are face-to-face discussion, online real-time live teacher instruction and feedback, Skype conversations, and chat rooms or virtual classrooms where everyone is online and working collaboratively at the same time. Since students are working collaboratively, synchronized learning helps students create an open mind because they have to listen and learn from their peers. Synchronized learning fosters online awareness and improves many students’ writing skills.

Asynchronous learning may use technologies such as e-mail, blogs, wikis, and discussion boards, as well as web-supported textbooks, hypertext documents, audio video courses, and social networking using web 2.0. At the professional educational level, training may include virtual operating rooms. Asynchronous learning is beneficial for students who have health problems or who have child care responsibilities. They have the opportunity to complete their work in a low stress environment and within a more flexible time frame. In *asynchronous* online courses, students proceed at their own pace. If they need to listen to a lecture a second time, or think about a question for a while, they may do so without fearing that they will hold back the rest of the class. Through online courses, students can earn their diplomas more quickly, or repeat failed courses without the embarrassment of being in a class with younger students. Students have access to an incredible variety of enrichment courses in online learning, and can participate in college courses, internships, sports, or work and still graduate with their class.

Linear Learning

Computer-based training (CBT) refers to self-paced learning activities delivered on a computer or handheld device such as a tablet or smartphone. CBT initially delivered

content via CD-ROM, and typically presented content linearly, much like reading an online book or manual. For this reason, CBT is often used to teach static processes, such as using software or completing mathematical equations. Computer-based training is conceptually similar to web-based training (WBT) which are delivered via Internet using a web browser. Assessing learning in a CBT is often by assessments that can be easily scored by a computer such as multiple choice questions, drag-and-drop, radio button, simulation or other interactive means. Assessments are easily scored and recorded via online software, providing immediate end-user feedback and completion status. Users are often able to print completion records in the form of certificates.

CBTs provide learning stimulus beyond traditional learning methodology from textbook, manual, or classroom-based instruction. CBTs can be a good alternative to printed learning materials since rich media, including videos or animations, can be embedded to enhance the learning. Help, CBTs pose some learning challenges. Typically, the creation of effective CBTs requires enormous resources. The software for developing CBTs (such as Flash or Adobe Director) is often more complex than a subject matter expert or teacher is able to use. The lack of human interaction can limit both the type of content that can be presented and the type of assessment that can be performed, and may need supplementation with online discussion or other interactive elements.

Collaborative Learning

Computer-supported collaborative learning (CSCL) uses instructional methods designed to encourage or require students to work together on learning tasks, allowing social learning. CSCL is similar in concept to the terminology, “e-learning 2.0” and “networked collaborative learning” (NCL). With Web 2.0 advances, sharing information between multiple people in a network has become much easier and use has increased. One of the main reasons for its usage states that it is “a breeding ground for creative and engaging educational endeavors.”

Learning takes place through conversations about content and grounded interaction about problems and actions. This collaborative learning differs from instruction in which the instructor is the principal source of knowledge and skills. The neo-logism “e-learning 1.0” refers to direct instruction used in early computer-based learning and training systems (CBL).

In contrast to that linear delivery of content, often directly from the instructor’s material, CSCL uses social software such as blogs, social media, wikis, podcasts, cloud-based document portals (such as Google Docs and Dropbox), and discussion groups and virtual worlds such as *Second Life*. This phenomenon has been referred to as Long Tail Learning. Advocates of social learning claim that one of the best ways to learn something is to teach it to others. Social networks have been used to foster online

learning communities around subjects as diverse as test preparation and language education. mobile-assisted language learning (MALL) is the use of handheld computers or cell phones to assist in language learning.

Collaborative apps allow students and teachers to interact while studying. Apps are designed after games, which provide a fun way to revise. When the experience is enjoyable the students become more engaged. Games also usually come with a sense of progression, which can help keep students motivated and consistent while trying to improve. Examples of educational games are Dragon Box, Mind Snacks, Code Spells and many more.

Classroom 2.0 refers to online multi-user virtual environments (MUVEs) that connect schools across geographical frontiers. Known as “eTwinning”, computer-supported collaborative learning (CSCL) allows learners in one school to communicate with learners in another that they would not get to know otherwise, enhancing educational outcomes and cultural integration.

Further, many researchers distinguish between collaborative and cooperative approaches to group learning. For example, Roschelle and Teasley (1995) argue that “cooperation is accomplished by the division of labour among participants, as an activity where each person is responsible for a portion of the problem solving”, in contrast with collaboration that involves the “mutual engagement of participants in a coordinated effort to solve the problem together.”

Flipped Classroom

This is an instructional strategy in which computer-assisted teaching is integrated with classroom instruction. Students are given basic essential instruction, such as lectures, before class instead of during class. This frees up classroom time for teachers to more actively engage with learners.

Media

Educational media and tools can be used for:

- Task structuring support: help with how to do a task (procedures and processes),
- Access to knowledge bases (help user find information needed)
- Alternate forms of knowledge representation (multiple representations of knowledge, *e.g.*, video, audio, text, image, data).

Numerous types of physical technology are currently used: digital cameras, video cameras, interactive whiteboard tools, document cameras, electronic media, and LCD projectors. Combinations of these techniques include blogs, collaborative software, ePortfolios, and virtual classrooms.

The current design of this type of applications includes the evaluation through tools of cognitive analysis that allow to identify which elements optimize the use of these platforms.

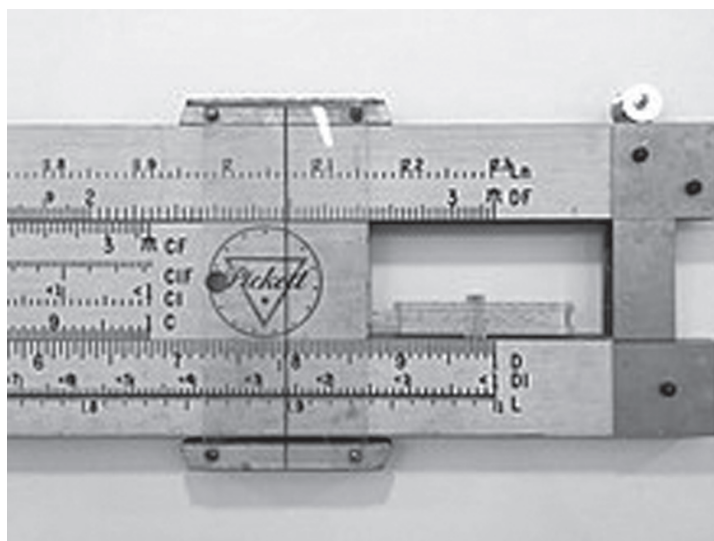


Fig. A 2.5m teaching slide rule compared to a normal sized model

Audio and Video

Radio offers a synchronous educational vehicle, while streaming audio over the internet with webcasts and podcasts can be asynchronous. Classroom microphones, often wireless, can enable learners and educators to interact more clearly. Video technology has included VHS tapes and DVDs, as well as on-demand and synchronous methods with digital video via server or web-based options such as streamed video from YouTube, Teacher Tube, Skype, Adobe Connect, and webcams. Telecommuting can connect with speakers and other experts. Interactive digital video games are being used at K-12 and higher education institutions.

Computers, Tablets and Mobile Devices

Collaborative learning is a group-based learning approach in which learners are mutually engaged in a coordinated fashion to achieve a learning goal or complete a learning task. With recent developments in smartphone technology, the processing powers and storage capabilities of modern mobiles allow for advanced development and use of apps. Many app developers and education experts have been exploring smartphone and tablet apps as a medium for collaborative learning.

Computers and tablets enable learners and educators to access web sites as well as programmes such as Microsoft Word, PowerPoint, PDF files, and images. Many mobile devices support m-learning. Mobile devices such as clickers and smartphones can be used for interactive audience response feedback. Mobile learning can provide performance support for checking the time, setting reminders, retrieving worksheets, and instruction manuals.



Fig. Teaching and learning online

OpenCourseWare (OCW) gives free public access to information used in undergraduate and graduate programmes. Participating institutions are MIT, Harvard, Yale, Princeton, Stanford, University of Pennsylvania, and University of Michigan. Google Classroom allows instructors to create, administer, and grade assignments. While Google Classroom ultimately strives to create a paperless learning environment, there are many different types of learner; a learning environment like the one that Google Classroom projects does not work for everyone.

Social Networks

Group webpages, blogs, wikis, and Twitter allow learners and educators to post thoughts, ideas, and comments on a web site in an interactive learning environment. Social networking sites are virtual communities for people interested in a particular subject to communicate by voice, chat, instant message, video conference, or blogs. The National School Boards Association found that 96% of students with online access have used social networking technologies, and more than 50% talk online about schoolwork. Social networking encourages collaboration and engagement and can be a motivational tool for self-efficacy amongst students.

Webcams

Webcams and webcasting have enabled creation of virtual classrooms and virtual learning environment. Webcams are also being used to counter plagiarism and other forms of academic dishonesty that might occur in an e-learning environment.



Fig. Combination whiteboard and bulletin board

Whiteboards

There are three types of whiteboards. The initial whiteboards, analogous to blackboards, date from the late 1950s. The term whiteboard is also used metaphorically to refer to virtual whiteboards in which computer software applications simulate whiteboards by allowing writing or drawing. This is a common feature of groupware for virtual meeting, collaboration, and instant messaging. Interactive whiteboards allow learners and instructors to write on the touch screen. The screen markup can be on either a blank whiteboard or any computer screen content. Depending on permission settings, this visual learning can be interactive and participatory, including writing and manipulating images on the interactive whiteboard.



Fig. Interactive whiteboard in 2007

Screencasting

Screencasting allows users to share their screens directly from their browser and make the video available online so that other viewers can stream the video directly. The presenter thus has the ability to show their ideas and flow of thoughts rather than simply explain them as simple text content. In combination with audio and video, the educator can mimic the one-on-one experience of the classroom. Learners have an ability to pause and rewind, to review at their own pace, something a classroom cannot always offer.

Virtual Classroom

A virtual learning environment (VLE), also known as a learning platform, simulates a virtual classroom or meetings by simultaneously mixing several communication technologies. For example, web conferencing software such as GoToTraining, WebEx Training or Adobe Connect enables students and instructors to communicate with each other via webcam, microphone, and real-time chatting in a group setting. Participants can raise hands, answer polls or take tests. Students are able to whiteboard and screencast when given rights by the instructor, who sets permission levels for text notes, microphone rights and mouse control.

A virtual classroom provides the opportunity for students to receive direct instruction from a qualified teacher in an interactive environment. Learners can have direct and immediate access to their instructor for instant feedback and direction. The virtual classroom provides a structured schedule of classes, which can be helpful for students who may find the freedom of asynchronous learning to be overwhelming. In addition, the virtual classroom provides a social learning environment that replicates the traditional “brick and mortar” classroom. Most virtual classroom applications provide a recording feature. Each class is recorded and stored on a server, which allows for instant playback of any class over the course of the school year. This can be extremely useful for students to retrieve missed material or review concepts for an upcoming exam. Parents and auditors have the conceptual ability to monitor any classroom to ensure that they are satisfied with the education the learner is receiving.

In higher education especially, a virtual learning environment (VLE) is sometimes combined with a management information system (MIS) to create a managed learning environment, in which all aspects of a course are handled through a consistent user interface throughout the institution.

Physical universities and newer online-only colleges offer select academic degrees and certificate programmes via the Internet. Some programmes require students to attend some campus classes or orientations, but many are delivered completely online. Several universities offer online student support services, such as online advising and registration, e-counseling, online textbook purchases, student governments and student newspapers.

Augmented reality (AR) provides students and teachers the opportunity to create layers of digital information, that includes both virtual world and real world elements, to interact with in real time. There are already a variety of apps which offer a lot of variations and possibilities.

Media psychology involves the application of theories in psychology to media and is a growing specialty in learning and educational technology.

Learning Management System

A learning management system (LMS) is software used for delivering, tracking and managing training and education. For example, an LMS tracks attendance, time on task, and student progress. Educators can post announcements, grade assignments, check on course activity, and participate in class discussions. Students can submit their work, read and respond to discussion questions, and take quizzes. An LMS may allow teachers, administrators, students, and permitted additional parties (such as parents if appropriate) to track various metrics. LMSs range from systems for managing training/educational records to software for distributing courses over the Internet and offering features for online collaboration. The creation and maintenance of comprehensive learning content requires substantial initial and ongoing investments of human labour. Effective translation into other languages and cultural contexts requires even more investment by knowledgeable personnel.

Internet-based learning management systems include Canvas, Blackboard Inc. and Moodle. These types of LMS allow educators to run a learning system partially or fully online, asynchronously or synchronously. Blackboard can be used for K-12 education, Higher Education, Business, and Government collaboration. Moodle is a free-to-download Open Source Course Management System that provides blended learning opportunities as well as platforms for distance learning courses. Eliademy is a free cloud-based Course Management System that provides blended learning opportunities as well as platforms for distance learning courses.

Learning Content Management System

A learning content management system (LCMS) is software for author content (courses, reusable content objects). An LCMS may be solely dedicated to producing and publishing content that is hosted on an LMS, or it can host the content itself. The Aviation Industry Computer-Based Training Committee (AICC) specification provides support for content that is hosted separately from the LMS. A recent trend in LCMSs is to address this issue through crowdsourcing (cf. SlideWiki).

Computer-aided Assessment

Computer-aided assessment (e-assessment) ranges from automated multiple-choice tests to more sophisticated systems. With some systems, feedback can be geared towards

a student's specific mistakes or the computer can navigate the student through a series of questions adapting to what the student appears to have learned or not learned. Formative assessment sifts out the incorrect answers, and these questions are then explained by the teacher. The learner then practices with slight variations of the sifted out questions. The process is completed by summative assessment using a new set of questions that only cover the topics previously taught.

Electronic Performance Support System

An electronic performance support system (EPSS) is, according to Barry Raybould, “a computer-based system that improves worker productivity by providing on-the-job access to integrated information, advice, and learning experiences”.

TRAINING MANAGEMENT SYSTEM

A training management system or training resource management system is a software designed to optimize instructor-led training management. Similar to an enterprise resource planning (ERP), it is a back office tool which aims at streamlining every aspect of the training process: planning (training plan and budget forecasting), logistics (scheduling and resource management), financials (cost tracking, profitability), reporting, and sales for-profit training providers.

For example, a training management system can be used to schedule instructors, venues and equipment through graphical agendas, optimize resource utilization, create a training plan and track remaining budgets, generate reports and share data between different teams. While training management systems focus on managing instructor-led training, they can complete an LMS. In this situation, an LMS will manage e-learning delivery and assessment, while a training management system will manage ILT and back-office budget planning, logistic and reporting.

LEARNING OBJECTS

Content

Content and design architecture issues include pedagogy and learning object re-use. One approach looks at five aspects:

- Fact – unique data (*e.g.*, symbols for Excel formula, or the parts that make up a learning objective)
- Concept – a category that includes multiple examples (*e.g.*, Excel formulas, or the various types/theories of instructional design)
- Process – a flow of events or activities (*e.g.*, how a spreadsheet works, or the five phases in ADDIE)
- Procedure – step-by-step task (*e.g.*, entering a formula into a spreadsheet, or the steps that should be followed within a phase in ADDIE)

- Strategic principle – task performed by adapting guidelines (*e.g.*, doing a financial projection in a spreadsheet, or using a framework for designing learning environments)

Pedagogical Elements

Pedagogical elements are defined as structures or units of educational material. They are the educational content that is to be delivered. These units are independent of format, meaning that although the unit may be delivered in various ways, the pedagogical structures themselves are not the textbook, web page, video conference, Podcast, lesson, assignment, multiple choice question, quiz, discussion group or a case study, all of which are possible methods of delivery.

Learning Objects Standards

Much effort has been put into the technical reuse of electronically based teaching materials and in particular creating or re-using learning objects. These are self-contained units that are properly tagged with keywords, or other metadata, and often stored in an XML file format. Creating a course requires putting together a sequence of learning objects. There are both proprietary and open, non-commercial and commercial, peer-reviewed repositories of learning objects such as the Merlot repository. Sharable Content Object Reference Model(SCORM) is a collection of standards and specifications that applies to certain web-based e-learning. Other specifications such as Schools Framework allow for the transporting of learning objects, or for categorizing metadata (LOM).

SETTINGS

Preschool

Various forms of electronic media are a feature of preschool life. Although parents report a positive experience, the impact of such use has not been systematically assessed. The age when a given child might start using a particular technology such as a cellphone or computer might depend on matching a technological resource to the recipient's developmental capabilities, such as the age-anticipated stages labeled by Swiss psychologist, Jean Piaget. Parameters, such as age-appropriateness, coherence with sought-after values, and concurrent entertainment and educational aspects, have been suggested for choosing media.

K–12

E-learning is utilized by public K–12 schools in the United States as well as private schools. Some e-learning environments take place in a traditional classroom, others allow students to attend classes from home or other locations. There are several states that are utilizing virtual school platforms for e-learning across the country that continue

to increase. For example: Digital technology is becoming increasingly commonplace in K-12 education, and many researchers argue that it will save money and transform schools into more effective institutions. With technology having such a heavy influence in our society it is no doubt that schools are looking into technology to teach students. Meanwhile, despite the debate over the effectiveness of computerized education, all-online K-12 schools are proliferating nationwide, and enrollment in online courses is soaring. Technology can make anyone's life much easier and it is proven that it can help students in an effective way.

But in the digitized world of 21st-century education, computers are increasingly taking on the teachers' role. Computers can now "hear" students speak, for example, correct their pronunciation and evaluate their progress over time, says Michael L. Kamil, a professor emeritus at the Stanford University School of Education. Many experts claim that digital education is the future. "If we want our kids to be prepared for life after school in the 21st century, we need to consider technology a basic element of public education," said New York's Deputy Chancellor of Education John White. And many schools are making it a requirement to take an online class. Digital learning has been getting a boost in localities across the nation this year. For example, Idaho became the first state to require high-school students to complete two or more online courses to receive a diploma.



Fig. Teacher showing primary school students how to work a programme at a primary school in Santa Fe, Mexico City.

It is no doubt that digital education is increasing in our educational system. Virtual school enables students to log into synchronous learning or asynchronous learning courses anywhere there is an internet connection. E-learning is increasingly being utilized by students who may not want to go to traditional brick and mortar schools due to severe allergies or other medical issues, fear of school violence and school bullying and

students whose parents would like to homeschool but do not feel qualified. Online schools create a haven for students to receive a quality education while almost completely avoiding these common problems. Online charter schools also often are not limited by location, income level or class size in the way brick and mortar charter schools are.

E-learning also has been rising as a supplement to the traditional classroom. Students with special talents or interests outside of the available curricula use e-learning to advance their skills or exceed grade restrictions. Some online institutions connect students with instructors via web conference technology to form a digital classroom.

National private schools are also available online. These provide the benefits of e-learning to students in states where charter online schools are not available. They also may allow students greater flexibility and exemption from state testing. Virtual education in K-12 schooling often refers to virtual schools, and in higher education to virtual universities. Virtual schools are “cybercharter schools” with innovative administrative models and course delivery technology.

Higher Education

Online college course enrollment has seen a 29% increase in enrollment with nearly one third of all college students, or an estimated 6.7 million students are currently enrolled in online classes. In 2009, 44 percent of post-secondary students in the USA were taking some or all of their courses online, which was projected to rise to 81 percent by 2014.

Although a large proportion of for-profit higher education institutions now offer online classes, only about half of private, non-profit schools do so. Private institutions may become more involved with on-line presentations as the costs decrease. Properly trained staff must also be hired to work with students online. These staff members need to understand the content area, and also be highly trained in the use of the computer and Internet. Online education is rapidly increasing, and online doctoral programmes have even developed at leading research universities. Although massive open online courses (MOOCs) may have limitations that preclude them from fully replacing college education, such programmes have significantly expanded.

MIT, Stanford and Princeton University offer classes to a global audience, but not for college credit. University-level programmes, like edX founded by Massachusetts Institute of Technology and Harvard University, offer wide range of disciplines at no charge, while others permit students to audit a course at no charge but require a small fee for accreditation. MOOCs have not had a significant impact on higher education and declined after the initial expansion, but are expected to remain in some form.

Private organizations also offer classes, such as Udacity, with free computer science classes, and Khan Academy, with over 3,900 free micro-lectures available via YouTube. Distributed open collaborative course (DOCC) sees itself as a counter-movement to MOOC, emphasizing decentralized teaching. University of the People is a non-profit

accredited online university. Coursera offers online courses. According to *Fortune* magazine, over a million people worldwide have enrolled in free online courses.

CORPORATE AND PROFESSIONAL

Companies with spread out distribution chains use e-learning for staff training and development and to bring customers information about the latest product developments. Continuing professional development (CPD) can deliver regulatory compliance updates and staff development of valuable workplace skills. For effectiveness and competitive learning performance, scoring systems are designed to give live feedback on decision-making in complex (mobile) learning scenarios.

Public Health

There is an important need for recent, reliable, and high-quality health information to be made available to the public as well as in summarized form for public health providers. Providers have indicated the need for automatic notification of the latest research, a single searchable portal of information, and access to grey literature. The Maternal and Child Health (MCH) Library is funded by the U.S., Maternal and Child Health Bureau to screen the latest research and develop automatic notifications to providers through the MCHAlert. Another application in public health is the development of mHealth (use of mobile telecommunication and multimedia into global public health). MHealth has been used to promote prenatal and newborn services, with positive outcomes. In addition, “Health systems have implemented mHealth programmes to facilitate emergency medical responses, point-of-care support, health promotion and data collection.” In low and middle income countries, mHealth is most frequently used as one-way text messages or phone reminders to promote treatment adherence and gather data.

ADHD

There has also been a growing interest in e-learning as a beneficial educational method for students with attention deficit hyperactivity disorder (ADHD). With the growing popularity in e-learning among K-12 and higher education, the opportunity to take online classes is becoming increasingly important for students of all ages. However, students with ADHD and special needs face different learning demands compared to the typical developing learner. This is especially significant considering the dramatic rise in ADHD diagnoses in the last decade among both children and adults. Compared to the traditional face-to-face classroom, e-learning and virtual classrooms require a higher level of executive functions, which is the primary deficit associated with ADHD. Although ADHD is not specifically named in the Rehabilitation Act of 1973, students with ADHD who have symptoms that interfere with their learning or ability may be eligible for assistive technology. Some examples of the resources that may help interest

students and adults with ADHD consist of, computer software, brain games, timers, calendars, voice recognition devices, screen magnifiers, and talking books.

Wolf lists 12 executive function skills necessary for students to succeed in postsecondary education: plan, set goals, organize, initiate, sustain attention/effort, flexibility, monitor, use feedback, structure, manage time, manage materials, and follow through. These skills, along with strong independent and self-regulated learning, are especially pronounced in the online environment and as many ADHD students suffer from a deficit in one or more of these executive functions, this presents a significant challenge and accessibility barrier to the current e-learning approach.

Some have noted that current e-learning models are moving towards applying a constructivism learning theory that emphasizes a learner-centered environment and postulates that everyone has the ability to construct their own knowledge and meaning through a process of problem solving and discovery. However, some principles of constructivism may not be appropriate for ADHD learners; these principles include active learning, self-monitoring, motivation, and strong focus.

Despite the limitations, students with special needs, including ADHD, have expressed an overall enthusiasm for e-learning and have identified a number e-learning benefits, including: availability of online course notes, materials and additional resources; the ability to work at an independent pace and spend extra time formulating thoughtful responses in class discussions; help in understanding course lecture/content; ability to review lectures multiple times; and enhanced access to and communication with the course instructor.

Disabilities

The design of e-learning platforms in ways that enable universal access has received attention from several directions, including the World Wide Web Consortium's Web Accessibility Initiative (WAI). WAI provides universal formatting standards for web sites so they can remain accessible to people with disabilities. For example, developing or adopting e-learning material can enable accessibility for people with visual impairment. The Perkins School for the Blind offers learning resources tailored for the visually impaired, including webcasts, webinars, downloadable science activities, and an online library that has access to over 40, 000 resource materials on blindness and deaf blindness.

Online education may appear to be a promising alternative for students with physical and sensory disabilities because they get to work at their own pace and in their own home. However, not all online programmes are equal when it comes to their resources for students with disabilities. Students with disabilities who wish to enroll in online education must either be able to advocate for themselves and their own rights or have a person who is willing to advocate for them. The American with Disabilities Act states that online programmes must provide appropriate accommodations for students with

disabilities, but has not specifically defined what that means. “Once students with disabilities are accepted into an online programme, they should prepare to be direct and open about what they need to succeed, experts say” (Haynie).

Identity Options

Educational technology, particularly in online learning environments, can allow students to use real identity, pseudonym, or anonymous identity during classroom communication. Advantages in anonymizing race, age, and gender are increased student participation and increased cross-cultural communication. Risks include increased cyberbullying, and aggressive or hostile language.

BENEFITS

Effective technology use deploys multiple evidence-based strategies concurrently (*e.g.*, adaptive content, frequent testing, immediate feedback, etc.), as do effective teachers. Using computers or other forms of technology can give students practice on core content and skills while the teacher can work with others, conduct assessments, or perform other tasks. Through the use of educational technology, education is able to be individualized for each student allowing for better differentiation and allowing students to work for mastery at their own pace.

Modern educational technology can improve access to education, including full degree programmes. It enables better integration for non-full-time students, particularly in continuing education, and improved interactions between students and instructors. Learning material can be used for long distance learning and are accessible to a wider audience. Course materials are easy to access. In 2010, 70.3% of American family households had access to the internet. In 2013, according to Canadian Radio Television and Telecommunications Commission Canada, 79% of homes have access to the internet. Students can access and engage with numerous online resources at home. Using online resources such as Khan Academy or TED Talks can help students spend more time on specific aspects of what they may be learning in school, but at home. Schools like MIT have made certain course materials free online. Although some aspects of a classroom setting are missed by using these resources, they are helpful tools to add additional support to the educational system. The necessity to pay for transport to the educational facility is removed. Students appreciate the convenience of e-learning, but report greater engagement in face-to-face learning environments.

According to James Kulik, who studies the effectiveness of computers used for instruction, students usually learn more in less time when receiving computer-based instruction and they like classes more and develop more positive attitudes towards computers in computer-based classes. Students can independently solve problems. There are no intrinsic age-based restrictions on difficulty level, *i.e.*, students can go at their own pace. Students editing their written work on word processors improve the quality

of their writing. According to some studies, the students are better at critiquing and editing written work that is exchanged over a computer network with students they know. Studies completed in “computer intensive” settings found increases in student-centric, cooperative and higher order learning, writing skills, problem solving, and using technology. In addition, attitudes towards technology as a learning tool by parents, students and teachers are also improved.

Employers’ acceptance of online education has risen over time. More than 50% of human resource managers SHRM surveyed for an August 2010 report said that if two candidates with the same level of experience were applying for a job, it would not have any kind of effect whether the candidate’s obtained degree was acquired through an online or a traditional school. Seventy-nine percent said they had employed a candidate with an online degree in the past 12 months. However 66% said candidates who get degrees online were not seen as positively as a job applicant with traditional degrees.

The use of educational apps generally has positive effect on learning. Pre- and post-tests reveal that the use of apps on mobile devices reduces the achievement gap between struggling and average students. Some educational apps improve group work by allowing students to receive feedback on answers and promoting collaboration in solving problems, examples of these apps can be found in the third paragraph. The benefits of app-assisted learning have been exhibited in all age groups. Kindergarten students that use iPads show much higher rates of literacy than non-users. Medical students at University of California Irvine that utilized iPad academically have been reported to score 23% higher on national exams than previous classes that did not.

DISADVANTAGES

Many US states spend large sums of money on technology. However, as of 2013, none were looking at technology return on investment (ROI) to connect expenditures on technology with improved student outcomes.

New technologies are frequently accompanied by unrealistic hype and promise regarding their transformative power to change education for the better or in allowing better educational opportunities to reach the masses. Examples include silent film, broadcast radio, and television, none of which have maintained much of a foothold in the daily practices of mainstream, formal education. Technology, in and of itself, does not necessarily result in fundamental improvements to educational practice. The focus needs to be on the learner’s interaction with technology—not the technology itself. It needs to be recognized as “ecological” rather than “additive” or “subtractive”. In this ecological change, one significant change will create total change.

According to Branford et al., “technology does not guarantee effective learning” and inappropriate use of technology can even hinder it. A University of Washington study of infant vocabulary shows that it is slipping due to educational baby DVDs. Published in the *Journal of Pediatrics*, a 2007 University of Washington study on the

vocabulary of babies surveyed over 1, 000 parents in Washington and Minnesota. The study found that for every one hour that babies 8–16 months of age watched DVDs and Videos they knew 6-8 fewer of 90 common baby words than the babies that did not watch them. Andrew Meltzoff, a surveyor in this study states that the result makes sense, that if the baby's 'alert time' is spent in front of DVDs and TV, instead of with people speaking, the babies are not going to get the same linguistic experience. Dr. Dimitri Chistakis, another surveyor reported that the evidence is mounting that baby DVDs are of no value and may be harmful.



Adaptive instructional materials tailor questions to each student's ability and calculate their scores, but this encourages students to work individually rather than socially or collaboratively. Social relationships are important but high-tech environments may compromise the balance of trust, care and respect between teacher and student.

Massively open online courses (MOOCs), although quite popular in discussions of technology and education in developed countries (more so in US), are not a major concern in most developing or low-income countries. One of the stated goals of MOOCs is to provide less fortunate populations (*i.e.*, in developing countries) an opportunity to experience courses with US-style content and structure. However, research shows only 3% of the registrants are from low-income countries and although many courses have thousands of registered students only 5-10% of them complete the course. MOOCs also implies that certain curriculum and teaching methods are superior and this could eventually wash over (or possibly washing out) local educational institutions, cultural norms and educational traditions. With the Internet and social media, using educational apps makes the students highly susceptible to distraction and sidetracking. Even though proper use has shown to increase student performances, being distracted would be detrimental. Another disadvantage is increased potential for cheating. Smartphones can be very easy to hide and use inconspicuously, especially if their use is normalized in the classroom. These disadvantages can be managed with strict rules and regulations on mobile phone use.

Over-stimulation

Electronic devices such as cellphones and computers facilitate rapid access to a stream of sources, each of which may receive cursory attention. Michel Rich, an associate professor at Harvard Medical School and executive director of the center on Media and Child Health in Boston, said of the digital generation, “Their brains are rewarded not for staying on task, but for jumping to the next thing. The worry is we’re raising a generation of kids in front of screens whose brains are going to be wired differently.” Students have always faced distractions; computers and cellphones are a particular challenge because the stream of data can interfere with focusing and learning. Although these technologies affect adults too, young people may be more influenced by it as their developing brains can easily become habituated to switching tasks and become unaccustomed to sustaining attention. Too much information, coming too rapidly, can overwhelm thinking.

Technology is “rapidly and profoundly altering our brains.” High exposure levels stimulate brain cell alteration and release neurotransmitters, which causes the strengthening of some neural pathways and weakening of others. This leads to heightened stress levels on the brain that, at first, boost energy levels, but, over time, actually augment memory, impair cognition, lead to depression, alter the neural circuitry of the hippocampus, amygdala and prefrontal cortex. These are the brain regions that control mood and thought. If unchecked, the underlying structure of the brain could be altered. Over-stimulation due to technology may begin too young. When children are exposed before the age of seven, important developmental tasks may be delayed, and bad learning habits might develop, which “deprives children of the exploration and play that they need to develop.” Media psychology is an emerging specialty field that embraces electronic devices and the sensory behaviours occurring from the use of educational technology in learning.

SOCIOCULTURAL CRITICISM

According to Lai, “the learning environment is a complex system where the interplay and interactions of many things impact the outcome of learning.” When technology is brought into an educational setting, the pedagogical setting changes in that technology-driven teaching can change the entire meaning of an activity without adequate research validation. If technology monopolizes an activity, students can begin to develop the sense that “life would scarcely be thinkable without technology.”

Leo Marx considered the word “technology” itself as problematic, susceptible to reification and “phantom objectivity”, which conceals its fundamental nature as something that is only valuable insofar as it benefits the human condition. Technology ultimately comes down to affecting the relations between people, but this notion is obfuscated when technology is treated as an abstract notion devoid of good and evil.

Langdon Winner makes a similar point by arguing that the underdevelopment of the philosophy of technology leaves us with an overly simplistic reduction in our discourse to the supposedly dichotomous notions of the “making” versus the “uses” of new technologies, and that a narrow focus on “use” leads us to believe that all technologies are neutral in moral standing. These critiques would have us ask not, “How do we maximize the role or advancement of technology in education?”, but, rather, “What are the social and human consequences of adopting any particular technology?”

Winner viewed technology as a “form of life” that not only aids human activity, but that also represents a powerful force in reshaping that activity and its meaning. For example, the use of robots in the industrial workplace may increase productivity, but they also radically change the process of production itself, thereby redefining what is meant by “work” in such a setting. In education, standardized testing has arguably redefined the notions of learning and assessment. We rarely explicitly reflect on how strange a notion it is that a number between, say, 0 and 100 could accurately reflect a person’s knowledge about the world. According to Winner, the recurring patterns in everyday life tend to become an unconscious process that we learn to take for granted. Winner writes,

By far the greatest latitude of choice exists the very first time a particular instrument, system, or technique is introduced. Because choices tend to become strongly fixed in material equipment, economic investment, and social habit, the original flexibility vanishes for all practical purposes once the initial commitments are made. In that sense technological innovations are similar to legislative acts or political foundings that establish a framework for public order that will endure over many generations. (p. 29)

When adopting new technologies, there may be one best chance to “get it right.” Seymour Papert (p. 32) points out a good example of a (bad) choice that has become strongly fixed in social habit and material equipment: our “choice” to use the QWERTY keyboard. The QWERTY arrangement of letters on the keyboard was originally chosen, not because it was the most efficient for typing, but because early typewriters were prone to jam when adjacent keys were struck in quick succession. Now that typing has become a digital process, this is no longer an issue, but the QWERTY arrangement lives on as a social habit, one that is very difficult to change.

Neil Postman endorsed the notion that technology impacts human cultures, including the culture of classrooms, and that this is a consideration even more important than considering the efficiency of a new technology as a tool for teaching. Regarding the computer’s impact on education, Postman writes (p. 19): What we need to consider about the computer has nothing to do with its efficiency as a teaching tool. We need to know in what ways it is altering our conception of learning, and how in conjunction with television, it undermines the old idea of school.

There is an assumption that technology is inherently interesting so it must be helpful in education; based on research by Daniel Willingham, that is not always the case. He

argues that it does not necessarily matter what the technological medium is, but whether or not the content is engaging and utilizes the medium in a beneficial way.

Digital Divide

The concept of the digital divide is a gap between those who have access to digital technologies and those who do not. Access may be associated with age, gender, socio-economic status, education, income, ethnicity, and geography.

Data Protection

According to a report by the Electronic Frontier Foundation, large amounts of personal data on children is collected by electronic devices that are distributed in schools in the United States. Often far more information than necessary is collected, uploaded and stored indefinitely. Aside name and date of birth, this information can include the child's browsing history, search terms, location data, contact lists, as well as behavioural information. Parents are not informed or, if informed, have little choice. According to the report, this constant surveillance resulting from educational technology can "warp children's privacy expectations, lead them to self-censor, and limit their creativity".

TEACHER TRAINING

Since technology is not the end goal of education, but rather a means by which it can be accomplished, educators must have a good grasp of the technology and its advantages and disadvantages. Teacher training aims for effective integration of classroom technology.

The evolving nature of technology may unsettle teachers, who may experience themselves as perpetual novices. Finding quality materials to support classroom objectives is often difficult. Random professional development days are inadequate.

According to Jenkins, "Rather than dealing with each technology in isolation, we would do better to take an ecological approach, thinking about the interrelationship among different communication technologies, the cultural communities that grow up around them, and the activities they support." Jenkins also suggested that the traditional school curriculum guided teachers to train students to be autonomous problem solvers. However, today's workers are increasingly asked to work in teams, drawing on different sets of expertise, and collaborating to solve problem. Learning styles and the methods of collecting information have evolved, and "students often feel locked out of the worlds described in their textbooks through the depersonalized and abstract prose used to describe them". These twenty-first century skills can be attained through the incorporation and engagement with technology. Changes in instruction and use of technology can also promote a higher level of learning among students with different types of intelligence.

ASSESSMENT

There are two distinct issues of assessment: the assessment *of* educational technology and assessment *with* technology. Assessments *of* educational technology have included the *Follow Through* project. Educational assessment *with* technology may be either formative assessment or summative assessment. Instructors use both types of assessment to understand student progress and learning in the classroom. Technology has helped teachers create better assessments to help understand where students who are having trouble with the material are having issues.

Formative assessment is more difficult, as the perfect form is ongoing and allows the students to show their learning in different ways depending on their learning styles. Technology has helped some teachers make their formative assessments better, particularly through the use of classroom response systems (CRS). A CRS is a tool in which the students each have a handheld device that partners up with the teacher's computer. The instructor then asks multiple choice or true or false questions and the students answer on their device. Depending on the software used, the answers may then be shown on a graph so students and teacher can see the percentage of students who gave each answer and the teacher can focus on what went wrong. Some examples of CRSs are Quizzler, Turning Systems, and the quiz aspect of the Mastering Programmes (for example Mastering Physics or Mastering Chemistry). Summative assessments are more common in classrooms and are usually set up to be more easily graded, as they take the form of tests or projects with specific grading schemes. One huge benefit to tech-based testing is the option to give students immediate feedback on their answers. When students get these responses, they are able to know how they are doing in the class which can help push them to improve or give them confidence that they are doing well. Technology also allows for different kinds of summative assessment, such as digital presentations, videos, or anything else the teacher/students may come up with, which allows different learners to show what they learned more effectively. Teachers can also use technology to post graded assessments online for students to have a better idea of what a good project is. Electronic assessment uses information technology. It encompasses several potential applications, which may be teacher or student oriented, including educational assessment throughout the continuum of learning, such as computerized classification testing, computerized adaptive testing, student testing, and grading an exam. E-Marking is an examiner led activity closely related to other e-assessment activities such as e-testing, or e-learning which are student led. E-marking allows markers to mark a scanned script or online response on a computer screen rather than on paper.

There are no restrictions to the types of tests that can use e-marking, with e-marking applications designed to accommodate multiple choice, written, and even video submissions for performance examinations. E-marking software is used by individual educational institutions and can also be rolled out to the participating schools of awarding exam organisations. e-marking has been used to mark many well known high stakes

examinations, which in the United Kingdom include A levels and GCSE exams, and in the US includes the SAT test for college admissions. Ofqual reports that e-marking is the main type of marking used for general qualifications in the United Kingdom. In 2014, the Scottish Qualifications Authority (SQA) announced that most of the National 5 question papers would be e-marked. In June 2015, the Odisha state government in India announced that it planned to use e-marking for all Plus II papers from 2016.

ANALYTICS

The importance of self-assessment through tools made available on Educational Technology platforms has been growing. Self-assessment in education technology relies on students analyzing their strengths, weaknesses and areas where improvement is possible to set realistic goals in learning, improve their educational performances and track their progress. One of the unique tools for self-assessment made possible by education technology is Analytics. Analytics is data gathered on the student's activities on the learning platform, drawn into meaningful patterns that leads to a valid conclusion, usually through the medium of data visualization such as graphs.

EXPENDITURE

The five key sectors of the e-learning industry are consulting, content, technologies, services and support. Worldwide, e-learning was estimated in 2000 to be over \$48 billion according to conservative estimates. Commercial growth has been brisk. In 2014, the worldwide commercial market activity was estimated at \$6 billion venture capital over the past five years, with self-paced learning generating \$35.6 billion in 2011. North American e-learning generated \$23.3 billion in revenue in 2013, with a 9% growth rate in cloud-based authoring tools and learning platforms.

CAREERS

Educational technologists and psychologists apply basic educational and psychological research into an evidence-based applied science (or a technology) of learning or instruction. In research, these professions typically require a graduate degree (Master's, Doctorate, Ph.D., or D.Phil.) in a field related to educational psychology, educational media, experimental psychology, cognitive psychology or, more purely, in the fields of educational, instructional or human performance technology or instructional design. In industry, educational technology is utilized to train students and employees by a wide range of learning and communication practitioners, including instructional designers, technical trainers, technical communication and professional communication specialists, technical writers, and of course primary school and college teachers of all levels. The transformation of educational technology from a cottage industry to a profession is discussed by Shurville et al.

Theories of Organisation and Management in Education

Contemporary theories of management tend to account for and help interpret the rapidly changing nature of today's organizational environments. As before in management history, these theories are prevalent in other sciences as well.

MANAGEMENT OF SCHOOL BUILDING

During The Construction Period: Liaison with the local authority and the contractor:

- Before construction activities commence on site, it is important for the school to be aware of the project communication arrangements between the local authority project manager and the contractor. Regular site meetings are likely to be held, and it may be appropriate for the school to be represented, or to be briefed on progress.
- In many projects, day-to-day communication should be encouraged between the school co-ordinator and the contractor's site manager regarding 'housekeeping' matters which will affect the running of the school.
- As well as having a clear understanding of the design and contract programme, the school should also be aware of what terms and conditions, if any, were included in the contract regarding the contractor's working restrictions, health and safety, security and so on. The school itself will have no authority to enforce the terms of the contract. It is therefore essential to establish a single point of contact with the local authority for all communication regarding the project.

Keeping Staff and Pupils Informed

- The flow of information to and from school users is very important throughout the project. This is usually an element of the school co-ordinator's role.
- Existing communication arrangements within the school such as regular departmental and whole staff meetings should be used wherever possible rather than setting up separate project meetings. However, staff should be able to raise individual issues directly with the school co-ordinator.
- Pupils are usually kept informed about the project through school assemblies and feedback is often provided through the pupil council or through 'house structures' where these operate in secondary schools.
- The project is also likely to feature heavily on the agenda of any School Board and PTA meetings. The HT and school co-ordinator, as well as the contractor, may well be expected to make regular progress reports at these meetings.
- *One secondary school undergoing a major refurbishment set up a specific project users group which met monthly throughout the duration of the project. The group consisted of members representing staff, pupils, parents and others from the school community. Meetings were chaired by the HT and were attended by the local authority project manager and the contractor.*

Managing Disruptions

- A certain degree of inconvenience is unavoidable during a major construction project. The level of disruption likely to be experienced by the school will depend upon a variety of factors such as the scale and type of building operations, their proximity to the school activities, the time at which the works are carried out, and the constraints of the existing buildings and site.
- The following examples are the most common disruptions reported by schools during construction projects:
 - *Dust and dirt:* Apart from being particularly uncomfortable, airborne dust and dirt can give rise to medical complaints leading to staff and pupil absences. Locating classrooms which require natural ventilation away from construction activities, and insisting on the constructing and maintaining of seals in affected areas can help to minimise the ingress of dust and dirt. In some cases, it may be appropriate to arrange additional cleaning for the school during the project.
 - *Noise:* Health and Safety regulations ensure that noise levels will not be hazardous to health, but they may still be extremely distracting. Where construction works are in close proximity to the school, the contractor may be excluded from undertaking certain noisy activities during particular periods such as exams. It may also be possible for the school to request the contractor, on an informal basis, to reduce noise levels for short periods from time to time.

- *Distractions from increased traffic, both vehicular and personnel:* In order to avoid continuing distraction it may be possible to locate particularly sensitive school classes and activities away from the main site access and construction works. Ensure site activities are appropriately screened and avoid allowing contractor's staff access through pupil areas.
- *Frequent changes to access points and circulation routes:* The need to advise school users of continuing changes in access arrangements can be disruptive and resource intensive. This should be considered when agreeing the sequence of phasing of the construction works.
- Planned and unplanned interruptions to water, gas, power, ICT services; prior consultation and contingency planning with staff about the consequences of a particular service failure or disconnection will allow the school to better manage these situations when they occur.
- *Reduced playground space, loss of playing fields and car parking:* The loss of amenities during the construction process is often an unavoidable source of inconvenience to school users. However, early consultation with those affected, provides the opportunity to investigate and implement alternatives.

Managing Hazards and Disruptions

- The contractor will manage the health and safety and security aspects of the construction activities in any project. However, this in itself does not prevent the school from having to consider a number of health and safety matters or having to prepare at certain times for considerable disruption to the normal day-to-day running of the school.

Health and Safety/security

- The school will need to monitor its health and safety management procedures throughout the project and carry out additional risk assessments where appropriate. In particular, any alterations to fire escape routes and gathering points, access arrangements and site boundaries should be clearly identified. Arrangements should be clearly displayed in the school with signage amended as appropriate. Fire drills should be undertaken each time escape routes are changed.
- Prior to construction commencing on site, all school users should be briefed on the health and safety arrangements for the project. Presentations to school assemblies by the contractor, often incorporating protective clothing and some basic statistics about construction site safety, are generally considered to have more impact than if these were delivered by the school management team.
- As school holidays approach further advice should be issued to pupils regarding the hazards of building sites. Security monitoring will be provided by the

contractor but in some instances it may also be appropriate to alert the local police who may patrol the site at high risk periods.

- Where work needs to be carried out in occupied buildings, contractor's staff may require to gain access to parts of the construction site through operational parts of the school. These situations generally place greater responsibilities on the contractor's staff and school users. These responsibilities need to be understood by all.
- It may be considered appropriate in these situations for the contractor's staff to wear agreed forms of identification and to be prepared to be challenged by staff on school premises. It may also be considered necessary to caution contractor's staff to avoid initiating contact with pupils. It has also been known for school pupils to abuse construction workers, so the conduct of school users must also be considered and managed.
- Despite contractual agreements, breaches in health and safety procedures can still occur in these situations. For example contractor's staff may leave doors unlocked or materials and equipment unattended in circulation routes. School users should be particularly vigilant about possible hazards and procedures should be in place to report such incidents.

CONTINGENCY THEORY

Basically, contingency theory asserts that when managers make a decision, they must take into account all aspects of the current situation and act on those aspects that are key to the situation at hand. Basically, it's the approach that "it depends." For example, the continuing effort to identify the best leadership or management style might now conclude that the best style depends on the situation. If one is leading troops in the Persian Gulf, an autocratic style is probably best (of course, many might argue here, too). If one is leading a hospital or university, a more participative and facilitative leadership style is probably best.

SYSTEMS THEORY

Systems theory has had a significant effect on management science and understanding organizations. First, let's look at "what is a system?"

A system is a collection of part unified to accomplish an overall goal. If one part of the system is removed, the nature of the system is changed as well.

For example, a pile of sand is not a system. If one removes a sand particle, you've still got a pile of sand. However, a functioning car is a system. Remove the carburetor and you've no longer got a working car. A system can be looked at as having inputs, processes, outputs and outcomes. Systems share feedback among each of these four aspects of the systems. Let's look at an organization.

Inputs would include resources such as raw materials, money, technologies and people. These inputs go through a process where they're planned, organized, motivated and controlled, ultimately to meet the organization's goals. Outputs would be products or services to a market.

Outcomes would be, *e.g.*, enhanced quality of life or productivity for customers/clients, productivity. Feedback would be information from human resources carrying out the process, customers/clients using the products, etc. Feedback also comes from the larger environment of the organization, *e.g.*,

influences from government, society, economics, and technologies. This overall system framework applies to any system, including subsystems (departments, programmes, etc.) in the overall organization. Systems theory may seem quite basic. Yet, decades of management training and practices in the workplace have not followed this theory. Only recently, with tremendous changes facing organizations and how they operate, have educators and managers come to face this new way of looking at things. This interpretation has brought about a significant change (or paradigm shift) in the way management studies and approaches organizations.

The effect of systems theory in management is that writers, educators, consultants, etc. are helping managers to look at the organization from a broader perspective. Systems theory has brought a new perspective for managers to interpret patterns and events in the workplace. They recognize the various parts of the organization, and, in particular, the interrelations of the parts, *e.g.*, the coordination of central administration with its programmes, engineering with manufacturing, supervisors with workers, etc. This is a major development. In the past, managers typically took one part and focused on that. Then they moved all attention to another part. The problem was that an organization could, *e.g.*, have a wonderful central administration and wonderful set of teachers, but the departments didn't synchronize at all.

Chaos Theory

As chaotic and random as world events seem today, they seem as chaotic in organizations, too. Yet for decades, managers have acted on the basis that organizational events can always be controlled.

A new theory (or some say "science"), chaos theory, recognizes that events indeed are rarely controlled. Many chaos theorists (as do systems theorists) refer to biological systems when explaining their theory. They suggest that systems naturally go to more complexity, and as they do so, these systems become more volatile (or susceptible to cataclysmic events) and must expend more energy to maintain that complexity. As they expend more energy, they seek more structure to maintain stability. This trend continues until the system splits, combines with another complex system or falls apart entirely. Sound familiar? This trend is what many see as the trend in life, in organizations and the world in general.

MANAGEMENT IN INDIA

For some period, much white collar work from the Western world has been exported to the Republic of India. This work has generally been in the field of information technology (IT). Since IT is an essential component to any business these days, it is quite important that US managers have knowledge of how their managerial theories can be implemented in a country like India. In a study by Richman and Copen (1972), it was demonstrated that managerial practices developed in the United States and other industrialized countries were easily transferable to managers in India. The similarity is attributed to a number of variables including: the increased managerial education in India, the higher status being attributed to managers, the similarity between technology and management of technology, as well as India's trend towards the decentralization of authority. The only major difference comes about in the emphasis on certain managerial facets.

Earlier, the individualism-collectivism scale was discussed with regards to the United States and China. Somewhere in the middle of the bipolar extremes of the US and China lies India. While Indian society does reflect certain characteristics of collectivism as do many Asian cultures, the basic cultural orientation of India is rooted in the ideas of Buddhism, Yogic psychology, and self-discipline. Although Indian managers and workers will exhibit collectivist traits in family and other social interactions, in business management practices they tend to mix these collectivist traits with Western management philosophy.

For example, managers in India maintain close personal relationships with their peers and subordinates, which reflects their collectivistic attribute. However, in their business activities, Western management methods such as agendas and constant feedback are used. Leadership and communication also rank quite high in the evaluation of management in India, as they do in the United States. Modern Indian management practices have evolved from the British system prior to India's independence, and India has adopted similar practices in the management agency system since independence in 1947. However, Indian management practices have developed a distinct style, which is different from the British system. For example, Indian firms are still run very paternalistically with the personal relationships playing a key role in the management of the organization.

Besides minor differences in style, most characteristics regarding the evaluation of managerial performance tend to be similar between the United States and India. This suggests that India may be a lot closer to the US on the aforementioned individualism-collectivism scale. All in all, culture plays an important role in global business, and to achieve success in cross-cultural management, US managers should understand both themselves and others. This includes knowledge of one's self and the culture of those whom he or she is managing or working with. The competitive advantage for the firm

comes about from the proper alignment between the external environment and the firm's internal strategies, structure, and management practices.

The Government seeks to follow an integrated approach to literacy. "This means, the Total literacy Campaign and the Post Literacy Programme will operate under one literacy Project. This approach would enable the enormous illiteracy problem to be tackled in a holistic manner. By treating the imparting of functional literacy as a continuum rather than as a one of benefit for the illiterate person, progress of literacy efforts would be made goaldirected.

Literacy campaigns would continue to run in those areas where there are large pools of residual illiteracy. At the same time, for those who have crossed the basic learning phase, programmes of consolidation, remediation, vocational skills, integration with life skills (emphasis added) and such other aspects would be considered the basic unit". The characterization of life skills is not available; but speaking to MHRD officials; it appears that they follow the standard international definitions in relation to NFE.

Some NGO NFE providers ordered the life skills differently for the Indian context in a 2001 UNESCO-supported study conducted by an Indian NGO, REMEDIA Trust, on life skills education within the NFE sector; with a view to address issues related to drug abuse and HIV/AIDS prevention.

According to the NGOs participating in the study, 'self awareness is the first necessary life skill. By using the Socratic Method, the students can be helped to debate the issue of an individual's place in a family, a social group, a community, and a country. The young people can be made aware of their rights and duties and their responsibilities to themselves and society. The need to preserve their physical and mental health has to be realized. The twin essence of self awareness is the life skill of empathy. Awareness of self should be counterbalanced by awareness of others, their different thinking, feelings, desires and wishes. This requires some imagination and fellow feeling. It is part of a process of socialization and self control. Indian culture emphasizes empathy with human beings, animals and the nature around us. This skill has to be cultivated carefully, especially among street and slum children as they often exhibit an aggressive stance in self protection. Critical thinking and creative thinking is the next pair of life skills. Training in thinking is very rare in formal as well as non-formal classrooms in India; and today as the media sends out a constant stream of messages, it is important to begin critical listening and asking the right questions. Training young people to ask for cause-effect relationship and then for rational thinking is essential, if they are going to withstand pressure.

Decision making and problem solving comes somewhat easier to children who are on the streets rather than to children who are protected by the family. Since, in India, the group rather the individual are given importance, these skills may be fostered by debates where individual differences can emerge and can be defended by logical arguments.

Coping with emotions and stress first requires admitting emotions and stress. In the Indian society, emotions and stress are regarded as shameful and stoicism is encouraged. Therefore, the discipline of standing aloof from one's emotions, recognising them and coping with them is an exercise that has to be carefully taught so that the extreme positions of either denying emotions or totally giving in to emotions are avoided. Effective communication and interpersonal relations skills are important to young people in order to maintain both their independence and their friendships. In India, friendship between young people of the same sex has been more highly regarded than elsewhere. Even in India, family bonds, though strong, need careful nurture. Drugs do not tempt whose family bonds are strong. Families offer the young protection from drugs and stress. Two major life skills necessary are spatial orientation and orientation in time.

The non-verbal spatial skill is needed to practice arts and crafts, which are part of the Indian way of life. Most vocational programmes depend on spatial skills. Orientation to time helps the young to plan for their future. It has been discovered that the disadvantaged do not have this orientation, so they tend to live in the present and do not save or invest money they earn for their morrow. If the young who come to the NFE centres begin to understand the time-line and the cause and effect relationship, they will begin to appreciate how their effort of today will bear fruit tomorrow. Many young people do not have a clear notion of how yesterday, today and tomorrow are linked. They have a vague notion of weekdays and months but if they are going to participate in the modern world of technology, they must have orientation to both space and time.

Moreover, through the Open Universities and distance learning initiatives, mechanisms are in place to upgrade skills at regular intervals and develop new competencies. People's needs of lifelong learning are constantly expanding. The Open University System was initiated in the country to augment opportunities for higher education as an instrument of democratising education and also to make it a lifelong process. The first Open University in the country was established by the state government of Andhra Pradesh in 1982. In 1985, the central government established the Indira Gandhi National Open University (IGNOU).

MANAGEMENT FUNCTIONS AND CHANGE

In view of the likely emergence of changes in management systems and the need to equip students with the ability to cope with them, effective mechanisms will be devised to understand the nature and direction of change per se and to develop the important skill of managing change. In view of the integrated nature of the task, the Ministry of Human Resource Development will coordinate the balanced development of engineering, vocational and management education as well as the education of technicians and craftsmen. Professional societies will be encouraged and enabled to perform their due role in the advancement of technical and management education. The All India Council for Technical Education will be vested with statutory authority for planning, formulation

and the maintenance of norms and standards, accreditation, funding of priority areas, monitoring and evaluation, maintaining parity of certification and awards and ensuring the coordinated and integrated development of technical and management education. Mandatory periodic evaluation will be carried out by a duly constituted Accreditation Board. In the interests of maintaining standards and for several other valid reasons, the commercialisation of technical and professional education will be curbed. An alternative system will be devised to involve private and voluntary effort in this sector of education, in conformity with accepted norms and goals.

THE MANAGEMENT OF EDUCATION

An overhaul of the system of planning and the management of education will receive high priority.

The guiding considerations will be:

- Evolving a long-term planning and management perspective of education and its integration with the country's developmental and manpower needs;
- Decentralisation and the creation of a spirit of autonomy for educational institutions;
- Giving pre-eminence to people's involvement, including association of non-governmental agencies and voluntary effort;
- Inducting more women in the planning and management of education;
- Establishing the principle of accountability in relation to given objectives and norms.

NATIONAL LEVEL

The Central Advisory Board of Education will play a pivotal role in reviewing educational development, determining the changes required to improve the system and monitoring implementation. It will function through appropriate Committees and other mechanisms created to ensure contact with, and coordination among, the various areas of Human Resource Development. The Departments of Education at the Centre and in the States will be strengthened through the involvement of professionals.

Indian Education Service

A proper management structure in education will entail the establishment of the Indian Education Service as an All-India Service. It will bring a, national perspective to this vital sector. The basic principles, functions and procedures of recruitment to this service will be decided in consultation with the State Governments.

State Level

State Governments may establish State Advisory Boards of Education on the lines of CABE. Effective measures should be taken to integrate mechanisms in the various

State departments concerned with Human Resource Development. Special attention will be paid to the training of educational planners, administrators and heads of institutions. Institutional arrangements for this purpose should be set up in stages.

District and Local Level

District Boards of Education will be created to manage education up to the higher secondary level. State Governments will attend to this aspect with all possible expedition. Within a multi-level framework of educational development, Central, State, District and Local level agencies will participate in planning, coordination, monitoring and evaluation. A very important role must be assigned to the head of an educational institution. Heads will be specially selected and trained. School complexes will be promoted on a flexible pattern so as to serve as networks of institutions and synergic alliances to encourage professionalism among teachers, to ensure observance of norms of conduct and to enable the sharing of experiences and facilities. It is expected that a developed system of school complexes will take over much of the inspection functions in due course. Local communities, through appropriate bodies, will be assigned a major role in programmes of school improvement.

Voluntary Agencies and Aided Institutions

Non-government and voluntary effort including social activist groups will be encouraged, subject to proper management, and financial assistance provided. At the same time, steps will be taken to prevent the establishment of institutions set up to commercialise education.

Resources and Review

The Education Commission of 1964-66, the National Education Policy of 1968 and practically all others concerned with education have stressed that the egalitarian goals and the Practical, development-oriented objectives of Indian society can be realised only by making investments in education of an order commensurate with the nature and dimensions of the task. Resources, to the extent possible, will be raised by mobilising donations, asking the beneficiary communities to maintain school buildings and supplies of some consumables, raising fees at the higher levels of education and effecting some savings by the efficient use of facilities. Institutions involved with research and the development of technical and scientific manpower should also mobilize some funds by levying access or charge on the user agencies, including Government departments, and entrepreneurs. All these measures will be taken not only to reduce the burden on State resources but also for creating a greater sense of responsibility within the educational system. However, such measures will contribute only marginally to the total funding. The Government and the community in general will find funds for such programmes as: the universalisation of elementary education; liquidating illiteracy; equality of access to educational opportunities to all parts throughout the country;

enhancing the social relevance, quality and functional effectiveness of educational programmes; generating knowledge and developing technologies in scientific fields crucial to self-sustaining economic development; and creating a critical consciousness of the values and imperatives of national survival.

The deleterious consequences of non-investment or inadequate investment in education are indeed very serious. Similarly, the cost of neglecting vocational and technical education and of research is also unacceptable. Sub-optimal Performance in these fields could cause irreparable damage to the Indian economy. The network of institutions set up from time to time since Independence to facilitate the application of science and technology would need to be substantially and expeditiously updated, since they are fast becoming obsolete. In view of these imperatives, education will be treated as a crucial area of investment for national development and survival. The National policy on Education, 1968, had laid down that the investment on education be gradually increased to reach a level of expenditure of 6 per cent of the national income as early as possible. since the actual level of investment has remained far short of that target, it is important that greater determination be shown now to find the funds for the programmes laid down in this Policy. While the actual requirements will be Computed from time to time on the basis of monitoring and review, the outlay on education will be shapped up to the extent essential for policy implementation in the Seventh plan. It will be ensured that from the Eighth Five Year Plan onwards it will uniformly exceed to 6 per cent of the National income.

Review

The implementation of the various parameters of the New Policy must be reviewed every five years. Appraisals at short intervals will also be made to ascertain the progress of implementation and the trends emerging from time to time.

The Future

The future shape of education in India is too complex to envision with precision. Yet, given our tradition which has almost always put a high premium on intellectual and spiritual attainment, we are bound to succeed in achieving our objectives. The main task is to strengthen the base of the pyramid, which might come close to a billion people at the turn of the century. Equally, it is important to ensure that those at the top of the pyramid are among the best in the world. our cultural well-springs had taken good care of both ends in the past; the skew set in with foreign domination and influence. It should now be possible to further intensify the nation-wide effort in Human Resource Development, with Education playing its multifaceted role.

Theory of Intelligence

According to Gardner, all human beings possess at least eight intelligences. Thus, intelligence is not a single entity; rather distinct types of intelligences exist. Each type of intelligence is independent one.

It means, if a person exhibits one type of intelligence, it does not necessarily indicate being high or low on other types of intelligences. Gardner also put forth that different types of intelligences interact and work together to find a solution to a problem.

Eight types of intelligence (Gardner):

1. *Linguistic (skills involved in the production and use of language):* It is the capacity to use language fluently and flexibly to express one's thinking and understands others.
2. *Logical-Mathematical (skills in scientific thinking and problem solving):* It is the ability think logically and critically.
3. *Spatial (skills in forming visual images and patterns):* It refers to the abilities involved in forming, using, and transforming mental images.
4. *Musical (sensitivity to musical rhythms and patterns):* It is the capacity to produce, create and manipulate musical patterns.
5. *Bodily-Kinaesthetic (using whole or portions of the body flexibly and creatively):* This consists of the use of the whole body or portions of it for display or construction of products and problem solving.
6. *Naturalistic (sensitivity to the features of the natural world):* This involves complete awareness of our relationship with the natural world.

7. *Interpersonal (sensitivity to subtle aspects of others' behaviours)*: This is the skill of understanding the motives, feelings and behaviours of other people so as to bond into a comfortable relationship with others.
8. *Intrapersonal (awareness of one's own feelings, motives, and desires)*: This refers to the knowledge of one's internal strengths and limitations and using that knowledge to effectively relate to others.

INTELLIGENCE AS A PROCESS

Sternberg's triarchic theory suggests that intelligent behaviour is the product of applying thinking strategies, handling new problems creatively and quickly, and adapting to contexts by selecting and reshaping our environment. Sternberg believes that intelligence is comprised of three separate, though interrelated abilities: analytical, creative, and practical.

Sternberg's view of Intelligence:

- *Componential (analytical) Intelligence*: It is the ability to think abstractly, process information and determine what needs to be done. This intelligence has three components, each serving a different function. First is the knowledge acquisition component, which is responsible for learning and acquisition of the ways of doing things. The second is the Meta or a higher order component, which involves planning concerning what to do and how to do. The third is the performance component, which involves actually doing things.
- *Experiential (creative) Intelligence*: It is the ability to formulate new ideas and combine unrelated facts. It is involved in using past experiences creatively to solve novel problems. It is reflected in creative performance. Persons high on this aspect integrate different experiences in an original way to make new discoveries and inventions.
- *Contextual (practical) Intelligence*: It is the ability to adapt to a changing environment and to shape one's world to optimise opportunities. It may be called '*street smartness*' or '*business sense*'.

NATURE OF INTELLIGENCE

Intelligence is the *manipulation of motor mechanisms* to derive happiness for the organism – inclusive of surviving – and entails the gathering of information external to the organism so by non-verbal communication and wise *motor manipulation*, it can maximize happiness. And, at root, all living organisms are nothing more than a physical organization of their genotype: the initial genetic blueprint of an organism.

Ultimately, at its root, the genotype is actually nothing more than immaterial energy configuring atoms and molecules into genetic code. As it is immaterial – as a thought or idea is immaterial – the genotype is more like a “thought” or “idea” interacting with the

other “thoughts” and “ideas” and set to interact in harmony with its environment – also a “thought.” Interaction is a “*thought being thought-out.*” This idea is not new nor original. Aristotle theorized that God was pure Thought and prior to the Greeks, the Judeo-Christian idea of the Logos, and Egyptian idea of their God, Amun. The physical is not the blueprint – the immaterial electromagnetic is.

The immaterial is idea-like – a living *thought*. As thinking is mediated by plus and minus ions (NaCl^- , KCl^- , Na^+ , K^+ , Cl^-) moving in and out of the nerve fibres of our brain and nervous system, *thought* and *thinking* is the sophisticated summation of the flows and interplay of positive and negative energies. Both the medium for thinking (*i.e.* the brain) and the ability of thinking itself exist in “seed-like” form in the genotype and is “species-specific.”

The nature of the genotype is that it is a *hand-glove model* of its environment both in the sense that all features of a given animal’s environment are modelled in the genotype’s blueprint and in that the genotype creates the animal’s body and behaviour to fit *hand-glove* into its unique environment. In essence, in tiny cells, genotypes taken as a whole – at some snapshot of geological time – serve as a perfect multimedia picture-record written, imprinted in genetic code, of the heavens, the skies, the waters, and the living earth.

Paul Davies, writing of the ability of birds to use stars in their astro-navigation, comments that, “The necessary astro-navigational information is built in DNA... given a sufficient understanding of the nature of DNA, one could ‘decode’ this information and reconstruct a map of the stars.” DNA is built on an immaterial blueprint. In the case of you and I, we are, in essence, just the immaterial blueprint of the one-celled genotype that became us: eyes, brain, charming smile, wit, and great brilliance included.

Language ability is nothing more than the ability of the genotype to associate random sounds with contemporary objects of evolution (*i.e.* today’s world is different from the first human’s world). Reading and writing is nothing less than the ability of the genome to devise and associate random signs (otherwise known as written alphabets, words, and languages) with contemporary objects of evolution such as the words “ice cream” and the “Internet.” Rats evolve right along with us finding great comfort in living structures and types of food the prehistoric world had no imagination would ever exist. Insects have also kept pace with human evolution. Even viruses: computer viruses.

We are living witnesses of co-evolution. We can observe the ability of organisms worldwide to harmoniously co-evolve ever upwards: *e.g.* in the microscopic-level arm’s race, viral innovations against medicines force humans to develop better medicines. Harmonious tit for tat co-evolution finds origins in the organizational marvel of the immaterial dimension. One unfortunate effect of imbalance is that it creates new “harmonious” balances and the imbalances the human has produced have eliminated many life forms from the environment that man is destroying in 200 years that took 2

million years to create. Nevertheless, for living creatures, harmony and balance with nature are written and achieved in genotypes and together, genotypes make the whole living picture of life on earth – each specie of genotype one piece of the puzzle fitting perfectly with the rest. With the right equipment, the genotype could play an organism's life like a video in a VCR. Through life – via the interaction of living things – Nature manifests on yet another level, the balance and harmony found throughout the universe. Balance is found, for instance, in every animal having a strength of defence against others, but a weakness making it available as fodder and the strong have the greatest likelihood of surviving, but death takes an old life away to make way for new.

While from the perspective of the individual, intelligence serves the purpose of striving for, realizing, and maintaining individual happiness, taken as a whole intelligence serves the greater purpose of bringing the whole of the living world and nature into harmonious oneness. Let us remember several things: living things have not the foggiest idea of the technologies that make their organs and forms and make them interact harmoniously within the organism and with the whole environment from starlight to raindrops.

Differences in Intelligence

In this module, we will consider the extremes of intelligence—from the very bright to the very dull. We will also look at a few of the many studies that have been done with IQ scores, correlating many things from sense of humour (Brumbaugh, 1939) to racial background. A few of these correlation studies and their results will be described in this module. As you read the text, try to answer the following questions.

- When is a child considered gifted? Retarded?
- What are the degrees of mental retardation?
- What were Terman's findings in his study of gifted children?
- How does intelligence correlate with sex? Race? Socioeconomic level?

SOCIAL INTELLIGENCE

Social intelligence just as to the original definition of Edward Thorndike, is “the ability to understand and manage men and women, boys and girls, to act wisely in human relations”. It is equivalent to interpersonal intelligence, one of the types of intelligences identified in Howard Gardner's Theory of multiple intelligences, and closely related to theory of mind. Some authors have restricted the definition to deal only with knowledge of social situations, perhaps more properly called social cognition or social marketing intelligence, as it pertains to trending socio-psychological advertising and marketing strategies and tactics. Sean Foleno, Social intelligence is a person's competence to comprehend his or her environment optimally and react appropriately for socially successful conduct.

SCIENTIFIC STUDIES

There are various types of intelligence. As society became more complex, intellectual competences became more sophisticated. This competence is social intelligence and can be defined as the intelligence that lies behind group interactions and This type of intelligence is closely related to cognition and emotional intelligence, and can also be seen as a first level in developing systems intelligence.

Research psychologists studying social cognition and social neuroscience have discovered many principles which human social intelligence operates. In early work on this topic, psychologists Nancy Cantor and John Kihlstrom outlined the kinds of concepts people use to make sense of their social relations and the rules they use to draw inferences and plan actions.

In 2005, business writer Karl Albrecht proposed a five-part model of social intelligence in his book *Social Intelligence: the New Science of Success*, presented with the acronym “S.P.A.C.E.”—(1) Situational awareness, (2) Presence, (3) Authenticity, (4) Clarity, and (5) Empathy. More recently, popular science writer Daniel Goleman has drawn on social neuroscience research to propose that social intelligence is made up of social awareness and social facility. Psychotherapy often involves helping people to modify their patterns of social intelligence, particularly those that cause them problems in their interpersonal relations.

Some efforts are also underway to use computer-based interventions to help people develop their own social intelligence. Paul Ekman, for example, has created the MicroExpression Training Tool, to allow people to practice identifying the brief emotional expressions that flit across people’s faces. Educational researcher Raymond H. Hartjen asserts that expanded opportunities for social interaction enhances intelligence. Traditional classrooms do not permit the interaction of complex social behaviour. Instead children in traditional settings are treated as learners who must be infused with more and more complex forms of information. Few educational leaders he adduces have taken this position as a starting point to develop a school environment where social interaction could flourish. If we follow this line of thinking then children must have an opportunity for continuous every day interpersonal experiences in order to develop a keen well developed ‘inter-personal psychology’. As schools are structured today very few of these skills, critical for survival in the real world, are allowed to develop.

Because we so limit the development of the skills of “natural psychologist” in traditional schools our students as graduates, enter the job market handicapped to the point of being incapable of surviving on their own. In contrast those students that have had an ability to develop their skills as a “natural psychologist” in multiage classrooms and at democratic settings rise head and shoulders over their less socially skilled peers. They have a good sense of self, know what they want out of life and have the skills necessary to begin their quest.

MEASURING SOCIAL INTELLIGENCE

Social IQ is a measure of social intelligence compared to other people of their age. Like IQ, Social IQ is based on the “100 point” scale, in which 100 is the average score. Scores of 140 or above are considered to be very high. Social IQ has until recently been measured by techniques such as question and answer sessions. These sessions assess the person’s pragmatic abilities to test eligibility in certain special education courses, however some tests have been developed to measure social intelligence. One of these is the EQ test. This test can be used when diagnosing autism spectrum disorders, including autism and Asperger syndrome. Other, non-autistic or semi-autistic conditions such as semantic pragmatic disorder or SPD, schizophrenia, dyssemia and ADHD, are also of relevance. People with low social IQ will be considered “child like” and immature, even at the adult age group. This test can also be used when assessing people that might have some sort of a disorder such as schizophrenia or ADHD.

A good way to measure Social IQ is to use the basic IQ system, adapted for social skills. Most people have social IQs from 85-115, but many exceed these limits. People with social IQs below 80 may have an autism spectrum disorder, such as Asperger syndrome or a severe case of some other disorder such as schizophrenia or maybe just generally low in general intelligence. These people may have trouble with making friends, and with communication, and might need some social skill training or extra support from specialists.

These people may not be hired as quickly as other people for employment, since they may not have the require interpersonal communication and social skills for success in the workforce. These people may work well in an office desk job, stay at home job or jobs that don’t require a lot of interaction, such as construction. People with social IQs over 120 are considered very socially skilled and well adjusted, and could work well with jobs that involve direct contact and communication with people.

The following example chart shows how a person’s social age can be higher or lower based on scores in the social IQ test.:

Social IQ	Social Age
120 (above average - socially mature for age)	20.4
110	18.7
100 (average)	17
90	15.3
80	13.6
70 (below this level, help is recommended)	11.9
60	10.2
50	8.5
40	6.8
30	5.1
20	3.4

EMOTIONAL INTELLIGENCE

Emotional intelligence is the ability, capacity, skill; or, in the case of the trait EI model, a self-perceived ability to identify, assess, and control the emotions of oneself, of others, and of groups. Different models have been proposed for the definition of EI and there is disagreement about how the term should be used. Despite these disagreements, which are often highly technical, the ability-EI and trait-EI models enjoy support in the literature and have successful applications in various domains.

HISTORY

The earliest roots of emotional intelligence can be traced to Darwin's work on the importance of emotional expression for survival and second adaptation. In the 1900s, even though traditional definitions of intelligence emphasized cognitive aspects such as memory and problem-solving, several influential researchers in the intelligence field of study had begun to recognize the importance of the non-cognitive aspects.

For instance, as early as 1920, E.L. Thorndike used the term social intelligence to describe the skill of understanding and managing other people. Similarly, in 1940 David Wechsler described the influence of non-intellective factors on intelligent behaviour, and further argued that our models of intelligence would not be complete until we can adequately describe these factors.

In 1983, Howard Gardner's *Frames of Mind: The Theory of Multiple Intelligences* introduced the idea of multiple intelligences which included both interpersonal intelligence and intrapersonal intelligence. In Gardner's view, traditional types of intelligence, such as IQ, fail to fully explain cognitive ability. Thus, even though the names given to the concept varied, there was a common belief that traditional definitions of intelligence are lacking in ability to fully explain performance outcomes.

The first use of the term "emotional intelligence" is usually attributed to Wayne Payne's doctoral thesis, *A Study of Emotion: Developing Emotional Intelligence from 1985*. However, prior to this, the term "emotional intelligence" had appeared in Leuner.

Greenspan also put forward an EI model, followed by Salovey and Mayer and Goleman. The distinction between trait emotional intelligence and ability emotional intelligence was introduced in 2000.

DEFINITIONS

Substantial disagreement exists regarding the definition of EI, with respect to both terminology and operationalizations. There has been much confusion about the exact meaning of this construct. The definitions are so varied, and the field is growing so rapidly, that researchers are constantly re-evaluating even their own definitions of the construct.

Currently, there are three main models of EI:

- Ability EI model
- Mixed models of EI
- Trait EI model

Different models of EI have led to the development of various instruments for the assessment of the construct. While some of these measures may overlap, most researchers agree that they tap different constructs.

Ability Model

Salovey and Mayer's conception of EI strives to define EI within the confines of the standard criteria for a new intelligence. Following their continuing research, their initial definition of EI was revised to "The ability to perceive emotion, integrate emotion to facilitate thought, understand emotions and to regulate emotions to promote personal growth." The ability-based model views emotions as useful sources of information that help one to make sense of and navigate the social environment. The model proposes that individuals vary in their ability to process information of an emotional nature and in their ability to relate emotional processing to a wider cognition. This ability is seen to manifest itself in certain adaptive behaviours.

The model claims that EI includes four types of abilities:

1. *Perceiving emotions:* The ability to detect and decipher emotions in faces, pictures, voices, and cultural artifacts—including the ability to identify one's own emotions. Perceiving emotions represents a basic aspect of emotional intelligence, as it makes all other processing of emotional information possible.
2. *Using emotions:* The ability to harness emotions to facilitate various cognitive activities, such as thinking and problem solving. The emotionally intelligent person can capitalize fully upon his or her changing moods in order to best fit the task at hand.
3. *Understanding emotions:* The ability to comprehend emotion language and to appreciate complicated relationships among emotions. For example, understanding emotions encompasses the ability to be sensitive to slight variations between emotions, and the ability to recognize and describe how emotions evolve over time.
4. *Managing emotions:* The ability to regulate emotions in both ourselves and in others. Therefore, the emotionally intelligent person can harness emotions, even negative ones, and manage them to achieve intended goals.

The ability EI model has been criticized in the research for lacking face and predictive validity in the workplace.

Measurement of the Ability Model

The current measure of Mayer and Salovey's model of EI, the Mayer-Salovey-Caruso Emotional Intelligence Test is based on a series of emotion-based problem-solving

items. Consistent with the model's claim of EI as a type of intelligence, the test is modeled on ability-based IQ tests. By testing a person's abilities on each of the four branches of emotional intelligence, it generates scores for each of the branches as well as a total score. Central to the four-branch model is the idea that EI requires attunement to social norms.

Therefore, the MSCEIT is scored in a consensus fashion, with higher scores indicating higher overlap between an individual's answers and those provided by a worldwide sample of respondents. The MSCEIT can also be expert-scored, so that the amount of overlap is calculated between an individual's answers and those provided by a group of 21 emotion researchers. Although promoted as an ability test, the MSCEIT is most unlike standard IQ tests in that its items do not have objectively correct responses. Among other problems, the consensus scoring criterion means that it is impossible to create items that only a minority of respondents can solve, because, by definition, responses are deemed emotionally "intelligent" only if the majority of the sample has endorsed them. This and other similar problems have led cognitive ability experts to question the definition of EI as a genuine intelligence.

In a study by Føllesdal, the MSCEIT test results of 111 business leaders were compared with how their employees described their leader. It was found that there were no correlations between a leader's test results and how he or she was rated by the employees, with regard to empathy, ability to motivate, and leader effectiveness. Føllesdal also criticized the Canadian company Multi-Health Systems, which administers the MSCEIT test. The test contains 141 questions but it was found after publishing the test that 19 of these did not give the expected answers. This has led Multi-Health Systems to remove answers to these 19 questions before scoring, but without stating this officially.

Mixed Models

The model introduced by Daniel Goleman focuses on EI as a wide array of competencies and skills that drive leadership performance.

Goleman's model outlines four main EI constructs:

1. *Self-awareness*: The ability to read one's emotions and recognize their impact while using gut feelings to guide decisions.
2. *Self-management*: Involves controlling one's emotions and impulses and adapting to changing circumstances.
3. *Social awareness*: The ability to sense, understand, and react to others' emotions while comprehending social networks.
4. *Relationship management*: The ability to inspire, influence, and develop others while managing conflict.

Goleman includes a set of emotional competencies within each construct of EI. Emotional competencies are not innate talents, but rather learned capabilities that must be worked on and can be developed to achieve outstanding performance. Goleman

posits that individuals are born with a general emotional intelligence that determines their potential for learning emotional competencies. Goleman's model of EI has been criticized in the research literature as mere "pop psychology".

Measurement of the Emotional Competencies (Goleman) Model

Two measurement tools are based on the Goleman model:

1. The Emotional Competency Inventory which was created in 1999, and the Emotional and Social Competency Inventory which was created in 2007.
2. The Emotional Intelligence Appraisal, which was created in 2001 and which can be taken as a self-report or 360-degree assessment.

Bar-On Model of Emotional-Social Intelligence

Bar-On defines emotional intelligence as being concerned with effectively understanding oneself and others, relating well to people, and adapting to and coping with the immediate surroundings to be more successful in dealing with environmental demands. Bar-On posits that EI develops over time and that it can be improved through training, programming, and therapy. Bar-On hypothesizes that those individuals with higher than average EQs are in general more successful in meeting environmental demands and pressures. He also notes that a deficiency in EI can mean a lack of success and the existence of emotional problems. Problems in coping with one's environment are thought, by Bar-On, to be especially common among those individuals lacking in the subscales of reality testing, problem solving, stress tolerance, and impulse control. In general, Bar-On considers emotional intelligence and cognitive intelligence to contribute equally to a person's general intelligence, which then offers an indication of one's potential to succeed in life.

Measurement of the ESI Model

The Bar-On Emotional Quotient Inventory is a self-report measure of EI developed as a measure of emotionally and socially competent behaviour that provides an estimate of one's emotional and social intelligence. The EQ-i is not meant to measure personality traits or cognitive capacity, but rather the mental ability to be successful in dealing with environmental demands and pressures. One hundred and thirty three items are used to obtain a Total EQ and to produce five composite scale scores, corresponding to the five main components of the Bar-On model. A limitation of this model is that it claims to measure some kind of ability through self-report items. The EQ-i has been found to be highly susceptible to faking.

Trait EI Model

Petrides and colleagues proposed a conceptual distinction between the ability based model and a trait based model of EI. Trait EI is "a constellation of emotional self-

perceptions located at the lower levels of personality”. In lay terms, trait EI refers to an individual’s self-perceptions of their emotional abilities. This definition of EI encompasses behavioural dispositions and self perceived abilities and is measured by self report, as opposed to the ability based model which refers to actual abilities, which have proven highly resistant to scientific measurement. Trait EI should be investigated within a personality framework. An alternative label for the same construct is trait emotional self-efficacy.

The conceptualization of EI as a personality trait leads to a construct that lies outside the taxonomy of human cognitive ability. This is an important distinction in as much as it bears directly on the operationalization of the construct and the theories and hypotheses that are formulated about it.

Measurement of the Trait EI Model

There are many self-report measures of EI, including the EQ-i, the Swinburne University Emotional Intelligence Test and the Schutte EI model. None of these assess intelligence, abilities, or skills but rather, they are limited measures of trait emotional intelligence. One of the more comprehensive and widely researched measures of this construct is the Trait Emotional Intelligence Questionnaire which is an open-access measure that was specifically designed to measure the construct comprehensively and is currently available in many languages. The TEIQue provides an operationalization for the model of Petrides and colleagues, that conceptualizes EI in terms of personality. The test encompasses 15 subscales organized under four factors: Well-Being, Self-Control, Emotionality, and Sociability. The psychometric properties of the TEIQue were investigated in a study on a French-speaking population, where it was reported that TEIQue scores were globally normally distributed and reliable.

The researchers also found TEIQue scores were unrelated to non-verbal reasoning which they interpreted as support for the personality trait view of EI. As expected, TEIQue scores were positively related to some of the Big Five personality traits as well as inversely related to others. A number of quantitative genetic studies have been carried out within the trait EI model, which have revealed significant genetic effects and heritabilities for all trait EI scores.

Two recent studies involving direct comparisons of multiple EI tests yielded very favourable results for the TEIQue. Alexithymia and EI. Alexithymia from the Greek words *lexis* and *thumos* is a term coined by Peter Sifneos in 1973 to describe people who appeared to have deficiencies in understanding, processing, or describing their emotions. Viewed as a spectrum between high and low EI, the alexithymia construct is strongly inversely related to EI, representing its lower range. The individual’s level of alexithymia can be measured with self-scored questionnaires such as the Toronto Alexithymia Scale or the Bermond-Vorst Alexithymia Questionnaire or by observer rated measures such as the Observer Alexithymia Scale.

CRITICISMS OF THE THEORETICAL FOUNDATION OF EI

EI cannot be Recognized as a Form of Intelligence

Goleman's early work has been criticized for assuming from the beginning that EI is a type of intelligence. Eysenck writes that Goleman's description of EI contains assumptions about intelligence in general, and that it even runs contrary to what researchers have come to expect when studying types of intelligence:

- “[Goleman] exemplifies more clearly than most the fundamental absurdity of the tendency to class almost any type of behaviour as an ‘intelligence’... If these five ‘abilities’ define ‘emotional intelligence’, we would expect some evidence that they are highly correlated; Goleman admits that they might be quite uncorrelated, and in any case if we cannot measure them, how do we know they are related? So the whole theory is built on quicksand: there is no sound scientific basis.”

Similarly, Locke claims that the concept of EI is in itself a misinterpretation of the intelligence construct, and he offers an alternative interpretation: it is not another form or type of intelligence, but intelligence—the ability to grasp abstractions—applied to a particular life domain: emotions. He suggests the concept should be re-labeled and referred to as a skill.

The essence of this criticism is that scientific enquiry depends on valid and consistent construct utilization, and that before the introduction of the term EI, psychologists had established theoretical distinctions between factors such as abilities and achievements, skills and habits, attitudes and values, and personality traits and emotional states. Thus, some scholars believe that the term EI merges and conflates such accepted concepts and definitions.

EI has little Predictive Value

Landy claimed that the few incremental validity studies conducted on EI have shown that it adds little or nothing to the explanation or prediction of some common outcomes. Landy suggested that the reason why some studies have found a small increase in predictive validity is a methodological fallacy, namely, that alternative explanations have not been completely considered:

- “EI is compared and contrasted with a measure of abstract intelligence but not with a personality measure, or with a personality measure but not with a measure of academic intelligence.”

Similarly, other researchers have raised concerns about the extent to which self-report EI measures correlate with established personality dimensions. Generally, self-report EI measures and personality measures have been said to converge because they both purport to measure personality traits. Specifically, there appear to be two dimensions of the Big Five that stand out as most related to self-report EI—neuroticism and

extroversion. In particular, neuroticism has been said to relate to negative emotionality and anxiety. Intuitively, individuals scoring high on neuroticism are likely to score low on self-report EI measures. The interpretations of the correlations between EI questionnaires and personality have been varied. The prominent view in the scientific literature is the Trait EI view, which re-interprets EI as a collection of personality traits.

CRITICISMS OF MEASUREMENT ISSUES

Ability EI Measures Measure Conformity, Not Ability

One criticism of the works of Mayer and Salovey comes from a study by Roberts, which suggests that the EI, as measured by the MSCEIT, may only be measuring conformity. This argument is rooted in the MSCEIT's use of consensus-based assessment, and in the fact that scores on the MSCEIT are negatively distributed.

Ability EI Measures Measure Knowledge

Further criticism has been offered by Brody who claimed that unlike tests of cognitive ability, the MSCEIT “tests knowledge of emotions but not necessarily the ability to perform tasks that are related to the knowledge that is assessed”. The main argument is that even though someone knows how he should behave in an emotionally laden situation, it doesn't necessarily follow that he could actually carry out the reported behaviour.

Self-Report Measures are Susceptible to Faking

More formally termed socially desirable responding, faking good is defined as a response pattern in which test-takers systematically represent themselves with an excessive positive bias. This bias has long been known to contaminate responses on personality inventories, acting as a mediator of the relationships between self-report measures. It has been suggested that responding in a desirable way is a response set, which is a situational and temporary response pattern.

This is contrasted with a response style, which is a more long-term trait-like quality. Considering the contexts some self-report EI inventories are used in, the problems of response sets in high-stakes scenarios become clear. There are a few methods to prevent socially desirable responding on behaviour inventories. Some researchers believe it is necessary to warn test-takers not to fake good before taking a personality test. Some inventories use validity scales in order to determine the likelihood or consistency of the responses across all items.

Claims for the Predictive Power of EI are too Extreme

Landy distinguishes between the “commercial wing” and “the academic wing” of the EI movement, basing this distinction on the alleged predictive power of EI as seen by the two currents. Landy, the former makes expansive claims on the applied value of

EI, while the latter is trying to warn users against these claims. As an example, Goleman asserts that “the most effective leaders are alike in one crucial way: they all have a high degree of what has come to be known as emotional intelligence....emotional intelligence is the sine qua non of leadership”.

In contrast, Mayer cautions “the popular literature’s implication—that highly emotionally intelligent people possess an unqualified advantage in life—appears overly enthusiastic at present and unsubstantiated by reasonable scientific standards.” Landy further reinforces this argument by noting that the data upon which these claims are based are held in “proprietary databases”, which means they are unavailable to independent researchers for reanalysis, replication, or verification. Thus, the credibility of the findings cannot be substantiated in a scientific way, unless those datasets are made public and available for independent analysis. In an academic exchange, Antonakis and Ashkanasy/Dasborough mostly agreed that researchers testing whether EI matters for leadership have not done so using robust research designs; therefore, currently there is no strong evidence showing that EI predicts leadership outcomes when accounting for personality and IQ. Antonakis argued that EI might not be needed for leadership effectiveness. A recently-published meta-analysis seems to support the Antonakis position: In fact, Harms and Credé found that overall, EI measures correlated only $r = .11$ with measures of transformational leadership. Interestingly, ability-measures of EI fared worst; the WLEIS did a bit better and the Bar-On measure better still. However, the validity of these estimates does not include the effects of IQ or the big five personality, which correlate both with EI measures and leadership. In a subsequent document analysing the impact of EI on both job performance and leadership, Harms and Credé found that the meta-analytic validity estimates for EI dropped to zero when Big Five traits and IQ were controlled for.

EI, IQ and Job Performance

Research of EI and job performance show mixed results: a positive relation has been found in some of the studies, in others there was no relation or an inconsistent one. This led researchers Cote and Miners to offer a compensatory model between EI and IQ, that posits that the association between EI and job performance becomes more positive as cognitive intelligence decreases, an idea first proposed in the context of academic performance. The results of the former study supported the compensatory model: employees with low IQ get higher task performance and organizational citizenship behaviour directed at the organization, the higher their EI.

DISTRIBUTION OF INTELLIGENCE

People who score above 130 on intelligence tests are usually considered gifted. An important study of gifted children was undertaken by who devised the Stanford-Binet

test. In the course of developing that test, Terman and his associates tested many thousands of children, and then did further research on those who had IO's of 140 or more. This group of more than 1500 children was in the top 1% of the population. Terman followed most of these children from 1921 until his death in 1956. Some of his associates are still continuing this detailed longitudinal study.

The study revealed an important fact related to the home environments of the children. Approximately one-third were children of professional people and about half were children of persons in business occupations. Only seven percent came from "working classes." These percentages do not agree at all with the distribution of social class membership in the general population. They indicate that relatively more gifted children come from the higher socioeconomic classes, although they do not tell us why.

Another discovery in Terman's study was that gifted children are more successful in later life. Twenty-five years after the study began, about 700 of the original group were still in contact with Terman. Of these, 150 were considered very successful. This distinction was based on such criteria as being listed in Who's Who or having received recognition for outstanding intellectual or professional achievement. Most of the others were much more successful than people of average intelligence. On the other hand, some persons in the gifted group had committed crimes, dropped out of school early, or been unsuccessful at a number of jobs.

Those least successful were more poorly adjusted emotionally and less motivated than the others. On the whole, though, the most striking fact revealed in the study was that gifted children made generally outstanding contributions to society in the form of intellectual achievements. In their early years, Terman's gifted children were above average in height, weight, and physical development, and they were also better adjusted.

It had been generally believed that very gifted children were likely to be maladjusted and socially backward. Terman's study definitely disproved this notion.

The gifted child is liable to encounter serious difficulties in school. He may be bored by children of his own age and his knowledge may even exceed that of his teacher. Teachers may consider him rude or a show-off. He may become a real problem simply because he is bored and loses interest in a class designed for "average" children. Some schools today are facing this problem by planning appropriate programmes for gifted children.

INTELLIGENCE TESTS AS CLINICAL INSTRUMENTS

INTELLIGENCE

Intelligence has been defined in several different methods including logic, abstract thought, understanding, self-awareness, communication, learning, having emotional

knowledge, retaining, planning, and problem solving. Intelligence is most widely studied in humans, but has also been observed in other animals and in plants. Artificial intelligence is the simulation of intelligence in machines.

Within the discipline of psychology, several approaches to human intelligence have been adopted. The psychometric approach is especially familiar to the general public, as well as being the most researched and through distant the most widely used in practical settings.

Definitions

The definition of intelligence is controversial. Some groups of psychologists have suggested the following definitions:

From “Mainstream Science on Intelligence” (1994), an editorial statement through fifty-two researchers: A very general mental capability that, among other things, involves the skill to cause, plan, solve problems, think abstractly, comprehend intricate ideas, learn quickly and learn from experience. It is not merely book learning, a narrow academic ability, or test-taking smarts. Rather, it reflects a broader and deeper capability for comprehending our surroundings—“catching on,” “making sense” of things, or “figuring out” what to do.

From “Intelligence: Known’s and Unknowns” (1995), a report published through the Board of Scientific Affairs of the American Psychological Association: Individuals differ from one another in their skill to understand intricate ideas, to adapt effectively to the environment, to learn from experience, to engage in several forms of reasoning, to overcome obstacles through taking thought. Although these individual differences can be substantial, they are never entirely constant: a given person’s intellectual performance will vary on different occasions, in different domains, as judged through different criteria. Concepts of “intelligence” are attempts to clarify and organize this intricate set of phenomena. Although considerable clarity has been achieved in some areas, no such conceptualization has yet answered all the significant questions, and none commands universal assent. Indeed, when two dozen prominent theorists were recently asked to describe intelligence, they gave two dozen, somewhat different, definitions.

Theories

Intelligence theorists fall into two categories. In one group are those who argue for a “general intelligence” that characterizes a person’s actions and thinking in all areas. Their critics consider that intelligence is composed of several separate types of aptitudes and abilities, and that a person who excels in one area will not necessarily excel in all areas.

Early Theories: Spearman and Thurstone

Spearman whispered that intelligence is general: People who are bright in one area are bright in other areas as well. Thurstone disagreed: He whispered that intelligence

encompasses seven mental abilities that are relatively independent of one another. In contrast, Cattell divided mental abilities into two clusters. The first is crystallized intelligence, or abilities such as reasoning and the verbal and numerical skills that are stressed in school. The second is fluid intelligence, or skills such as spatial and visual imagery, the skill to notice visual details, and rote memory.

Contemporary Theories: Sternberg and Gardner

In the mid-1980s, Yale psychologist Robert Sternberg proposed a triarchic theory of intelligence that comprises a much broader range of skills and abilities. According to this theory, intelligence consists of three overarching characteristics: componential intelligence, the traditional mental processes or skills accentuated through earlier theories of intelligence, such as the skill to acquire new knowledge and perform tasks efficiently; experiential intelligence, characterized through insight and creative adaptability as well as efficient and quick processing of information without conscious thought; and contextual intelligence, marked through responsiveness to the environment. Intelligent people, according to Sternberg, are adept at making the most of their strengths and compensating for their weaknesses. Howard Gardner has proposed his theory of multiple intelligences, which asserts that what we refer to as intelligence actually consists of several separate abilities, each of which is relatively independent of the others.

Formal theories of intelligence serve as the foundation for the design and administration of intelligence tests. And because experts do not view intelligence in exactly the same method that non-experts do, it is understandable that most tests of intelligence do not contain items that several non-experts think of as part of intelligence.

Measures of Intelligence

Alfred Binet and Theodore Simon, commissioned through the French government to devise a test that would differentiate flunked by children who could not learn because of low skill and those who would not learn because of poor motivation, created the first intelligence test in 1905. Binet did not really have a theory of mental skill, but he thought that intelligence could be best expressed in and assessed through intricate cognitive tasks. The first intelligence test consisted of items that assessed memory, knowledge, and reasoning skills arranged in a sequence of increasing difficulty and grouped into age levels, or when the “average” child could accomplish them. This first test has undergone numerous revisions and is now recognized as the Stanford-Binet Intelligence Scale, Fifth Edition. Currently, the most widely-used intelligence tests are the Wechsler Scales. Intended through David Wechsler, a clinical psychologist who used the test to help identify cognitive strengths and weaknesses in his clients and assist in diagnoses, the tests were based on his definition of intelligence as “the aggregate or global capability of the individual to act purposefully, to think rationally, and to deal effectively with his environment”.

Because the test seemed to be a valid measure of general intelligence, its use expanded to other populations.

There are three separate Wechsler scales:

- Wechsler Preschool and Primary Scale of Intelligence, 3rd ed. (WPPSI-III)
- Wechsler Intelligence Scale for Children, 4th ed. (WISC-IV)
- Wechsler Adult Intelligence Scale, 3rd ed. (WAIS-III)
- The WISC-IV is intended for use with children and adolescents ages 6 to 16 years 11 months and will be the focus of this section. Sample items from many of the 10 core subtests are presented below:
- *Similarities*: In what method are wool and cotton alike?
- *Vocabulary*: What does corrupt mean?
- *Comprehension*: Why do people buy fire insurance?
- *Digit Span*: I am going to say some numbers. Listen cautiously and when I am through, say the numbers right after me: 734186
- *Coding*: In the top row, each figure is shown with a number. Fill in the number that goes with each figure in the second row
- *Matrix Reasoning*: For each item the child looks at a partial matrix and selects the missing portion from five response options.
- *Picture Concepts*: The child identifies objects that share some common property.

Some of the items in this test require children and adolescents to arrange materials rather than talk with the test administrator. This procedure allows non-English-speaking children, as well as youth with speech or language disorders, to demonstrate their intellectual capabilities. Some professionals find the division flanked by verbal and performance items very useful for assessment purposes.

The items create up four index scores—Verbal Comprehension, Perceptual Organization, Freedom from Distractability, and Processing Speed—that are combined to provide an overall skill score. When the four index scores are added together, they produce a total skill score, or intelligence quotient (IQ). On the Wechsler scales, the mean total IQ is set at 100 with a standard deviation of 15. Most other major intelligence tests have set the same mean so it is easy to compare the IQ score from one intelligence test to another.

Determines

Heredity

Historically, research on the determinants of intelligence has focused on identical twins—some reared together; others reared separately in separate households. The correlation flanked by the IQs of all identical twins is usually very high, indicating that their identical genetic inheritance is a more powerful determinant of intelligence than their experiences. But critics of this research create many strong points: (1) It is hard to

find identical twins who have been separated at birth, so that there are only a few such studies; (2) identical twins tend to be placed in households similar in socioeconomic background to those of their biological parents; and (3) even twins separated at birth have had almost identical prenatal experiences.

Environment

Research on rats as well as on humans strengthens the case for environment as a factor in the development of superior intellectual skill. Therefore, even though certain mental abilities are inherited, without the necessary stimulation a child's intelligence will not develop. This finding is significant because lower-income families don't have access to the kinds of resources that other families do. Significantly, when they are placed in more stimulating environments, economically deprived children show an improvement in their level of intelligence. For instance, lower-income children raised in middle-class homes display important gains in IQ compared with their counterparts rising up in low-income households. Similarly, children who participate in intervention programmes such as Head Start regularly exhibit improvements in cognitive abilities, although the long-term effects of such programmes have yet to be confirmed.

The IQ Debate: A Continuing Controversy

Accounting for group differences in IQ poses a vexing problem in psychology. A milestone in this debate was the 1969 publication of an article through psychologist Arthur Jensen, claiming that overall differences in IQ scores flanked by the races are largely inherited. Jensen's article raised a storm of controversy, which swelled up again in 1994 with the publication of a book on this topic through Richard Herrnstein and Charles Murray. Significantly, most participants in this debate agree that both heredity and environment affect IQ scores.

Testing for Special Aptitudes

Aptitude

An aptitude is a component of a competency to do a certain kind of work at a certain level, which can also be measured "talent". Aptitudes may be physical or mental. Aptitude is not knowledge, understanding, learned or acquired abilities (skills) or attitude. The innate nature of aptitude is in contrast to achievement, which represents knowledge or skill that is gained.

Intelligence

Aptitude and intelligence quotient are related, and in some methods opposite views of human mental skill. Whereas intelligence quotient sees intelligence as being a single measurable characteristic affecting all mental skill, aptitude refers to one of several

different characteristics which can be independent of each other, such as aptitude for military flight, air traffic control, or computer programming. This is more similar to the theory of multiple intelligences.

Concerning a single measurable characteristic affecting all mental skill, analysis of any group of intelligence test scores will almost always show them to be highly correlated. The U.S. Department of Labor's General Learning Skill, for instance, is determined through combining Verbal, Numerical and Spatial aptitude subtests. In a given person some are low and others high.

In the context of an aptitude test the "high" and "low" scores are usually not distant separately, because all skill test scores tend to be correlated. Aptitude is better applied intra-individually to determine what tasks a given individual is more skilled at performing. Inter-individual aptitude differences are typically not very important due to IQ differences. Of course this assumes individuals have not already been pre-screened for aptitude through some other process such as SAT scores, GRE scores, or finishing medical school.

Combined Aptitude and Knowledge Tests

Tests that assess learned skills or knowledge are regularly described achievement tests. Though, certain tests can assess both types of constructs. An instance that leans both methods is the Armed Services Vocational Aptitude Battery (ASVAB), which is given to recruits entering the armed forces of the United States. Another is the SAT, which is intended as a test of aptitude for college in the United States, but has achievement elements. For instance, it tests mathematical reasoning, which depends both on innate mathematical skill and education received in mathematics.

Aptitude tests can typically be grouped according to the type of cognitive skill they measure:

- *Fluid intelligence:* the skill to think and cause abstractly, effectively solve problems and think strategically. It's more commonly recognized as 'street smarts' or the skill to 'quickly think on your feet'. Examples of what employers can learn from your fluid intelligence in relation to the your suitability for the role for which you are applying
- *Crystallized intelligence:* the skill to learn from past experiences and relevant learning, and to apply this learning to work-related situation. Work situations that require crystallized intelligence contain producing and analyzing written reports, comprehending work instructions, using numbers as a tool to create effective decisions, etc.

The Stanford-Binet Intelligence Scale

The Binet-Simon Scale, the first test of intelligence, was urbanized in France through Alfred Binet and Theodore Simon for testing children. Originally issued in 1908, it

consisted of 30 tests arranged in order of increasing difficulty. From the average scores of children, Binet urbanized the concept of mental age. The best-recognized Binet version, created through Stanford University's L. M. Terman in 1916, is the Stanford-Binet Intelligence Scale. Terman introduced the term intelligence quotient (IQ), which is a numerical value given to scores on an intelligence test (a score of 100 corresponds to average intelligence). The Stanford-Binet is intended to measure skills in four areas: verbal reasoning, abstract/visual reasoning, quantitative reasoning, and short-term memory.

The Wechsler Intelligence Scales

The Wechsler Adult Intelligence Scale-Third Edition (WAIS-III) was urbanized through David Wechsler especially for adults. The test measures both verbal and performance abilities. Wechsler also created the Wechsler Intelligence Scale for Children-Third Edition (WISC-III), which is meant to be used with school-aged children. It measures verbal and performance abilities separately, though it also yields an overall IQ score.

Group Tests

Group tests are administered through one examiner to several people at one time. Group tests are most commonly used through schools. The California Test of Mental Maturity (CTMM) and the SAT are group tests. Group tests aim to overcome the problems of time and expense associated with individual tests and to eliminate bias on the part of the examiner. Though, in a group setting the examiner is less likely to notice whether an individual test taker is tired, ill, or confused through the directions. Emotionally disturbed children and people who have less experience taking tests usually do better on individual tests than on group tests.

Performance and Culture-Fair Tests

Some intelligence tests may discriminate against members of certain cultural or ethnic groups. Performance tests are intelligence tests that do not involve language, so they can be useful for testing people who lack a strong command of English. The Seguin Form Board, the Porteus Maze, and the Bayley Scale of Infant Development are performance tests. Culture-fair tests are intended to eliminate cultural bias through minimizing skills and values that vary from one culture to another. The Good enough-Harris Drawing Test and the Progressive Matrices are examples of culture-fair tests.

MOTIVATION

Motives as Inference

There is some reference (be it conscious or unconscious) underlying our behaviour as it is not random or purposeless. So, motivation plays a role of energizing behaviour

from inside. A motivated individual initiates and accomplishes several responses in the environment, and achieves his/her goals faster, more vigorously, and more persistently than does one who is less motivated. Motivation is noted to be long lasting. It accounts for performed behaviours and explains the why of behaviour. We can so say that motivation has direct impact on behaviour as it creates behaviour much more goal and purpose oriented.

Through nature, most motives go cyclically. There is usually a start point, which ends to become itself after some time laps. (Here, note that there are few motives that do not go in cycles). In explaining the cycle of motives, let us first describe motivation. “Motivation refers to states within a person or animal that drive behaviour towards some goal.” In this definition, there are three points that need emphasis;

Motivating state- (this state could also be described the ‘drive.’) It arouses the motivating behaviour. The arousal could start to function mainly through three factors:

- Internal bodily needs,
- External sensory stimulation, and
- Cognitive experiences

The behaviour that is aroused through the drive - this one is instrumental in satisfying our drive. This behaviour is a means to an end. It is going to settle the unsettled feelings in us. The goal (end) - depending on the drive that activates (incites) the present motivating state, this is the reduction of the drive. This one is an end through itself hence it is the satisfaction of the need we experienced at the first stage.

Depending on the motivating state that is moving us at a certain point in time, goals could be positive or negative. We approach positive goals and try to satisfy them. On the other hand, we avoid or escape from negative goals that frustrate or endanger us. Both satisfy a certain motive though we approach the positive as pleasant and avoid the negative as unpleasant.

Theories of Motivation

- *Drive-reduction theory (‘push’ theory):* States that the reduction of biological drives, like hunger, thirst, oxygen, pain avoidance, ..., is pleasure producing. A drive is an internal states of tension that motivates (pushes) an organism to engage in activities that should reduce this tension. This theory is simple with several limitations since the source of pleasure is not always associated with biological drive reduction. But it has significance in explaining the fact that when we lack something that is significant for our survival, there is the ‘drive’ that motivates us to satisfy it.
- *Incentive (‘pull’) theory:* This is described ‘pull’ theory because the goal objects pull behaviour towards them. In contrast from the drive theory, this theory says that the response to what is motivating us at a certain point in time is valued more than anything in the environment. This is so because people expect pleasure

from the attainment of what are described positive incentives, and from avoidance of negative incentives. So, the response is acting as a goal (incentive) to our motivating behaviour. Experiments on the proof of this theory show that organisms tend to like the goal they achieved when they really wanted it and when it is something they like more.

- *Arousal Theory*: This theory tells us in relation to the things arouse (excite) organisms to respond in a certain manner to their nearby. It says that both high and low arousal states do not create our performance effective. So, to become effective, it is good to be in an intermediate state of arousal.

There are four general sources of arousal;

- Physiological (internal) drives and incentives are arousing,
- Intensity of environmental stimuli affect our arousal,
- Surprising or new events arouse curiosity,
- Drugs arouse people to act in an excited manner.

Self-Actualization Theory

A. Maslow put what drive human beings from the bottom to the higher in order of hierarchy. The most significant ones are found at the base. Human beings necessity first satisfies their needs for survival – named *physiological needs*. After being able to eat, to drink, to sleep, ..., (after securing survival), people continue to care in relation to their.

Safety needs that are avoidance of accidents and painful sensations. They contain concern for long-term survival, job security, pension, buying insurance, saving money, and the like.

Love and belongingness (social) needs continue through having someone to love and through affiliating with others as a contributing member in the society. Here, an individual is able to identify his/her personal being.

Esteem (ego) needs are the fourth elements in the hierarchy that Maslow described human beings get motivated through. People want to report their success to get prestige. This is the feeling of self-worth through knowing that others are aware of one's competence and value. At the top of Maslow's hierarchy comes the last of the motivating states for human beings, *i.e. self-actualization*. Through this we explore and understand our potentialities to the full and influence the world. This is the state of self-fulfillment. Getting motivated through the self-actualization needs ultimately goes in making us actualize who we really are. At this stage, people feel satisfied of their current state of affairs. This fifth need is highly influenced through the society we live in. This is because a urbanized society helps one get facilitated through the materials (the infra-structure) one can use in the process of actualizing his/her own self (inner potentialities), while less urbanized societies are not conducive places to satisfy this need. Of these five hierarchies, Maslow labels the last three as psychological states while the first two are basic physiological ones.

Biological Motives

Biological Motives to a large extent, rooted in the physique (body). Since they are innate and internal, their arousal is basically unlearned. Most of them necessity be met for survival. They may show up in our behaviour starting from birth, or they may come in relation to the maturation. The most significant primary motives contain hunger, pain avoidance, a need for oxygen, sleep, elimination of wastes, and regulation of body temperature. Several of these motives are triggered through departures from balanced state of equilibrium described homeostasis in their internal physiological processes. Homeostasis, body equilibrium, works automatically to maintain balance among internal physiological circumstances.

Social Motives

These are motives that are learned and satisfied in the context of others. As they are the wellsprings of several of human actions, they are intricate motive states. These human motives can be looked upon as general states that lead to different kinds of scrupulous behaviours. Not only do they help to determine much of what a person does, they persist, never fully satisfied, over the years. No sooner is one goal reached than the motive is directed to another one. These motives are significant components of personality. As they are learned, they also have different strengths from one person to another.

Motives to Know and to be Effective

Motives to know and to be effective just like the physiological drives, these ones are also innate or unlearned. They are persistent and seem to exist to one degree or another in everyone. But their root is not the physique. Drives under this category highly contribute to the normal behavioural development of human beings. In some instances, there could be some interconnection flanked by the physiological needs and the ones in this classification. These motives, though, are motives that drive a person to seek diversity in stimulation, to process information in relation to the nearby world, to explore, and to be effective in mastering challenges from the environment. So, they seek out sensory stimulation through continual interaction with the environment.

INTEGRITY INVENTORY

The *Integrity Inventory* (stylized as *I*), is a nationally normed entry-level personnel selection tool that incorporates employment integrity testing. It was developed by industrial organizational psychologist Fred Rafilson, Ph.D., CEO and Founder of I/O Solutions, Inc. or Industrial/Organizational Solutions Inc, referred to as IOS in the 2009 United States Supreme Court case, Ricci v. DeStefano. In the United States, vocations within the public safety sector, (*i.e.*, firefighter, sheriff and police officer, correctional officer, emergency medical services including emergency medical technician); and

employment in for-profit retail and wholesale business, and non-profit sectors often require Industrial and Organizational Psychology employment testing, for initial employment and advancement throughout the ranks.

The Integrity Inventory, when used as a pre-employment screening tool, predicts individuals' likelihood of engaging in counterproductive workplace behaviours including assessing: ethics and moral character, work attitudes, theft attitudes, potential for substance abuse (*i.e.*, alcohol or drug use), emotional stability, turnover intentions, and/or behaviours that are hazardous and place the civilian population and coworkers at heightened risks. Qualified individuals with a reduced risk of such counterproductive workplace behaviours are selected for employment, leading to a more productive workforce. Counterproductive workplace behaviours lead to real financial losses for business; in the public safety and private sector, such behaviours are hazardous and place the civilian population and coworkers at heightened risk. As such, the prediction of counterproductive workplace behaviours constitutes a business necessity as outlined in the U.S. Equal Employment Opportunity Commission's Uniform Guidelines on Employee Selection Procedures. Psychological testing is a standard and growing practice for human resources departments throughout the United States who seek high integrity employees.

This exam has the sensitivity needed to accurately assess the individuals that will be charged with protecting our lives, families, property, and businesses. The Integrity Inventory displays no adverse impact against protected classes and allows you to select and rank-order candidates most likely to succeed based on their integrity and honesty.

INTERNATIONAL TEST COMMISSION

The International Test Commission (ITC) is an association of national psychological associations, test commissions, organizations and individuals, who promote "the proper development, evaluation and uses" of educational and psychological tests. The ITC is a non-profit organization, affiliated to the International Association of Applied Psychology (IAAP) and the International Union of Psychological Science (IUPsyS). ITC stimulates international cooperation on research projects relevant to a scientifically and ethically sound use of tests. ITC develops guidelines on the use of tests; holds a biennial international conference; publishes a peer-reviewed journal (*International Journal of Testing*) and a newsletter (*Testing International*).

The founding of the ITC was due to the efforts of Jean Cardinet (Switzerland), who presented his concerns with the ethical use of psychological tests to the General Assembly of the Swiss Psychological Society in 1968. In 1971, the Swiss Professional Association of Applied Psychology published a set of regulations to promote the quality and prevent the abuse of tests. Cardinet initiated a project to create national Test Commissions in all countries using psychological tests. His idea was approved by the International Association of Applied Psychology (IAAP) in 1971; and by 1974, at least 15 national

test commissions existed. In 1975, an advisory Council of the emerging ITC met to draft a constitution and agree on first initiatives, including a public survey of test attitudes. The ITC was officially “born” in 1976, during the Congress of the International Union of Psychological Sciences (IUPsyS), where its constitution was provisionally approved.

The draft Constitution defined two membership categories:

- Full members, consisting of national test commissions recognized by the psychological association in their respective country, and
- Affiliate members, who were either international associations with an interest in testing, or national associations from countries not full members of the ITC².
- *The membership structure was amended several times:* first in 1998, when affiliate membership was opened to any organization interested in testing (including test publishers or universities, who had not been accepted previously), and then in 2000, when the individual membership category was created.

Today, the ITC membership covers most of the European Countries and North America, as well as some countries in the Middle and Far East, South America and Africa¹. The International Journal of Testing (IJT) is the official journal of the ITC published since 1999 by Routledge (now Taylor and Francis Group). There are four issues per year. The IJT publishes original articles discussing theoretical issues, methodological approaches, and empirical research in the area of tests and testing.

Having no policy enforcement powers for the standards in psychological testing at national levels, the ITC promotes good practice in test construction and use through the development of best practice guidelines.

To date, the ITC has published the following guidelines:

- ITC Guidelines on Adapting Tests
- ITC Guidelines on Test Use
- ITC Guidelines on Computer-Based and Internet-delivered Testing
- ITC Guidelines on Quality Control in Scoring, Test Analysis and Reporting of Test Scores
- ITC Guidelines on the Security of Tests, Examinations, and Other Assessments

The guidelines may be downloaded in English and various other languages from the ITC web site.

The newsletter “Testing International” disseminates information of international relevance to the ITC membership twice a year. Past editions are available to the public from the ITC web site. The ITC organizes a biannual scientific conference, which usually brings together around 400-500 delegates from around 40 countries. The details of the past and the forthcoming meetings are announced at the ITC web site.

RISK INTELLIGENCE

Risk intelligence is a concept that generally means “beyond risk management”, though it has been used in different ways by different writers. The term is being used more

frequently by business strategists when discussing integrative business processes related to governance, risk, and compliance.

Definitions

The first non-definitive usage of the phrase Risk Intelligence appears in the 1980s and aligns to the definition of *intelligence* as being information from an enemy (for example, regarding credit risk.) The topic of balancing risk and innovation using information and the cognitive processes involved also appears at this time. Recent usage is more aligned to *intelligence* as understanding and problem solving.

The US business writer David Apgar defines it as the capacity to learn about risk from experience.

Management consultant and risk advisor Rick Funston defines risk intelligence as a dynamic approach to protect and create value amid uncertainty. It is an enterprise wide process integrating people, processes (systems), and tools to increase information available to decision makers for improved decision making.

The UK philosopher and psychologist Dylan Evans defines it as “a special kind of intelligence for thinking about risk and uncertainty”, at the core of which is the ability to estimate probabilities accurately. Evans includes a risk intelligence test (RQ) in his book and on his web site (below) analogous to IQ or EQ.

American financial executive, author, and Columbia University professor Leo Tilman defined risk intelligence as “The organizational ability to think holistically about risk and uncertainty, speak a common risk language, and effectively use forward-looking risk concepts and tools in making better decisions, alleviating threats, capitalizing on opportunities, and creating lasting value.” He has argued that risk intelligence is essential to survival, success, and relevance of companies and investors in the post-crisis world.

Comparison with Business Intelligence

As an emerging concept, risk intelligence shares characteristics with other topics such as business intelligence and competitive intelligence. As such, there are some in those camps who believe that risk intelligence is the set of processes for the transformation of risk data into meaningful and useful information for risk analysis, treatment and planning purposes.

MAYER-SALOVEY-CARUSO EMOTIONAL INTELLIGENCE TEST

The Mayer-Salovey-Caruso Emotional Intelligence Test (MSCEIT) is an ability-based measure of emotional intelligence. The test was constructed by academics John D. Mayer, Peter Salovey, and David R. Caruso at Yale and the University of New Hampshire in cooperation with Multi-Health Systems Inc. The test measures emotional intelligence through a series of questions and tests the participant’s ability to perceive, use, understand, and regulate emotions. Using questions based on everyday scenarios,

the MSCEIT measures how well people respond to social tasks, read facial expressions, and solve emotional problems. The MSCEIT is used in corporate, educational, research, and therapeutic settings.

Test Structure

MSCEIT measures four aspects of emotional intelligence:

Ability	Question types	How the ability may be used	Test section
Accurately identify emotions of people and symbolized by objects in pictures	Identify and read emotions in people, landscape and designs	"Read" people's mood for feedback	Faces, pictures
Using emotions and solve problems with the emotion	Compare and match emotions to sensations such as color, light and temperature	Create the right feeling to assist in problem solving, communicate a vision, lead people	Facilitation, sensations
Understand the causes of emotion	Multiple choice emotion vocabulary (scenario presented) questions	Be able to predict how people will emotionally react	Changes, Blends
Being open to emotions and fuse emotions with thinking	Answer which emotional strategy would be best in social relationships as well as managing one's self.	Integrate emotion and thought to make effective decisions	Emotion management, Emotional relationships

AMMONS QUICK TEST

Ammons Quick Test (QT) is an intelligence test that was designed in 1962 by Robert B. Ammons and his wife Carol H. Ammons. This test has been used for many years to help assess premorbid intelligence. It is a passive response picture-vocabulary test. The Quick Test (QT) can be administered to the physically handicapped as well as individuals with attention span deficits. The QT results correlate well with the Wechsler Adult Intelligence Scale (WAIS) Full Scale IQ. The Quick Test raw score of 46 translates to a WAIS IQ score of 110, which is in the high average range of intellectual functioning for an adult.

ARMY ALPHA

The Army Alpha is a group-administered test developed by Robert Yerkes and six others in order to evaluate the many U.S. military recruits during World War I. It was first introduced in 1917 due to a demand for a systematic method of evaluating the intellectual and emotional functioning of soldiers. The test measured “verbal ability, numerical ability, ability to follow directions, and knowledge of information”. Scores on the Army Alpha were used to determine a soldier’s capability of serving, his job classification, and his potential for a leadership position. Soldiers who were illiterate or foreign speaking would take the Army Beta, the non-verbal equivalent of the exam. Army Beta The Development of the beta test and of the performance test for the

examination of the foreign speaking and illiterate presented special problems. The use of demonstration charts and mime to convey the instructions to the persons being examined proved successful. The new type of the test in the beta, using geometrical designs, mutilated pictures, etc., required different principles in its construction. The individual performance tests also involved additional and peculiar standards of construction and evaluation.

The important purpose of these supplementary tests was, of course, to give to those handicapped by language difficulties a real opportunity to show their ability. In addition, two definite aims were planned in the use of all forms of testing: first, to point out the feeble-minded and those incapable of military service because of mental deficiency; and second, to find those of unusual or special ability. The arrangement of each test, in both group and individual examinations, was therefore checked against the sources of men in institutions for the feeble-minded.

If no score had meant low mentality, the first task would have been solved; but it had been shown that literacy was an important factor in the alpha test. The beta test practically eliminated this factor and was thus a step further in selecting those of low intelligence.

To prove conclusively that a man was weak-minded and not merely indifferent or malingering, the performance test was added.

The individual examinations as finally used in the U.S. Army were, therefore, primarily checks on the group examinations. No person was reported as feeble-minded until a detailed individual psychological examination had been made. Many cases of mental disorder were discovered and referred to the psychiatrists for examination. Disciplinary cases referred to the psychologists were always given individual examinations, as were referred cases of men having difficulty with drill or those who failed to improve in the YMCA schools and elsewhere. Both the Army Alpha and Army Beta tests were discontinued after World War I.

Relationship of Scores and Errors

In any psychological aptitude test, the person scoring the test has to take into consideration any error that the examinee will possibly make while taking the test. C. R. Atwell did a small study on the relationship of scores and errors based on the results of administration of the "Army Alpha".

C. R. Atwell (1937, p. 451) wrote, "The number of errors made by a subject on a test should be indicative of his approach to the test, whether he works in a hurriedly and rashly or slowly and cautiously.

Considered alone, however, the number of errors is a relatively meaningless figure, since more errors would be expected with lower scores. If for a given score wide deviations occur in the number of errors, the error score of a subject should be of value in giving additional information about him".

Purpose

Yerkes outlined seven purposes for administering the “Army Alpha” test:

- “Classify soldiers according to their mental ability, thus supplementing personnel records of occupational qualifications and assisting with assignment in the Army
- Supply a mental rating for each soldier which shall assist personnel officers in building organizations of equal or of appropriate strength
- Assist in regimental, company and medical officers by careful examination and report on men who are not responding satisfactorily to training, or are otherwise troublesome
- Assist officers of development battalion with classification, grading, training, and ultimate assignment of men
- Assist in discovering men of superior mental ability who should be selected for officers’ training camps, for promotion, or for assignment to special tasks
- Assist in discovering and properly placing men of marked special skill, as for example, observers or scouts for intelligence service
- Assist in discovering men who are mentally inferior and who in accordance with degree of defectiveness should be recommended for discharge, development battalions, labour organizations or regular military training”

Methods and Results

The administration of the “Army Alpha” was carried out in the following manner:

- The obviously illiterate were segregated from the literate
- Administration of the “Army Alpha” (for literates). The time of the test was 40 to 50 minutes and there were 100 to 200 men in a group.
- Administration of the “Army Beta” (for illiterates and men who failed the “Army Alpha”). The time of the test was 50 to 60 minutes and there were up to 60 men in a group.
- Men who failed the “Army Beta” were subject to individual examinations. Literates who failed the “Army Alpha” and “Army Beta” are given the Stanford-Binet and/or the Point-Scale, depending on the subject. Illiterates who failed the “Army Beta” were given a Performance Test

On November 11, 1918, the psychological personnel consisted of about 120 officers and 350 enlisted men. Over five hundred additional clerks were used in the examining service in the 35 different camps in which psychological examining had been established. The army intelligence examination had been given to 1, 726, 966 men; of these 41, 000 were officers. Approximately 83, 000 individual examinations had been given. Over 7, 800 men had been recommended for immediate discharge; 10, 014 had been recommended for labour battalions or other service organizations; 9, 487 had been recommended for development battalions for further observation and preliminary training. Nearly 30 percent of the 1, 556, 011 men for whom statistics are available

were found to be unable to “read and understand newspapers and write letters home”, and were given a special examination prepared for illiterates, the Army Beta. The methods originally prepared for use in the Army were subjected to repeated revisions, in the light of results, for increase in reliability and military value. The procedure finally adopted and used throughout the Army consists of two chief types of examination: the group examination and the individual examination. The former was necessitated by the demand for speed of examination and report, the latter by the desire for reliability and fairness to the individual.

Men were examined in groups as large as five hundred. Every man was supplied with a pencil and an examination blank. He then, under military discipline, follows directions to the best of his ability. The examination required approximately fifty minutes. It demands almost no writing since responses were indicated by underscoring, crossing out, or checking. The examination papers were quickly scored by means of stencils, and mental ratings recorded for prompt report. To avoid, within reasonable limits, the risk of coaching, several duplicate forms of the examination had been made available. Each test of examination alpha consisted of a number of parts arranged in order of difficulty low to high. It was therefore possible for low-grade subjects to make a start on each test, and, at the same time, practically impossible for highly intelligent subjects to complete the tests within the time allowed. The test were varied in character and undoubtedly sample the most important types of intellectual process.

Structure

The general procedure of examining, which was developed to meet military requirements, is briefly describable as follows: A group of draftees, the size of which is determined by the seating capacity of examining room (it varied from one hundred to five hundred men) is reported to the psychological examining building for mental testing. The first essential step is the segregation of illiterates. This is accomplished by having all men who cannot read and write their own letters and those who have not proceeded beyond the fifth grade in school step out of the original group. The remaining men are sent to the alpha room. Naturally, among them there are likely to be several who will subsequently have to take the Army Beta examination. The illiterates are sent directly to the Army Beta room. Men who fail in the alpha were sent to the Army Beta in order that injustice by reason of relative unfamiliarity with English may be avoided. Men who fail in Army Beta are referred for individual examination by means of what may appear to be the most suitable and altogether appropriate procedure among the varied methods available. This reference for careful individual examination is yet another attempt to avoid injustice either by reason of linguistic handicap or accidents incident to group examining. The “Army Alpha” is divided into 8 tests. There are also five forms of the test—forms 5, 6, 7, 8, 9 and the questions asked are ordered in ascending difficulty. Army recruits were asked to answer the questions as quickly as possible.

They tested arithmetic, “common sense”, vocabulary, word order, number patterns, analogies, and sentence completion.

Grading

The Army Beta yield numerical scores or intelligence scores which for practical military purposes are translated into letter grades.

The several intelligence letter grades used in the Army, with their score-equivalents and appropriate definitions are presented here:

Grade	Intelligence	Score (alpha)	Score (beta)	% of total	Description
A	Very superior	135-212	100-118	4% to 5%	Men of marked intellectuality. They are of high officer type when they are also endowed with leadership and other necessary qualities.
B	Superior	105-134	90-99	8% to 10%	Less exceptional than “A”. Many men of the commissioned officer type and a large amount of non-commissioned officer material.
C+	High average	75-104	80-89	15% to 18%	A large amount of non-commissioned officer material with occasionally a man whose leadership and power to command fit him for commissioned rank.
C	Average	45-74	65-79	About 25%	Excellent private type with a certain amount of fair non-commissioned officer material.
C-	Low average	25-44	45-64	About 20%	Below average intelligence, but usually good privates and satisfactory in work of a routine nature.
D	Inferior	15-24	20-44	About 15%	Likely to be fair soldiers, but usually slow in learning and rarely go above the rank of private. Short on initiative and so need more than the usual amount of supervision. Many are illiterate or foreign.
D-	Very inferior	0-14	0-19		Very inferior in intelligence

				but considered fit for regular service.
E	Very inferior	0-14	0-19	Men whose mental inferiority justifies their recommendation for development battalion, special service organization, rejection, or discharge.

The majority of “D-” and “E” men are below ten years in “mental age”.

History

The “Army Alpha’s” origin was based on intelligence tests created by Alfred Binet. As World War I began to unfold, some men realized that the proper use of manpower, and more particularly of mind or brain power, would assure ultimate victory. The “Army Alpha” was created by a group of psychologists that consisted of: Robert Yerkes, W. V. Bingham, Henry H. Goddard, T. H. Haines, Lewis Terman, G. M. Whipple, F. L. Wells. Each of them brought to the committee a large amount of material that was sifted to produce the group test and individual examining material “Examiners Guide”. These seven experts in practical mental measurement were organized in summer 1917 and called together to prepare or select suitable methods. They worked almost continually for a month, devising, selecting, and adapting methods. Another month was spent thoroughly testing the methods in military stations so that their value might be definitely established before they should be recommended to the Medical Department of the Army. The results were gratifying and the methods were therefore recommended to the Surgeon General of the Army in August 1917, and promptly accepted for official trial. The official medical inspector formulated these statements and recommendations:

“The purposes of psychological testing are:

- To aid in segregating the mentally incompetent,
- To classify men according to their mental capacity,
- To assist in selecting competent men for responsible positions.”

The original purposes of the committee in preparing methods for intelligence testing were less important than the uses made of the results. The committee intended as stated above to prepare an examination that would indicate the drafted men who were too low-grade mentally to make satisfactory privates in the Army; it was desired also to indicate, if possible, those who were mentally unstable or who might prove incorrigible so far as army discipline was concerned. Also, the committee hoped to be able to pick out exceptional types of men who could be used for special tasks that demanded a high degree of intelligence. In interesting contrast with these original purposes of mental examining, stand the results actually achieved.

- Assigning an intelligence rating to every soldier on the basis of systematic examination.

- Designating and selecting men whose superior intelligence indicates the desirability of advancement or special assignment.
- Promptly selecting and recommending for development battalions of men who are so inferior intellectually as to be unsuited for regular military training.
- Providing measurements of mental ability which enable assigning officers to build organizations of uniform mental strength or in accordance with definite specifications concerning intelligence requirements.
- Selecting men for various types of military duty or for special assignment, as for example, to military training schools, colleges, or technical schools.
- Providing data for forming special training groups within the regiment or battery in order that each man may receive instruction suited to his ability to learn.
- Quickly finding and recommending for elimination men whose intelligence is so inferior that they cannot be used to advantage in any line of military service.

“Examination A” was the pilot version of the “Army Alpha”. 140, 000 prospects were administered “Examination A” in the fall of 1917. There were actually multiple different pilot versions created. The different pilot versions include; Examination A, B, C, D, and E. “Examination A” is the most talked about because it was the first pilot administered and virtually the beginning of the testing of the “Army Alpha”. Robert Yerkes and his members asked the question, “How should the results of widely distributed testing of this sort be interpreted?” As a result of these different forms of pilot tests, Robert Yerkes and the other members found decided that, “in this connection it may be emphasized again that the group examination used in the Army was interpreted entirely in terms of military need”. They made necessary changes to the test to measure an individual’s abilities to place them in the right field of the Army.

This is more about the history of how the test was constructed. “The test to be devised for army use the committee believed should:

- First, be adaptable for group use for the examining of large numbers rapidly.
- Second, it should be a high degree of validity as a measure of intelligence.
- Third, the range of intelligence measured by the tests should be wide; that is, the test should be made difficult enough to measure the higher levels of intelligence and at the same time be an adequate measure of the extremely low levels that would probably be found in the Army.
- Fourth, as far as possible, it should be arranged for objectivity of scoring and the elimination of personal judgement concerning correct answers; thus the results of scoring in one camp would be strictly comparable with those obtained in another.
- Fifth, the test should be so arranged that the scoring could be done rapidly and with the least chance of error. Also, this arrangement should be so simple that relatively inexpert assistance could be used in scoring the large numbers of papers.

- Sixth, there must be either different forms or alternative tests of equal difficulty to prevent coaching.
- Seventh, it was necessary also to obtain clues which would enable examiners to detect malingering in connection with the examination.
- Eighth, cheating must also be avoided.
- Ninth, the test must be made as completely independent of schooling and educational advantages as possible.
- Tenth, the arrangement should be such as to allow a minimum of writing in recording answers.
- Eleventh, the tests must consist of material which would arouse interest in the subjects.
- Twelfth, the different tests used should be arranged to yield an accurate measure of intelligence in a reasonably short time.” With these criteria in mind the committee set to work on the materials available to produce what is now known as the army mental test.

Originally, there were 13 tests. These were rated by the psychologists present on the basis of their validity as measures of intelligence. These were given to selected groups and the results compared with the criteria laid down. As a result, certain tests were eliminated because they failed to meet the requirements. For each test a series of sample items correctly answered was given. Psychologists spent over two months in the study of results and in the revision methods. From this study of the results, the non-verbal group examination beta was prepared for examiners to make a rapid survey of the 30 percent who either could not read English or read it too slowly. The Stanford-Binet and the Point Scale were adapted for army use at this time and the individual examination for foreigners and illiterates was prepared. The “Army Alpha”: First Nebraska Edition was a revision that used four earlier forms of the “Army Alpha”. It was revised in 1937. The most diagnostic items were selected and items referring unnecessarily to military affairs or depending upon out-of-date information were eliminated. The items are arranged in order of difficulty as empirically determined.”

“The make-up and scoring follow the plan of the original Alpha. The directions are given orally for each item in the directions test, also separately timed, and oral directions precede each of the subtests. This has often seemed to be one of the inconvenient features of the Alpha, especially when used with college and adult subjects”.

“Norms were apparently not recalculated from an administration of the present revision, as “the average level of difficulty for each test is the same as that for the original forms of the Army Alpha, and so the customary norms should, therefore, apply”. Norms are given in terms of percentiles for the adult population. An estimated mental age and IQ may also be obtained from the same table. The norms for IQ are derived from a comparison of the percentile norms with those offered by Terman and Merrill for the distribution of IQ’s in the American white population”.

“The manual contains no data on reliability or validity, presumably as high or higher than the original. The test should serve as a profitable research device or as a preferred substitute for the earlier Alphas when given in the original manner.”

The Army Group Examination Alpha: Schrammel-Brannin Revision was published in 1936. “Three of the original five forms of the five forms of the army test, with modifications and the introduction of new items. The items are equal from form to form. The eight subtests of the original Alpha have been retained. The scoring is easier, the responses being digits or a plus or minus placed in parentheses to the left of each item. Each item is not timed individually. Oral Directions were eliminated, and the test is self-administered. The examiner records the time for the group at five-minute intervals, allowing a later return to earlier items not completed if time allows. The items are presumably in order of difficulty. Scoring was done by cardboard keys supplied with the test.”

“Grade, age, and a few occupational norms are supplied. The grade norms extend from the fourth grade through the elementary, high school, and college level. The number of cases for each grade averages around 700, with over 2, 000 college freshmen and about 300 for each of the other college years. Age norms extend from 8 to 25 and above. There is progressive increase in the median score up to age 17, after which the increase is irregular. The norms would seem to be best from ages 9 to 17, both on this point and on the number of cases. The tentative norms are based almost entirely on school and college groups, a fact evidently to be taken into consideration if the subjects to be tested are not students.” The occupational norms were tentatively for Civilian Conservation Corps camps, job applicants to GE, police officers, student nurses, and public school teachers.

“The mimeographed manual gives no data on reliability, which perhaps would be high, nor on validity, which would probably also be fairly good for the measurement of abstract intelligence or scholaptitude. The elimination of oral directions, the existence of three forms, and the modification for ease of scoring are appealing features of the test.”

Criticism

One of the most obvious criticisms of the “Army Alpha” is that testing has evolved so much since World War I that the information purposed in the demographics of the intelligence test is out of date. Although, a lot of the formats, even some of the questions, used from the “Army Alpha” are still around today.

There were even criticisms when the test was originally accepted to be used. Yerkes and Yoakum (1920) wrote, “When psychological examining was originally accepted by the Medical Department for Official trial, there was extreme and widely prevalent skepticism even among psychologists themselves concerning the reliability of the measurements of intelligence which could be secured and still more concerning their

practical value to the Army". Another problem that arose during its creation and administration, was that psychologist examiners had a difficult time basing a soldier's future with the Army solely on a test. Yerkes and Yoakum (1920) state, "It was repeatedly stated and emphasized by psychological examiners that a man's value to the service should not be judged by his intelligence alone, but that instead temperamental characteristics, reliability, ability to lead and to "carry on" under varied conditions should be taken into account. Even after the feasibility of securing a fairly reliable measure of every soldier's intelligence or mental alertness had been demonstrated, it remained uncertain whether these measurements would correlate positively with military value to a sufficient degree to render them useful. Data which have become available during the past year settle this question definitely by indicating a relatively high correlation between officers's judgements of military value and the intelligence ratings".

BLOCK DESIGN TEST

A block design test is a subtest on many IQ test batteries used as part of assessment of human intelligence. It is thought to tap spatial visualization ability and motor skill. The test-taker uses hand movements to rearrange blocks that have various colour patterns on different sides to match a pattern. The items in a block design test can be scored both by accuracy in matching the pattern and by speed in completing each item. David Wechsler adapted a block design subtest for his Wechsler-Bellevue test, the predecessor of his WAIS (Wechsler Adult Intelligence Scale), from the Kohs block design test developed in 1923 at Stanford University by Samuel Calmin Kohs (1890–1984). A later revision by Hutt of the Kohs test incorporated the time taken to complete each item into the scoring of the test. Wechsler followed that practice in making both accuracy and speed factors in scoring the test. Good performance on the block design test is indicative of appropriate functioning of the parietal and frontal lobes. Head injury, Alzheimer's disease, and stroke can severely reduce the performance of an individual on the block design test. Additional evidence suggests impairment in block design performance among schizophrenic and bipolar disorder patient populations, though this represents only preliminary findings. The Block Design test is also a relatively accurate measure of spatial ability and Spatial Visualization Ability used in daily life. The Block Design test is considered one of the best measures of spatial ability, although it is subject to certain problems of administration, such as anxiety or over-cautious responding. Linda Kreger Silverman has proposed the block design subtest as the best putative measure of spatial ability among the Wechsler subtests.

Uta Frith, in her book *Autism: Explaining the Enigma*, addresses the superior performance of autistic individuals on the block design test. This was also addressed in this earlier paper. A particularly interesting article demonstrates the differences in construction time in the performance of the block design task by individuals with and without Asperger syndrome. An essential point here is that in an unsegmented version

of the task, people with Asperger syndrome performed significantly faster than neurotypical individuals. Recent research has demonstrated a connection between spatial ability and math and science proficiency at the highest levels. Of particular interest, a recent study in the *Lancet* (2002) demonstrated that high spatial ability was related to the performance of surgery. Additionally, although this is somewhat speculative, Simon Baron-Cohen has shown the grandfathers and fathers of autistics were more likely to be engineers, and since it is known that autistics have an ability peak in block design, it is possible that an inherited ability for block design performance may be responsible for the increased number of engineers and scientists among the relatives of autistic individuals. Dror et al., 1993 found that pilots' performance was superior to non-pilots on a test of the speed of mental rotation. Although the block design test is characterized as a test of spatial visualization, not mental rotation, spatial visualization ability as measured by the block design test is highly correlated to mental rotation ability. As performance on the Block Design test has been suggested as a predictive measure for performance in fields such as engineering and physics, Felder, at North Carolina State University, has developed a learning style questionnaire that attempts to assess spatial ability in an educational context.

BRACKEN SCHOOL READINESS ASSESSMENT

The Bracken School Readiness Assessment ("BSRA") is an individual cognitive test designed for children, pre-K through second grade. The BSRA was developed by Bruce A. Bracken and first published in 2002 by The Psychological Corporation. Raw scores can be converted to percentile rank scores and standard scores. The publisher has supporting guides for converting standard scores into criterion categories (*e.g.* "advanced" or "delayed"). Reliability measures were within expected ranges given the age of the students. Validity was assessed using other published assessments (*e.g.* WPPSI-R, DAS). The BSRA Spanish Edition was developed, normed, and validated independently. The BSRA can be used with children as young as 2.6 years of age. The BSRA is composed of the first six subtests from the revised Bracken Basic Concepts Scale (BBCS-R).

The test assesses six basic skills:

- Colours Student must identify common colours by name.
- Letters Students must identify upper-case and lower-case letters.
- Numbers | Counting Student must identify single- and double-digit numerals, and must count objects.
- Sizes Student must demonstrate knowledge of words used to depict size (*e.g.*, tall, wide, etc.)
- Comparisons Student must match or differentiate objects based on a specific characteristic.
- Shapes Student must identify basic shapes by name.

CATTELL CULTURE FAIR III

The Culture Fair Intelligence Test (CFIT) was constructed by Raymond B. Cattell, PhD, DSc in an attempt to produce a measure of cognitive abilities that accurately estimated intelligence devoid of sociocultural and environmental influences. Scholars have subsequently concluded that the attempt to construct measures of cognitive abilities devoid of the influences of experiential and cultural conditioning is a challenging one. Cattell proposed that general intelligence (g) comprises both Fluid Intelligence (Gf) and Crystallized Intelligence (Gc). Whereas Gf is biologically and constitutionally based, Gc is the actual level of a person's cognitive functioning, based on the augmentation of Gf through sociocultural and experiential learning (including formal schooling). In addition to Gf and Gc, as Gregory John Boyle commented (p. 267), "At the second-stratum level, several major factors (Gf, Gc, Gm, Gps, Gr, Gv, Ga) have been well replicated in numerous studies....Gf is the constitutionally and endowed (innate) intellectual potential which interacts with environmental experience...to produce Gc."

Unlike the most widely used individual tests of cognitive abilities, such as the current editions of the Wechsler Adult Intelligence Scale and the Stanford–Binet Intelligence Scale, which report cognitive ability scores as "deviation IQs" with 15 IQ points corresponding to one standard deviation above or below the mean, Cattell built into the CFIT a standard deviation of 24 IQ points, thereby allowing for a greater spread of scores and concomitantly greater discrimination. This means that about 68.2 percent of test-takers would obtain IQ scores on the CFIT that lie somewhere between 76 and 124, while scoring between 85 and 115 on most other IQ tests.

Crystallized intelligence (Gc) refers to that aspect of cognition in which initial intelligent judgements have become crystallized as habits. Fluid intelligence (Gf) is in several ways more fundamental and is particularly evident in tests requiring responses to novel situations. Before biological maturity individual differences between Gf and Gc will be mainly a function of differences in cultural opportunity and interest. Among adults, however, these discrepancies will also reflect differences with increasing age because the gap between Gc and Gf will tend to increase with experience which raises Gc, whereas Gf gradually declines as a result of declining brain function. The Culture Fair tests consist of three scales with non-verbal visual puzzles. Scale I includes eight subtests of mazes, copying symbols, identifying similar drawings and other non-verbal tasks. Both Scales II and III consists of four subtests that include completing a sequence of drawings, a classification subtest where respondents pick a drawing that is different from other drawings, a matrix subtests that involves completing a matrix of patterns and conditions subtests which involve which out of several geometric designs fulfill a specific given condition. The Cattell Culture Fair Intelligence Test (like the Raven's Progressive Matrices) is not completely free from the influence of culture and learning. Some high-IQ societies, such as The Triple Nine Society, accept high scores on the CFIT-III as one of a variety of old and new tests for admission to the society. A combined

minimum *raw score* of 85 on Forms A and B is required for admission. The tests are used by many including Mensa, which offers a place in their society to anyone scoring in the top 2% IQ scores. Direct concept validity (sometimes called construct validity) refers to the degree to which a certain scale correlates with the concept or construct (*i.e.*, source trait) which it purports to measure. Concept validity is thus measured by correlating the scale with the pure factor and this can only be carried out by performing a methodologically sound factor analysis. The relatively high loading of the Culture Fair Intelligence Test on the fluid intelligence factor indicates that the CFIT does, in fact, have a reasonably high direct concept validity with respect to the concept of fluid intelligence. The Culture Fair Intelligence Test was found to load more highly on a “General Intelligence” factor than on an “Achievement” factor, which is consistent with the concept that the CFIT is a measure of “fluid” rather than “crystallized” intelligence. Convergent Validity is the extent to which the Culture Fair Intelligence Test correlates with other tests of intelligence, achievement, and aptitude. The intercorrelations between the Culture Fair Intelligence Test and some other intelligence tests have been reported, as shown in the Table below.

EDUCATIONAL QUOTIENT

An educational quotient, or EQ, is a score designed to assess a subject’s level of general education. Though related to intelligence quotient, there is no direct correlation between the two. A person of high IQ, may have a low EQ, and vice versa. A person’s EQ is generally found by dividing the results of their Wide Range Achievement Test by their IQ and multiplying by 100, or by dividing their Education Age by their Chronological Age and multiplying by 100.

FIGURE REASONING TEST

Figure Reasoning Test (FRT) is an intelligence test created by John Clifford Daniels.

G FACTOR (PSYCHOMETRICS)

The *g* factor (also known as general intelligence, general mental ability or general intelligence factor) is a construct developed in psychometric investigations of cognitive abilities and human intelligence. It is a variable that summarizes positive correlations among different cognitive tasks, reflecting the fact that an individual’s performance on one type of cognitive task tends to be comparable to that person’s performance on other kinds of cognitive tasks. The *g* factor typically accounts for 40 to 50 percent of the between-individual performance differences on a given cognitive test, and composite scores (“IQ scores”) based on many tests are frequently regarded as estimates of individuals’ standing on the *g* factor. The terms IQ, general intelligence, general cognitive ability, general mental ability, or simply intelligence are often used interchangeably to refer to this common core shared by cognitive tests. The *g* factor targets a particular

measure of *general intelligence*. The existence of the *g* factor was originally proposed by the English psychologist Charles Spearman in the early years of the 20th century. He observed that children's performance ratings, across seemingly unrelated school subjects, were positively correlated, and reasoned that these correlations reflected the influence of an underlying general mental ability that entered into performance on all kinds of mental tests. Spearman suggested that all mental performance could be conceptualized in terms of a single general ability factor, which he labeled *g*, and a large number of narrow task-specific ability factors. Today's factor models of intelligence typically represent cognitive abilities as a three-level hierarchy, where there are a large number of narrow factors at the bottom of the hierarchy, a handful of broad, more general factors at the intermediate level, and at the apex a single factor, referred to as the *g* factor, which represents the variance common to all cognitive tasks.

Traditionally, research on *g* has concentrated on psychometric investigations of test data, with a special emphasis on factor analytic approaches. However, empirical research on the nature of *g* has also drawn upon experimental cognitive psychology and mental chronometry, brain anatomy and physiology, quantitative and molecular genetics, and primate evolution. While the existence of *g* as a statistical regularity is well-established and uncontroversial, there is no consensus as to what causes the positive correlations between tests.

Research in the field of behavioural genetics has established that the construct of *g* is highly heritable. It has a number of other biological correlates, including brain size. It is also a significant predictor of individual differences in many social outcomes, particularly in education and employment. The most widely accepted contemporary theories of intelligence incorporate the *g* factor. However, critics of *g* have contended that an emphasis on *g* is misplaced and entails a devaluation of other important abilities, as well as supporting an unrealistic reified view of human intelligence.

Cognitive Ability Testing

Cognitive ability tests are designed to measure different aspects of cognition. Specific domains assessed by tests include mathematical skill, verbal fluency, spatial visualization, and memory, among others. However, individuals who excel at one type of test tend to excel at other kinds of tests, too, while those who do poorly on one test tend to do so on all tests, regardless of the tests' contents. The English psychologist Charles Spearman was the first to describe this phenomenon. In a famous research paper published in 1904, he observed that children's performance measures across seemingly unrelated school subjects were positively correlated. This finding has since been replicated numerous times. The consistent finding of universally positive correlation matrices of mental test results (or the "positive manifold"), despite large differences in tests' contents, has been described as "arguably the most replicated result in all psychology". Zero or negative correlations between tests suggest the presence of sampling error or restriction

of the range of ability in the sample studied. Using factor analysis or related statistical methods, it is possible to compute a single common factor that can be regarded as a summary variable characterizing the correlations between all the different tests in a test battery. Spearman referred to this common factor as the *general factor*, or simply *g*. (By convention, *g* is always printed as a lower case italic.) Mathematically, the *g* factor is *a source of variance among individuals*, which entails that one cannot meaningfully speak of any one individual's mental abilities consisting of *g* or other factors to any specified degrees. One can only speak of an individual's standing on *g* (or other factors) compared to other individuals in a relevant population.

Different tests in a test battery may correlate with (or "load onto") the *g* factor of the battery to different degrees. These correlations are known as *g* loadings. An individual test taker's *g* factor score, representing his or her relative standing on the *g* factor in the total group of individuals, can be estimated using the *g* loadings. Full-scale IQ scores from a test battery will usually be highly correlated with *g* factor scores, and they are often regarded as estimates of *g*.

For example, the correlations between *g* factor scores and full-scale IQ scores from David Wechsler's tests have been found to be greater than .95. The terms IQ, general intelligence, general cognitive ability, general mental ability, or simply intelligence are frequently used interchangeably to refer to the common core shared by cognitive tests.

The *g* loadings of mental tests are always positive and usually range between .10 and .90, with a mean of about .60 and a standard deviation of about .15. Raven's Progressive Matrices is among the tests with the highest *g* loadings, around .80. Tests of vocabulary and general information are also typically found to have high *g* loadings. However, the *g* loading of the same test may vary somewhat depending on the composition of the test battery. The complexity of tests and the demands they place on mental manipulation are related to the tests' *g* loadings. For example, in the forward digit span test the subject is asked to repeat a sequence of digits in the order of their presentation after hearing them once at a rate of one digit per second. The backward digit span test is otherwise the same except that the subject is asked to repeat the digits in the reverse order to that in which they were presented. The backward digit span test is more complex than the forward digit span test, and it has a significantly higher *g* loading. Similarly, the *g* loadings of arithmetic computation, spelling, and word reading tests are lower than those of arithmetic problem solving, text composition, and reading comprehension tests, respectively.

Test difficulty and *g* loadings are distinct concepts that may or may not be empirically related in any specific situation. Tests that have the same difficulty level, as indexed by the proportion of test items that are failed by test takers, may exhibit a wide range of *g* loadings. For example, tests of rote memory have been shown to have the same level of difficulty but considerably lower *g* loadings than many tests that involve reasoning.

Theories

While the existence of g as a statistical regularity is well-established and uncontroversial among experts, there is no consensus as to what causes the positive intercorrelations. Several explanations have been proposed.

Mental Energy or Efficiency

Charles Spearman reasoned that correlations between tests reflected the influence of a common causal factor, a general mental ability that enters into performance on all kinds of mental tasks. However, he thought that the best indicators of g were those tests that reflected what he called *the eduction of relations and correlates*, which included abilities such as deduction, induction, problem solving, grasping relationships, inferring rules, and spotting differences and similarities. Spearman hypothesized that g was equivalent with “mental energy”. However, this was more of a metaphorical explanation, and he remained agnostic about the physical basis of this energy, expecting that future research would uncover the exact physiological nature of g .

Following Spearman, Arthur Jensen maintained that all mental tasks tap into g to some degree. According to Jensen, the g factor represents a “distillate” of scores on different tests rather than a summation or an average of such scores, with factor analysis acting as the distillation procedure. He argued that g cannot be described in terms of the item characteristics or information content of tests, pointing out that very dissimilar mental tasks may have nearly equal g loadings. Wechsler similarly contended that g is not an ability at all but rather some general property of the brain. Jensen hypothesized that g corresponds to individual differences in the speed or efficiency of the neural processes associated with mental abilities. He also suggested that given the associations between g and elementary cognitive tasks, it should be possible to construct a ratio scale test of g that uses time as the unit of measurement.

Sampling Theory

The so-called sampling theory of g , originally developed by E.L. Thorndike and Godfrey Thomson, proposes that the existence of the positive manifold can be explained without reference to a unitary underlying capacity. According to this theory, there are a number of uncorrelated mental processes, and all tests draw upon different samples of these processes. The intercorrelations between tests are caused by an overlap between processes tapped by the tests. Thus, the positive manifold arises due to a measurement problem, an inability to measure more fine-grained, presumably uncorrelated mental processes.

It has been shown that it is not possible to distinguish statistically between Spearman’s model of g and the sampling model; both are equally able to account for intercorrelations among tests. The sampling theory is also consistent with the observation that more complex mental tasks have higher g loadings, because more complex tasks are expected

to involve a larger sampling of neural elements and therefore have more of them in common with other tasks. Some researchers have argued that the sampling model invalidates *g* as a psychological concept, because the model suggests that *g* factors derived from different test batteries simply reflect the shared elements of the particular tests contained in each battery rather than a *g* that is common to all tests. Similarly, high correlations between different batteries could be due to them measuring the same set of abilities rather than *the* same ability.

Critics have argued that the sampling theory is incongruent with certain empirical findings. Based on the sampling theory, one might expect that related cognitive tests share many elements and thus be highly correlated. However, some closely related tests, such as forward and backward digit span, are only modestly correlated, while some seemingly completely dissimilar tests, such as vocabulary tests and Raven's matrices, are consistently highly correlated. Another problematic finding is that brain damage frequently leads to specific cognitive impairments rather than a general impairment one might expect based on the sampling theory.

Mutualism

The "mutualism" model of *g* proposes that cognitive processes are initially uncorrelated, but that the positive manifold arises during individual development due to mutual beneficial relations between cognitive processes. Thus there is no single process or capacity underlying the positive correlations between tests. During the course of development, the theory holds, any one particularly efficient process will benefit other processes, with the result that the processes will end up being correlated with one another. Thus similarly high IQs in different persons may stem from quite different initial advantages that they had. Critics have argued that the observed correlations between the *g* loadings and the heritability coefficients of subtests are problematic for the mutualism theory.

Factor Structure of Cognitive Abilities

Factor analysis is a family of mathematical techniques that can be used to represent correlations between intelligence tests in terms of a smaller number of variables known as factors. The purpose is to simplify the correlation matrix by using hypothetical underlying factors to explain the patterns in it. When all correlations in a matrix are positive, as they are in the case of IQ, factor analysis will yield a general factor common to all tests. The general factor of IQ tests is referred to as the *g* factor, and it typically accounts for 40 to 50 percent of the variance in IQ test batteries.

Charles Spearman developed factor analysis in order to study correlations between tests. Initially, he developed a model of intelligence in which variations in all intelligence test scores are explained by only two kinds of variables: first, factors that are specific to each test (denoted *s*); and second, a *g* factor that accounts for the positive correlations

across tests. This is known as Spearman's two-factor theory. Later research based on more diverse test batteries than those used by Spearman demonstrated that g alone could not account for all correlations between tests. Specifically, it was found that even after controlling for g , some tests were still correlated with each other. This led to the postulation of *group factors* that represent variance that groups of tests with similar task demands (*e.g.*, verbal, spatial, or numerical) have in common in addition to the shared g variance.

Through factor rotation, it is, in principle, possible to produce an infinite number of different factor solutions that are mathematically equivalent in their ability to account for the intercorrelations among cognitive tests. These include solutions that do not contain a g factor. Thus factor analysis alone cannot establish what the underlying structure of intelligence is. In choosing between different factor solutions, researchers have to examine the results of factor analysis together with other information about the structure of cognitive abilities.

There are many psychologically relevant reasons for preferring factor solutions that contain a g factor. These include the existence of the positive manifold, the fact that certain kinds of tests (generally the more complex ones) have consistently larger g loadings, the substantial invariance of g factors across different test batteries, the impossibility of constructing test batteries that do not yield a g factor, and the widespread practical validity of g as a predictor of individual outcomes. The g factor, together with group factors, best represents the empirically established fact that, on average, overall ability differences *between* individuals are greater than differences among abilities *within* individuals, while a factor solution with orthogonal factors without g obscures this fact. Moreover, g appears to be the most heritable component of intelligence. Research utilizing the techniques of confirmatory factor analysis has also provided support for the existence of g .

A g factor can be computed from a correlation matrix of test results using several different methods. These include exploratory factor analysis, principal components analysis (PCA), and confirmatory factor analysis. Different factor-extraction methods produce highly consistent results, although PCA has sometimes been found to produce inflated estimates of the influence of g on test scores.

There is a broad contemporary consensus that cognitive variance between people can be conceptualized at three hierarchical levels, distinguished by their degree of generality. At the lowest, least general level there are a large number of narrow first-order factors; at a higher level, there are a relatively small number – somewhere between five and ten – of broad (*i.e.*, more general) second-order factors (or group factors); and at the apex, there is a single third-order factor, g , the general factor common to all tests. The g factor usually accounts for the majority of the total common factor variance of IQ test batteries. Contemporary hierarchical models of intelligence include the three stratum theory and the Cattell–Horn–Carroll theory.

“Indifference of the Indicator”

Spearman proposed the principle of the *indifference of the indicator*, according to which the precise content of intelligence tests is unimportant for the purposes of identifying *g*, because *g* enters into performance on all kinds of tests. Any test can therefore be used as an indicator of *g*. Following Spearman, Arthur Jensen more recently argued that a *g* factor extracted from one test battery will always be the same, within the limits of measurement error, as that extracted from another battery, provided that the batteries are large and diverse. According to this view, every mental test, no matter how distinctive, calls on *g* to some extent. Thus a composite score of a number of different tests will load onto *g* more strongly than any of the individual test scores, because the *g* components cumulate into the composite score, while the uncorrelated non-*g* components will cancel each other out. Theoretically, the composite score of an infinitely large, diverse test battery would, then, be a perfect measure of *g*. In contrast, L.L. Thurstone argued that a *g* factor extracted from a test battery reflects the average of all the abilities called for by the particular battery, and that *g* therefore varies from one battery to another and “has no fundamental psychological significance.” Along similar lines, John Horn argued that *g* factors are meaningless because they are not invariant across test batteries, maintaining that correlations between different ability measures arise because it is difficult to define a human action that depends on just one ability.

To show that different batteries reflect the same *g*, one must administer several test batteries to the same individuals, extract *g* factors from each battery, and show that the factors are highly correlated. This can be done within a confirmatory factor analysis framework. Wendy Johnson and colleagues have published two such studies. The first found that the correlations between *g* factors extracted from three different batteries were .99, .99, and 1.00, supporting the hypothesis that *g* factors from different batteries are the same and that the identification of *g* is not dependent on the specific abilities assessed. The second study found that *g* factors derived from four of five test batteries correlated at between .95–1.00, while the correlations ranged from .79 to .96 for the fifth battery, the Cattell Culture Fair Intelligence Test (the CFIT). They attributed the somewhat lower correlations with the CFIT battery to its lack of content diversity for it contains only matrix-type items, and interpreted the findings as supporting the contention that *g* factors derived from different test batteries are the same provided that the batteries are diverse enough. The results suggest that the same *g* can be consistently identified from different test batteries.

Population Distribution

The form of the population distribution of *g* is unknown, because *g* cannot be measured on a ratio scale. (The distributions of scores on typical IQ tests are roughly normal, but this is achieved by construction, *i.e.*, by normalizing the raw scores.) It has been argued

that there are nevertheless good reasons for supposing that g is normally distributed in the general population, at least within a range of ± 2 standard deviations from the mean. In particular, g can be thought of as a composite variable that reflects the additive effects of a large number of independent genetic and environmental influences, and such a variable should, according to the central limit theorem, follow a normal distribution.

Spearman's Law of Diminishing Returns

A number of researchers have suggested that the proportion of variation accounted for by g may not be uniform across all subgroups within a population. *Spearman's law of diminishing returns* (SLODR), also termed the *cognitive ability differentiation hypothesis*, predicts that the positive correlations among different cognitive abilities are weaker among more intelligent subgroups of individuals. More specifically, SLODR predicts that the g factor will account for a smaller proportion of individual differences in cognitive tests scores at higher scores on the g factor.

SLODR was originally proposed by Charles Spearman, who reported that the average correlation between 12 cognitive ability tests was .466 in 78 normal children, and .782 in 22 "defective" children. Detterman and Daniel rediscovered this phenomenon in 1989. They reported that for subtests of both the WAIS and the WISC, subtest intercorrelations decreased monotonically with ability group, ranging from approximately an average intercorrelation of .7 among individuals with IQs less than 78 to .4 among individuals with IQs greater than 122.

SLODR has been replicated in a variety of child and adult samples who have been measured using broad arrays of cognitive tests. The most common approach has been to divide individuals into multiple ability groups using an observable proxy for their general intellectual ability, and then to either compare the average interrelation among the subtests across the different groups, or to compare the proportion of variation accounted for by a single common factor, in the different groups. However, as both Deary et al. (1996), and Tucker-Drob (2009) have pointed out, dividing the continuous distribution of intelligence into an arbitrary number of discrete ability groups is less than ideal for examining SLODR. Tucker-Drob (2009) extensively reviewed the literature on SLODR and the various methods by which it had been previously tested, and proposed that SLODR could be most appropriately captured by fitting a common factor model that allows the relations between the factor and its indicators to be non-linear in nature. He applied such a factor model to a nationally representative data of children and adults in the United States and found consistent evidence for SLODR. For example, Tucker-Drob (2009) found that a general factor accounted for approximately 75% of the variation in seven different cognitive abilities among very low IQ adults, but only accounted for approximately 30% of the variation in the abilities among very high IQ adults.

A recent meta-analytic study by Blum and Holling also provided support for the differentiation hypothesis. As opposed to most research on the topic, this work made it

possible to study ability and age variables as continuous predictors of the g saturation, and not just to compare lower- vs. higher-skilled or younger vs. older groups of testees. Results demonstrate that the mean correlation and g loadings of cognitive ability tests decrease with increasing ability, yet increase with respondent age. SLODR, as described by Charles Spearman, could be confirmed by a g -saturation decrease as a function of IQ as well as a g -saturation increase from middle age to senescence. Specifically speaking, for samples with a mean intelligence that is two standard deviations (*i.e.*, 30 IQ-points) higher, the mean correlation to be expected is decreased by approximately .15 points. The question remains whether a difference of this magnitude could result in a greater apparent factorial complexity when cognitive data are factored for the higher-ability sample, as opposed to the lower-ability sample. It seems likely that greater factor dimensionality should tend to be observed for the case of higher ability, but the magnitude of this effect (*i.e.*, how much more likely and how many more factors) remains uncertain.

Practical Validity

The practical validity of g as a predictor of educational, economic, and social outcomes is more far-ranging and universal than that of any other known psychological variable. The validity of g is greater the greater the complexity of the task.

A test's practical validity is measured by its correlation with performance on some criterion external to the test, such as college grade-point average, or a rating of job performance. The correlation between test scores and a measure of some criterion is called the *validity coefficient*. One way to interpret a validity coefficient is to square it to obtain the variance accounted by the test. For example, a validity coefficient of .30 corresponds to 9 percent of variance explained. This approach has, however, been criticized as misleading and uninformative, and several alternatives have been proposed. One arguably more interpretable approach is to look at the percentage of test takers in each test score quintile who meet some agreed-upon standard of success. For example, if the correlation between test scores and performance is .30, the expectation is that 67 percent of those in the top quintile will be above-average performers, compared to 33 percent of those in the bottom quintile.

Academic Achievement

The predictive validity of g is most conspicuous in the domain of scholastic performance. This is apparently because g is closely linked to the ability to learn novel material and understand concepts and meanings.

In elementary school, the correlation between IQ and grades and achievement scores is between .60 and .70. At more advanced educational levels, more students from the lower end of the IQ distribution drop out, which restricts the range of IQs and results in lower validity coefficients. In high school, college, and graduate school the validity coefficients are .50–.60, .40–.50, and .30–.40, respectively. The g loadings of IQ scores

are high, but it is possible that some of the validity of IQ in predicting scholastic achievement is attributable to factors measured by IQ independent of *g*. According to research by Robert L. Thorndike, 80 to 90 percent of the *predictable* variance in scholastic performance is due to *g*, with the rest attributed to non-*g* factors measured by IQ and other tests.

Achievement test scores are more highly correlated with IQ than school grades. This may be because grades are more influenced by the teacher's idiosyncratic perceptions of the student. In a longitudinal English study, *g* scores measured at age 11 correlated with all the 25 subject tests of the national GCSE examination taken at age 16. The correlations ranged from .77 for the mathematics test to .42 for the art test. The correlation between *g* and a general educational factor computed from the GCSE tests was .81.

Research suggests that the SAT, widely used in college admissions, is primarily a measure of *g*. A correlation of .82 has been found between *g* scores computed from an IQ test battery and SAT scores. In a study of 165,000 students at 41 U.S. colleges, SAT scores were found to be correlated at .47 with first-year college grade-point average after correcting for range restriction in SAT scores (the correlation rises to .55 when course difficulty is held constant, *i.e.*, if all students attended the same set of classes).

Job Attainment

There is a high correlation of .90 to .95 between the prestige rankings of occupations, as rated by the general population, and the *average* general intelligence scores of people employed in each occupation. At the level of individual employees, the association between job prestige and *g* is lower – one large U.S. study reported a correlation of .65 (.72 corrected for attenuation). Mean level of *g* thus increases with perceived job prestige. It has also been found that the dispersion of general intelligence scores is smaller in more prestigious occupations than in lower level occupations, suggesting that higher level occupations have minimum *g* requirements.

Job Performance

Research indicates that tests of *g* are the best single predictors of job performance, with an average validity coefficient of .55 across several meta-analyses of studies based on supervisor ratings and job samples. The average meta-analytic validity coefficient for performance in job *training* is .63. The validity of *g* in the highest complexity jobs (professional, scientific, and upper management jobs) has been found to be greater than in the lowest complexity jobs, but *g* has predictive validity even for the simplest jobs. Research also shows that specific aptitude tests tailored for each job provide little or no increase in predictive validity over tests of general intelligence. It is believed that *g* affects job performance mainly by facilitating the acquisition of job-related knowledge. The predictive validity of *g* is greater than that of work experience, and increased experience on the job does not decrease the validity of *g*.

In a recent meta-analysis, researchers found that general cognitive ability (GCA) predicted job performance better than personality (Five factor model) and three streams of emotional intelligence. They examined the relative importance of these constructs on predicting job performance and found that cognitive ability explained most of the variance in job performance. Other studies suggested that GCA and emotional intelligence have a linear independent and complementary contribution to job performance. Côté and Miners (2015) found that these constructs are interrelated when assessing their relationship with two aspects of job performance: organisational citizenship behaviour (OCB) and task performance. Emotional intelligence is a better predictor of task performance and OCB when GCA is low and vice versa. For instance, an employee with low GCA will compensate his/her task performance and OCB, if emotional intelligence is high.

Although these compensatory effects favour emotional intelligence, GCA still remains as the best predictor of job performance. Several researchers have studied the correlation between GCA and job performance among different job positions. For instance, Ghiselli (1973) found that salespersons had a higher correlation than sales clerk. The former obtained a correlation of 0.61 for GCA, 0.40 for perceptual ability and 0.29 for psychomotor abilities; whereas sales clerk obtained a correlation of 0.27 for GCA, 0.22 for perceptual ability and 0.17 for psychomotor abilities. Other studies compared GCA – job performance correlation between jobs of different complexity. Hunter and Hunter (1984) developed a meta-analysis with over 400 studies and found that this correlation was higher for jobs of high complexity (0.57). Followed by jobs of medium complexity (0.51) and low complexity (0.38).

Job performance is measured by objective rating performance and subjective ratings. Although the former is better than subjective ratings, most of studies in job performance and GCA have been based on supervisor performance ratings. This rating criteria is considered problematic and unreliable, mainly because of its difficulty to define what is a good and bad performance. Rating of supervisors tends to be subjective and inconsistent among employees. Additionally, supervisor rating of job performance is influenced by different factors, such as halo effect, facial attractiveness, racial or ethnic bias, and height of employees. However, Vinchur, Schippmann, Switzer and Roth (1998) found in their study with sales employees that objective sales performance had a correlation of 0.04 with GCA, while supervisor performance rating got a correlation of 0.40. These findings were surprising, considering that the main criteria for assessing these employees would be the objective sales.

In understanding how GCA is associated job performance, several researchers concluded that GCA affects acquisition of job knowledge, which in turn improves job performance. In other words, people high in GCA are capable to learn faster and acquire more job knowledge easily, which allow them to perform better. Conversely, lack of ability to acquire job knowledge will directly affect job performance. This is due to low

levels of GCA. Also, GCA has a direct effect on job performance. In a daily basis, employees are exposed constantly to challenges and problem solving tasks, which success depends solely on their GCA. These findings are discouraging for governmental entities in charge of protecting rights of workers. Because of the high correlation of GCA on job performance, companies are hiring employees based on GCA tests scores. Inevitably, this practice is denying the opportunity to work to many people with low GCA. Previous researchers have found significant differences in GCA between race/ ethnicity groups. For instance, there is a debate whether studies were biased against Afro-Americans, who scored significantly lower than white Americans in GCA tests. However, findings on GCA-job performance correlation must be taken carefully. Some researchers have warned the existence of statistical artifacts related to measures of job performance and GCA test scores. For example, Viswesvaran, Ones and Schmidt (1996) argued that is quite impossible to obtain perfect measures of job performance without incurring in any methodological error. Moreover, studies on GCA and job performance are always susceptible to range restriction, because data is gathered mostly from current employees, neglecting those that were not hired. Hence, sample comes from employees who successfully passed hiring process, including measures of GCA.

Income

The correlation between income and g , as measured by IQ scores, averages about .40 across studies. The correlation is higher at higher levels of education and it increases with age, stabilizing when people reach their highest career potential in middle age. Even when education, occupation and socioeconomic background are held constant, the correlation does not vanish.

Other Correlates

The g factor is reflected in many social outcomes. Many social behaviour problems, such as dropping out of school, chronic welfare dependency, accident proneness, and crime, are negatively correlated with g independent of social class of origin. Health and mortality outcomes are also linked to g , with higher childhood test scores predicting better health and mortality outcomes in adulthood.

Genetic and Environmental Determinants

Heritability is the proportion of phenotypic variance in a trait in a population that can be attributed to genetic factors. The heritability of g has been estimated to fall between 40 and 80 percent using twin, adoption, and other family study designs as well as molecular genetic methods. It has been found to increase linearly with age. For example, a large study involving more than 11, 000 pairs of twins from four countries reported the heritability of g to be 41 percent at age nine, 55 percent at age twelve, and 66 percent at age seventeen. Other studies have estimated that the heritability is as high

as 80 percent in adulthood, although it may decline in old age. Most of the research on the heritability of g has been conducted in the United States and Western Europe, but studies in Russia (Moscow), the former East Germany, Japan, and rural India have yielded similar estimates of heritability as Western studies.

Behavioural genetic research has also established that the shared (or between-family) environmental effects on g are strong in childhood, but decline thereafter and are negligible in adulthood. This indicates that the environmental effects that are important to the development of g are unique and not shared between members of the same family.

The genetic correlation is a statistic that indicates the extent to which the same genetic effects influence two different traits. If the genetic correlation between two traits is zero, the genetic effects on them are independent, whereas a correlation of 1.0 means that the same set of genes explains the heritability of both traits (regardless of how high or low the heritability of each is). Genetic correlations between specific mental abilities (such as verbal ability and spatial ability) have been consistently found to be very high, close to 1.0.

This indicates that genetic variation in cognitive abilities is almost entirely due to genetic variation in whatever g is. It also suggests that what is common among cognitive abilities is largely caused by genes, and that independence among abilities is largely due to environmental effects. Thus it has been argued that when genes for intelligence are identified, they will be “generalist genes”, each affecting many different cognitive abilities. The g loadings of mental tests have been found to correlate with their heritabilities, with correlations ranging from moderate to perfect in various studies. Thus the heritability of a mental test is usually higher the larger its g loading is. Much research points to g being a highly polygenic trait influenced by a large number of common genetic variants, each having only small effects. Another possibility is that heritable differences in g are due to individuals having different “loads” of rare, deleterious mutations, with genetic variation among individuals persisting due to mutation–selection balance.

A number of candidate genes have been reported to be associated with intelligence differences, but the effect sizes have been small and almost none of the findings have been replicated. No individual genetic variants have been conclusively linked to intelligence in the normal range so far. Many researchers believe that very large samples will be needed to reliably detect individual genetic polymorphisms associated with g . However, while genes influencing variation in g in the normal range have proven difficult to find, a large number of single-gene disorders with mental retardation among their symptoms have been discovered.

Several studies suggest that tests with larger g loadings are more affected by inbreeding depression lowering test scores. There is also evidence that tests with larger g loadings are associated with larger positive heterotic effects on test scores. Inbreeding depression and heterosis suggest the presence of genetic dominance effects for g .

Neuroscientific Findings

g has a number of correlates in the brain. Studies using magnetic resonance imaging (MRI) have established that g and total brain volume are moderately correlated ($r \sim .3-.4$). External head size has a correlation of $\sim .2$ with g . MRI research on brain regions indicates that the volumes of frontal, parietal and temporal cortices, and the hippocampus are also correlated with g , generally at .25 or more, while the correlations, averaged over many studies, with overall grey matter and overall white matter have been found to be .31 and .27, respectively. Some but not all studies have also found positive correlations between g and cortical thickness. However, the underlying reasons for these associations between the quantity of brain tissue and differences in cognitive abilities remain largely unknown. Most researchers believe that intelligence cannot be localized to a single brain region, such as the frontal lobe. Brain lesion studies have found small but consistent associations indicating that people with more white matter lesions tend to have lower cognitive ability. Research utilizing NMR spectroscopy has discovered somewhat inconsistent but generally positive correlations between intelligence and white matter integrity, supporting the notion that white matter is important for intelligence.

Some research suggests that aside from the integrity of white matter, also its organizational efficiency is related to intelligence. The hypothesis that brain efficiency has a role in intelligence is supported by functional MRI research showing that more intelligent people generally process information more efficiently, *i.e.*, they use fewer brain resources for the same task than less intelligent people.

Small but relatively consistent associations with intelligence test scores include also brain activity, as measured by EEG records or event-related potentials, and nerve conduction velocity.

g in Non-Humans

Evidence of a general factor of intelligence has also been observed in non-human animals. Studies have shown that g is responsible for 47% of the individual variance in primates and between 55% and 60% in mice. While not able to be assessed using the same intelligence measures used in humans, cognitive ability can be measured with a variety of interactive and observational tools focusing on innovation, habit reversal, social learning, and responses to novelty. Non-human models of g such as mice are used to study genetic influences on intelligence and neurological developmental research into the mechanisms behind and biological correlates of g .

g (or c) in Human Groups

Similar to g for individuals, a new research path aims to extract a general collective intelligence factor c for groups displaying a group's general ability to perform a wide range of tasks. Definition, operationalization and statistical approach for this c factor

are derived from and similar to g . Causes, predictive validity as well as additional parallels to g are investigated.

Other Biological Associations

Height is correlated with intelligence ($r \sim .2$), but this correlation has not generally been found within families (*i.e.*, among siblings), suggesting that it results from cross-assortative mating for height and intelligence. Myopia is known to be associated with intelligence, with a correlation of around .2 to .25, and this association has been found within families, too.

Group Similarities and Differences

Cross-cultural studies indicate that the g factor can be observed whenever a battery of diverse, complex cognitive tests is administered to a human sample. The factor structure of IQ tests has also been found to be consistent across sexes and ethnic groups in the U.S. and elsewhere. The g factor has been found to be the most invariant of all factors in cross-cultural comparisons. For example, when the g factors computed from an American standardization sample of Wechsler's IQ battery and from large samples who completed the Japanese translation of the same battery were compared, the congruence coefficient was .99, indicating virtual identity. Similarly, the congruence coefficient between the g factors obtained from white and black standardization samples of the WISC battery in the U.S. was .995, and the variance in test scores accounted for by g was highly similar for both groups.

Most studies suggest that there are negligible differences in the mean level of g between the sexes, and that sex differences in cognitive abilities are to be found in more narrow domains.

For example, males generally outperform females in spatial tasks, while females generally outperform males in verbal tasks. Another difference that has been found in many studies is that males show more variability in both general and specific abilities than females, with proportionately more males at both the low end and the high end of the test score distribution.

Consistent differences between racial and ethnic groups in g have been found, particularly in the U.S. A 2001 meta-analysis of millions of subjects indicated that there is a 1.1 standard deviation gap in the mean level of g between white and black Americans, favouring the former. The mean score of Hispanic Americans was found to be .72 standard deviations below that of non-Hispanic whites. In contrast, Americans of East Asian descent generally slightly outscore white Americans. Several researchers have suggested that the magnitude of the black-white gap in cognitive ability tests is dependent on the magnitude of the test's g loading, with tests showing higher g loadings producing larger gaps. It has also been claimed that racial and ethnic differences similar to those found in the U.S. can be observed globally.

Relation to Other Psychological Constructs

Elementary Cognitive Tasks

Elementary cognitive tasks (ECTs) also correlate strongly with g . ECTs are, as the name suggests, simple tasks that apparently require very little intelligence, but still correlate strongly with more exhaustive intelligence tests. Determining whether a light is red or blue and determining whether there are four or five squares drawn on a computer screen are two examples of ECTs. The answers to such questions are usually provided by quickly pressing buttons. Often, in addition to buttons for the two options provided, a third button is held down from the start of the test. When the stimulus is given to the subject, they remove their hand from the starting button to the button of the correct answer. This allows the examiner to determine how much time was spent thinking about the answer to the question (reaction time, usually measured in small fractions of second), and how much time was spent on physical hand movement to the correct button (movement time). Reaction time correlates strongly with g , while movement time correlates less strongly. ECT testing has allowed quantitative examination of hypotheses concerning test bias, subject motivation, and group differences. By virtue of their simplicity, ECTs provide a link between classical IQ testing and biological enquiries such as fMRI studies.

Working Memory

One theory holds that g is identical or nearly identical to working memory capacity. Among other evidence for this view, some studies have found factors representing g and working memory to be perfectly correlated. However, in a meta-analysis the correlation was found to be considerably lower. One criticism that has been made of studies that identify g with working memory is that “we do not advance understanding by showing that one mysterious concept is linked to another.”

Piagetian Tasks

Psychometric theories of intelligence aim at quantifying intellectual growth and identifying ability differences between individuals and groups. In contrast, Jean Piaget’s theory of cognitive development seeks to understand qualitative changes in children’s intellectual development. Piaget designed a number of tasks to verify hypotheses arising from his theory. The tasks were not intended to measure individual differences, and they have no equivalent in psychometric intelligence tests. For example, in one of the best-known Piagetian conservation tasks a child is asked if the amount of water in two identical glasses is the same. After the child agrees that the amount is the same, the investigator pours the water from one of the glasses into a glass of different shape so that the amount appears different although it remains the same. The child is then asked if the amount of water in the two glasses is the same or different. Notwithstanding the

different research traditions in which psychometric tests and Piagetian tasks were developed, the correlations between the two types of measures have been found to be consistently positive and generally moderate in magnitude. A common general factor underlies them. It has been shown that it is possible to construct a battery consisting of Piagetian tasks that is as good a measure of *g* as standard IQ tests.

Personality

The traditional view in psychology is that there is no meaningful relationship between personality and intelligence, and that the two should be studied separately. Intelligence can be understood in terms of what an individual *can* do, or what his or her *maximal* performance is, while personality can be thought of in terms of what an individual *will typically* do, or what his or her general tendencies of behaviour are. Research has indicated that correlations between measures of intelligence and personality are small, and it has thus been argued that *g* is a purely cognitive variable that is independent of personality traits.

In a 2007 meta-analysis the correlations between g and the “Big Five” personality traits were found to be as follows:

- Conscientiousness -.04
- Agreeableness.00
- Extraversion.02
- Openness.22
- Emotional stability.09

The same meta-analysis found a correlation of .20 between self-efficacy and *g*. Some researchers have argued that the associations between intelligence and personality, albeit modest, are consistent. They have interpreted correlations between intelligence and personality measures in two main ways. The first perspective is that personality traits influence performance on intelligence *tests*. For example, a person may fail to perform at a maximal level on an IQ test due to his or her anxiety and stress-proneness. The second perspective considers intelligence and personality to be *conceptually* related, with personality traits determining how people apply and invest their cognitive abilities, leading to knowledge expansion and greater cognitive differentiation.

Creativity

Some researchers believe that there is a threshold level of *g* below which socially significant creativity is rare, but that otherwise there is no relationship between the two. It has been suggested that this threshold is at least one standard deviation above the population mean. Above the threshold, personality differences are believed to be important determinants of individual variation in creativity. Others have challenged the threshold theory. While not disputing that opportunity and personal attributes other than intelligence, such as energy and commitment, are important for creativity, they argue that *g* is positively

associated with creativity even at the high end of the ability distribution. The longitudinal Study of Mathematically Precocious Youth has provided evidence for this contention. It has showed that individuals identified by standardized tests as intellectually gifted in early adolescence accomplish creative achievements (for example, securing patents or publishing literary or scientific works) at several times the rate of the general population, and that even within the top 1 percent of cognitive ability, those with higher ability are more likely to make outstanding achievements. The study has also suggested that the level of *g* acts as a predictor of the *level* of achievement, while specific cognitive ability patterns predict the *realm* of achievement.

THEORY OF INTELLIGENCES: SINGLE, TWO FACTOR AND MULTIFACTOR THEORIES

The theory of multiple intelligences was proposed by Howard Gardner in 1983, to more accurately define the concept of intelligence and address whether methods which claim to measure intelligence (or aspects thereof) are truly scientific.

Gardner's theory argues that intelligence, particularly as it is traditionally defined, does not sufficiently encompass the wide variety of abilities humans display. In his conception, a child who masters multiplication easily is not necessarily more intelligent *overall* than a child who struggles to do so. The second child may be stronger in another *kind* of intelligence, and therefore may best learn the given material through a different approach, may excel in a field outside of mathematics, or may even be looking through the multiplication learning process at a fundamentally deeper level that hides a potentially higher mathematical intelligence than in the one who memorizes the concept easily.

DEVELOPMENT OF THE THEORY

“Howard Gardner, multiple intelligences and education” by given field (child prodigies, autistic savants);

- Neurological evidence for areas of the brain that are specialized for particular capacities (often including studies of people who have suffered brain damage affecting a specific capacity);
- The evolutionary relevance of the various capacities;
- Psychometric studies; and
- Susceptibility to encoding in a symbol system or notation (*e.g.* written language, musical notation, choreography).
- An identifiable core operation or set of operations.
- A distinctive development history, along with a definable set of ‘end-state’ performances.

Gardner originally identified seven core intelligences: linguistic, logical-mathematical, spatial, bodily-kinesthetic, musical, interpersonal and intrapersonal.

In 1997 at the symposium “MIND 97” (Multiple Intelligences New Directions) he added an eighth, the “Naturalist” Intelligence. Investigation continues on whether there are Existentialist (existential) and Spiritualist (spiritual) Intelligences.

The theory has been widely criticized in the psychology and educational theory communities.

The most common criticisms argue that Gardner’s theory is based on his own intuition rather than empirical data and that the intelligences are just other names for talents or personality types.

Despite these criticisms, the theory has enjoyed a great deal of popularity amongst educators over the past twenty years.

There are several schools which espouse MI as a pedagogy, and many individual teachers who incorporate some or all of the theory into their methodology. Many books and educational materials exist which explain the theory and how it may be applied to the classroom.

Teacher Education: Quality Assurance and its Indicators

‘Quality’ became very common word nowadays. What actually quality means. It is very familiar term but its meanings and uses vary widely. In general term quality is the position of a ‘product’ or a ‘process’ attribute on good –bad scale. It is often associated or linked with defects and deficiencies in products or process.

DEFINITIONS

“Quality is the capability of products or services to knowingly satisfy those preconceived composite wants of the user(s) that are intelligently related to the characteristics of performance, and do not cause major overt or covert reactions or actions by other people.”

Johnson –1987 “Fitness of use.” Juran –1974 “Conformance to requirements.” Crosby – 1984 From the above concepts of quality when two aspects combined together may help in understanding the concept in an optimum way.

- Relates to the features and attributes of the product or service.
- Absence of deficiencies in the products.

Thus quality can be defined as the totality of features and characteristics of a product or service that bear on its ability to satisfy stated or implied needs. Alexander Astin (1993) pointed out in the discussion on quality in education system,

You – cannot – define – it – but – you – know – it – when – you – see – it.

According to him there are four views on excellence in quality higher education-

- Excellence in reputation
- Excellence in resources
- Excellence in content
- Excellence in outcome.

Educational institutions have a tendency to measure their quality in terms of students' academic achievement. The assumption is that if the results of an institution are improving, its quality must be good.

Besides, academic achievement is not a competent measure of educational quality since it depends on many other factors such as students innate intelligence, study habits, learning styles, motivation, parental socio-economic status and home environment. Hence we have to identify relevant indicators of educational quality and measure it in a reliable and valid manner.

In quality management system total quality management starts with-

Quality Inspection

This is very initial step and through this, defects are observed and communicated for quality improvement.

Quality Improvement

It is the commitment and approach used to continuously improve every process in every part of an organization. Quality improvement is long term approach, established through incremental steps. Quality improvement is a way of working that facilitates change and influences teaching and learning by quality providing the stimulus and support for further assurance of quality.

Quality Assurance

Once quality is assured at reasonable level it needs to be controlled from deviations to ensure quality management. In the TQM the goods or services are expected to be with zero defects and when require further modification. Due to social or professional reasons the same may be communicated at the appropriate level for compliance. Thus quality assurance acts a backbone in the total system of quality management and hence requires more concern.

The subject of quality assurance has become a important word in the sphere of higher education. Among higher education teacher education occupies central stage due to its being the caretaker of macro educational system.

The teacher education has assign the task of producing the teachers for primary and secondary education and also build solid foundation of the system. Due to its role and significance, the teacher education has vast amount of growth during last decade. Education of teachers is not only responsible for improvement of school education but

also for preparing competent, committed and professionally well qualified teachers who can meet the demands of a system.

A teacher's content knowledge, verbal skills or enthusiasm for learning necessarily constitutes high mark for being a high quality teacher. Viewed in this perspective, a high quality teacher must imbibe the four pillars of education as mentioned in the Jacques Delor's report, "Learning the Treasure Within" (1996). These four pillars of education are,

Learning to Know

The teachers must have an urge for learning to learn. It can be achieved by combining a sufficiently broad general knowledge with the opportunity to work in depth on a small number of subjects.

***Learning to do**

Emphasizes acquisition not only of an occupational skill but also of the competence to deal with many situations and work in teams.

***Learning to Live Together**

Requires development of an understanding of other people and an appreciation of interdependence in a spirit of respect for the values of pluralism, mutual understanding and peace.

***Learning to be**

Will support the development of one's personality and ability to act with greater autonomy, judgement and personal responsibility. Thus these four pillars are considered as a founding pillars in the preparation of high quality teacher.

Also high quality teacher must possess three C's,

***Competence**

It is concerned with identifying the basic competencies in the teacher which will have the potential and promise for ensuring results and delivering the goods.

***Commitment**

It will imply a lot of self-governance and self-initiative of the teacher.

***Confidence**

The teacher perceives instrumentality and their utilization for improving the day-to-day situation.

However with the expansion of teacher education institutes the assurance of quality has been challenged which is the basic requirement in knowledge society. Due emergence

of knowledge society the need of quality assurance has further multiplied its intensity. The application of quality assurance in industrial sector helps us designing quality assurance strategies for teacher education. i) Maintain ii) Enhancing iii) Ownership iv) Checking quality As a whole quality assurance in teacher education may be ensured by studying its customer than assuring quality at every stage because all these parts are interrelated. i) The education service (Value added learning) The value added learning in teacher education is complex of inputs to the teaching and learning, teaching process, service quality, and institutional arrangement to assure quality in the inputs the discipline may use national curricula with inbuilt flexibility suiting to regional requirements. The disseminator of teacher education is a teacher and preparing the teachers where he/she is in the direct contact with the end users.

The close interaction between the service provider and end user also allow multiple opportunities for feedback and evaluation directing at satisfaction level of the end users. However due to involvement of repeated human interactions the possibility of human errors also enhances. Therefore, for the purpose of assuring quality in teacher education we need to infuse soft indicators of quality assurance *i.e.* care. Courtesy, concern, friendliness, and helpfulness to delivery superior service quality. ii) Primary external customer (learners) They are future teachers *i.e.* nation builders hence entry norm should be made more reliable. Once student has been enrolled in the teacher education programme again it is the teachers who can make him feel the real worth of the course.

The trainee teachers should be prepared socially responsible teachers with high values and morality. After completion of degree students should enjoy feeling of self- sufficacy and fulfillment. iii) Secondary external customers (parents, society) Parents are preliminary school for children and the prime decision maker in career choice. They should send their ward in teacher education if they convinced that he/she can be a real teacher and can do justice in the field. There should be mindset and valuing the teachers in the society. iv) Internal customer (teachers) In the modern society, the satisfaction of internal customer is as important as that of external customer. In order to get value from outside any system first should be valued from within because poor internal relationship will definitely prevent the institution from working properly and ultimately external customer suffers. Indicators of quality assurance i) Organizational mission The mission of an institution is its basic purpose and reason for existence. The mission of an institution should be articulated in a mission statement, understood by all the teachers. Valued by them and used consistently to guide all plans, goals and action. For example, mission of a ashram school could be empowerment of vulnerable section of society. ii) Continuous improvement Critical thinking and innovations, openness and flexibility. iii) Student orientation Quality implies fulfilling and exceeding the expectations of students in particular and the larger society in general. iv) Leadership commitment Commitment is in terms of emphasis on the value and importance of quality, allocation of resources

and support for quality and building commitment among all the staff members towards quality. v) Empowerment Empowerment is helping teachers to use their personal, professional or situational powers that they possess. vi) Collaboration Collaboration teams include members from different departments, subjects levels etc. so as to utilize divers viewpoints, knowledge and skills of such people. vii) Focus on processes Failures in quality are the result of faulty processes rather than faulty human beings. processes are a set of actions carried out to achieve predetermined results. Processes are generally complex and are linked to preceding and following actions. *i.e.*, processes are linked to inputs (preceding actions) and outputs (following action) of an institution. For enhancing the quality of processes, it is necessary to ensure the consistency, validity, reliability, usability and cost-effectiveness of the processes.

According to national curriculum framework 2005, list vision for teacher education.

Teacher education must become more sensitive to the emerging demands from the school system. For this it must prepare the teacher:

1. To care for children, and should love to be with them.
2. To understand children within social, cultural and political contexts.
3. Be receptive and be constantly learning.
4. View learning as a search for meaning out of personal experience, and knowledge generation as a continuously evolving process of reflective learning.
5. View knowledge not as an external reality embedded in textbooks, but as constructed in the shared context of teaching-learning and personal experience.
6. Own responsibility towards society, and work to build a better world.
7. Appreciate the potential of productive work and hands on experience as a pedagogic medium both inside and outside the classroom.
8. Analyze the curricular framework, policy implications and texts.

TEACHER

In education, a teacher is a person who provides schooling for others. A teacher who facilitates education for an individual student may also be described as a personal tutor. The role of teacher is often formal and ongoing, carried out by way of occupation or profession at a school or other place of formal education. In many countries, a person who wishes to become a teacher at state-funded schools must first obtain professional qualifications or credentials from a university or college. These professional qualifications may include the study of pedagogy, the science of teaching. Teachers will have to continue their education after they receive their degree from a college or university. Teachers may use a lesson plan to facilitate student learning, providing a course of study which helps cover a standardized test curriculum. A teacher's role may vary between cultures. Teachers teach literacy and numeracy, or some of the other school subjects. Other teachers may provide help or certain education instruction in

craftsmanship or vocational training, the Arts, religion or spirituality, civics, community roles, or life skills. In some countries, formal education can take place through home schooling. Informal learning may be assisted by a teacher occupying a transient or ongoing role, such as a parent or sibling or within a family, or by anyone with knowledge or skills in the wider community setting. Religious and spiritual teachers, such as gurus, mullahs, rabbis, pastors/youth pastors and lamas may teach religious texts such as the Quran, Torah or Bible.

PROFESSIONAL EDUCATORS

Teaching may be carried out informally, within the family which is called home schooling or the wider community. Formal teaching may be carried out by paid professionals. Such professionals enjoy a status in some societies on a par with physicians, lawyers, engineers, and accountants. A teacher's professional duties may extend beyond formal teaching. Outside of the classroom teachers may accompany students on field trips, supervise study halls, help with the organization of school functions, and serve as supervisors for extracurricular activities. In some education systems, teachers may have responsibility for student discipline.

Around the world teachers are often required to obtain specialized education, knowledge, codes of ethics and internal monitoring. There are a variety of bodies designed to instill, preserve and update the knowledge and professional standing of teachers. Around the world many governments operate teacher's colleges, which are generally established to serve and protect the public interest through certifying, governing and enforcing the standards of practice for the teaching profession. The functions of the teacher's colleges may include setting out clear standards of practice, providing for the ongoing education of teachers, investigating complaints involving members, conducting hearings into allegations of professional misconduct and taking appropriate disciplinary action and accrediting teacher education programmes. In many situations teachers in publicly funded schools must be members in good standing with the college, and private schools may also require their teachers to be college peoples. In other areas these roles may belong to the State Board of Education, the Superintendent of Public Instruction, the State Education Agency or other governmental bodies. In still other areas Teaching Unions may be responsible for some or all of these duties.

PEDAGOGY AND TEACHING

In education, teachers facilitate student learning, often in a school or academy or perhaps in another environment such as outdoors. A teacher who teaches on an individual basis may be described as a tutor. The objective is typically accomplished through either an informal or formal approach to learning, including a course of study and lesson plan that teaches skills, knowledge and/or thinking skills. Different ways to teach are often referred to as pedagogy. When deciding what teaching method to use

teachers consider students' background knowledge, environment, and their learning goals as well as standardized curricula as determined by the relevant authority. Many times, teachers assist in learning outside of the classroom by accompanying students on field trips. The increasing use of technology, specifically the rise of the internet over the past decade, has begun to shape the way teachers approach their roles in the classroom. The objective is typically a course of study, lesson plan, or a practical skill. A teacher may follow standardized curricula as determined by the relevant authority.

The teacher may interact with students of different ages, from infants to adults, students with different abilities and students with learning disabilities. Teaching using pedagogy also involve assessing the educational levels of the students on particular skills. Understanding the pedagogy of the students in a classroom involves using differentiated instruction as well as supervision to meet the needs of all students in the classroom. Pedagogy can be thought of in two manners. First, teaching itself can be taught in many different ways, hence, using a pedagogy of teaching styles. Second, the pedagogy of the learners comes into play when a teacher assesses the pedagogic diversity of his/her students and differentiates for the individual students accordingly. Perhaps the most significant difference between primary school and secondary school teaching is the relationship between teachers and children.

In primary schools each class has a teacher who stays with them for most of the week and will teach them the whole curriculum. In secondary schools they will be taught by different subject specialists each session during the week and may have 10 or more different teachers. The relationship between children and their teachers tends to be closer in the primary school where they act as form tutor, specialist teacher and surrogate parent during the course of the day. This is true throughout most of the United States as well. However, alternative approaches for primary education do exist. One of these, sometimes referred to as a "platoon" system, involves placing a group of students together in one class that moves from one specialist to another for every subject. The advantage here is that students learn from teachers who specialize in one subject and who tend to be more knowledgeable in that one area than a teacher who teaches many subjects. Students still derive a strong sense of security by staying with the same group of peers for all classes. Co-teaching has also become a new trend amongst educational institutions. Co-teaching is defined as two or more teachers working harmoniously to fulfill the needs of every student in the classroom. Co-teaching focuses the student on learning by providing a social networking support that allows them to reach their full cognitive potential. Co-teachers work in sync with one another to create a climate of learning.

RIGHTS TO ENFORCE SCHOOL DISCIPLINE

Throughout the history of education the most common form of school discipline was corporal punishment. While a child was in school, a teacher was expected to act as

a substitute parent, with all the normal forms of parental discipline open to them. In past times, corporal punishment was one of the most common forms of school discipline throughout much of the world. Most Western countries, and some others, have now banned it, but it remains lawful in the United States following a US Supreme Court decision in 1977 which held that paddling did not violate the US Constitution. 30 US states have banned corporal punishment, the others have not. It is still used to a significant degree in some public schools in Alabama, Arkansas, Georgia, Louisiana, Mississippi, Oklahoma, Tennessee and Texas.

Private schools in these and most other states may also use it. Corporal punishment in American schools is administered to the seat of the student's trousers or skirt with a specially made wooden paddle. This often used to take place in the classroom or hallway, but nowadays the punishment is usually given privately in the principal's office. Official corporal punishment, often by caning, remains commonplace in schools in some Asian, African and Caribbean countries. Currently detention is one of the most common punishments in schools in the United States, the UK, Ireland, Singapore and other countries. It requires the pupil to remain in school at a given time in the school day; or even to attend school on a non-school day, *e.g.*, "Saturday detention" held at some US schools. During detention, students normally have to sit in a classroom and do work, write lines or a punishment essay, or sit quietly.

A modern example of school discipline in North America and Western Europe relies upon the idea of an assertive teacher who is prepared to impose their will upon a class. Positive reinforcement is balanced with immediate and fair punishment for misbehaviour and firm, clear boundaries define what is appropriate and inappropriate behaviour. Teachers are expected to respect their students, and sarcasm and attempts to humiliate pupils are seen as falling outside of what constitutes reasonable discipline.

Whilst this is the consensus viewpoint amongst the majority of academics, some teachers and parents advocate a more assertive and confrontational style of discipline. Such individuals claim that many problems with modern schooling stem from the weakness in school discipline and if teachers exercised firm control over the classroom they would be able to teach more efficiently. This viewpoint is supported by the educational attainment of countries—in East Asia for instance—that combine strict discipline with high standards of education.

It's not clear, however that this stereotypical view reflects the reality of East Asian classrooms or that the educational goals in these countries are commensurable with those in Western countries. In Japan, for example, although average attainment on standardized tests may exceed those in Western countries, classroom discipline and behaviour is highly problematic. Although, officially, schools have extremely rigid codes of behaviour, in practice many teachers find the students unmanageable and do not enforce discipline at all. Where school class sizes are typically 40 to 50 students, maintaining order in the classroom can divert the teacher from instruction, leaving

little opportunity for concentration and focus on what is being taught. In response, teachers may concentrate their attention on motivated students, ignoring attention-seeking and disruptive students. The result of this is that motivated students, facing demanding university entrance examinations, receive disproportionate resources, while the rest of the students are allowed, perhaps expected to, fail. Given the emphasis on attainment of university places, administrators and governors may regard this policy as appropriate.

OBLIGATION TO HONOUR STUDENTS RIGHTS

Sudbury model democratic schools claim that popularly based authority can maintain order more effectively than dictatorial authority for governments and schools alike. They also claim that in these schools the preservation of public order is easier and more efficient than anywhere else. Primarily because rules and regulations are made by the community as a whole, thence the school atmosphere is one of persuasion and negotiation, rather than confrontation since there is no one to confront. Sudbury model democratic schools' experience shows that a school that has good, clear laws, fairly and democratically passed by the entire school community, and a good judicial system for enforcing these laws, is a school in which community discipline prevails, and in which an increasingly sophisticated concept of law and order develops, against other schools today, where rules are arbitrary, authority is absolute, punishment is capricious, and due process of law is unknown.

TEACHER ENTHUSIASM

Since teachers can affect how students perceive the course materials, it has been found that teachers who showed enthusiasm towards the course materials and students can affect a positive learning experience towards the course materials. On teacher/course evaluations, it was found that teachers who have a positive disposition towards the course content tend to transfer their passion to receptive students. Teachers cannot teach by rote but have to find new invigoration for the course materials on a daily basis. Teachers have to keep in mind that they are teaching new minds every term or semester. Otherwise, teachers will fall into the trap of having done this material again and start feeling bored with the subject which in turn bore the students as well. Students who had enthusiastic teachers tend to rate them higher than teachers who didn't show much enthusiasm for the course materials.

Teachers that exhibit enthusiasm can lead to students who are more likely to be engaged, interested, energetic, and curious about learning the subject matter. Recent research has found a correlation between teacher enthusiasm and students' intrinsic motivation to learn and vitality in the classroom. Controlled, experimental studies exploring intrinsic motivation of college students has shown that non-verbal expressions of enthusiasm, such as demonstrative gesturing, dramatic movements which are varied,

and emotional facial expressions, result in college students reporting higher levels of intrinsic motivation to learn. Students who experienced a very enthusiastic teacher were more likely to read lecture material outside of the classroom. There are various mechanisms by which teacher enthusiasm may facilitate higher levels of intrinsic motivation.

Teacher enthusiasm may contribute to a classroom atmosphere full of energy and enthusiasm which feed student interest and excitement in learning the subject matter. Enthusiastic teachers may also lead to students becoming more self-determined in their own learning process. The concept of mere exposure indicates that the teacher's enthusiasm may contribute to the student's expectations about intrinsic motivation in the context of learning. Also, enthusiasm may act as a "motivational embellishment"; increasing a student's interest by the variety, novelty, and surprise of the enthusiastic teacher's presentation of the material.

Finally, the concept of emotional contagion, may also apply. Students may become more intrinsically motivated by catching onto the enthusiasm and energy of the teacher. Research shows that student motivation and attitudes towards school are closely linked to student-teacher relationships. Enthusiastic teachers are particularly good at creating beneficial relations with their students.

Their ability to create effective learning environments that foster student achievement depends on the kind of relationship they build with their students. Useful teacher-to-student interactions are crucial in linking academic success with personal achievement. Here, personal success is a student's internal goal of improving himself, whereas academic success includes the goals he receives from his superior. A teacher must guide his student in aligning his personal goals with his academic goals. Students who receive this positive influence show stronger self-confidence and greater personal and academic success than those without these teacher interactions. Students are likely to build stronger relations with teachers who are friendly and supportive and will show more interest in courses taught by these teachers.

Teachers that spend more time interacting and working directly with students are perceived as supportive and effective teachers. Effective teachers have been shown to invite student participation and decision making, allow humor into their classroom, and demonstrate a willingness to play. The way a teacher promotes the course they are teaching, the more the student will get out of the subject matter. The three most important aspects of teacher enthusiasm are enthusiasm about teaching, enthusiasm about the students, and enthusiasm about the subject matter.

A teacher must enjoy teaching. If they do not enjoy what they are doing, the students will be able to tell. They also must enjoy being around their students. A teacher who cares for their students is going to help that individual succeed in their life in the future.

The teacher also needs to be enthusiastic about the subject matter they are teaching. For example, a teacher talking about chemistry needs to enjoy the art of chemistry and

show that to their students. A spark in the teacher may create a spark of excitement in the student as well. A enthusiastic teacher has the ability to be very influential in the young students life.

MISCONDUCT

Misconduct by teachers, especially sexual misconduct, has been getting increased scrutiny from the media and the courts. A study by the American Association of University Women reported that 0.6 per cent of students in the United States claim to have received unwanted sexual attention from an adult associated with education; be they a volunteer, bus driver, teacher, administrator or other adult; sometime during their educational career. A study in England showed a 0.3 per cent prevalence of sexual abuse by any professional, a group that included priests, religious leaders, and case workers as well as teachers. It is important to note, however, that the British study referenced above is the only one of its kind and consisted of “a random... probability sample of 2,869 young people between the ages of 18 and 24 in a computer-assisted study” and that the questions referred to “sexual abuse with a professional,” not necessarily a teacher. It is therefore logical to conclude that information on the percentage of abuses by teachers in the United Kingdom is not explicitly available and therefore not necessarily reliable.

The AAUW study, however, posed questions about fourteen types of sexual harassment and various degrees of frequency and included only abuses by teachers. “The sample was drawn from a list of 80,000 schools to create a stratified two-stage sample design of 2,065 8th to 11th grade students” Its reliability was gauged at 95 per cent with a 4 per cent margin of error. In the United States especially, several high-profile cases such as Debra LaFave, Pamela Rogers, and Mary Kay Latourneau have caused increased scrutiny on teacher misconduct. Chris Keates, the general secretary of National Association of Schoolmasters Union of Women Teachers, said that teachers who have sex with pupils over the age of consent should not be placed on the sex offenders register and that prosecution for statutory rape “is a real anomaly in the law that we are concerned about.” This has led to outrage from child protection and parental rights groups.

UNIVERSITY

A university is an institution of higher education and research, which grants academic degrees in a variety of subjects. A university is a corporation that provides both undergraduate education and postgraduate education. The word university is derived from the Latin *universitas magistrorum et scholarium*, roughly meaning “community of teachers and scholars.” The definition of a university varies widely even within some countries. For example, there is no nationally standardized definition of the term in the United States although the term has traditionally been used to designate research institutions and was once reserved for research doctorate-granting institutions. Some

states, such as Massachusetts, will only grant a school “university status” if it grants at least two doctoral degrees. In the United Kingdom, an institution can only use the term if it has been granted by the Privy Council, under the terms of the Further and Higher Education Act 1992. In India, a new tag deemed universities was created a few years ago, by the cabinet minister Arjun Singh during his tenure as the Minister for Human Resource Development. Through this provision many universities sprung up in India, which are commercial in nature and have been established just to exploit the demand of higher education.

A METHODOLOGICAL APPROACH TO INTEGRATING WEB-BASED LEARNING INTO TRADITIONAL TEACHING.

Most educational organizations have to address the imperative for change. The circumstances that have unfolded over the last decade, including the use of new computer and communication technologies together with a variety of social, demographic and economic changes, challenge educationalists to reflect upon traditional methods of teaching. Included among these challenges are strong theoretical challenges concerning the process of learning in students and, now, given greater technological advances a contextual challenge as to how and where real learning takes place and the nature of its facilitation. The technological conditions in which universities operate have changed in very significant ways. The real challenge is to discover the ways to improve learning. In order to do this we need to state what set of initial conditions might be imagined to improve learning using new technologies?

New computer and communications technologies present, in Popperian terms, opportunities to create bold new ideas concerning the problem of how to improve student learning. In less than a decade, the facility for the storage and rapid communication of knowledge and ideas plus the potential for mass assessment to these has improved almost beyond expectation.

This potentiality is now being applied to problems of inclusiveness, and Life Long Learning. New technologies are applied and pedagogic strategies invented that encompass the term, e-Learning..

One analogy is that of the ecological niche. A new niche has opened up to allow educationalists and learners to achieve their ends. The conjecture is that bold new approaches to learning mediated via computers and communication technologies can improve learning by removing impediments. For this conjecture to be tested it must be capable of being falsified. If the conjecture is falsified, as difficult as it may be, then the logic compels the enquirer to look for alternative hypotheses to improve learning. Just at the moment, the theory that new technologies can improve learning have as much a sporting chance of not being refuted, as do other competing hypotheses such as a change

in educational policy or finance. Yet, Popper's methodology is compels us to realistically distinguish good from bad ideas. This is an important heuristic for developing realist solutions to improving learning by means of eliminating of error. We ought to relinquish new theory if it a is mistaken.

Popper's problem solving approach for improvement (ameliorism) is a combination of theory creation and elimination of falsity. It is a dynamic, socratic dialogue between a transformationalist and a sceptical stance. The function of creativity is to build new imaginative ideas. The function of criticism is to eliminate bad ideas and severely check them against reality in which brute facts may kill beautiful theories.

In examining the validity of Popper's philosophy in education much can be gained from situational analysis. This approach consists of constructing a fallible reconstruction of the learning situation in an attempt to make sense of events. Briefly, an attempt is made at situational analysis in order to provide a fairly rough model of the typical learning situation, including a summary of the aims, knowledge and means to achieving the aims. The approach attempts to select the facts that have an important bearing and reject those that have none.

What we seek to discover and remove are the impediments to learning. We can do this by asking ourselves the simple question: what stops learning? By adopting this approach, it might be possible to design new learning situations or re-design and improve upon orthodox learning situations from which the impediments are removed. The innovator by incorporating new learning approaches such as web-based learning formulates a horizon of expectations of what ought to happen in the improved situation. Principle objectives might include improving learning by providing greater responsibility to learners, or by giving them more control.

A methodology that can be most appropriately adopted is Popper's problem solving approach. Below I have attempted to list a number of questions for consideration when examining new theories and ideas. The questions are based upon ideas about the process of discovery in Popper (1972,265) and Polya (1973).

- Are you clear about the problem?
- Has the problem changed and become more definite?
- Are you clear about the theories you hold? What are they ?
- Are you holding some theories unconsciously or taking them for granted?
- Might any of them be false theories?
- Have you made several attempts to formulate your theories - for instance by writing them out of discussing them and then tried to criticise them?
- Try to construct alternative theories, this may help you understand your current theory more clearly.
- Be wary if you believe your theory to be the only possible one, perhaps you haven't fully understood it, or the problem you have to solve.

- If not in practice, then in theory, perhaps in written form, think of experiments that may test your theory and perhaps overthrow it.
- If an experiment or observation appears to support a theory, remember this may mean it really weakens some alternative theory - perhaps one you may not have thought of before.
- Attempt to refute your own theories, it is better than defending them and leaving it to others to refute them.
- A good defence of a theory against criticism is a necessary part of any fruitful discussion since only by defending it can we find out its strengths and the strength of criticism directed against it.
- Try to put your theory in its strongest form and strongly argue against it.

The theme of improvement by self-criticism is very apparent in Popper's methodology, but it is not unique to Popper. For instance, a similar but more elaborate approach is employed by the mathematician Polya (1973) and many more. A common feature of the problem solving methodology is a generate and test cycle. For instance, Newell and Simon (1972) use a general problem solver approach in which the requirements are a knowledge of current situations, the desired situation (goals) and the difference between the two. The actions change the situation to bring the situation near to achieving the goals. The problem solving process must select actions likely to reduce or remove difficulties. It is illuminating to review the theory and formulate problems using this approach. I have argued that teachers solve a multitude of problems on a day-to-day basis. They are continually put in situations in which they are required to offer tentative solutions to mundane problems, yet they rarely reflect upon the value and necessity of the problem solving approach. It was from this position of necessity that I began to experiment with web based learning. The theory was simple: web-based learning when employed in conjunction with traditional approaches such as lectures and student seminars should remove impediments normally found in traditional approaches. For instance, one advantage in employing web-based learning might lie in the much greater student access to course materials. The explanation for improved learning, if it happens, might be that students who naturally have a wide variety of learning approaches gain more benefit from greater access to web-based learning resources. For instance, by allowing the students greater control of the pace and timing improved qualities and efficiencies may emerge. In reality, the problem solving approach can provide a table of connections, impediments and actions: a whole series of P_n 's. The approach gives a means of evaluating progress by trying alternative paths. Actions taken may change the situation to bring the situation nearer to the goals. For instance, it may be speculated that by employing web-based learning several problems are avoided such as:

- The lack of access to intellectual material relevant to the learning situation
- The lack of durability and adaptability of the current learning situations *i.e.* university scheduled lectures and seminars

- Over dependence on lecturers
- The lack of use of new technologies
- A lack of active learning and motivation.

At the moment, these conjectures remain as somewhat fuzzy, yet interesting, speculations. They need much tighter formulation and testing.

THE METHOD OF CASE STUDIES- DEALING WITH PROTO-SCIENTIFIC IDEAS IN EDUCATION

A brief word can be said about the methodological value of case studies. A method is needed for the description and evaluation of unique situations. During the early phases of implementing a new idea, it is not always possible to severely test a scientific hypothesis using classical experiments which normally involve manipulation, control, replicated observations and randomization. In biological and social disciplines even quasi-experimental conditions are difficult to produce, even if they are ethically defensible. In reality, when dealing with innovation in particular, there are unknowable boundary conditions, unknown bias, a lack of knowledge of underlying phenomena and poor data.

When case studies are employed, it is possible to focus on specific processes or phenomena. Further enquiry can result in discovery and investigation rather than a collaboration or refutation of a hypothesis, although the logic of Popper's ideas inevitably take us down that path. Case studies can aim at testing clarifying, amending, evaluating examples and making recommendations.

There are different methods of approach including writing a comprehensive literature review, seeking multiple sources of evidence, creating a data or evidentiary base and chasing an initial question to some kind of conclusion. Inferences can be made about significant phenomena, about what caused events and how typical they are. The aim is to make sense of a particular situation. All of this requires practical insight into the situation, discerning judgement, the evaluation of alternatives and the examination of competing explanations.

The case study might be the only methodological option to begin with, even though its major weakness is bias. So although there are no general theories for precise predictions, it is possible to accomplish practical tasks, rough generalisations, lower level theories and possibly some level of control. Using this method, it is possible to report upon exemplars as a focus for debate, and test out ideas by uncovering flaws and, also, to formulate hypotheses despite uncertainty and grossly imperfect evidence. More systematic and more organised studies may ensue as more crucial details are revealed. Of course, we should make the tightest possible case for our ideas, and give our critical preference for the approaches that guide action, yet these should be formulated in such a way that they can openly be criticized in theory and practice and capable of refutation and improvement by ourselves or others.

A CASE STUDY CONCERNING THE LEARNING AND TEACHING OF ECOLOGY: CONTEXT, IMPEDIMENTS AND ASPIRATIONS

In conjunction with conventional teaching approaches, it was considered that several problems could be solved using web-based learning in two modules in the undergraduate B.A. programme. Both modules, although quite different in approach, sought to encourage students to think more scientifically, to engage more fruitfully in active learning and to take a more critical, or even philosophical, approach to environmental problems. The targeted group of students, about 20-25 in number, had a limited science background, yet they had a demonstrated interest in environmental studies. I speculated that this group, with their confidences and back ground, had much to gain from a combination of web-based learning and more conventional approaches. The latter included short introductory lectures, group discussions and student presentations. Most of these students had a good level of computer literacy and have demonstrated a confident and studious approach towards the use of the inter-net for learning.

Each module began with an introduction to the value of using web pages for learning. A tour through the web site was given. Much emphasis was placed on the potential of the web pages for learning, yet criticism and suggestions for improvement were requested and received from users. Each web page was designed to help overcome common learning difficulties such as those involved with trying to think scientifically, including formulating and testing ideas, writing up reports, writing intellectual journals etc. For many years, students had found thesis skills more difficult than others and had lost marks as a result. The evidence could be seen in their written work.

In a nutshell, the theory to be tested was this. If clear and encouraging web-based learning resources were made accessible on the internet then some of the impediments to learning would be removed resulting in an improved learning situation. Students might gain an advantage in their learning approaches by more open access to learning materials on the web. This, in turn, would result in greater motivation and efficiencies possibly as a result of more student control over the construction and pacing of learning. Hopefully, this was placed within the context of what characterizes effective teaching including: setting clear learning goals, appropriate assessment and workloads, and emphasis on independent learning.

A family of structured web pages written in HTML and entitled Ecospace was created. These pages are continually adapted and improved as teaching progresses and the module is reviewed from year to year. The aim was to harness the tremendous power of the web to allow more effective communication. Web pages are very accessible, durable, interoperable and re-usable. Each web page is designed to solve a problem that results from or cannot be solved by the traditional approaches of lecturing and whole class or group discussions. For example, a web page may consist of a clear exposition of a concept that most students find difficult. Students, if so motivated may prepare from this material for discussion during contact time. Small exercises can be presented that extend

vocabulary or mathematical prowess. These are enrichment exercises. Other web pages may be designed to encourage more active learning in the form of issuing problems or encouraging the use of search engines or electronic discussion groups. The problems encountered in the unique learning situation will dictate or influence the manner in which web pages can be most suitable put to good use. A few examples are given in the following table:

Problem (P1)	Tentative solution (TT)	Evidence
Lecture/transmission approach tends to stifle individual approaches to learning.	Design web based materials for continuous open access by student. Invite a critical response rather than a descriptive one	Improved learning as evidenced by more individual and critical responses in the examination.
Lack of depth of understanding.	Create web pages for a clearer identification of core problems and theories with hyperlinks to relevant external web sites for enrichment.	A demonstration of more independent enquiry and understanding including wider reading.
Lack of active learning	Design a web page to facilitate more active learning using search engines and discussion groups	Evidence of wider reading and participation in discussions including fresh and original thinking.
Limited vocabulary and powers of expression.	Design a web page for focussing on refining and extending vocabulary.	A richer and felicitous vocabulary

The durability of web pages, unmatched by text books and even lecture summaries and hand-outs, has to be a strength. This strength if sensitively employed can produce a quite dynamic and exciting aspect to teaching. Using Frontpage, which is Microsoft's publishing team for creating, managing and publishing on the world wide web, learning resources can be quickly created and published on the server in response to the need of the learners. During discussions and presentations, the group or class may identify weaknesses in their knowledge. Using a web-based approach the lecturer may responsively alert the class to a new paper or guide them more precisely in their reading. Eventually, the web pages may even become a very effective collaborative learning exercise between lecturing staff and students.

When experimenting with new ideas, it is necessary to spot the emergence of bad learning approaches and experiences. For instance, over reliance on web-based learning may de-humanise and inhibit the vital social exchanges required for a healthy approach towards learning. These unintended consequences or 'life's little ironies' are most likely to emerge and can be avoided if a piecemeal approach is adopted.

WEB BASED LEARNING: NEW TECHNOLOGIES WITH DIFFERENT PEDAGOGIES

One of the strongest arguments for adopting a web-based learning approach springs from Popper's idea that as learners we create ideas before we test or evaluate them:

according to E.H. Gombrich, learners 'make and match'. A learner may grasp a problem and suggest a solution. The solution may or may not be a realist one. The world wide web improves access to information and ideas and is becoming an important learning resource which is potentially accessible to most people in the western world. It is a huge source ideas of varying quality. For the learner to effectively use this resource they need to become more autonomous in their learning. In one sense, the world wide web challenges teachers and learners to adopt a new pedagogy, one less dependent on traditional approaches involving the transmission of knowledge from teachers to learner. This new pedagogy is one in which the learner has far more independence and control.

H.J. Perkinson has considered how the lessons of Popper's philosophy of science may be applied to the classroom or lecture theatre. In a series of books, Perkinson shows how the popperian approach to learning, the system of conjectures and refutations and problem-solving, can be fashioned into a non-inductive model of education. This approach views the teacher not as someone who injects knowledge into passive minds by a process of transmission, but as one who elicits students responses, creates environments in which ideas are criticized freely and mistakes are made without humiliation and corrected. The student is an active creator of ideas and solver of problems.

The role of the teacher or lecturer shifts away from a more orthodox one involving the effective transmission of ideas, facts and figures to a role that encourages learners to actively and critically engage with and, perhaps, improve new and old ideas. Active engagement by the learners is compelled by the logic of the fallibility of knowledge. The posing of ideas and questioning is more familiarly known as the Socratic approach.

Popper's methodology, the approach of creating solutions to solve educational problems, a critical realism, and its use as a heuristic for a learning is an approach that sharply contrast with many applied in education. The methodology has been and still is misunderstood, or ignored, yet there are encouraging signs that this approach is being used, both in theory and practice, for improving learning. The challenge implied by Popper's approach and the relentless surge of technological innovation and the potential for improving learning, or otherwise, is one that educators would ill advised to ignore. The severer we test these opportunities the more generous the opportunities for eliminating false hopes.

QUALITIES OF EFFECTIVE TEACHERS

We know intuitively that these highly effective teachers can have an enriching effect on the daily lives of children and their lifelong educational and career aspirations. We now know empirically that these effective teachers also have a direct influence in enhancing student learning. Years of research on teacher quality support the fact that effective teachers not only make students feel good about school and learning, but also that their work actually results in increased student achievement. Studies have

substantiated that a whole range of personal and professional qualities are associated with higher levels of student achievement. For example, we know that verbal ability, content knowledge, pedagogical knowledge, certification status, ability to use a range of teaching strategies skillfully, and enthusiasm for the subject characterize more successful teachers.¹ The following are some of the key qualities of effective teachers:

- Have formal teacher preparation training.
- Hold certification of some kind (standard, alternative, or provisional) and are certified within their fields.
- Have taught for at least three years.
- Are caring, fair, and respectful.
- Hold high expectations for themselves and their students.
- Dedicate extra time to instructional preparation and reflection.
- Maximize instructional time via effective classroom management and organization.
- Enhance instruction by varying instructional strategies, activities, and assignments.
- Present content to students in a meaningful way that fosters understanding.
- Monitor students' learning by utilizing pre- and postassessments, providing timely and informative feedback, and reteaching material to students who did not achieve mastery.
- Demonstrate effectiveness with the full range of student abilities in their classrooms, regardless of the academic diversity of the students.

Not only does a reasonable consensus exist on what effective teachers do to enhance student learning, but also meta-analyses by researchers such as Marzano, Pickering, and Pollock (2001) have begun to quantify the average effects of specific instructional strategies.

When properly implemented, instructional strategies such as identifying similarities and differences, summarizing and note taking, and reinforcing effort and providing recognition can result in percentile gains of 29–45 points in student achievement.² Such an increase would mean that the score of an average student at the 50th percentile might rise to the 79th or even the 95th percentile with the effective use of selected instructional strategies.

While teaching undeniably will remain an art, there is also a science to it that we are only beginning to aggressively apply to practice. As observed by Mike Schmoker, author of *Results: The Key to Continuous School Improvement*, “when we begin to more systematically close the gap between what we know and what we do, we will be on the cusp of one of the most exciting epochs in the history of education.”³ With state standards and federal legislation, such as No Child Left Behind, more explicitly defining accountability, the time has arrived for a systematic application of our research-based knowledge.

IMPACT OF TEACHER EFFECTIVENESS ON STUDENT ACHIEVEMENT

The work of Bill Sanders, formerly at the University of Tennessee's Value-Added Research and Assessment Center, has been pivotal in reasserting the importance of the individual teacher on student learning.⁴ One aspect of his research has been the additive or cumulative effect of teacher effectiveness on student achievement. Over a multi-year period, Sanders focused on what happened to students whose teachers produced high achievement versus those whose teachers produced low achievement results. He discovered that when children, beginning in 3rd grade, were placed with three high-performing teachers in a row, they scored on average at the 96th percentile on Tennessee's statewide mathematics assessment at the end of 5th grade. When children with comparable achievement histories starting in 3rd grade were placed with three low-performing teachers in a row, their average score on the same mathematics assessment was at the 44th percentile,⁵ an enormous 52-percentile point difference for children who presumably had comparable abilities and skills. Elaborating on this body of research, Dr. Sanders and colleagues reported the following:

- ... the results of this study well document that the most important factor affecting student learning is the teacher. In addition, the results show wide variation in effectiveness among teachers. The immediate and clear implication of this finding is that seemingly more can be done to improve education by improving the effectiveness of teachers than by any other single factor. Effective teachers appear to be effective with students of all achievement levels, regardless of the level of heterogeneity in their classrooms.

Further analysis of the Tennessee data indicated that the effects on achievement of both strong and weak teachers persisted over three years: subsequent achievement was enhanced or limited by the experiences in the classrooms of strong or weak teachers, respectively.

In other words, learning gains realized by students during a year in the classroom of an effective teacher were sustained over later years and were compounded by additional years with effective teachers. Conversely, depressed achievement results resisted improvement even after a student was placed with an effective teacher, and the negative impact was discernible statistically for approximately three subsequent years. Given results like these, it's no wonder that the researchers found that "a major conclusion is that teachers make a difference."

In a comparable study by researchers in Dallas, Texas, similar results were found in both math and reading during the early grades. When 1st grade students were fortunate enough to be placed with three high-performing teachers in a row, their average performance on the math section of the Iowa Tests of Basic Skills increased from the 63rd percentile to the 87th, in contrast to their peers with similar scores whose performance decreased from the 58th percentile to the 40th, a percentile difference of 42 points. A similar analysis in reading found a percentile difference of 44 percentile

points. The studies in Tennessee and Texas produced strikingly similar findings: Highly effective teachers are able to produce much greater gains in student achievement than their less effective counterparts.

While the numbers help to summarize the cumulative academic effects of less effective teachers, we can only imagine the sense of failure and hopelessness that these children and their parents experienced during the years in these classrooms. Undoubtedly, the children wondered what was wrong with them when, in reality, it was the quality of their instruction. A common yet misguided bit of folk wisdom has been that adversity, in the guise of an ineffective teacher, builds character and that a student can catch up the following year. The research indicates otherwise.

Based on the findings from the Dallas Public Schools' Accountability System, the negative effects of a poor-performing teacher on student achievement persist through three years of high-performing teachers. The good news is that if students have a high-performing teacher one year, they will enjoy the advantage of that good teaching in future years. Conversely, if students have a low-performing teacher, they simply will not outgrow the negative effects of lost learning opportunities for years to come. Further exacerbating the negative effects of poor-performing teachers, the Dallas research shows that "lower-achieving students are more likely to be put with lower effectiveness teachers... Thus, the negative effects of less effective teachers are being visited on students who probably need the most help."

Summarizing the findings from studies of the Dallas and Tennessee Value-Added Assessment Systems, Mendro states:

Research... has demonstrated the effects of teachers on student achievement. They [the researchers] show that there are large additional components in the longitudinal effects of teachers, that these effects are much larger than expected, and that the least effective teachers have a long-term influence on student achievement that is not fully remediated for up to three years later.

In straightforward terms, these residual effects studies make it clear that not only does teacher *quality* matter when it comes to how much students learn, but also that, for better or worse, a teacher's *effectiveness* stays with students for years to come.

HIGHLY QUALIFIED VERSUS HIGHLY EFFECTIVE

Given the growing body of knowledge about the impact of effective teachers on children, it seems that educational policy is beginning to acknowledge the importance of classroom teachers in addition to curriculum standards and assessments. A case in point is the federal No Child Left Behind Act of 2001, which has introduced both the concepts of "adequate yearly progress," based on annual testing, and "highly qualified teacher," based on teacher credentials, as strategies to improve U.S. education. According to the legislation, "highly qualified" teachers are defined as those who hold at least a bachelor's degree, are fully licensed or certified by the state in the subjects they teach,

and can demonstrate competence in the subjects they teach. While licensure or certification is a significant indicator of teacher quality, these factors alone are insufficient for teacher effectiveness. As discussed earlier, teacher effectiveness is characterized by a far more complex set of qualities than one's professional preparation. It includes dispositions and an array of planning, organizational, instructional, and assessment skills. Effective teachers are able to envision instructional goals for their students, then draw upon their knowledge and training to help students achieve success. A "highly qualified" teacher is certainly a good starting point, but most of us would want our child to have a highly effective teacher whose teaching effort yields high rates of student learning.

Promoting Teacher Effectiveness

How do we support and cultivate effective teachers for all our schools and all our children? It is our belief that teachers want and need feedback, not only on the *act* of teaching, but also on the *results* of teaching. Timely, informative feedback is vital to any improvement effort. For instance, consider the role of a track coach, fitness trainer, or weight counselor. These individuals provide guidance on *how* to perform better, but the evidence of their effectiveness as professionals manifests in tangible *results*: improved running time, weight lifted, or pounds lost. It is evident that "people work more effectively, efficiently, and persistently... while gauging their efforts against results."

Teacher evaluation systems are often intended to serve the purpose of providing feedback and guidance for improving professional practice. In fact, most authors identify the fundamental purposes of teacher evaluation as improving performance and documenting accountability. The *performance improvement function* relates to the personal growth dimension and involves helping teachers learn about, reflect on, and improve their practice. The improvement function generally is considered *formative* in nature and suggests the need for continuous professional growth and development. The *accountability function*, on the other hand, reflects a commitment to the important professional goals of competence and quality performance. Accountability is typically viewed as *summative* and relates to judging the effectiveness of educational services.

Teacher evaluation traditionally has been based on the *act* of teaching and documented almost exclusively through the use of classroom observations. In a study conducted by the Educational Research Service, 99.8 percent of U.S. public school administrators used direct classroom observation as the primary data collection technique. However, primary reliance on formal observations in evaluation poses significant problems (*e.g.*, artificiality, small sample of performance) for teacher evaluation. Even under the best of circumstances, when principals might visit a classroom three or four times in a given year, the observation

- Can be artificial by nature,
- Suggests an inspection approach to supervision,

- Has limited validity based on the skill of the observer,
- Is narrow in scope (*i.e.*, restricted to instructional skills only), and
- Involves a small sample of the teacher's actual work with students (*e.g.*, four hours of observation would equal less than 1/2 of 1 percent of a teacher's time teaching during a given year).

Despite these substantial drawbacks to the traditional evaluation process, the truly fundamental flaw in such an approach is the assumption that the presence of good practice during the observation equates to the academic success of students. If student learning is our ultimate goal, then it should be measured directly and not extrapolated from limited observations of classroom instruction. A more balanced approach to teacher evaluation would involve an assessment of the *act* of teaching as well as the *results* of teaching. We don't suggest throwing out the use of classroom observation to foster teacher improvement; rather we advocate that teacher effectiveness be judged and demonstrated by both classroom instruction *and* the learning gains of students.

Assessing Teacher Effectiveness

Most educators would agree that they are responsible for student learning, but the profession as a whole has avoided evaluations based on measures of student learning, sometimes with good reason, given the unfair approaches that have been proposed. The solution, however, is not to continue with traditional strategies simply because they are benign and comfortable, but rather to develop fair and reasonable means of assessing teacher success with students. A number of school systems and educational programmes, to be discussed later in this book, have explored innovative ways of capturing valid and reliable data on student learning to inform the teacher evaluation process. Developing fair approaches for the assessment of teacher effectiveness requires an unflinching look at both the legitimate concerns that have driven the avoidance of a results orientation in the past, and the promising possibilities that make it more attractive in today's climate of greater accountability for student learning outcomes.

Concerns

Concerns about assessing teacher quality based on measures of student learning have clustered around what Schalock terms the collective and conditional nature of accountability, as well as the strategies for measuring student learning. Often, accountability efforts in schools are reduced to simplistic mandates for students to reach specified achievement goals at certain points in time. While gratifying as a bottom line, these expectations ignore the complex interdependencies of the learning enterprise. We must consider questions such as, "accountability by whom, with what resources, and as measured by what?"

Collective nature of accountability: Responsibility by all stakeholders. Accountability should be thought of as a collective responsibility for supporting learning by parents,

principals, superintendents, school board members, *and* teachers, to say nothing of the students themselves. Holding teachers accountable for student achievement without recognition of the roles played by these other partners in the educational process is patently unfair and can amount to scapegoating. Likewise, requiring students to attend summer school, or retaining them due to limited progress, avoids the collective nature of accountability if school systems have not provided the quality of instruction necessary for students to meet grade level expectations.

Ultimately, learning is a phenomenon that occurs as a result of the interactions between a teacher and student. Teachers cannot be solely responsible for student learning because it is an internally controlled activity. However, teachers are expected to optimize the conditions for learning. It is what they were hired to do and it is their professional obligation. As Schalock notes, “educator accountability for student progress in learning goes hand-in-hand with the social contract that assigns responsibility for education to schools.”

Conditional nature of accountability: Resources and student needs. Just as many actors affect the educational process, many variables affect the learning process within a classroom and are beyond the control of the individual teacher. These external variables include the level of support provided by the community and state, the availability of books for every child, the number of computers, sufficient instructional supplies, the support of curriculum specialists, and so forth. Within the classroom, the number and type of students can have dramatic effects on the level of academic achievement experienced by the class. Class size does make a difference, especially when a teacher is expected to work with a large number of at-risk students, whether they are disabled, limited in their English, or poor.

Measurement of student learning. One additional concern about the use of student learning assessments in the teacher evaluation process is the way in which learning is assessed. The traditional use of grades or standardized achievement scores is certainly suspect for a variety of reasons, including the

- Accuracy of grading procedures,
- Alignment of achievement tests with the curriculum,
- Diagnostic value of either approach for instructional improvement, and
- Single-point-in-time nature of these indicators.

In the absence of meaningful pre-test data, grades or achievement test scores at the end of the year are hardly valid measures of a teacher’s influence during a given year; indeed, they reflect the cumulative effects of what students have learned at home and school over preceding years. A much more accurate measure of what a student has learned would be reflected by an assessment that is curriculum-aligned and administered both at the beginning and end of the year. When such learning gains are averaged over a whole class of students, we have a general indication of the magnitude of learning that took place with that group of children.

As has become evident, the interplay of factors affecting student learning is multifaceted and quite challenging. It is also difficult to reach consensus on how best to measure student learning. Given these complexities, many educators have avoided being too explicit or public about tracking student learning for the purpose of improving instruction or evaluating performance. However, the current context of high-stakes accountability for students and schools found in most states, and which is being developed as a result of No Child Left Behind, provides an impetus and urgency for examining ways to assess teacher quality that are fair and realistic. Today, superintendents, principals, teachers, and students are being held accountable for higher levels of student achievement. Teachers are being pressured to produce results, yet often lack the necessary information and support to make data-driven instructional decisions. The use of approaches such as those suggested in this book can offer feedback on how to improve instruction in a balanced and meaningful manner.

Possibilities

Two primary purposes of teacher evaluation, as noted earlier, are professional growth and accountability. The use of data on student learning in the teacher evaluation process offers a potential tool for both improvement and for refocusing teacher evaluation on the accomplishments of teachers versus stylistic issues or their political standing. Too often, personal opinions or biases contaminate the evaluation process and undermine the credibility and trust necessary for meaningful dialogue about instruction. Reliable and valid information on student learning helps to align the evaluation process with the fundamental concerns of schooling. There are numerous advantages to this approach.

More objective measure of teacher effectiveness. The importance of objective data in the evaluation process becomes more striking in a story from one principal in Dallas. As she entered the new school to which she was assigned, the outgoing principal informed her of two problematic teachers for whom she would need to begin laying the groundwork to dismiss.

One teacher tended to be scattered in her approach to tasks and had a somewhat disorganized room. Her students were often talking and moving around the room at will as they worked. The other teacher was brusque with her students, rigid with her class rules, and worked the students hard. They were polar opposites in terms of style, but at the end of the year, when the new principal received the test data on the teachers in her building, she found that both of these teachers were top performers in terms of gains in student achievement. She decided that she could tolerate individual personality differences if children were being well served by these teachers. This story offers a compelling message: an evaluation approach that examines both the *act* of teaching and the *results* of teaching provides a more balanced and realistic appraisal of teacher effectiveness.

Meaningful feedback for instructional improvement. Objective feedback in the form of assessment data also offers an invaluable tool for supervision. As Barbara Howard and Wendy McColskey note, “evaluation that leads to professional growth requires teachers to look honestly at their weaknesses and strengths.” Self-assessment can be limited because of a lack of objectivity. Feedback from colleagues or supervisors based on a few classroom visits is equally limited because of the narrow sampling of behaviour it provides. Assessment data of student learning over a marking period or even half a year can provide substantive feedback on students’ cumulative mastery of material. It provides a broader and richer sampling of the teacher’s impact on students and permits the identification of specific patterns in the learning of content and skills that were taught.

The evidence from schools that have been successful in increasing the achievement level of students, particularly those serving high-poverty and high-minority populations, has been that better use of data is a key ingredient in their success. Data analysis has been used as a means of monitoring success and ensuring accountability for the identified goals of schools and school systems. In a recent study of 32 schools in the San Francisco Bay area, the frequency with which teachers collected, interpreted, and analyzed data for instructional improvement was found to differ among schools that were closing the achievement gap versus those that were not. “Two-thirds of the teachers surveyed in the gap-closing schools said they used test and other data at least several times a month to understand their students’ skills gaps, and sometimes several times a week.” Instructional responsiveness to student assessments is a powerful tool for increased student achievement.

Barometer of success and motivational tool. In addition to providing meaningful feedback for instructional improvement, student achievement data can provide encouragement and a sense of gratification.

As Schmoker observes, “Data and results can be a powerful force for generating an intrinsic desire to improve.” Credible data on the results of teaching efforts inform instructors on what to change to improve their performance and gauge their success in doing so. Without concrete feedback on the results of their work, teachers can hardly hope to improve them. “Data make the invisible visible, revealing strengths and weaknesses that are easily concealed. Data promote certainty and precision, which increases teachers’ confidence in their abilities.”

Assessment is an integral facet of instruction. More than 30 years ago, Lortie noted that “the monitoring of effective instruction is the heart of effective instruction.” Truly effective teachers monitor student learning on an ongoing basis and use the information to improve their teaching.

How do we encourage all teachers to embrace this practice to benefit their teaching and the learning of their students? The purpose of *Linking Teacher Evaluation and Student Learning* is to present methodologies that have attempted to balance the

competing demands of fairness, diagnostic value for professional growth, and accountability for student learning. The details of implementation are daunting; each methodology reflects years of careful consideration of the myriad issues that influence student learning and its assessment. All the models presented in this book have both advantages and disadvantages, but they have a proven track record for connecting teacher evaluation to student learning.

Role of Indian Society

One characteristic that is mainly often noticed in relation to the India is its unity in diversity. This overworked cliché has become a part of India's self-identity. India is a country of sub-continental proportions. From north to south, east to west, people from diverse backgrounds have mixed and cultures have intermingled in excess of centuries. Nevertheless, there has been an underlying stability in identity. There are very few countries which have such an enormous cultural diversity that India have to offer.

Beneath the bewildering diversity of religion, language and customs of this vast country, the underlying unity is extra ordinary. The thought of unity is traced back through scholars to ancient times. The underlying cultural unity was strengthened further with the administrative unity brought in relation to the throughout the British rule and with the construction of India as a contemporary self-governing nation after the independence. The enduring nature of Indian unity has always been fascinating.

Indian unity is the product of sure historical factors that are present in several meadows of Indian social life. It appears as if the inhabitants from the Himalayas in the north to Kanyakumari in the south and Kutch in the west to Arunachal in the east are woven jointly into a beautiful tapestry. In the procedure of its development, Indian society has acquired a culture characterised through stable patterns of pluralism. Though, the acceptance of cultural pluralism does not detract us from the thought of promoting economic, political and social integration.

European Sociology conceptualises unity in a society in conditions of linguistic nationality or in conditions of political sovereignty. Therefore, the primary foundation of unity belongs to a nation. Just as to several Indian sociologists, though, unity in India

and the whole of South Asia, in information, has been civilization, going back to ancient times and continuing to the present day. Therefore, at the civilization stage there is unity in South Asia, but this South Asian civilization is divided today into several nations like India, Pakistan, Bangladesh, Sri Lanka and Nepal. In conditions of social institutions like the family, caste and lifestyles there is a fundamental unity in the dissimilar societies and nations of South Asia.

A significant source of unity in traditional India was rooted in the procedures of cultural communication and interaction. Sociologists have recognised the role of traders, storytellers, crafts-people and artists, for instance potters, musicians, dancers in traditional India, in structure general cultural traditions. The institutions of pilgrimage, fairs and festivals give yet another link for cultural unity. In addition to these agencies, the social structure and economy forged linkages of reciprocity and interaction flanked by regions, groups and cultural traditions. Accommodation without assimilation has been the feature of Indian civilization. Accommodation is a social procedure through which dissimilar elements of a society are integrated without losing their separate identity.

Assimilation on the other hand is a kind of integration where the earlier identity of the elements is dissolved. In Indian history and culture, additions of new components have not meant the discarding of old ones. The accommodation of diversity has been the underlying ideology and numerous social and cultural factors have contributed to the enduring nature of Indian unity. Order and stability was maintained not primarily through the state but through social, cultural, moral and technological values and institutions of Indian civilization. Indian civilization gave enough freedom for the practice of any method of life although dissimilar customary methods were ranked in a hierarchy. This created inequality and integration as well.

SOCIETY AND GENDER ROLES

Psychologists such as Sandra Bem, one cognitive process that seems nearly inevitable in humans is to divide people into groups. We can partition these groups on the basis of race, age, religion, and so forth. However, most of the times we split humanity on the basis of gender. The first thing we instantly determine, when meeting someone new, is their gender. This process of categorizing others in terms of gender is both habitual and automatic. It's nearly impossible to suppress the tendency to split the world in half, using gender as the great divider. When we divide the world into two groups, males and females, we tend to consider all males similar, all females similar, and the two categories of "males" and "females" very different from each other. In real life, the characteristics of women and men tend to overlap. Unfortunately, however, gender polarization often creates an artificial gap between women and men and gender roles that are very difficult to change in time.

GENDER STEREOTYPES FOR MALES AND FEMALES

Stereotypes are representative of a society's collective knowledge of customs, myths, ideas, religions, and sciences. It is within this knowledge that an individual develops a stereotype or a belief about a certain group. Social psychologists feel that the stereotype is one part of an individual's social knowledge. As a result of their knowledge, or lack of knowledge, the stereotype has an effect on their social behaviour. Stereotypic behaviour can be linked to the way that the stereotype is learned, transmitted, and changed and this is part of the socialization process as well.

The culture of an individual influences stereotypes through information that is received from indirect sources such as parents, peers, teachers, political and religious leaders, and the mass media. In order to understand stereotyping, an individual must first be made knowledgeable about the definition of a stereotype. Stereotyping is how we perceive each other, especially individuals outside our group. What we believe to be "normal" is associated with who we are hanging out with.

Which are usually our friends and social networks. Gender stereotypes are related to cognitive processes because we have different expectations for female and male behaviour and the traditional gender roles help to sustain gender stereotypes, such as that males are supposed to be adventurous, assertive aggressive, independent and task-oriented, whereas females are seen as more sensitive, gentle, dependent, emotional and people-oriented. Here as suggested, deal with the opposite male dominance and feeling superior to women. Of course, not all men have power and arrogantly dominate women; indeed, according to Miller, many men are dominated by "the system" and considered disposable. Also, women are given certain advantages and "protected" in many ways that men do not enjoy. Clearly, each sex has and utilizes power in certain ways and we are getting more equal, but, clearly, the sexes aren't equal yet. The most recent suggestion to solve this problem is to completely disassociate gender from all personality traits. Within the two career families of today, the women-are-inferior attitude is muted and concealed, but the archaic sex role expectations are still subtly there.

The old rules still serve to "put down women and keep them in their place." By nature, men and women have some biological differences, but it is life experience that reinforces or contradicts those differences. The truth lies in differential socialization, which claims that males and females are taught different appropriate behaviours for their gender.

GENDER SOCIALISATION

Socialisation is the process, through which the child becomes an individual respecting his or her environment laws, norms and customs. Gender socialisation is a more focused form of socialisation, it is how children of different sexes are socialised into their gender roles and taught what it means to be male or female. Gender socialisation begins the moment we are born, from the simple question "is it a boy or a girl?".

We learn our gender roles by agencies of socialisation, which are the “teachers” of society. The main agencies in Western society are the family, peer groups, schools and the media. In respect with gender socialisation, each of the agencies could reinforce the gender stereotypes. Gender differences result from the socialization process, especially during our childhood and adolescence.

The classical example of gender socialisation is the experiment done with babies that were introduced as males to half of the study subjects and as females to the other half. The results are interesting and quite disturbing at the same time. The participants behave differently according to the sex they had been told. These findings show that other people contribute a lot to how we see ourselves only on the basis of gender.

THE FAMILY AS GENDERED RELATIONSHIPS: INFLUENCES ON GENDER SOCIALIZATION PROCESS

It is said before that parents are the primary influence on gender role development in the early years of one’s life. With regard to gender difference, the family in fact, unlike other groups, is characterized by a specific way of living and constructing gender differences through a process that is surely biological, but also relational and social. The family is “the social and symbolic place in which difference, in particular sexual difference, is believed to be fundamental and at the same time constructed”. In particular, in the family the gender characterization reflects the individualities of the parents. The family is therefore a “gender relation”.

In the family, the relation with the father and the mother assumes therefore one fundamental importance in the definition of the gender belonging, because it’s the first experience of relation with males and females. Gender identities and the expectations towardss male and female roles are socialized within the parents-children relationship; such expectations are today various and new compared with the past. The models from which fathers and mothers take inspiration need to be verified because “the crisis of the paternal authority has given more space to the father in shaping the educational relation with the child.

They think that the important thing is to converse and to build convincing representations of the world”. The gender socialization inside the familiar relations evidences therefore also the temporal dimension of the transmission of styles and expectations between parents and children. The parents’ generation, in comparison with the child’s one, can highlight marked differences too. Parents today probably have different expectations from those their parents had, and their children have even more different expectations. We must go deeper into the matter on how transmission of gender differences happens today and how the gender belonging is constructed..

If such differences seem to diminish on the one side, on the other instead they move on different areas in comparison with the past. Between children in fact the sexual difference “produces various models of belongings and continuity”, and they are today

completely different from those of the previous generations. In the past, families had different educational demands for their sons and daughters after puberty, they then tended to differentiate them in the sense to promote the autonomy of the males and the dependency of the females.

It was implicit that the boy should realise himself, even if against familiar ties, while the girl had, in some ways, to accept and to conserve them. This difference has always favoured the fact that young women lived their desire of autonomy with a sense of guilt and of independency with intolerance.

A child's parents are the first socialization agents he or she will come into contact with. Parents teach stereotypes through different ways and behaviour: "the way they dress their children, the way they decorate their children's rooms, the toys they give their children to play with, their own attitudes and behaviour".

A RELATIONAL APPROACH TO GENDER ROLES SOCIALIZATION

The most important aspect of the sociological reflection is the ability to use the concepts elaborated in the theoretical debate at an empirical level, realizing "a hermeneutic" connection between the interpretative framework and social life. Gender socialization can be read like a "relational process". It is unavoidable that in the transformation a simplification is put into effect, a reduction of the complexity of the terms in game, because you need to lead back to the factors that explain a social phenomenon to one more rigid pattern of reality: in order not to fall into the trap of the merely casual interpretation it is necessary to always place, to the centre of attention, the relation between different factors that concur to see the phenomena from more points of view, in a multidimensional perspective. The relational model is assumed like the point of observation to verify the hypotheses in order to characterize those that are the gender socializing outcomes in the contemporary society. Within a risky society the relational model considers every phenomenon as the outcome of a process in which the challenges and the resources are put implicitly or explicitly in comparison.

The risk therefore is given from the relation of adequacy/inadequacy between challenges and resources. That appears clear if it is believed that every choice is linked to multidimensional situations, which are relational contexts, in which the phenomena are networks of phenomena and every node represents interlaces of challenges, ties and resources. Speaking about challenges and resources in gender socialization simplifies reality and circumscribes a point of view from which to observe a phenomenon, but it always takes into account that is a relational phenomenon, in which more dimensions are intersected. Consequently the gender socialization process is divided into two orders of factors, one leads the challenges and the other the resources, in the hypothesis that behind every phenomenon there are however the intentions of the actors who arrange in a more or less balanced way, with reference to the context of options that delimits the action, objects to reach and strategies of participation.

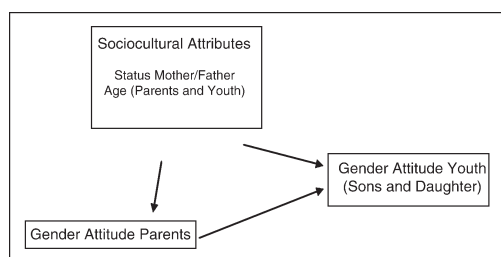


Fig. Analytical Framework for the Relationships between Soci-Demographic Attributes, Gender Attitude Parents and Gender Attitudes Youth.

FEMINIST EFFECTS ON SOCIETY

The feminist movement has effected change in Western society, including women's suffrage; greater access to education; more equitable pay with men; the right to initiate divorce proceedings; the right of women to make individual decisions regarding pregnancy (including access to contraceptives and abortion); and the right to own property. Harvard Psychology Professor Steven Pinker argues that feminism has reduced domestic violence, especially against men as their likelihood of being killed by a female intimate partner has decreased six-fold.

WOMEN'S RIGHTS

From the 1960s on, the women's liberation movement campaigned for women's rights, including the same pay as men, equal rights in law, and the freedom to plan their families. Their efforts were met with mixed results. Issues commonly associated with notions of women's rights include, though are not limited to, the right to bodily integrity and autonomy; to vote (universal suffrage); to hold public office; to work; to fair wages or equal pay; to own property; to education; to serve in the military; to enter into legal contracts; and to have marital, parental, and religious rights. Feminists have worked to protect women and girls from child sexual abuse, challenging the prior belief that girls caused men to have sex with them even when the girls were very young.

In the U.K., a public groundswell of opinion in favour of legal equality gained pace, including during the modern movement, partly through the extensive employment of women in men's traditional roles during both world wars. By the 1960s, the legislative process was being readied, tracing through MP Willie Hamilton's select committee report, his Equal Pay for Equal Work Bill, the creation of a Sex Discrimination Board, Lady Sear's draft sex anti-discrimination bill, and a government Green Paper of 1973, until 1975 when the first British Sex Discrimination Act, an Equal Pay Act, and an Equal Opportunities Commission came into force. With encouragement from the U.K. government, the other countries of the EEC soon followed suit with an agreement to ensure that discriminatory laws would be phased out across the European Community.

In the U.S.A., the National Organization for Women (NOW) was created in 1966 with the purpose of bringing about equality for all women. NOW was one important group that fought for the Equal Rights Amendment (ERA). This amendment stated that “equality of rights under the law shall not be denied or abridged by the United States or any state on account of sex.” But there was disagreement on how the proposed amendment would be understood. Supporters believed it would guarantee women equal treatment. But critics feared it might deny women the right be financially supported by their husbands. It died in 1982 because not enough states had ratified it. ERAs have been introduced in subsequent Congresses, but have failed to be passed. Nonetheless, various laws advancing women’s rights were promulgated, although many issues remained to be resolved.

In the final three decades of the 20th century, Western women knew a new freedom through birth control, which enabled women to plan their adult lives, often making way for both career and family. The movement had been started in the 1910s by U.S. pioneering social reformer Margaret Sanger and in the U.K. and internationally by Marie Stopes.

Publication of data and advocacy relevant to rights increased in recent decades. For example, the *United Nations Human Development Report 2004* estimated that, when both paid employment and unpaid household tasks are accounted for, on average women work more than men. In rural areas of selected developing countries women performed an average of 20% more work than men, or an additional 102 minutes per day. In the OECD countries surveyed, on average women performed 5% more work than men, or 20 minutes per day. At the UN’s Pan Pacific Southeast Asia Women’s Association 21st International Conference in 2001, it was stated that “in the world as a whole, women comprise 51% of the population, do 66% of the work, receive 10% of the income and own less than one percent of the property”.

INTERNATIONAL LAW

CEDAW

The *Convention on the Elimination of All Forms of Discrimination Against Women* (CEDAW) is an international convention adopted by the United Nations General Assembly. Described as an international bill of rights for women, it came into force on 3 September 1981. Several countries have ratified the Convention subject to certain declarations, reservations, and objections. Iran, Sudan, Somalia, Qatar, Nauru, Palau, Tonga, and the United States have not ratified CEDAW. Expecting a U.S. Senate vote, NOW has encouraged President Obama to remove U.S. reservations and objections added in 2002 before the vote. The CEDAW does not protect men from any form of discrimination.

Reproductive Rights

In the U.S., a major focus of political activism has centred on reproductive rights, including for (and, among opponents, against) the decision by the U.S. Supreme Court in the case of *Roe v. Wade* enunciating a Constitutional right for a woman to choose whether to carry a pregnancy to term.

Gender-neutral Language

Gender-neutral English is a description of language usages which do not recognize gender and are aimed at minimizing assumptions regarding the biological sex of human referents. The advocacy of gender-neutral language reflects, at least, two different agendas: one to clarify the *inclusion* of both sexes or genders (gender-inclusive language); the other that gender, as a category, is rarely worth marking in language (gender-neutral language). Gender-neutral language is sometimes described as *non-sexist language* by advocates and *politically correct language* by opponents.

Heterosexual Family Relationships

The increased entry of women into the workplace beginning in the twentieth century has affected gender roles and the division of labour within households. Sociologist Arlie Russell Hochschild in *The Second Shift* and *The Time Bind* presented evidence that, in two-career couples, men and women, on average, spend about equal amounts of time working, but women still spend more time on housework. Feminist writer Cathy Young responds to Hochschild's assertions by arguing that, in some cases, women may prevent the equal participation of men in housework and parenting.

Feminist criticisms of men's contributions to child care and domestic labour in the Western middle class are typically centred around the idea that it is unfair for women to be expected to perform more than half of a household's domestic work and child care when both members of the relationship also work outside the home. Several studies provide statistical evidence that the financial income of married men does not affect their rate of attending to household duties. In *Dubious Conceptions*, Kristin Luker discusses the effect of feminism on teenage women's choices to bear children, both in and out of wedlock. She says that, as childbearing out of wedlock has become more socially acceptable, young women, especially poor young women, while not bearing children at a higher rate than in the 1950s, now see less of a reason to get married before having children. Her explanation for this is that the economic prospects for poor men are slim, hence poor women have a low chance of finding a husband who will be able to provide reliable financial support.

Although research suggests that, to an extent, both women and men perceive feminism to be in conflict with romance, studies of undergraduates and older adults have shown that feminism has positive impacts on relationship health for women and sexual satisfaction for men, and found no support for negative stereotypes of feminists.

Feminist Religion and Theology

Feminist theology reconsiders the traditions, practices, texts, and theologies of religions from a feminist perspective. Its goals include increasing the role of women among the clergy and religious authorities, reinterpreting male-dominated imagery and language about the deity or deities, determining women's place in relation to career and motherhood, and studying images of women in the religion's sacred texts. Difference feminism offers compatibility with gender-differentiating teachings of many major theologies, although difference feminism, when essentialist, is itself controversial.

Christian feminism is a branch of feminist theology which seeks to interpret and understand Christianity in light of the equality of women and men. Because this equality has been historically ignored, Christian feminists believe their contributions are necessary for a complete understanding of Christianity. While there is no standard set of beliefs among Christian feminists, most agree that God does not discriminate on the basis of biologically determined characteristics such as sex. Their major issues are the ordination of women, male dominance in Christian marriage, and claims of moral deficiency and inferiority of abilities of women compared to men. They also are concerned with the balance of parenting between mothers and fathers and the overall treatment of women in the church. New feminism is a branch of difference feminism situated within Catholicism.

Islamic feminism is concerned with the role of women in Islam and aims for the full equality of all Muslims, regardless of gender, in public and private life. Islamic feminists advocate women's rights, gender equality, and social justice grounded in an Islamic framework. Although rooted in Islam, the movement's pioneers have also utilized secular and Western feminist discourses and recognize the role of Islamic feminism as part of an integrated global feminist movement. Advocates of the movement seek to highlight the deeply rooted teachings of equality in the Quran and encourage a questioning of the patriarchal interpretation of Islamic teaching through the Quran, *hadith* (sayings of Muhammad), and *sharia* (law) towards the creation of a more equal and just society.

Jewish feminism is a movement that seeks to improve the religious, legal, and social status of women within Judaism and to open up new opportunities for religious experience and leadership for Jewish women. Feminist movements, with varying approaches and successes, have opened up within all major branches of Judaism. In its modern form, the movement can be traced to the early 1970s in the United States. According to Judith Plaskow, who has focused on feminism in Reform Judaism, the main issues for early Jewish feminists in these movements were the exclusion from the all-male prayer group or *minyan*, the exemption from positive time-bound *mitzvot*, and women's inability to function as witnesses and to initiate divorce.

The Dianic Wicca or Wiccan feminism is a female-focused and Goddess-centred Wiccan faith that is also known as a feminist religion that teaches witchcraft as every woman's right. It is also one faith of the many practiced in Wicca.

Atheist feminism advocates the equality of men and women within atheism. Atheist feminists also oppose religion, being a major source of female oppression and inequality, believing that all religions are sexist and oppressive to women.

Feminist Theology

Feminist theology, sometimes referred to as the Goddess movement, is a movement found in several religions to reconsider the traditions, practices, scriptures, and theologies of those religions from a feminist perspective. Some of the goals of feminist theology include increasing the role of women among the clergy and religious authorities, reinterpreting male-dominated imagery and language about the deity or deities, determining women's place in relation to career and motherhood, and studying images of women in the religion's sacred texts. In Wicca, "the Goddess" is a deity of prime importance, along with her consort the Horned God. In the earliest Wiccan publications, she is described as a tribal goddess of the witch community, neither omnipotent nor universal, and it was recognised that there was a greater "Prime Mover", although the witches did not concern themselves much with this being.

MIDDLE AGES WOMEN IN ISLAMIC SOCIETY

The rise and expansion of Islam, after the foundation by Mohammed, is an amazing story. Ultimately, the Muslims, as the believers of Islam are called, started a world-wide faith that today makes up the world's second largest religious group after Christians. The role and status of women within Islam is one of essential contribution to its origins and continuation, but ever-changing roles.

Mohammed could not have founded Islam alone. Marriage to his employer, the wealthy widow Khadija, allowed him the leisure and financial backing to pursue his new religious journey. Born in Mecca, Mohammed was raised by his grandfather and uncle when his parents died. At the age of twenty-five he married Khadija. Mecca was an important trading centre for caravan trade between Medina, the Near East and India. An integral part of this caravan and commercial trade was raiding, and the Arabs were excellent warriors. These characteristics facilitated Islam to spread via the Muslim's holy war or Jihad within a few decades beyond the Arabian Peninsula, first to the Near East, and then Northern Africa and parts of southern Europe, including Spain. As Islam came in contact with other societies, both monotheistic and polytheistic, there was much cultural integration. Consequentially, there was room for confusion and conflict as to the expectations and responsibilities of women's lives. As in Christianity and the Bible, so too in Islam and the Koran (the Muslims' holy book), the interpretation of the written versus the exegesis explanations of succeeding centuries confounded the historical record and oral transmissions. Mohammed's marriage to Khadija took place when he was twenty-five and she was forty. As his employer, Khadija hired Mohammed

to oversee her caravan trade between Mecca and Syria. She then proposed to him, and she was his only wife until her death about twenty-five years later. After about fifteen years of marriage to Khadija Mohammed no longer needed to work regularly, and he was free to lead a life of contemplation. While meditating in surrounding caves, Mohammed was visited by the angel Gabriel, who informed him that he was to be the messenger of Allah, the Arabic word for God. This meeting with Gabriel occurred in the year 610, and thereafter Mohammed continued to experience similar visions with specific religious instructions from Allah. Convinced that he was now the prophet of Allah, Khadija became his first convert. In general, however, Mohammed's ideas met with general skepticism by the people of Mecca. This turned into active persecution after Mohammed denounced polytheistic worship, centred at the Kaaba, as idolatry, and he and Khadija were forced to flee Mecca for Medina. Welcomed in Medina, Mohammed established his tenets and practices for Islam. The five pillars of Islam were straight forward, facilitating people's conversion: 1. Announcing there is no God but Allah and Mohammed is his Prophet; 2. Prayer five times per day; 3. Fasting during the month of Ramadan; 4. Alms to the poor and ill; and 5. if one could afford it, pilgrimage to Mecca at least once in a person's lifetime. Mohammed did not claim his ideas to be a new religion, but a culmination of beliefs from Judaism and Christianity. Muslims always called Jews and Christians as people of the same book. In other words, the Koran, compiled by Mohammed's followers after his death, recorded the messages he had received from Allah, and stated that they were the final words of Allah.

After Khadija's death, the vision came from Allah that it was acceptable to have more than one wife. His next spouse was Aisha, a child bride of nine or ten. She was immediately succeeded by many more wives, usually widows or daughters, whose husbands and fathers had lost their lives in the course of spreading Islam. The Quranic (Koranic) verse detailing this is "Marry other women as may be agreeable to you, two or three or four." At the time of Mohammed's polygamy, disapproval of this was espoused by women from Medina, but it is thought that as Islam spread and encountered polygamy acceptance in other cultures, then multiple wives became an accepted practice for Muslims.

Veiling and seclusion are two other practices that were added after Khadija's death. Apparently, after Mohammed married Aisha and others, guests would stay too long in his wife or wives' rooms at a wedding feast or other occasions, leading to Mohammed's revelation from Allah that veiling for women was necessary. Veiling, a custom from ancient times, was prevalent in the higher classes in the Near East dating back to the ancient Assyrians or even Mesopotamians. The ancient Greeks and Romans had adopted this custom too. Then when Christianity was established and starting spreading, veiling was deemed mandatory for respectable women. Muslims say that Koranic verses justify veiling, but it only tells women to veil their bosoms and hide their ornaments, not language specific to the head. As in other cultures and religions, interpretations vary

throughout the Muslim world, and today there are different degrees of veiling. Mohammed died in his wife Aisha's room in Medina, where he is buried. This is now the second most sacred spot in Islam after the Kaaba in Mecca.

Over time as Islam won new followers, certain customs concerning women were established and written down in the Hadith, a further elucidation of the Koran. Mohammed's second wife, Aisha, is given much credit for many of these writings, even those not pertaining to women. Aisha is credited with 2,210 traditions in the Hadith, and as she lived about forty years beyond Mohammed's death, she was in a position to transmit events and words from the Prophet. Aisha was not alone in becoming one of the major sources for the Hadith. Mohammed's other wives were important too, and indicates the high esteem in which his wives were held.

Arab marriage practices before Islam were both polyandrous and polygamous. Matrilineal marriages were common. A wife remained with her tribe after marriage, and the husband would visit or reside with her. The children belonged to the mother's tribe. When polygamy was practiced, the husband still visited his wives, not *vice versa*. Before Islam, adultery by women was not seen as serious, and it was not a crime. In one case, a prominent woman of the same tribe as Mohammed's, was divorced by one of her husbands on a charge of adultery. While embarrassing for her, her life was not in danger. Her father referred the matter to a soothsayer, who proclaimed her innocent of any wrongdoing. This was the common custom at the time.

When Islam came into being, many of these positive practices for women in marriage were slowly eroded, but it was always stated that Islam was an improvement for women. Mohammed himself sought to improve the treatment of women, but he thought that women were to be obedient to men. Muslim women in general though were clearly perceived as subordinate to men. Several examples will elucidate this improvement versus subordination of women in Islam. With the establishment of Islam, polyandry was now prohibited. Outlawing of female infanticide was clearly spelled out in the Koran. Apparently before Islam, infanticide on baby girls was practiced even as late as five or six years of age. The inscription in the Koran states "When the buried infant shall be asked for what sin she was slain."

A significant bride-price was paid to the bride's family or the bride herself, part of the arranged marriage procedures. As Mohammed had daughters, it is thought this last part supports that. If a man could not afford the brideprice, then he was allowed to marry a slave to keep from engaging in wrongful acts. Husbands were made responsible for financial support of their wife or wives, but wives were not obligated to share their wealth with their husbands. In Islam, divorce was restricted generally to husbands only, although divorced was not taken lightly, and the Koran encouraged reconciliation. Upon the divorce, the husband had to return the full bridal gift and any other goods his wife brought to the marriage. Since it was expensive for men to marry, polygamy was often favoured over divorce to avoid repaying the bride-price. Divorce did not interfere with

remarriage for women. After Mohammed's death, one of his successors, Umar (reigned 634-44), many harsh customs regarding women were inaugurated in both public and private lives. Apparently Umar was physically abusive to his own wives. Stoning for adultery was begun according to the Shariah, (the Muslim law) although it is not specified in the Koran. In the Koran, which bans adultery, it orders that offenders be whipped one hundred lashes. To prove adultery or fornication, the Koran is most clear. Four eyewitnesses are necessary, although this is rarely followed. After Umar's death, some lessening of onerous restrictions on women were invoked, but ultimately the direction of a more restrictive life for women was occurring. Although in the Koran it stated that a husband in exchange for his financial support of his wife, was allowed to punish a disobedient wife by admonishing her, sleeping in a separate bed, or beating her. Keeping apart from his wife, is an indication that Islam accepted women's sexual needs, so different from the ideas of Christianity.

As Islam conquests brought in more wealth, slavery was increased as well. It is thought that the majority of Muslim slaves were women and children. Many women were placed in the harems of rulers, and the ever-increasing number of slaves and concubines will become legendary. As in ancient Greece, where the courtesans or hetaerae came from the slave class, so too in Islam. Some of the best education and training in music, singing, and literature were reserved for the female slaves, who would command the highest prices. Even philosophical discourse was open to these slaves. For Abassid times, Khayzuran was the most celebrated slave, who eventually became the beloved wife of the caliph al-Mahdi (775-785), Two of her sons became caliphs. It is thought that she was active in affairs of state for both her husband and sons. One of her sons was the illustrious Harun al-Rashid, a contemporary of Charlemagne's. When a female slave give birth to a child by her master, she then could secure her freedom during her master's lifetime or after his death. Supposedly Aisha's brother-in-law left one thousand slaves and one thousand concubines when he died in 656. Guarded by eunuchs, as Islam spread to the Persians and Indians, large harems were popular. Jealousy and intrigue in harems became commonplace, leading to the idea of harem politics, whereby women resorted to poison and other intrigues to ensure their own son's political birthright. Consequently, throughout history harem politics as played out by women have been used against women to prove their instability and ineptitude for any positive change to their lives.

While women could neither inherit nor claim a share of the spoils won on the battlefield, they could own and inherit property. Inheritance laws in the Koran stipulate that female children receive half that of male children.

During the Abbasid Caliphate 750-1250, public appearances and clothing of women were increasingly regulated by government decrees. Part of this change was signaled when, Caliph Mansour 754-775, ordered a separate bridge built for women over the Euphrates River in his new capital of Baghdad. Women were to be kept separate from

men. Whatever were the original tenets of Islam, as it spread to other societies, local cultural conventions were interwoven with Muslim beliefs and there alterations became accepted as orthodox. The development of the Shariah acted as a guide to proper lifestyles for Muslims in these new surroundings. While women in Mecca and Medina had not been veiled, seclusion of women in the Middle East was linked to the development of urban life, not the nomadic culture of Arabia. As economic growth made it possible for domestic female slaves to be hired, this made it easier for middle and upperclass women to become increasingly confined to their homes. As in ancient Greece and Rome, only domestic servants went to market. Respectable women stayed at home or went out in secluded litters, fully veiled. Gradually veiling spread to the countryside, but it is not known when this occurred.

CULTURAL AND INTELLECTUAL PURSUITS OF MUSLIM WOMEN

Muslim women participated in a wide range of cultural and intellectual pursuits. During the Umayyad Caliphate (661-750), women provided salons where scholars, poets and other educated people could gather, centuries before the famous female salonnières of Enlightenment times in France. Prominent among such women was the great granddaughter of Mohammed, Sukaina. As in pre-Islamic society, women continued to write poetry, and professionally sing elegies at funerals, which they probably composed. Women were not only students at schools, they taught as well as the men. Law, theology, and medicine were studied by women. When Islam came to Spain, women were professors at the universities of Cordoba and Valencia. The political role of women in medieval Islam varied substantially. Some played active part in politics Mohammed's daughter Fatimah, was the wife of the Caliph who succeeded when Mohammed died. Subsequent successors to Mohammed through his daughter are followers of Islam called the Shites or Shias. The followers of the other branch of Islam which allows the most promising person to be a successor of Mohammed's, are called Sunnis. The Sunnis make up the majority of the Muslims today. Wife get name of Ummayad Caliph Al Walid I (705-715) encouraged her husband to build large mosques, and acted as political advisor. One of the greatest of the Abbasid Caliphs, Harum Al-Rashid (786-809), apparently owed his position to his wealthy mother, Khayzuran. She was a former slave, who intrigued to gain the throne for Harum at the expense of her other son. It is thought that Khayzuran was the model for Scherezad in the Arabain Nights Tales. Harum's wife, Zubaydah, was a prominent patron for the urban communities of Mecca and Baghdad (check this out for right town She sponsored hospitals, schools, aqueducts, and she was a major role model for other aristocratic women's civic contributions.

Role of Women in Islam

Women's role in the religion of Islam is still being researched, although during Mohammed's lifetime, communal religious activities were undertaken by women. At

first Meccans were not receptive to Mohammed and the Muslim faith. Many battles were fought to keep Islam out and later to bring Islam in. Women were on the battlefield as well, both for and against Islam. This was a continuation of the pre-Islamic practice of women accompanying their menfolk on military campaigns. Compared to Christianity, where the first martyr was a man, St. Stephen, in Islam it was a woman, a slave named Sumayyah bint Khubbat, who was killed for her faith. Hind bint Utbah, a prominent woman in the region, became known as the “liver-eater,” when she avenged her father’s death. At the battle of Uhud in 625, when the Prophet’s uncle Hamz was killed, Hind ripped open his chest, pulled out his liver, and bit into it. Finally, three years later she converted to the Muslim faith. As the Koran accepts the tenets of the Old Testament, Eve was their ancestor too. However, her eating of the forbidden fruit was not an original sin, but the acquisition of knowledge. Mohammed’s granddaughter, Zaynab displayed enormous courage on the battlefield in the 7th century. Revolutionary women’s groups in Iran use Zaynab as their role model.

Women transmitted hadiths, which were statements relating to the Prophet’s actions and sayings. Difficult passages of the Koran were elucidated in the hadiths. Theology and law utilized these hadiths continually. Women’s role in this transmission is so different than in the early centuries of Christianity, where men dominated. Women as well as men collected and memorized the Koran, still an important part of the religious belief for Muslims today. Women attended the mosques and participated in religious festivals. Umar, political and religious head of Islam, ruling in the first half of the 7th century, sought to prevent women from attending prayers at mosques, but while he was not able to accomplish this extremely restrictive measure, he was able to institute segregated prayers, which have remained the standard today. We do know that as in other religions, a mystic element of Islam arose in the 8th century, entitled Sufism. Women were prominent in these observances as they were in Christian mysticism. Sufi Muslim saints emerged as in Christianity, and one of the foremost sufi saints was a slave woman, Rabiah. Veneration of sufi saints was not limited to ones that had died. Pilgrimages to the tombs of these saints were popular and still is among women and men. Sufis sought to seek union with Allah, by leading a simple life of devotion. As in Christianity, sufism developed into particular schools that descended from individual sufi masters. While most sufis were celibate, many married. Both convents and monasteries were established to support their spiritual life. When a woman converted to sufism, this allowed her freedom from domestic and social obligations. Many women rose to prominence as spiritual guides.

During Mohammed’s lifetime and in the decades after his death, women’s status was much higher than it was later. As the Muslims conquered and converted cultures of different areas, profound changes occurred for women. In the crusading mentality of a new faith, Islam like Christianity, practices were allowed that did not always keep faith with the Koran and the traditions and commentaries regarding Mohammed’s beliefs

and sanctions. When the Abbasids came to power in 750, this date is usually seen as the pivotal point of change for women. As its capital was located in ancient Mesopotamia or modern-day Iraq, local guidelines were used and sanctioned through the Shariah or traditions. Over time the Shariah came to have almost the same religious regulations as the Koran, and it was thought to be inspired by Allah too. It is therefore necessary to explore what these middle eastern cultures were like to see why women's status was becoming more restrictive.

RELIGION IN TRIBAL SOCIETIES

CHIEF FEATURE OF TRIBAL SOCIETIES: EASY FORM OF RELIGION

Every religious system consists of a set of three essential elements:

- Belief in the subsistence of a superhuman world;
- Human's connection to it; and
- Practice of ritual as an instrument of establishing connection.

Here:

- Refers to the belief system,
- To the value system, and
- To the action system.

The form and meaning of these elements, *i.e.*, religious belief, religious value and religious action, may differ from one order to another. Religion of a tribe is easy insofar as it is expressed in everyday language and experienced in everyday life. It is descriptive, demonstrative and readily discernible. In the middle of the tribe's religious myth, belief, religious value and religious action are not treated as something separately from other types of belief and behaviour, as followed in social, economic and political contexts. Yet, the meaning of beliefs and behaviour of the tribal's appears mysterious to the outsiders. This is precisely because theirs is a religion without explanation. Though, tribal religion is no less complete than the highly urbanised form of intricate religion to the extent that its implicit philosophy recognises the similar universal truth.

Let us now explain to you all these characteristics of tribal religion, its implicit philosophy and the worldview. This we will do through discussing tribal lifestyle, tribal ritual intricate and tribal worldview.

Tribal Lifestyle

Religion pervades all characteristics of tribal life that is why for understanding the tribal lifestyle we need to first understand the tribe's religious notions. To explain this, we will take an instance of the rustic Toda of the Nilgiri in South India. Through their own explanation, the Toda and their extensive horned buffaloes were created on the high massif of the Nilgiri Hills through the great goddess Teikirzi. Their resolution is

marked through the unique barrel-vaulted homes and dairy structures. The dairy structures are temples.

The Toda are socially organised into two endogamous sections, named *Tarthoral* and *Teivaliol*. The division is based on ritual specialisation. The *Tarthar* people alone can own the mainly sacred dairy temples with their associated herds; the *Teivaliol* men, described 'the servants of the gods' fulfil the highest priestly tasks associated with them. Their economic and social lives are centred on the buffaloes. The greater part of their religious observance is also focused on this animal. The buffaloes are ritually guarded. Every task of the dairyman, every substance and lay associated with the herds has received the impress of ritual. The Toda observe a number of special ceremonies related to the dairies, the buffaloes and the pastures. There are rites through which they honour the dairy, purify it when it has been defiled, and create offerings to it. There are rites associated with rebuilding and rethatching. After that are the rites for naming a buffalo, milking a temple buffalo for the first time, giving salt to the herds and moving them to the new pastures. There is also the rite of lighting the gods' fire to ensure the fertility of the pastures.

Life's major events are highly ritualised in the middle of the Toda. The first pregnancy of a Toda woman is seen as ritually contaminating and hence in almost fifth month she spends a complete lunar month in a temporary 'pollution hut' outside her hamlet. In her seventh month of pregnancy, the husband ritually provides her a bow-and-arrow to set up the social paternity of the unborn child. If a child were born to a woman who had not been given a bow, it would be a bastard with no patrician affiliation. In former days when polyandry was the norm, the many husbands of a woman took turns to assume the paternity of her children. Childbirth is a polluting event. It is not only the mother and her child who are ritually defiled, but also all other people and things which approach in secure get in touch with them. Purificatory rite is performed on or presently before the day of the new moon after birth. The 'face-uncovering' ceremony is performed some time flanked by one and three months after the birth of a child. It is only then the Toda child is recognised as a social being.

The name-giving ceremony follows it. Marriage is initiated in childhood and completed at maturity. Meanwhile the ear-piercing ceremony is performed for all boys. For a girl there are two separate rites supposed to precede her entry into womanhood. The two rites essentially are symbolic and an actual defloration is performed before a girl's first menstruation. When the partners in a marriage alliance reach maturity, arrangements are made for their cohabitation and for the subsequent payment of dowry. On this occasion the girl's father performs 'the girl-sending' ceremony. The passage of a Toda man or woman into middle or old age is not marked through ritual. As death approaches, the elders can look forward to an afterlife not radically dissimilar from their present one. As the goddess Teikirzi rules in excess of the livelihood Toda and their buffaloes, so does her brother, On, who reigns in excess of the dead. Death generates

the greatest profusion of ritual. The corpse of the deceased is cremated at the end of the first funeral ceremony throughout which buffaloes are sacrificed so that they may accompany the spirit to the afterworld. A second ceremony is held, usually months after the first, in which a relic of the deceased is cremated.

What comes through this account of the Toda passage through life cycle, as also the organisation of their territory, society and economy, is the great importance of religion. You have seen how the intricate web of ritual is woven approximately the comparatively easy tasks of the husbandry man. One might, so, conclude that tribal life and society cannot be fully understood without understanding their religion. Though, as Winter has pointed out, in some societies religion plays a much more significant role in group structure than it does in others. This has been illustrated with reference to African tribes. The social structure of the Iroquois cannot be described without taking religion into explanation; that of the Amaba can be described in such a manner.

Tribal Ritual Intricate

For a religious human, the superhuman world is the real world, which is ritually replicated in legroom and time. The believer communicates with this world through a symbolic code which Saraswati calls the ritual language, consisting of:

- Esoteric word, *mantra*,
- Gesture, *mudra*,
- Sound, *dhvani*,
- Everyday language of interaction, *vyavahariki*.

The tribal ritual intricate may be described beside this row. Here we will first speak of ritual legroom and then of ritual time. After that, we will talk about ritual language in conditions of *mantra*, *mudra*, *dhvani*, and *vyavahariki*.

Ritual Legroom

In the tribal perception the legroom is filled with spiritual beings. There are two classes of legroom: cosmic legroom and physical, or substance legroom. The Santal consider that the creator *Thakur Jiu* dwells up in the sky; the *Sin Bonga* or the Sun God also occupies the cosmic legroom. There is also the *Hanapuri*, or the land of the dead. The spirits inhabit concrete objects in and approximately their resolution. Such spaces are named after the spirits associated with them. For instance, the spirit of the region, the spirit of the village boundary, the home spirit, the spring spirit, the mountain spirit, the forest spirit, the spirit of the rice field, the spirit of the threshing floor, cremation ground, etc.

The ritual legroom is dissimilar from the ordinary legroom inasmuch as it is sacred effective. But any ordinary legroom can become a ritual legroom, depending on its association with the spirit. Usually, it is allowed to retain its natural form. For instance, the *Jaher*, or the sacred grove of the Santal consists of a clump of sal tree with one

mahua standing close to. Three trees in a row are dedicated to Mama Buru, the spirit who dwelt with the first Santal, Jaher Era, the spirit of the sacred grove, and to More ko Turuc ko, the spirit described Five-Six. Another *sal* tree is dedicated to Pargana Bonga, the spirit of the region, while *Mahua* tree is reserved for Gosae Era, the spirit of the sacred home. At the base of each of these trees, a stone is embedded and all five are recognised as *Bonga* trees. An easy construction may also spot the sacred legroom. The Santal Manjhithan consists of smoothly cemented plinth of mud with two stories to symbolise the founder of the village, Manjhi and his wife, a centre pillar of *sal* wood, a second and smaller pillar supporting a pot of water and finally a roof of thatch or tiles.

Tribes, through and big, entertain no notion of hierarchy of the sacred spaces. But, as Walker has reported, the Toda dairy temples are graded into an intricate hierarchy just as to relative sanctity. The higher they are, the more elaborate is the ritual associated with the daily tasks of the dairyman, and the higher necessity is the purity in which the dairy, its appurtenances, and the dairyman himself are maintained.

Ritual Time

The tribes order their ritual time on the Nature's signal, *i.e.*, on the blossoming plants and trees, on the location of stars and moon, the biological time, and, if the circumstances so demand, through direct divination. The principal festival of the Oraon is described *Sarhul*, the Feast of the *Sal* Blossoms, also recognised as the Marriage of the Earth. Another festival recognised as *Kadlota*, is celebrated when rice granules have shaped in the rice plants. The Toda perform their rituals just as to the location of stars and new moon. The star described ket, the 18th *naksatra* or lunar mansion, is measured to have a malignant power. Mainly rituals are performed on or in relation to the new moon. The moon-time is followed rather strictly.

The Toda regard childbirth pollution particularly dangerous to the ritual condition of the sacred dairies of the hamlet. Therefore they insist that a new mother and her child should be taken, as soon as the woman could move, out of her hamlet to the pollution hut. The length of the time mother and child were compelled to remain in this hut depended upon whether or not the child was her first. In the case of the second or subsequent child, the era of seclusion would last until the new moon following the birth, perhaps presently a few days.

Therefore, if a woman had given birth to her first child presently after a new moon, she would have to endure approximately two months of seclusion. On the third day of the new moon the newly delivered mother receives butter and buttermilk from the dairy, which event marks the end of the restriction on her drinking milk and consuming milk products from the dairy. The 'social paternity rite' or 'bow-and-arrow ceremony' is performed on the day of the new moon. The first funeral rite is also held on the day of the new moon. Mainly tribes consider biological time as an occasion to perform ritual, such as the first menstruation, childbirth, and death.

Impact of other Religious Belief Systems

Several tribal religions have blended with other faiths of non-tribal groups. Of these Hinduism and Christianity are the major religious systems which have affected the tribal societies. Let us first consider the impact of Hinduism on tribal religious beliefs and practices.

Hinduism and Tribal Religions

Ethnographic studies of dissimilar tribal groups illustrate that the get in touch with of the tribal's with their neighbours varied from part-separation to complete assimilation. The Tharu and the Khasa, the two central Himalayan tribes in North India are a good instance of totally assimilated or Hinduised tribes. Through adopting Hindu caste names, wearing the sacred thread, establishing social links with the local Rajput and Brahmin groups, these tribal's have incorporated their identity with high caste Hindus.

Likewise, the Kshatriya model of Hinduism has been adopted in middle India through the Chero, Kharwar, Pahariya of Bihar and the Bhumij of Madhya Pradesh. The Bhumij Kshatriya Association, founded in 1935, showed wider implications of Hinduisation of the tribal's. The concept of tribal Rajput continuum was evolved in the course of historical studies of the Bhumij Raj of Birbhum. In Eastern India the Bauri of West Bengal accepted to observe the prescribed number of days of pollution for mourning, wear the sacred thread, go to pilgrimage and follow Vaishnavism. They now claim to belong to the Brahmin caste. Adhering to beliefs in the Hindu concepts of Karma, pollution, merit (*punya*) and observing Hindu life cycle rituals, the Mahali of West Bengal have adopted and assimilated the elements of local belief and practices of the Hindus. Likewise, in several of the Oraon villages of Chotanagpur, Hindu gods and goddesses are worshipped; Hindu priests are employed to carry out ritual performances throughout life-cycle ceremonies. In Western India also, we discover that several of the Hindu deities are worshipped through the tribal's. For instance, Mahadev is the main deity of the Bhil tribe. Shah has shown that in Gujarat the Dubla, Naika, Gamit and Dhanaka are quite Hinduised and employ a Brahmin to perform the rituals.

Looking at the situation in South India, we discover that beliefs and practices of the neighbouring Hindu castes have been adopted through the Chenchu, Kadar and Muthuvan. Hindu gods and goddesses like Aiyappan, Maruti and Kali are worshipped through the Kadar. Presiding deities of Madurai temple, Palaniandi and Kadavallu are treated as their chief deities through the Muthuvan.

Here we would like to also mention that contacts flanked by the tribals and their Hindu neighbours have not only resulted in the impact of Hinduism on tribal beliefs and practices, "We discover also the impact of tribal religions on the practices of sure Hindu group's livelihood in tribal villages. For instance, the procedure of triplication in the Bastar region of Madhya Pradesh reflected in acceptance through high caste Hindus of tribal morals, rituals and belief. Majumdar mentions the concept of trans-culturation

in the middle of the Ho of Singhbhum, signifying reciprocal impact of tribal culture on local Hindu castes. Hutton observed that Hinduism and tribal religions share a general base, while Bose is of the opinion that the tribal population of India has contributed to the creation of Hinduism.

It is not out of lay here to mention that the tribal groups have also been affected through their neighbours. In upper Lahaul and the Ladakh region, the Bhot tribals are mainly Buddhists. Likewise, the Gujjar of the North-Western Himalayan region and the Bhil of Rajasthan have secure contacts with Muslim groups and are affected through their beliefs and practices. We have basically mentioned in relation to the neighbouring groups' impact on the tribal groups. We have not discussed the procedure of these changes. This point will be examined in the after that unit on *Tribes and Modernisation in India*. Let us now look at Christianity, which has affected several of the tribal groups in India.

Christianity and Tribal Religions

Beginning with the conversion of the Khasi of Assam in 1813, of the Oraon of Chotanagpur in 1850 and of the Bhil of Madhya Pradesh in 1880 through Christian missionaries, Christianity has brought in relation to the several changes in the cultural life of the tribals in India.

The missionaries attempted to convert numerically major tribes. As a result, minor tribes remained untouched through the new religion while mass conversion of major tribes gave the impression of the hold of Christianity on tribal India. Of the total population of Christians in India at least one sixth belong to tribal groups. Mainly of the tribal Christians are establish in the North-East Himalayan zone. Elwin estimated that ranging from half to approximately the whole population of the Mizo, Garo and Naga tribes professed Christianity. In middle India, two-thirds of the Khasi, one-fourth of the Munda and one fourth of the Oraon follow Christianity. In the tribal regions of Madhya Pradesh and Orissa, we discover little pockets of Christian tribals. The tribes like the Chero, Kharwar, Pahariya, Birhor, Baiga, Balhudi, Bedia, Karmali, do not have a single Christian convert.

In South India, Hill Pulaya, Malayarayan and Palliar tribals of Kerala have been proselytised and almost two-third of their population has accepted Christianity. Conversion to Christianity gave the tribals a model of westernisation. Here it will suffice to say that the Church organisation, western education, values and morals reached the tribals through Christianity. Their introduction implied a demand to provide up tribal belief and practices. In some cases, traditional festivals were reinterpreted in conditions Christianised myths. For instance, origin of the festival Sarhul of the Munda was, after conversion, associated through them to the fight flanked by Alexander and King Porus in 400 B.C. Sahay has concluded that the Oraon of Chainpur in Ranchi district of Jharkhand gave up their faith in traditional Sarna religion and adopted Christian faith.

This resulted in considerable changes in celebration of festivals, village organisation, economic life and other characteristics of their culture. Therefore, some scholars have viewed Christianity as a source of disintegration of tribal religion.

We can also mention the element of fusion with Christianity introduced in the middle of the tribal groups. Under the rubric of one denomination of Christianity some of the previously separated tribal groups came jointly and even accepted marital dealings crossways tribal boundaries. Under the Church organisational network, several tribal groups scattered in excess of a wider region came jointly and built contacts not only with the provincial and national but also international Church bodies. Sahay has studied the procedure of Christianisation of the Oraon tribals and recognised a set of five procedures, signifying dissimilar shapes of interaction flanked by the tribal Christian norms and values. Under the patronage of British rule, conversion to Christianity had establish a favourable environment while in self-governing India, several revivalist movements in the middle of the tribal groups have led the tribals to go back to their traditional religious beliefs and practices.

Socio-Religious Movements

All in excess of the world, socio-religious movements have evolved in approximately all religions. These range from cargo cults to millenarian movements. The word millenarian means, literally, a thousands years and refers to the belief that the world order is soon going to end, giving method to a new and perfect society. Ethnographic material is well researched and documented to explain the emergence of such movements. Here we will look at the emergence of some of the socio-religious movements in the middle of the tribal groups as products of change within the group, of get in touch with the out-group, and of reinterpretation of the sacred realm. Through focusing on these movements in the middle of the tribal groups, we do not mean to say that such movements do not happen in the middle of other groups.

In information, several such movements have been recorded in the middle of the untouchable and middle castes in India. Examples of such movements in the middle of the tribals are the Munda Rebellion, Jatra Bhagat Movement and Kharwar Movement. All these examples illustrate that the tribals (in the middle of whom the movements appeared) were never totally in accessibled from the main currents of Indian society. Secondly, the exploitative forces (against whom the movements were addressed) were not only colonisers, but also the non-tribal upper castes. Thirdly, the influencing get in touch with the out-group did not approach only from Christian missionaries but also from Hindus and Muslims.

We may have to enquire ourselves the question as to why it is that no socio-religious movement had appeared in the North-Eastern Boundary Agency while several did in the middle of the tribals of Chotanagpur. If we go through the theory of deprivation as one of the causes of emergence of such movements, it may appear that perhaps the

tribals of Chotanagpur were deprived to the worst extent and hence in the middle of them a multiplicity of such movements arose. We may also look at the question of duration of contacts flanked by dissimilar cultures. We may enquire, at what stage of get in touch with, does a socio-religious movement arises? Due to varying intensity of crises, several durations may be necessary in scrupulous groups for the movements to crystallise. Also, a movement may emerge, become active and then lie dormant. Again, a socio-religious movement may approach at the initial stage of get in touch with or it may approach after the completion of acculturation.

When one culture meets the other, one social order is affected through the other. Here, while re-emphasizing the similar, it can be said that mainly of the time meeting of cultures has been prompted through colonisation. The colonisers (be they foreigners or other cultural groups within the country) integrate their colonies into wider markets through introducing dissimilar shapes of economy, through exporting the local products and raw materials. As a result rapid changes take lay and the existing social order breaks down. In the case of tribal India, big level changes were introduced through the missionaries. These changes, in turn, produced disturbance in the method society was previously organised in the middle of the tribals. The disturbances caused several a dilemmas for the people, leading to both psychological social deprivations. These are the situations, which triggered the path to socio-religious movements.

With this background of their emergence, let us now take two examples of socio-religious movements, namely, the Tana Bhagat Movement in the middle of the Oraon of Bihar (Jharkhand) and the Birsa Munda movement of the Munda in and approximately the districts of Ranchi and Singhbhum in Chotanagpur Division of present day Jharkhand. Both these movements are essentially religious in nature. Here, Jatra Bhagat and Birsa Munda, the tribal leaders of the respective movements were essentially fighting the foreign exploiters, like the landlords and contractors. The tribals, feeling deprived and inaccessibled, establish through these movements a sense of unity and a general identity. Yet, the two movements also differed from each other. Let us consider each of them separately.

Tana Bhagat Movement

In Tana Bhagat Movement, as in other Bhagat movements, an effort is made to emulate the method of life of the Hindu higher castes. Tana Bhagat movement is one type of the Bhagat movement that appeared in the middle of the Oraon of Chotanagpur, Jharkhand. There were other Bhagat movements like Nemha Bhagat and Bachidan Bhagat. The term Bhagat has been employed in several parts of Bihar and Jharkhand to refer to sorcerers and magicians. In the middle of the Oraon it is though applied to a separate section of the tribe which subscribes to the cult of Bhakti. The whole Bhagat movement, attempting as it did to raise the status of its members in the eyes of the nearby Hindu society, is characterised big level incorporation of Hindu belief-practices

into its ideology. The leader of Tana Bhagat movement was described Jatra Oraon who existed in village Beparinwatoli in Bishanpur Thana of Gumla sub-division of Ranchi district. In 1914, this person announced in the month of April that Dharmesh, the high god of the Oraon, had revealed to him that the people would have to provide up the worship of ghosts and spirits and the practice of exorcism. He told his people that they would have to refrain from animal sacrifice, meat eating, liquor drinking etc. Even farming through plough with the use of animal power was rejected. It was whispered that god had given to Jatra sure songs or spells through which fever, sties, and other ailments could be cured.

The leaders of this movement whispered that the tribal spirits and deities whom they had been worshipping were of no use to them since it did not help them alleviate the socioeconomic ills of their society. They also tried to prove that in information it was these deities who were responsible for their present state of misery and degradation. Through asserting that these tribal deities were of Munda origin, the founders of Tana Bhagat movement embarked on a programme of proselytisation and agitation for the exorcism of the foreign spirits and deities. The cult emphasized a return to the original Oraon religion. Its earliest manifestation was in the expulsion of evil spirits imported from the Munda and in the active rebellion against unfair landlords who exploited them. Even when their leader Jatra Bhagat was imprisoned, some of the cult members refused to pay rents to their landlords and ceased to cultivate their lands. Such and other rebellious behaviours like ghost-hunting drive and holding meetings through the followers of the Tana Bhagat movement were regarded with suspicion and branded as 'disloyal and illegal' gatherings.

This movement broke up into many smaller cults after sometime. Some of these cults are extremist and orthodox like Sibuh Bhagat. The rest are mixtures of tribal and Hindu religious beliefs and practices.

Birsa Munda Movement

Throughout the second half of the nineteenth century the whole of Chotanagpur underwent a tremendous change. The old Munda system of *khuntkatti* tenure gave method to a new and alien system of use through the landlords recognised as *Jagirdar* and *Thikadar*. Under their greed and cruelty the tribal population was squeezed out of their land and other possessions. Suffering economically and politically, the Munda were assigned low social status. Later the Christian missionaries tried to provide the Munda back their rights on their land. But this too had a price, which was conversion to Christianity. A big number of the Munda did convert to Christianity. But in due course they realised that this was not the solution. The missionaries had failed to redeem them from abject poverty and oppression.

At this time when the Munda society was seething with discontent, in 1895, Birsa Munda of Chalked started a movement. In him the Munda establish the embodiment of

their aspirations. He gave them leadership, a religion and a code of life. He held before them the prospect of Munda Raj in lay of a foreign rule. Under the power of a learned Satnami Pandit, Birsa Munda became a vegetarian and the religion that he preached had elements of both Hinduism and Christianity. His religion promised to end the misery of his society and so it was a means to an end. His religion had the element of charisma. Birsa came to be regarded as an embodiment of god '*Birsa Bhagwan*'; his people whispered that Birsa could bring the dead back to life. Whenever there was epidemic he visited his people and cured them. He had the magic touch and proclaimed himself to be the prophet of Sing-Bonga, the one and only God. So distant Birsa was seen as a provider of new and better life for his people, But later his movement assumed quasi-political and militaristic form. As he organised a force to fight oppressive landlords, Christian missionaries and British officials, he was imprisoned. Since his people whispered in Birsa's magical power, they did not enquire for his release, rather they wanted to go to jail with him. They whispered that within three days he will himself approach out of jail. He was released only after two years.

Out of jail, Birsa asked his people to ready their arms to fight injustice. He trained his army and became the politician leading his people to their goal of self-rule. He was though arrested again and died in jail. Yet the seeds of unrest were sown in the middle of his people and they sustained to fight against injustice. The erstwhile, 'Jharkhanda Movement', which ultimately culminated in the formation of the Jharkhand State has its roots in this movement. Our account of both the Tana Bhagat movement and Birsa Munda movement underlines the importance of a 'prophet-like' figure, who is the guiding spirit of the movement. He draws his legitimacy from the divine revelation. His prophecy appeals to the crisis situation. He becomes the epicentre of the communal expectations of his people. He suggests rejection of sure behaviours, rituals and customs. He incorporates new elements and adapts sure components of out-group culture to suit the needs of communal expectations. He reinterprets the myth, reformulates the ritual and prepares his people for communal action. We say that cult myths are dynamic characteristics of tribal religion as they express the hopes of the people. We can conclude here that through discussing some of the socio-religious movements in the middle of the tribals we have gained an added understanding of tribal religions.

ROLE OF CASTE IN INDIAN SOCIETY

The political process of any society is influenced by the nature of the society. To understand the nature of the society we study its social structure. India's social structure is best understood in terms of caste system wherein the cast is hierarchically arranged. Over the years, the caste system developed into an elaborate system to maintain socio-economic inequalities in the society. Individuals born in and belonging to the lower castes and the out-castes suffered from many disadvantages and were

oppressed and exploited by the upper castes. The conditions of the outcastis (Dalits) was particularly pathetic.

The practice of untouchability epitomized their conditions:

1. In the typical Varna-Vyavastha there are four Varnas: Brahmin (the priest and the intellectual class), Kshatriya (warrior and the ruling class), Vaisyas (the producing class - peasants and artisaas) and Shudra (those who performed menial and 'polluting' jobs. One must note here that the 'varna-vyavastha' provides more the theory than the actuality of the caste. In reality, there are not four but thousands of castes and jatis, in which the caste system is organized. It is possible, nevertheless to classify most of the jatis in accordance with the Varna distinctions, although it is easier to be done at the extreme ends of the social spectrum than at the middle ranges. In other words, Varna system is related to jati in that it gives a holistic frame-work to which any jati will fit.
2. Caste is a localized group having a traditional association with an occupation. The principle of birth forms the exclusive basis of membership in a caste group. Accordingly, the choice of occupation is not open but is determined on the basis of one's birth in a caste. In addition caste groups have rules governing food and marriage. The group defines rules of behaviour for its members and exercise some degree of authority over them including the right to expel those who defy its authority. Caste as group identity, however, got strengthened in the new context of modern ideas and institutions. This happened because it became one of the bases of political mobilization among the many castes and classes before, during the freedom struggle and afterwords. The socio-religious movements of the 19th century had made the lower castes conscious of their backward conditions and also of their rights that had been denied to them over the centuries. As a result, many amongst them were no longer prepared to accept their inferior status as divinely ordained. In the backdrop of this awakening, the introduction of democratic principle of governance, the emergence of party-centered politics and the attempt of the British rulers to mobilize the lower and the out-castes along with the Muslims in its support to thwart the growing national movement — all combined to prepare the ground for the politicization of the castes.

By the time India gained Independence, the Backward classes, because of politicization, had become a force to reckon with. Their claims and demands could no longer be ignored.

At the same time the nationalist leaders were also committed to the task of improving their conditions. In light of the above, the constitution makers enjoined upon the new state to take positive measures for bringing the backward classes at par with the rest of the society.

They realised that without the positive intervention by the state it would not be possible to remove their historically accumulated backwardness. The policy of reservation for the backward classes needs to be understood in this context. Backward classes refer to three categories of the people — the scheduled castes, scheduled tribes and the other backward classes (OBC).

SOCIETY FAMILY

Family organizations in foraging societies often revealed by their institutional components, the difficulty of living in marginal ecosystems. Whatever their original habitats might have been, contemporary foragers like the San and the Australian aborigines lived in desert-like regions, the Inuit in the frozen north, and the Batwa and Siriono in dense forests. Many of these groups may have been driven or retreated into the areas where they were encountered, and had subsequently adapted to local conditions. For example, many of the hunters on the Great Plains of North America are believed to have been river valley cultivators before they became equestrian hunters of the buffalo. Nevertheless, it is true, that the food producing repertoires used by foragers and consisting of fishing, hunting and gathering wild plant foods, were common to human beings for more than 99 percent of their existence. Moreover, such implements as the digging sticks, clubs, spears, bows and arrows, fishing devices, traps, fire and containers for food, permitted these families to survive. True, under normal circumstances, the sparse and often irregular sources of plant and animal food inhibited foragers from congregating in groups larger than fifty or one hundred persons, but fisher folk, on the Northwest Coast of North America, could, and did inhabit relatively large villages.

Families in foraging societies were usually exogamous, meaning that they married outside domestic groups normally composed of near kin. Levi-Strauss noted the frequency of what he termed “restricted exchange” in band societies. Men usually exchanged sisters as wives, or more usual, exchanged women between bands. Such marriages effectively supported incest prohibitions, but also extended the range for food collecting during periods of scarcity. Edward Tylor wrote that early human groups quickly learned that people had to “marry out or die out.” Band exogamy, when linked to frequent changes of residence and limited food supply may have facilitated the development of a bilateral kinship system where people traced their descent from both mothers and fathers, rather than almost exclusively either through males or females. The aboriginal Australians had complex rules of band exogamy and kinship terminologies, but this did not prevent persons from moving from place to place and finding relatives with whom they could live.

Monogamy was the usual marriage rule among foraging bands. In some cases widows were inherited as plural wives, and high status men occasionally had plural spouses. But limited resources precluded plural marriages. Residence was normally virilocal,

that is, with the husband, but as we have seen married persons often changed residence for economic or personal reasons. Without exception, the !Kung and San of South Africa required that all men spend several years providing “bride service” with the family of their brides. The reason they gave for this was that the parents of brides needed the proceeds from the hunting of their son-in-law. The Inuit were normally monogamous, but exchanged wives when specific women were needed to perform particular domestic chores.

While male dominance was usual in foraging societies, Kathleen Gough was basically correct when she asserted: “Especially lacking in hunting societies is the kind of male possessiveness and exclusiveness regarding women that leads to such institutions as savage punishments or guarding of female chastity and virginity, the denial of divorce to women, or the ban on a woman’s remarriage after her husband’s death.” Kung, San and Inuit men did not have the power to brutalize wives, and could be sanctioned by male relatives or by the entire community. Unreasonable Inuit husbands were known to be the butt of ridicule when they were defeated in “song duels.”

Children were valued in foraging societies as replacements for the old and deceased. Twins and children born when siblings were still suckling were often suppressed. Twins were considered unnatural beings who overburdened mothers who could not provide enough milk for them. Children in foraging bands learned by doing. Boys were carefully trained to participate in the hunt, and girls were taught how to harvest plants, roots and berries. Yet even on this level, children acquired special knowledge during initiation ceremonies that marked puberty. Among the hunters of “great plains” in North America, young males were required to undertake vision quests to test their fortitude and to acquire “guardian spirits.” Successful youths often returned to camp filled with ambition to become great hunters and members of police societies that monitored the hunt. Failure to receive guardian spirits often indicated to the community that some youths had different religious, sexual and gender orientations. Such young men sometimes became skilled “medicine men” or *berdaches* whose life choice as female was respected by the band. The most valuable lessons learned by children in many foraging societies was to respect and even support gerontocide, the killing of parents unwilling to permit illnesses to jeopardize the safety of the community.

Most foraging societies were egalitarian and did not possess the resources to permit stratification among families. The Kwakiutl of the Northwest Coast of North America was a much studied exception. Profiting from their ownership of clam-beds, sections of river banks and off-shore areas, and of hunting, berry and wild-root areas, some families were able to take titles and erect totem poles. Individual men and women were able to engage in elaborate gift-giving contests known as *potlatches*. Wealthy families often held chieftainships, kept slaves and boasted that they did not fight with weapons, but fought with property. Admittedly, this behaviour was rare among foraging societies, and may have been due to contact with the West. Yet, the Kwakiutl indicated the social

structure possibilities of foraging societies in favourable circumstances. Data about the beliefs of families of foragers, suggest people tended to identify with *zoomorphic* animals and plants. This was especially true among the Australians whose totemic beliefs and rituals linked families to mythical people and animals. As the worship of the sea goddess, Sedna, among the Inuit indicated, the hope was mostly to augment the food supply. But beliefs about “dream time” among the Australians may have represented an attempt to keep track of historical events? Shamans, among the !Kung or Inuit employed curing ceremonies on behalf of family members. The Batwa Pygmies of the Ituri forests of Zaire held Molimo ceremonies to appease the denizens of their forest homes.

Herding Society Family

Herding societies often referred to as “pastoral cultures” because they were often linked to plant cultivators, normally provided families with greater resources than did foragers. These families were usually embedded in larger social units known as lineages, clans or “tribes” based on actual or putative “genealogies.” Generally, lineages viewed themselves as corporate groups possessing rights in persons and property, but these attributes were usually anchored in lineage segments that lived together. For example, the Somali dia-paying groups used to pay the families of the victims of feuds), were said to be lineage-wide, but were really based on highly localized sublineages.

Undoubtedly, the genealogical structure of herding families was a function of their need to adapt to the transhumance needs of their livestock, and a corresponding need to protect their animals against both human and animal predators. Nomadic herders ranged far and wide to secure pasturage for their livestock, and they often developed protective or aggressive institutions to replenish herds depleted through raids and diseases.

Marriage alliances among herders were usually arranged by heads of extended families and lineage segments often without regard to the wishes of the persons to be married. Central to these marriages was the *bridewealth*, a customary gift from the husband and his group to the wife and her group, before, at, or after marriage. Ostensibly the gift was a compensation to the bride’s family for the loss of her companionship and labour. It was not a “brideprice” since the bride could not be subsequently sold. Bridewealth was very much a “progeny price” that gave a woman’s children full membership in her husband’s descent group, even though she remained a stranger to that group.

The bridewealth accomplished a number of ancillary functions. It tended to stabilize marriages and discourage divorce because persons who negotiated the bridewealth disliked haggling over refunding the gift in event of a divorce. Bridewealth also fortified group alliances because it was linked to the *levirate*, an institution in which the widow was inherited by her late husband’s male relatives. A companion institution was the *sororate* in which families replaced a woman who died before bearing children for their husband’s family or proved infertile, with another woman. It was not uncommon for men whose fathers had died before completely paying the gift, to finish the transaction

to legitimize themselves. Herding groups such as Arab or other Semitic pastoralists tended to be endogamous, that is, they approved marriages among persons with known genealogical relationships. Marriage with father's brother's daughter was not only permitted but was preferred. Such marriage practices undoubtedly served to preserve wealth within extended families or lineage segments, but they also solidified these units. The Somali and the Fulani of the Sudan further strengthened family bonds by adding a dowry from the bride's family to the new family. This gift included furnishings for the new household. Couples who received both parts of bridewealth and dowries, had an economic cushion when they established their households.

Families in herding societies were frequently stratified due to the exigencies of raiding and warfare, or the health and increase or decrease of their herds. For example, Mongol families were socially stratified on the basis of wealth gained or lost as a result of raids. Poor families became dependent upon wealthy ones, but wealthy families kept few chattel slaves and massacred captives deemed useless or dangerous since they could decamp with animals. The Tuaregs of the Sahara and Fulani, on the other hand, sought and held captives from cultivators and used them to produce plant products, and to perform household chores.

POSITION OF WOMEN IN HERDING SOCIETIES

The position of women in herding societies differed widely, and may have been conditioned by relations with agricultural societies with whom they were in contact. Wealthy men were permitted to have plural wives and concubines, but the exigencies of nomadic life precluded secluding women who often traveled with the herds. Tuareg women in the Western Sahara, had extraordinarily high statuses. Girls were permitted to indulge in a great deal of "petting" with young men, while friendships between married women and men were taken as a matter of course. How much of women's liberty was due their remaining at home while Tuareg men traveled widely, is open to debate.

While not as privileged as Tuareg women, Mongol women were fairly autonomous. They rode like men, and engaged in warfare when necessary. Each wife had her own "yurt" where she lived with her children and could acquire property on her own. The first wife in a Mongol's household was regarded as head of the establishment and took charge of her husband's herds while he was away. In marked contrast, the nomadic Arab family tended to be highly patriarchal with a corresponding diminution of the status of women. Men insisted upon of virginity of their brides at marriage, and often executed unchaste girls or an unfaithful wife and her lover.

There was a corresponding wide range of affective relations within the conjugal families of herders. Mongol boys were subject to their fathers and younger brothers to their elder brothers, but relations between Mongol mothers and their sons were close and persisted all their lives. They often formed an amiable conspiratorial group to

circumvent the father. In contrast, Arab fathers tended to be authoritarian within their families, and often invoked supernatural sanctions to control the sexual activities of their sons. It has been suggested that the combination of patriarchal suppression, and religious proscriptions often gave rise to homosexuality among Arab men, especially among boys and unmarried men. But this practice was usually replaced by heterosexual behaviour when men married and became independent of their fathers.

EDUCATION OF THE YOUNG IN HERDING SOCIETIES

The education of the young in herding societies was primarily by emulation. Girls learned milking techniques from women when custom permitted, and how to care for their households, including erecting tents. Boys learned herding techniques from their male kin. In addition there were usually special institutions to train young males to protect their herds and to participate in the public affairs of their communities. Chief among these was the “age-set” system formed among boys who had been initiated, and who had jointly experienced the trauma of circumcision. They often received special instructions and were provided with a name to enhance their esprit de corps. Normally, such age-sets were formed during the period throughout the society, and even across ethnic groups. These age-set went through stages known as “agegrades.” Thus a specific age-set would pass through the age-grade associated with herding, then through a grade known as junior warriors, then senior warriors, husbands, councilors, priests, and the ultimate, age-grade—the ancestors. Sometimes society-wide age-sets and those from neighbouring societies were organized into regiments for warfare. East African pastoralists had cycling age-sets, that were revived every sixty or more years permitting new groups of men to serve the age-grades of their societies. Some societies had age-set and age-grade systems for females, but most of these were auxiliary to those of the males. Maasai girls formed a cohort to frequent the bachelor quarters of initiated men where they engaged in heavy petting.

Religious practitioners such as shamans in pastoral societies tended to invoke the supernatural for protection of families and their herds and were not above invoking supernatural sanctions against evil-doers. Shamans among the reindeer Chuckchee of Siberia were often curers who used ventriloquism to impress their congregations.

AGRARIAN SOCIETY

Agrarian societies are societies in which the primary means of subsistence is the cultivation of crops using a mixture of human and non-human means. Agriculture is the process of producing food, feed, fibre, and other desired products by the cultivation of plants and the raising of domesticated animals. Agriculture can refer to subsistence agriculture or industrial agriculture. Subsistence agriculture is agriculture carried out for the production of enough food to meet just the needs of the agriculturalist and his/

her family. Subsistence agriculture is a simple, often organic, system using saved seed native to the ecoregion combined with crop rotation or other relatively simple techniques to maximize yield.

Historically most farmers were engaged in subsistence agriculture and this is still the case in many developing nations. In developed nations a person using such simple techniques on small patches of land would generally be referred to as a gardener; activity of this type would be seen more as a hobby than a profession. Some people in developed nations are driven into such primitive methods by poverty. It is also worth noting that large scale organic farming is on the rise as a result of a renewed interest in non-genetically modified and pesticide free foods.

In developed nations, a farmer or industrial agriculturalist is usually defined as someone with an ownership interest in crops or livestock, and who provides labour or management in their production. Farmers obtain their financial income from the cultivation of land to yield crops or the commercial raising of animals or both. Those who provide only labour but not management and do not have ownership are often called farmhands, or, if they supervise a leased strip of land growing only one crop, as sharecroppers.

Agriculture allows a much greater density of population than can be supported by hunting and gathering and allows for the accumulation of excess product to keep for winter use or to sell for profit.

The ability of farmers to feed large numbers of people whose activities have nothing to do with material production was the crucial factor in the rise of standing armies.

The agriculturalism of the Sumerians allowed them to embark on an unprecedented territorial expansion, making them the first empire builders. Not long after, the Egyptians, powered by effective farming of the Nile valley, achieved a population density from which enough warriors could be drawn for a territorial expansion more than tripling the Sumerian empire in area.

DEVELOPMENT OF HORTICULTURE AND AGRICULTURE

Horticulture and agriculture as types of subsistence developed among humans somewhere between 10,000 and 80,000 B.C.E. in the Fertile Crescent region of the Middle East. The reasons for the development of horticulture and agriculture are debated but may have included climate change and the accumulation of food surplus for competitive gift-giving. Most certainly there was a gradual transition from hunter-gatherer to agricultural economies after a lengthy period when some crops were deliberately planted and other foods were gathered from the wild.

In addition to the emergence of farming in the Fertile Crescent, agriculture appeared by at least 6,800 B.C.E. in East Asia and, later, in Central and South America. Small scale agriculture also likely arose independently in early Neo-lithic contexts in India and Southeast Asia. Full dependency on domestic crops and animals was not until the

Bronze Age. If the operative definition of agriculture includes large scale intensive cultivation of land, mono-cropping, organised irrigation, and use of a specialized labour force, the title “inventors of agriculture” would fall to the Sumerians, starting ca. 5,500 B.C.E. By the early 1800s agricultural practices, particularly careful selection of hardy strains and cultivars, had so improved that yield per land unit was many times that seen in the Middle Ages and before, especially in the largely virgin lands of North and South America.

AGRICULTURE TODAY

In the Western world, the use of gene manipulation, better management of soil nutrients, and improved weed control have greatly increased yields per unit area. At the same time, the use of mechanization has decreased labour requirements. The developing world generally produces lower yields, having less of the latest science, capital, and technology base. More people in the world are involved in agriculture as their primary economic activity than in any other, yet it only accounts for four per cent of the world's GDP.

The rapid rise of mechanization in the 20th century, especially in the form of the tractor, reduced the necessity of humans performing the demanding tasks of sowing, harvesting, and threshing. With mechanization, these tasks could be performed with a speed and on a scale barely imaginable before.

These advances have led to efficiencies enabling certain modern farms in the United States, Argentina, Israel, Germany and a few other nations to output volumes of high quality produce per land unit at what may be the practical limit. An example of the influence of technology can be seen in terms of output per farmer. In the early 1900s, one American farmer produced food for 2.5 people; today, a single farmer can feed over 130 people.

INDUSTRIAL

An industrial society is a society in which the primary means of subsistence is industry. Industry is a system of production focused on mechanized manufacturing of goods. Like agrarian societies, industrial societies increase food surpluses, resulting in more developed hierarchies and significantly more division of labour. The division of labour in industrial societies is often one of the most notable elements of the society and can even function to re-organize the development of relationships.

Whereas relationships in pre-industrial societies were more likely to develop through contact at one's place of worship or through proximity of housing, industrial society brings people with similar occupations together, often leading to the formation of friendships through one's work. When capitalised, Industrial Revolution refers to the first known industrial revolution, which took place in Europe during the 18th and 19th centuries.

What is some times referred to as The Second Industrial Revolution describes later, somewhat less dramatic changes resulting from the widespread availability of electric power and the internal-combustion engine.

Many developing nations began industrialisation under the influence of either the United States or the USSR during the Cold War.

POST-INDUSTRIAL

A post-industrial society is a society in which the primary means of subsistence is derived from service-oriented work, as opposed to agriculture or industry. It is important to note here that the term post-industrial is still debated in part because it is the current state of society; it is difficult to name a phenomenon while it is occurring.

Post-industrial societies are often marked by:

- An increase in the size of the service sector or jobs that perform services rather than creating goods
- Either the outsourcing of or extensive use of mechanization in manufacturing
- An increase in the amount of information technology, often leading to an Information Age
- Information, knowledge, and creativity are seen as the new raw materials of the economy

Post-industrial society is occasionally used critically by individuals seeking to restore or return to industrial development. Increasingly, however, individuals and communities are viewing abandoned factories as sites for new housing and shopping. Capitalists are also realizing the recreational and commercial development opportunities such locations offer.

STRUCTURE OF INDIAN SOCIETY

Tribe is one of the earliest identifiable ethnic cum social organizations. A fairly good number of tribes such as Munda, Ho, Oraon, Bhil, Gaddi, Santhal, Kol, Kandh, Khasi, Garo, Mizo, Naga exist in different parts of India. These have been clubbed together in a group in a schedule of the Constitution of India and are known as the Scheduled Tribes.

The pertinent question here is: Who is a tribal, and what differentiates a tribe from other groups based on caste or varna? The answer to this lies in the functioning of the institution. We can identify a tribe by the following features: (i) All members are related to each other by blood (ii) All members are equal in status (iii) All members believe that they have descended from a common ancestor (iv) All members have equal access to the resources (v) The sense of private property is least visible (vi) Social differentiation exists only on the basis of age and sex. Tribes are very often made up of more than one clan. A clan is an exogamous institution whereas a tribe is endogamous.

Such tribes existed during the Vedic period. The Bharatas, Yadus, Turvasas, Druhyus, Purus and Anus were some of the prominent Tribes. The head of the Tribe was known as Raja or King who was first among the equals. The major differentiation existed between people on the basis of 'varna' or colour. Thus, the people who did not belong to the vedic tribes were termed as dasa varna. They are said to have had an alien language, a dark complexion and worshiped different gods.

This initial differentiation later developed into a complex 'varna system', which in place of the Tribe had divided society into Brahmana, Kshatriya, Vaishya and Shudra categories. These categories emerged principally because of frequent inter-tribal and intra-tribal wars, subjugation of the vanquished, appropriation of loot by the war-leaders, and unequal distribution of wealth.

The members of the Brahman 'varna' usually functioned as priests. The Kshatriyas were associated with the political functions, the Vaishyas were mostly agriculturists and the Shudras were the servile class. The first two 'varnas' exploited the most numerous and productive Vaishya varna, whereas the three in unison exploited the Shudras. The three higher 'varnas' were also entitled to the sacred (upanayana) thread ceremony and were known as 'dvijas' (twice-born).

Many changes took place in the economy during the post-vedic period. These changes needed new ideological support which came in the form of numerous unorthodox religious movements such as Buddhism which ridiculed the Hindu social system. But Buddhism could not negate the caste system completely. The Kshatriyas were considered the highest 'varna' in the social hierarchy as seen by the Buddhists. The Vaishyas who were exploited under Brahmanism became members of these religions in large numbers in order to gain respectability. However, there was no caste hierarchy within the Buddhist and Jain monastic communities.

'Varna/Jati' system underwent further changes when new groups such as the Shakas, Kushanas, Parthians and Indo-Greeks people arrived in India. The economy was also undergoing changes. Growth of urbanisation, craft production, and trade resulted in the rise of guilds or 'shreni' which in later times became castes. The behaviour of guild members was controlled through a guild court. Customarily the guild (shreni-dharma) had the power of law. These guilds could act as bankers, financiers and trustees as well. Generally, these functions were carried out by a different category of merchants known as the 'shreshthins' (present day Seths of North India and the Chettis and Chettiyars of South India). Thus, the period between 500 BC and 500 AD saw the crystallisation of the caste system. The number of castes increased manifold because of the growth of a number of crafts, arrival of new elements in the population, inter-caste marriages (anuloma and pratiloma) and inclusion of many Tribes into caste hierarchy. The Dharmashastras and the Smritis tried to fix the duties of each caste. The relations between castes were generally governed by the rules of endogamy, commensuality and craft-exclusiveness.

Between the fifth and seventh centuries further changes occurred in the varna/caste organisation. A creation of landlords through landgrants resulted in the transformation of Vaishyas into servile peasants. Large numbers of tribal cultivators were enrolled as Shudras. Consequently the Shudras now emerged as farmers like the Vaishyas. Now, the Vaishyas and Shudras were lumped together. However, in regions other than the Ganga basin, the caste system took a different shape. In Bengal, South India and other border areas mainly two castes emerged *i.e.* the Brahmanas and the Shudras. The Rajputs emerged as a significant factor in the society of Northern India from the seventh century onwards. In this period caste system became so entrenched in the psyche of the people that even the plants were divided on the basis of the varna/caste system. One text prescribes varying sizes of houses according to the varnas.

An important development during medieval times, particularly in South India, was the division of the Shudras into 'pure' (sat) and 'impure' (asat) categories. Also typical of the south were the so called 'left-hand' (idangai) and 'right-hand' (velangai) castes. Manu noted 61 castes whereas a later text counted hundreds of mixed castes (varnasankara).

Other than the Rajputs, another caste that developed during this period was the Kayastha. The Kayasthas were traditionally scribes who got transformed into a separate caste as all types of scribes got clubbed together to form one endogamous group. Very often a large number of these castes claim origin from highly respected ancestors and explain their current lower status in terms of economic or other factors. The Khatri, an important caste in Northern India, claim that they were of Kshatriya origin, but took to commerce, which brought them the contempt of their caste fellows and they had to accept Vaishya status.

The Gurjars, Jats and Ahirs all claim Kshatriya origin, the status they came to lose later for various reasons. This process of claiming high caste origin continued till the 1950s when the Constitution of India provided for reservations for lower castes in most government jobs.

The caste system has its regional variations. The formation of regions and regional consciousness after the eighth century AD contributed to this variation. The North Indian Brahmanas are divided not only on the basis of gotra but also on the basis of their residence. Thus we have Kanyakubja, Sarayuparier and Maithila brahmanas belonging to Kanauj, Sarayu river and Mithila respectively. A large number of Rajput sub-castes carry their earlier tribal names such as, Tomaras, Kacchavahas, Hadas and Chauhanas. These subcastes also acquired gotras for themselves. Thus the number of castes and sub-castes as well as that of the gotras kept increasing in medieval times. The Marathas also became a caste.

In modern times the caste system acquired some new features. Thus, the idea of 'surnames' was also added. One particular caste or sub-caste came to acquire one, sometimes more, 'surnames'. However, ample variation is found in this matter. Caste

and anti-caste movements of the modern period resulted in the formation of further new castes. Thus the Brahma-samajis acquired status which is very similar to a caste.

The caste system, as can be seen from the above description, has a complex working method. Substantial regional variation is found and hence it has been impossible to define it in a precise manner region, sub-region, economic status, nearness to political authority, craft, vocation, following a particular deity all have come to play a role in its formation and subsequent changes.

The influence of caste system has been so great that even though the egalitarian religious reformers of the medieval period such as Basava, Ramanand and Kabir, tried to abolish caste among their followers, their sects soon took on the characteristics of new castes. The Sikhs, could not overcome caste feelings. Even the Muslims formed caste groups. Syrian Christians of Kerala, earlier divided into sections, took on a caste character. The Christian converts brought their caste prejudices with them and high caste converts consider themselves aloof from those of the lower orders.

UNTOUCHABILITY

The concept of purity and pollution in Indian society is best reflected in the formation of a category known as untouchables (antyajas). These people were considered as belonging outside the Brahmanical society. This notion of untouchability took roots during the last phase of the Vedic period and became a separate social category in the age of the Buddha. Sometimes they were called the fifth varna (panchamd). Chandala is the term used loosely for many types of untouchables. They were not allowed to reside in the caste villages and had to stay in special quarters outside the main settlements. Their main task was to carry and cremate the corpses. The law-books have prescribed that they should be dressed in the garments of the corpses they cremated, should eat food from broken vessels and should wear only iron ornaments. By the Gupta period their status fell so much that they were forced to further strike a wooden clapper on entering a town. The hunters (nishada), fishermen (kaivartas) leather workers (charmakaras), sweepers (kukkusa), and basket makers (vend) all became untouchables. 'Dom' and 'Domb' was a tribe which became an untouchable category after coming into contact with the caste divided groups. We also hear of domb kings apart from many Shudra Kings. Mlechchas were also considered untouchables. This untouchability has continued till recent times. Although the practice of untouchability is considered a crime but in rural areas it still continues. Mahatma Gandhi initiated a campaign against this practice. He preferred to call them harijans. The Government of India has enacted many laws against anyone practising or promoting untouchability.

Education and social movements has contributed towards bridging the large gulf between them and the others. It is hoped that this very inhuman practise is wiped out soon.

SLAVERY

Slavery as it existed in India was different both in form and essence, from the classical Greek and Roman slavery. That is why Megasthenes could not find slaves in India. And he was certainly wrong in his judgement. Slavery was an established institution and the legal relationship between the owner and slave was clearly defined. For example, if a female slave bore her master a son, not only was she legally free but the child was entitled to the legal status of the master's son. The 'Arthashastra' states that a man could be a slave by birth, by voluntarily selling himself, by being captured in war, or as a result of a judicial punishment. The sanskrit word for slave is 'dasa' which initially meant a member of the people captured by migrating Aryan hordes. Many such 'dasas' were reduced to bondage.

In India, the slaves were generally employed as domestic servants and personal attendants. The slave was, in fact, a subordinate member of his master's household. The masters had no rights over the lives of their slaves. There were no slave markets in the early periods, however in the early centuries of the Christian era, there was trade in slave-girls between India and the Roman empire in both directions, and slave markets existed in the 16th century Vijayanagara empire.

The Sultans of Delhi kept a large number of slaves (Bandagans). However, their situation was quite different. One of the slaves named Malik Kafur became the commander-in-chief of Alauddin Khilji. In the sixteenth century the Portuguese indulged in slave trade on a large scale. Linschoten has given harrowing accounts of the slave trade at Goa saying, "they drive slaves as we do horses here". A large number of slaves were used in Khalisa (crown land) land and in karakhanas (workshops). This kind of slavery continued during British domination and was attacked by Lord Cornwallis in a proclamation. The slave trade is now completely prohibited.

PURUSHARTHA, ASHRAMA AND SAMSKARA

The three concepts of 'purusharth', 'ashrama', and samskara' are inter linked. 'Purushartha' means aims of life which is divided into four 'ashramas' or stages of life. For each of the 'ashramas' there are prescribed 'samskaras' or rites that need to be performed. We often hear of varnashrama' – 'dharma' *i.e.*, the dharma of varna and stages of life. There-are four aims of life 'dharma' (right conduct), 'artha'(pursuits through which livelihood is earned), 'kama' (worldly desires) and 'moksha' (salvation). To enable a person to practise the four 'purusharthas', the life of a person is divided into four 'ashrama' or stages. Detailed rules of conduct to be observed in each 'ashrama' is also laid down. The four 'ashramas' are like the four steps of a ladder.

The first 'ashrama' is known as 'brahmacharya', which a person enters after the 'upanayana sanskara' or the sacred thread ceremony. During the period of this 'ashrama' he receives education and learns to discipline his will and emotions. He has to live with the guru at his 'ashrama' in the forest and learn to obey and serve him.

He begs alms for himself and his teacher and does odd jobs like fetching water and cleaning around the 'ashrama'. He practises the ideals of plain living and high thinking. Even a royal prince performed all these duties. The 'brahmacharya' stage thus prepared the individual to take on the responsibilities of a householder.

The 'grihastha ashrama' is the most important stage in a person's life. In this stage he practises artha and kama in accordance with the dharma. He gets married, begets children and earns livelihood to support his family and discharge his obligations towards the society.

After discharging all the duties of a householder, he enters the 'vanaprastha ashrama'. During this stage he leaves the family and retires to the forest, where he practises detachment from all worldly pursuits and interests. He had to live only on fruits and vegetables and wear clothes made of deer-skin or barks of a tree. He practices meditation and austerities as well, and if he dies in this stage he attains moksha. Otherwise, he enters the 'sanyasa ashrama' in which he practises complete renunciation. Breaking all bonds with the society, he lives like an ascetic striving constantly for the attainment moksha.

This scheme represents the ideal and not the real situation. Most men never passed through the first stage of life in the form laid down, while only a few went beyond the second. This could also be an attempt to find room for the conflicting claims of study, family life, and asceticism in a single life time. It is also possible that the system of the ashramas was evolved partly to counter the unorthodox sects such as Buddhism and Jainism which encouraged young men to take up asceticism and by-pass family life altogether, a practice which did not receive the approval of the orthodox, though in later times provision was made for it.

According to the scheme, the four stages of life began not with physical birth, but with sacred thread ceremony. Thus the child could become the full member of the society only after investiture with the sacred thread. This was one of the important ceremonies or 'samskara' at all the stages of life of a man from his conception to death. There are some forty such 'samskaras'. Some important ones are: 'garbhadana' (conception), 'pumsavana' (male child), 'simantonayana' (safety), 'jatakarma' (birth ceremony) 'nishkramana' (showing the sun) 'annaprashana' (first feeding of solid food), 'chudakarman' (tonsure) 'upanayana' (investiture with sacred thread), 'samavartana' (end of the first stage) 'vivaha' (marriage), 'antyeshti' (the last rites) etc. All these 'samskaras' are prescribed for the three upper varnas and not for the shudras and untouchables. In fact even the women of the higher varnas were not entitled to a large number of the 'samskaras'.

Modern Trends and New Dimensions in Education

VALUE BASED EDUCATION

The word for 'Education' in many Indian languages is vidya. The root vid, from which vidya is derived, represents a homology meaning, 'to know' and 'to exist' from which words like vidwan are derived nanya pantha vidyate anyanaya. Thus, the word vidya translated into English means 'To learn is to exists', 'Existence is knowledge or learning to be'. Every living organism is prewired for the capacity to learn, to remember and experience. Therefore, neither can there be life without education nor can there be education divorced form life.

Vidya becomes a-vidya when education initiates a process where wisdom is lost in knowledge and knowledge in information, where materialism divorced from spiritualism seeks pleasure and comfort which distort perception of reality and where complete lack or distorted vision of inter-connectedness leads to alienation, isolation and anomie. When this happens, one's responsibility to oneself, to one's neighbours, country and the world becomes the premium.

Melvin J. Lasky, in his book Utopla and Revolution points out how utopia ends in revolution, revolution turns into dogma, dogma provokes heresay which in turn triggers revolution. This cycle enslaves the minds of intellectuals in such a way that they become victims of a new cycle. In the words of Nietzsche ".....life no longer resides in the

whole. The word becomes sovereign and leaps out of the page, and the page comes to life at the expense of whole, the whole is no longer a whole”.

In ancient India, life was measured in terms of fullness. Since fullness is such a concept that the product of all the four mathematical operations is fullness there was no space for emptiness, isolation and alienation when Vasudhaiva Kutumbakam and Yadum ure yavarum Ke $\frac{1}{2}$ ir “the world is my village and every person my kinsman”, how can one become lonely? When God is so pervasive that one can accept and surrender, reject and deny, or doubt and question, how can one escape God? Thus, God being a presence even in refusal and rejection, a person cannot, but be aware of interconnectedness, environmental, social and cosmic, value education must, therefore, begin with awareness of one’s connection with the immediate eco-culture, with fellow beings in society and with cosmic laws and forces which bind the particular with the universal. The creative interdependence among the three has become all the more essential in face of modern science and technology, which is based on the triple principles of self destructive competition, materialistic acquisition and emphasis on commodity values.

Rabindranath Tagore made a distinction between Mukhos ‘mask’ and Mukhashree ‘natural glow of the face’. That distinction is all the more important to remember today, when education tends to teach the use of mask rather than helping the natural inner glow to be reflected. School is not an extension of home, it has become either a substitution or rejection of home. The school does not treat the child as a resource. The child is treated as an object to be fashioned in the image of the elders by knowing textbook lessons doing social work pre-determined by curriculum makers. There is no effort at relating knowing and doing with being and becoming. That explains why modernity is not rooted in tradition and seeking of status and affluence through grossly improper measures of excellence gets precedence over professional excellence and idealism to fight against untruth, injustice and inequality or to seek the causes of all of these.

The denial of the child and the refusal to treat the child as an independent layer of social science concern finds expression in the rejection of the child’s home language in formal schooling. The teacher’s lack of cognition of the processes of language acquisition and processes of reading and writing on the one hand and the teacher’s belief that there is a single standard and correct form of language is responsible for this rejection. The dialects and the minority languages are also rejected on this count. Whether it is the child or the non-standard ‘dialect’ speaker, (s)he is not perceived as a human being, but a human becoming. With the waves of educational theories since World War II, the focus of concern has moved like a pendulum from the subject matter to the child and vice versa, but the medium has been taken for granted. That is why curricular reform has meant change in textbooks and methods of approaching them, but has seldom concerned itself with modes of language use, communicability of languages used in textbooks and linkage between home language and school language on the one hand

and first, second and further language on the other. Intellect, emotion and will are the basic faculties of human psyche and all three are integrally related to language development. By rejecting, suppressing, supplanting, or denigrating the mother tongue, not only creativity and innovativeness is curbed, but the resultant intellectual mediocrity and emotional sterility distorts the perception of life as an integrated whole. Take for example English medium education for Indian language speaking children. Lack of words in English for the familiar flowers, fruits, plants, trees, birds, beasts, rains, winds results in an imbalanced relation between the child and the environment. Neutralisation of the three dimensional kin terms by terms like uncle, aunt and cousin result in distortion in the perception of societal relations. Lack of transmission of the myths and other cultural symbols leads to the creation of cultural perception blind spots which affect appreciation of literature, plastic and performing arts and architecture which use such myths and symbols. All these lead to disintegration of society and culture. All these erodes the values the culture holds high.

Value is not mask to be worn, but is a glow permeating culture. It is manifested in the personal, societal, psychological, cultural, educational, economic and political behaviour. As the seminar on the New Education Policy and Moral Education convened by the Bharatiya Vidya Bhavan, rightly observes, “A society wallowing in luxury, conspicuous consumption, obscenity, dissipation, corruption, disparities, exploitation, rivalries, hatred and violence can never achieve any real progress howsoever vast and well planned the efforts of Government may be for its economic development.” It is unfortunate that neither social scientists nor agencies engaged in the study of development have undertaken trend measurement in respect of values among the youth both in school and out of school.

Let me share with you my perception of changes in values which have taken place during my life time. When I was young, money was not the defining criteria of success and respectability. Family tradition was a major factor. A Complex set of factors explained respectability of family and cut across castes and class. Joint family was still the norm and naturally family ties were cohesive and dense which absorbed a lot of socio-economic shock and tension. Marriage was arranged by the family, where informally the son or daughter's consent was obtained, family traditions were checked and the whole society participated in celebrations. There was a work ethic. A person who did not work was criticised. Today a person who works is criticised. We were then told that, early to bed and early to rise, keeps a man healthy, wealthy and wise. The present day youth addicted to late night TV shows or nocturnal violence wakes up late from an alcoholic slumber. For him/her to be lazy, corrupt and inefficient is to be healthy, wealthy and wise.

In my young days, a person who did not take loan was respected. Now a person who does not take loan is considered foolish and one who gets away with the loan is considered clever. There has been a movement towards “deauthoritisation” as a result of which there is lessening of obedience to authority of law, the police, the government, the

principal in an educational institution and the boss in a work situation. In my young days a sense of patriotism led us to movement of disobedience to an alien authority. The present trend appears to have no respect of the traditional concept of patriotism. The un-critical acceptance of Marxian dogma that labour has no country and the capitalist dogma that capital has no country has made the Indian intellectuals rootless and abettors of ruthless exploitation. Uncritical acceptance of liberalised sex modes without reference to the individual and social values has created a class of people who are neither Indian nor Western. The present day Indian youth is taught in a dilemma. Their values are not a synthesis between what is good in traditional and in modern values, but an antithesis of traditional values. It is primarily due to ignorance of tradition and its distorted representation as seen through Western eyes. Ideologically, they are opposed to a purely economic society bound in daily routines and which has no care for others, but practically they seek security in money which leads them to support an undering society. Neither organized religion nor organized schooling has been able to present a synthesis of life which would help deal with opposites and steer a course which would make a person to be Indian as well as universal without losing many identities which link the local with the universal.

In the past Hinduism was a cover term for all those who practised their own religion, without disrespect for other religions. Thus Hinduism encompasses monism, dualism, triad, transcendentalism, immanence, worship of 23 millions of gods including nature, idea and object gods as well as those challenging the existence of god. This was not a religion in the narrow sense of the term, but Dharma which binds together apparently disparate elements. In a single thread by which one perceived one self as a part of cosmic whole. Unless the core value of respect for different is emphasized in all disciplines and in all modes of behaviour, the thirst of Western values incompatible with the India, will continue to erode the value base of the Indian society and threaten its very existence. At the time of doubt and despair, Buddha's message to his discipline, was Attadipo bhava. This has been variously translated as "make yourself a light" and 'look inwards of light', At a time when we are uncertain and afraid of our own identities and confused about the road to take, there is greater need to emphasize an integrated life and value based education. Therefore, all of us, young and old, who are victims of identity crisis, disbelief, dismay, and resultant paralysis of thought and action join together in involving the cosmic principle.

NEW APPROACHES AND TOPICS IN HIGHER EDUCATION

Genuinely interdisciplinary research spawns the new disciplines of tomorrow and is more likely to be relevant to industrial opportunities and the resolution of industrial or societal problems. One challenge is how to stimulate interdisciplinary research pro-

actively when institutions are normally organised on discipline-based departments and when much external evaluation reinforces this, by itself reviewing disciplines and implicitly discouraging interdisciplinary connections. The rigid disciplinary boundaries between the hard sciences and the social sciences and the humanities that limited the comprehension of the fundamental processes of nature and society are already breaking down in some fields.

There must be a stimulus to the creation of interdisciplinary programmes and resources must be ensured for collaborative research among different disciplines and involving groups around thematic projects. A further issue is the extent to which higher education institutions perceive themselves as having the optimum range of disciplines for their likely future needs. An individual institution may seek solution to its problem through strategic alliances with other institutions or with companies, through amalgamation with other institutions, or by growing its new specialisms itself. Many experts see the need for renewal of teaching and learning approaches and contents. They emphasise that higher education institutions must be more responsive in meeting the needs of employers and adapting to the generation of new knowledge in the various academic disciplines. It is important and urgent to carry out a series of case studies in the different regions on regional priorities, in which higher education institutions should play an important role.

The exercise prepared by the CRE for the Palermo European Regional Meeting with 20 institutional case studies from European higher education institutions is worthwhile looking into. In the case of Africa, case studies proposed in the Action Plan (1997) include: the type of leadership to be promoted, strategic management and planning, systemic interactions between primary, secondary, tertiary and continuing education, revision of programmes of education and training, the relative importance and feasibility of face-to-face and distance teaching programmes, strategies for ensuring improved participation of women in education and in decision making bodies, town and country planning, measures against the security problems of Africa (such as poverty, displaced populations, the trauma of war).

In connection with relevance, the Action Plan for Latin America proposes to elaborate projects on the problems of widening access to higher education, design of integrated educational systems and life-long education for all throughout life. Appropriate emphasis needs to be placed on renewal of curriculum; new approaches to both classroom and distance education curriculum, which traditionally focused on academic disciplines in arts and sciences and training for elite professions, now put considerable emphasis on applied science and technology, business and management studies and professional training in such fields as engineering, accounting and computer science. Frequently new programmes are begun within the framework of the international UNITWIN/UNESCO Chairs and form students on different subjects, such as the resolution of conflicts, regional development and the social integration of marginalised groups,

including refugees who constitute a grave problem in the African region. These formal initiatives must be complemented by community sensitivity activities such as public conferences and the creation of relevant groups such as those of NGOs and diplomatic and cooperation staff.

To overcome the limitations of higher education institutions, in the European university case studies some strategic issues and tensions as well as directions for action are proposed:

- Limited development of new fields of multi-disciplinary study from the stimulation of synergy between existing foci (Basel: ecological studies), especially where these correspond to market opportunities. This is seen as a means of sustaining critical masses in key subject areas.
- The addition of new disciplines perceived as critical to the university's contribution to society -a form of horizontal diversification (Jagiellonian University of Cracow, University College of South Stockholm).
- Accessing wider learning networks, through strategic alliances with other universities with complementary disciplines (Catalonia) or extensive use of Internet arrangements (Copenhagen Business School).
- The possibility of institutional mergers over the next decade to facilitate the creation of critical mass, as a natural regional consequence of the foregoing. The Australian experience has already been mentioned above.
- Accessing the knowledge base of industry, especially where professional/vocational training is a core element in institutional mission (Pau, Porto, Sheffield Hallam).

ADVANCEMENT OF KNOWLEDGE BY RESEARCH

To fulfil its mission to society, ideally every system of higher education should have the capacity to carry out research and have staff who are actively involved in research activities. At a minimum, this is desirable to support regular teaching activities and the staff development needs of academics, as well as to provide expertise in accessing the international body of research knowledge and assisting in technology transfer to local industry. At the same time, in many countries it is now recognised widely that not all higher education institutions need to be involved in research and not all academics need to be undertaking research activities on a regular basis in order to provide high quality teaching, especially at more junior levels. The World Bank has advised developing and newly industrialised countries, in the interests of economic efficiency, to strictly limit the establishment of new research universities and the number of students enrolled in them and to provide cheaper and more cost effective alternatives, such as junior colleges, technological institutes and short cycle institutions, to cater for a large proportion of the student population. Because of pressures on resources, in some countries research funding has been reduced over the past decade, but in others research funding

has increased appreciably and special efforts have been made to establish new research centres and build additional links between universities and industry.

A major policy dilemma for higher education systems and research universities is how scarce research resources should be distributed and what mechanisms should be employed to do this. A common trend is the allocation of research funds to institutions and individuals on a competitive basis depending on the research funds they have already attracted, publication output and the number of research higher degree completions. There is a policy debate about whether research and research infrastructure funds should be concentrated to a greater degree in order to develop a small group of stronger internationally recognised research universities, or whether the current method of competition will ensure a sufficient degree of concentration. Another policy dilemma relates to the balance between basic and applied research.

While effective applied research needs to be supported by basic research, the actual mix of basic and applied within a higher education system varies greatly, depending on government and institutional priorities and the respective roles in research of higher education and industry within the country. In China, the policy of the State Education Commission is 'that scientific research should be oriented to the economy, that training people and developing science and technology should be combined and complement each other and that different types of institutions should conduct scientific and technical activities in different ways'. It seems that there is a growing feeling that the distinction between basic, strategic and applied research is breaking down in the higher education institution which is genuinely oriented towards economic regeneration and societal modernisation. How this synthesis is accelerated and planned for is a genuine agenda item for the next century?

There is an exponential growth in knowledge, with very little direct contribution from some regions, among them the African region, as a consequence of insufficient attention to and insufficient resources for research, as well as lack of a long-term vision in the planning and management of teaching and research activities. Institutions of higher education in Africa should make special efforts to develop scientific and technological programmes to help meet the demands of the accelerated development of new technologies, especially new information and communication technologies.

These programmes should be supported by intensive research activities, from which will emerge the critical mass of the expertise needed for the region's development as it faces the pressures of globalisation.

Existing potentials of information and communications technologies should be boosted to give rise to virtual universities, which could considerably improve access, while at the same time proving world-class educational resources. Research should be made to bear a closer relation to the needs of African societies, so that basic research can be more closely linked with applied and development-oriented research stressing genuine partnerships with public and private institutions and the civil society. This

would be one way of ensuring the active involvement of higher institutions in societal development efforts.

Responsibilities to Other Educational Levels

One area where there is consensus that universities can make a major contribution that will have direct benefits in the longer term for higher education is direct involvement in the training of school teachers and staff development activities for teachers. The University of Campinas in Brazil is a good example of this, with a large responsibility in teacher training at the national level parallel to its concentration on research. Strengthening the quality of teaching in schools will lead in time to better prepared university students, who can become agents of their own education; promoting socio-educational research into such problems as early school drop-outs and students' repeating courses; and ensuring its contribution to the design of State policies in the field of education. It is widely recognised that higher education has an important contribution to make in community service and in assisting other education sectors in society. However, in some contexts there is some measure of disappointment with performance, especially with that of more traditional universities. This is unfortunate, since close links with the community and the other education sectors can do a great deal for universities in building wide community and political support, achieving renewal of the curriculum and sometimes generating new forms of financial support. The last point is particularly important since in the current environment of financial constraint there is often a strong tendency for universities to decrease or abandon much of their more traditional community service.

THE POTENTIAL AND THE CHALLENGE OF TECHNOLOGY

The rapid breakthroughs in new information and communication technologies will further change the way knowledge is developed, acquired and delivered.

It is also important to note that the new technologies offer opportunities to innovate on course content and teaching methods and to widen access to higher learning.

However, it should be borne in mind that new information technology does not reduce the need for teachers but changes their role in relation to the learning process and that the continuous dialogue that converts information into knowledge and understanding becomes fundamental. Higher educational institutions should lead in drawing on the advantages and potential of new information and communication technologies, ensuring quality and maintaining high standards for education practices and outcomes in a spirit of openness, equity and international cooperation by:

- a. Engaging in networks, technology transfer, capacity-building, developing teaching materials and sharing experience of their application in teaching, training and research, making knowledge accessible to all.

- b. Creating new learning environments, ranging from distance education facilities to complete virtual higher education institutions and systems, capable of bridging distances and developing high-quality systems of education, thus serving social and economic advancement and democratization as well as other relevant priorities of society, while ensuring that these virtual education facilities, based on regional, continental or global networks, function in a way that respects cultural and social identities.
- c. Noting that, in making full use of information and communication technology (ICT) for educational purposes, particular attention should be paid to removing the grave inequalities which exist among and also within the countries of the world with regard to access to new information and communication technologies and to the production of the corresponding resources.
- d. Adapting ICT to national, regional and local needs and securing technical, educational, management and institutional systems to sustain it.
- e. Facilitating, through international cooperation, the identification of the objectives and interests of all countries, particularly the developing countries, equitable access and the strengthening of infrastructures in this field and the dissemination of such technology throughout the society.
- f. Closely following the evolution of the 'knowledge society' in order to ensure high quality and equitable regulations for access to prevail.
- g. Taking the new possibilities created by the use of ICTs into account, while realizing that it is, above all, institutions of higher education that are using ICTs in order to modernize their work and not ICTs transforming institutions of higher education from real to virtual institutions.

Strengthening higher education management and financing:

- a. The management and financing of higher education require the development of appropriate planning and policy-analysis capacities and strategies, based on partnerships established between higher education institutions and state and national planning and coordination bodies, so as to secure appropriately streamlined management and the cost-effective use of resources. Higher education institutions should adopt forward-looking management practices that respond to the needs of their environments. Managers in higher education must be responsive, competent and able to evaluate regularly, by internal and external mechanisms, the effectiveness of procedures and administrative rules.
- b. Higher education institutions must be given autonomy to manage their internal affairs, but with this autonomy must come clear and transparent accountability to the government, parliament, students and the wider society.
- c. The ultimate goal of management should be to enhance the institutional mission by ensuring high-quality teaching, training and research and services to the community. This objective requires governance that combines social vision,

including understanding of global issues, with efficient managerial skills. Leadership in higher education is thus a major social responsibility and can be significantly strengthened through dialogue with all stakeholders, especially teachers and students, in higher education. The participation of teaching faculty in the governing bodies of higher education institutions should be taken into account, within the framework of current institutional arrangements, bearing in mind the need to keep the size of these bodies within reasonable bounds.

- d. The promotion of North-South cooperation to ensure the necessary financing for strengthening higher education in the developing countries is essential.

FROM 'BRAIN DRAIN' TO 'BRAIN GAIN'

The 'brain drain' has yet to be stemmed, since it continues to deprive the developing countries and those in transition, of the high-level expertise necessary to accelerate their socio-economic progress. International cooperation schemes should be based on long-term partnerships between institutions in the South and the North and also promote South-South cooperation. Priority should be given to training programmes in the developing countries, in centres of excellence forming regional and international networks, with short periods of specialized and intensive study abroad. Consideration should be given to creating an environment conducive to attracting and retaining skilled human capital, either through national policies or international arrangements to facilitate the return—permanent or temporary—of highly trained scholars and researchers to their countries of origin. At the same time, efforts must be directed towards a process of 'brain gain' through collaboration programmes that, by virtue of their international dimension, enhance the building and strengthening of institutions and facilitate full use of endogenous capacities. Experience gained through the UNITWIN/UNESCO Chairs Programme and the principles enshrined in the regional conventions on the recognition of degrees and diplomas in higher education are of particular importance in this respect.

Partnership and Alliances

Partnership and alliances amongst stakeholders—national and institutional policy-makers, teaching and related staff, researchers and students and administrative and technical personnel in institutions of higher education, the world of work, community groups—is a powerful force in managing change. Also, non-governmental organizations are key actors in this process. Henceforth, partnership, based on common interest, mutual respect and credibility, should be a prime matrix for renewal in higher education.

We, the participants in the World Conference on Higher Education, adopt this Declaration and reaffirm the right of all people to education and the right of access to higher education based on individual merit and capacity. We pledge to act together within the frame of our individual and collective responsibilities, by taking all necessary measures in order to realize the principles concerning higher education contained in

the Universal Declaration of Human Rights and in the Convention against Discrimination in Education.

We solemnly reaffirm our commitment to peace. To that end, we are determined to accord high priority to education for peace and to participate in the celebration of the International Year for the Culture of Peace in the year 2000. We adopt, therefore, this World Declaration on Higher Education for the Twenty-First Century: Vision and Action. To achieve the goals set forth in this Declaration and, in particular, for immediate action, we agree on the following Framework for Priority Action for Change and Development of Higher Education.

FINANCING OF HIGHER EDUCATION AS A PUBLIC SERVICE

The funding of higher education requires both public and private resources. The role of the state remains essential in this regard.

- a. The diversification of funding sources reflects the support that society provides to higher education and must be further strengthened to ensure the development of higher education, increase its efficiency and maintain its quality and relevance. Public support for higher education and research remains essential to ensure a balanced achievement of educational and social missions.
- b. Society as a whole must support education at all levels, including higher education, given its role in promoting sustainable economic, social and cultural development. Mobilization for this purpose depends on public awareness and involvement of the public and private sectors of the economy, parliaments, the media, governmental and non-governmental organizations, students as well as institutions, families and all the social actors involved with higher education.

Sharing knowledge and know-how across borders and continents:

- a. The principle of solidarity and true partnership amongst higher education institutions worldwide is crucial for education and training in all fields that encourage an understanding of global issues, the role of democratic governance and skilled human resources in their resolution and the need for living together with different cultures and values. The practice of multilingualism, faculty and student exchange programmes and institutional linkage to promote intellectual and scientific cooperation should be an integral part of all higher education systems.
- b. The principles of international cooperation based on solidarity, recognition and mutual support, true partnership that equitably serves the interests of the partners and the value of sharing knowledge and know-how across borders should govern relationships among higher education institutions in both developed and developing countries and should benefit the least developed countries in particular. Consideration should be given to the need for safeguarding higher education institutional capacities in regions suffering from conflict or natural

disasters. Consequently, an international dimension should permeate the curriculum and the teaching and learning processes.

- c. Regional and international normative instruments for the recognition of studies should be ratified and implemented, including certification of the skills, competences and abilities of graduates, making it easier for students to change courses, in order to facilitate mobility within and between national systems.

TECHNICAL DESCRIPTION OF COMPUTER EDUCATION

A computer is an electronic device that accepts data, performs operations on that data in sequence (decided by a programme) and output the results. Computers are termed according to their size and thus we come across terms like mainframe computers, mini-computers and micro computers. Besides size, specifications of the electronic devices (that differ from computer to computer) are also used to identify computer types. The specification includes details about the amount and type of storage, the capabilities of the central processing unit, and the type and nature of the peripheral equipment (such as disc storage) that can be connected to the computer. The advent of microcomputers in the 1970s at comparatively cheaper price made them accessible to many people and made the possibility of using them for educational purpose a reality. The basic micro computer system consists of an input device (*e.g.* the keyboard), the central processing unit of the computer and storage devices (*e.g.*, cassette tape player, floppy disc device) and the output device (*e.g.*, and TV) display unit, or monitor and printing terminals. The actual equipments of the computer system is called the hardware, while the term software refers to the instructions or the programmes that the computer carries out.

APPLICATIONS OF COMPUTER

Two types of applications of computer are popular in India. These are as:

1. Administrative Applications of Computers.
 2. Instructional Situations of Computer Application.
1. Administrative Applications of Computers
 - a. Budgeting and monitoring of finances.
 - b. Keeping student's records.
 - c. Maintaining Statistics.
 - d. Student Billing.
 - e. Preparation of payroll, bills and date ledger.
 - f. Mailing of score.
 - g. Monitoring materials production register and so on.
 2. Instructional Situations of Computer Application
 - i. Pre-testing of student's knowledge on enrolment.

- ii. Planning and printing individualized programmes
- iii. Monitoring students progress and
- iv. Compiling test and scores etc. A number of such data files can be stored in the computer.

ADVANTAGES OF COMPUTER

The computer applications described above have certain remarkable advantages over the traditional 'print' medium of instruction. These advantages can be briefly pointed out in the form of the following hints:

1. The novelty of working with a computer raises the learner motivation;
2. Colour, music and animated graphics can realism and appeal to drilling exercises, laboratory activities, simulations etc.
3. High-speed personalized response to learner actions yield a high rate of reinforcement;
4. Memory capacity allows student's past performance to be recorded and used in planning the subsequent steps;
5. Step, can present a more positive and effective climate, especially for slow learner.
6. The record keeping ability of the computer makes individualized instructional feasible individual prescriptions can be prepared for all students (particularly mainstreamed special students) and their progress can be monitored; and
7. The teacher's span of controls enlarged, as more information is put easily at the teacher's disposal, helping to keep control close to.

SCOPES OF COMPUTER

Certain characteristics of computer-interaction can make computers well suited for distance learning. The features listed below make the prospects of computer-use look more promising:

1. Access to experts and respected peers
2. One-to-one and one-many communications
3. Active learner participation
4. Linking of new learning to concrete on-the-job problems
5. Follow up, feedback and implementation support from peers or experts
6. Self-direction-control over stop or start, time, pace and place of learning or communication activity.

LIMITATIONS OF COMPUTER

In spite of the advantages and the promising prospects, computers do differ from some disadvantages too. Some of the major limitations are:

1. Despite the dramatic reduction in cost of computer and computer use,

computerized instruction is still relatively expensive. Careful consideration must be given to the costs and benefits of computers in education. Maintenance can also be a problem especially if the equipment is subjected to heavy use.

2. The design and production of computer specifically for instructional purposes has lagged being design and production for other purposes.
3. There is a lack of high quality direct-instruction matter for use with computers, especially for the micro-computers. Their use with computers, especially for the micro-computers. There is also a compatibility problem-often the software developed for one computer system cannot be used with another. The ease with which the software-especially cassettes can be duplicate without permission has inhibited commercial publishers and private entrepreneurs from producing and marketing instructional software (Heinich et al. 1982: 318).

TELECONFERENCING

Educational teleconferencing can be a valuable medium for interactive group communication by means of a two-way broadcast. Three main types of teleconferencing have been identified.

- i. Audio teleconferencing
- ii. Video teleconferencing and
- iii. Computer teleconferencing. We discuss audio teleconferencing being the most commonly used technique in distance education institution in detail.

Till the early 1980 audio teleconferencing was at an experienced stage and was used occasionally. But during the past few years it grew to be a routing part of the delivery strategies of some distance education institutions. It was found that with the use of means, the communication costs could be reduced and the quality of service to distant learners could be improved without compromising the academic integrity of the programmes the flexibility, and the low capital and operating costs of studio teleconferencing make it a communication means of special interest of institutions serving students who live in small and widely separated communities. Today, the dependence on this technology is accelerating. The interest in using audio and related visual data that can be transmitted over regular telephone lines appears to be related to the growth of education.

TECHNICAL DESCRIPTION OF TELECONFERENCING

Teleconferencing is an electronic means which can bring together three or four people in two or more locations to discuss or share the use of two-way and one-way video, both full motion and slow scan, electronic black boards, facsimile, computer graphic, radio, satellite and Videotex. However, the most essential part of all forums of teleconferencing is a good quality audio to help immediate interaction among the

participates for information exchange. Audio teleconferencing requires a multi-telephone line electronic switch or interconnection device called a bridge to which the user can attach a wide variety of data transmission devices and telephone. The normal practice is to connect device per line into the bridge. Audio equipments used with the bridge are the usual handsets, speaker phones, radiotelephones, and microphone speaker units (often called 'conveners').

Audio teleconferencing uses regular telephone lines provided by local public telephone companies. If the quality of standard business or household line is good enough virtually any line could be used.

EFFECTIVE CONTENT ENVIRONMENT

Occasional users normally purchase their service from the, local telephone company, teleconferencing consultants or major user that is willing to sell available time on its system. Any specialized terminal equipment is usually rented for including all necessary equipment long distance tool charge, and user training.

The costs for starting a university or college-based private audio teleconferencing system are not large if the local telephone system has:

- i. A relatively quiet line,
- ii. Ready accessibility, and
- iii. Acceptable local and long distance rates.

ADVANTAGES OF TELECONFERENCING

There have been a few studies on the effectiveness of audio teleconferencing from the point of view of students learning. These studies show that the telephone was as effective medium as face-to-face teaching. A study conducted in Canada (Robertson, 1986) found that upper level undergraduate statistics students taught on teleconference did as well as better than most campus based students and had a drop-out rate of zero. Traditional correspondence institution on the other hand, average 40-60 percent dropout rates. Satisfaction among the students was recorded high and remote learners through the course did not lack rigour or suitable contact.

During the 1984 International symposium on Teleconferencing there was evidence for strong support for educational audio teleconferencing. The support was mainly due to the consideration of the following advantages:

- i. *Effective support from remote learners:* Teleconferencing can be very useful when most of the potential students are widely scattered among communities that are far apart and when each centre has them learner population, say less than ten students, on a given course or programme.
- ii. *Cost effectiveness:* The cost for starting and operating an audio teleconferencing system is relatively low in comparison with other available methods of serving remote learners.

- iii. *Flexible system*: The system used can be adjusted quickly to serve large or small group.
- iv. *Familiar instruction mode*: The mode of instruction is simpler to that seminar with the instructor being in-charge of the discussion and able to stimulate multiplication interaction.
- v. *Easy scheduling adjustment*: A scheduling adjustment can be made almost as readily as for no-campus classrooms.
- vi. *Multi-vocational access control*: Access to the instruction or programmes can be controlled through a limited number of off-campus centers.
- vii. *Multi-vocational access control*: The quality of the instructional materials can be maintained high because of the need for careful and easy preparation.
- viii. *Immediate feedback*: The teleconferencing system provides the facility to the learners for immediate feedback and to convey their reactions to the tutors.

EXPERIMENTS IN TELECONFERENCING

The potential of teleconferencing as a viable educational medium is being experimented in institutions the world over. The UKOU has been experimenting with its system of teleconferencing called CYCLOPS. As an interactive audio-visual aid the system has helped the learners and the tutors of the experimental groups to write electronically to each other over the telephone and have their words and pictures reproduced at the other end on a TV screen. It was found during the pilot stage of this experiment that nine different groups of learner scattered hundreds of miles apart could be tutored simultaneously sharing a common picture display and linked by voice through loud speaking telephones. Experiments on teleconferencing are being conducted in India too. The Space Applications Center, Ahmedabad, has undertaken some experiments in this direction. The experiments oriented towards JI"ocess and educational purposes have been conducted using satellite and/or terrestrial circuits, and one-way TV and two way audio equipments.

The experiments abroad and within point of feasibility of using teleconferencing as an effective, viable device to promote education among widely scattered potential learners on low enrolment course.

VIDEODISC AS NON-PRINT MEDIA

Videodisc is comparatively a newer medium that has widened the scope of use of the television set. Videodisc players are available in two models:

- i. Consumer model, and
- ii. Educational industrial model.

The consumer model is basically a playback medium and is mostly used in homes. The educational/industrial model more versatile, it can provide interaction-individualized instruction. Sometimes, it is rendered more capable by means of a built-in

microprocessor. Videodisc can record materials from a variety of media, including film slides, audio and printed text. This makes rich the opportunities for individualized instruction for students learning at a distance.

Technology of Videodisc

A videodisc system is made up of a video disc on which the information is stored), a videodisc player television set. Videodisc is used as information storage medium. A videodisc rotates on the turntable of the videodisc player at a high speed and is read by a stylus or laser. Microscopic pits, of varying length and density, contain information on the disc. The pits are in tracks or grooves. The information read from these pits is displayed on a regular television set.

ADVANTAGES OF VIDEO DISC

Use of videodisc is inexpensive, when the discs are mass-produced. Besides, the discs require little storage room. These two factors make it possible to have an extensive library of courses on small discs, the discs also provide easy accessibility and retrieval of the material of our choice. The videodisc can be used to increase learner productivity through:

1. More information in attractive presentation modes.
2. Immediate access (forward and reverse) to different parts of the disc.
3. Facility for repetition to achieve mastery,
4. Higher motivation through stimulations of actual events in living colours,
5. Provision for immediate 'feedback',
6. Automatic frame stop, and
7. Provision for students.

Limitations of Videodisc

However, we have to bear in mind that the information on disc is permanently etched into the surface of the disc. One cannot record new material into a finished disc as one can with videotape.

APPLICATION OF VIDEO DISC

There are two major types of videodisc systems: capacitance and optical.

- i. **Capacitance Videodisc**—Capacitance is a technical term for the electrical charge that can be collected on a plate. It may vary from one to another. Thus, when the stylus rides over the disc (which contains the pits covered with a plastic coating to even out the surface), the distance between the disc and the stylus varies, causing changes in capacitance to occur. These changes are translated into the video and audio signals which are displayed by the television set.
- ii. **Optical Videodisc**—The optical videodisc is more useful than the capacitance

videodisc. The potential of the optical videodisc lies in its enormous storage capacity and optical disc can store 108,000 tracks (54,000 on each side). Each of the 54,000 frames on a side is addressable with a five-digit number and can be randomly accessed in 5 seconds or less. The entire Encyclopaedia Britannica could be stored on a single videodisc with room to spare.

In addition to the information stored onto the disc, there are two channels of accessible audio, the accessibility to the dual audio channels allows for use of a second language, *i.e.*, for the use of one language with the content oriented for a audience of a different linguistic background. It can allow for the addition of stereo sound too.

The videodisc system is capable of fast forward, fast reverse, slow motion and stop frame movement. Individual images can be displayed indefinitely in freeze-frame mode without a wear on the disc since there is no mechanical contact. But sound cannot be used in the freeze-frame mode.

- iii. *Intelligent Videodisc*: The intelligent videodisc system (videodisc connected with computer), which is particularly useful for educations. Consists of a videodisc player under the direct control of a computer. This system permits interactive instruction for individualized learning. The videodisc is used simply as a storage medium for both still frames and video. The external computer programme may be used interactively to compare a student's response with a programmed correct response and can automatically move to a resigned number of material depending on the response. Thus by combining two powerful technologies the computer and the television the videodisc can offer more individualized and interactive instruction for learners.

VIDEOTEX AS NON-PRINT MEDIA

Video is a system for providing an interactive information retrieval service. This particular technological development is one further step of the 'information revolution' which many industrialized countries are now experiencing. There are three Videotex prevalent today: British (Prestal), French (Antiope) and Canadian (Telidon).

Videotex allows the home television to function like a computer terminal that retrieves text information and graphics from a remote data. Thus, these technologies bring in the 'power into home at a reasonable cost.

Application and Advantages of Videotex

Most Videotex application is currently at initial phases, number of applications of Videotex has been tested which as work game teaching as second language, instruction to defeat etc. In distance education Videotex can be used for the transmission and to-way interaction on general or specific information courses.

A Videotex system is very useful in disseminating general information about courses and programmes available, in raising the level of awareness of potential learners about education opportunities available. The system with a substantial database can be very helpful to learners in remote areas as they could have access to the kind and the volume of information currently available to learners in areas with large libraries.

Limitations of Videotex

However, while the Videotex system provides graphic capabilities, there is no provision for including audio. So Videotex would work better if it is used as a part of a multi-media system in providing instructional material. As it is still at an experimental stage, its cost effectiveness could not be determined. Thus Videotex system provides means for delivering instructional material to those individual who have chose to learn at a distance.

However, there are, at present few instructional programmes, which are being delivered by this method. To date, most educational applications of Videotex and teletex (a system using a similar technology as that of Videotex and is discussed below) systems have as their information base is catalogue listing the courses available or services provided by the institution.

Technology of Videotex

The basic Videotex system consists of keypad or keyboard a TV display unit, a decoder, a telecommunication link and a central computer containing a database. Videotex uses television and telephone in order to provides access to the computerized information. The keypad, a small frame with number-buttons on it, is used to request pages of required information from a remote computer.

The television is connected to the computer by telephone and it is adapted by means of a decoder to display the recordings on Videotex when it is played. Thus the home television set is turned into a terminal. As a television set is already present in most of the homes of developed countries this device promises a wide range of social and educational utility.

The decoder can be wired directly into the television set during the manufacturing process or can be fitted on externally at a later stage. The decoder allows a TV set to display text and graphics from digital transmitted down the telephone line and to transmit back to information entered through keypad. A modern (modulator-de-modulator) enables the computer and the decoder to communicate in digital form over telephone lines designed for the speech.

Videotex offers advantages over other time-bound communication media like radio, television or telephone because it is free from a time schedule. If a keyboard could be added to the user terminal, a Videotex can offer computer assisted instruction also. Two-way interaction between the teacher and the taught is also possible with

alphanumeric keyboard (*i.e.*, a keyboard with number-buttons). This interaction resembles an electronic main and learner can address questions to the tutor and receive immediate feedback. It is definitely faster than the post and has an advantage over telephone because this kind of interaction does not have to wait for a mutually convenient time for discussion. In the long run it may also prove to be cost-effective.

TELETEXT—A SYSTEM RELATED TO VIDEOTEX

There are two systems that deliver words and graphics to a modified television set—one, Videotex which uses the telephone to connect the user to the database; two the broadcast version known as Teletext in which pages of information are broadcast in a cycle (usually 400 pages) and when the requested page comes up in the cycle, it is grabbed, and displayed on the screen.

With the Videotex the amount of material available for retrieval is limited only by the storage of the host computer. This is in contrast to the Teletext, which has limit on the amount of material that can be transmitted.

TELETEXT APPLICATION IN INDIA

The Teletext service in India was formally inaugurated by the Doordarshan, Delhi on November 14, 1985. It is known as 'INTEXT' (Meaning Indian teletex). Teletext uses the medium television for transmitting information. The datas are organized into pages, which contain text and graphic, symbols. Six colours can be used in creating these graphics. The information is poled and transmitted on a few pre-determined lines in vertical 'blanking' interval of television signals. The information is in the form of magazines, each of which contains about 100 pages. The details of contents of the magazine are given on the first page.

The news items, sports events, financial trends, timings of arrival and departure of important trains, weather forecast, city engagements. AIR and TV programmes to be telecast, etc., are given through Teletext service. The information is collected through hot-lines connecting Teletext information-room to:

- i. The meteorological department
- ii. The northern railway central enquiry,
- iii. The Indian Airlines enquiry (for international arrivals) and
- iv. The Air traffic control (for interactional arrivals)

The using of existing Television broadcast, additional information can be transmitted in the form of Teletext service. Efforts are being made to provide the Teletext service on a large scale. In addition, work has been undertaken to develop necessary software for creating pages with characters from different Indian languages. Non-effort has so far been made to use teletex service for educational purpose. In advanced countries also the educational application of telex text has been largely on experimental basis so far.

USE OF COMPUTER IN INDIA

Computers have been put to a widespread use for educational purpose in the advanced countries. Especially with the advance of the relatively cheap micro-computer, the technology has become more popular with the instruction (CAI) has already been used in Canada and at the Open University in Britain to provide diagnostic feedback remedial help; and revision aids in 1986. The projects in the area of Remedial English in Indian education are a new phenomenon but we have made a beginning in using computers in our school system.

DEMOGRAPHICS OF ONLINE INSTRUCTORS

Sixty-six per cent of the survey respondents held teaching positions (professors, instructors, or lecturers), while nearly one-fourth were administrators or instructional designers. Respondents represented institutions of various types: approximately half were employed by public, four-year colleges or universities; 23 per cent by community colleges or vocational institutes; and 16 per cent by private postsecondary institutions.

A large majority (87 per cent) said their institutions offer online courses, and about 70 per cent of them had taught online courses. Respondents' experience with online teaching varied from none to more than 10 years. Although not every respondent had online teaching experience, more than 95 per cent had experience integrating computer or Web technology into their face-to-face teaching. Survey results show that women appear to be teaching online in far greater numbers than just a few years ago. In fact, more than half of the respondents (53 per cent) were women. Such findings were surprising because a similar study conducted a few years earlier was dominated by male instructors who were full professors at tier-one universities. Perhaps female instructors had become more comfortable teaching and sharing activities online during the few years that elapsed between surveys, or perhaps support for instructors had improved on college campuses, or both.

EMERGING TECHNOLOGY

When asked about several emerging technologies for online education, 27 per cent of respondents predicted that use of course management systems (CMSs) would increase most drastically in the next five years. Those surveyed also said that video streaming, online testing and exam tools, and learning object libraries would find significantly greater use on campus during this time. Between 5 and 10 per cent of respondents expected to see increases in asynchronous discussion tools, videoconferencing, synchronous presentation tools, and online testing.

The survey also asked what technology would most impact the delivery of online learning during the next five years. Respondents could select one of 14 key technologies. About 18 per cent of respondents predicted that reusable content objects and wireless

technologies would have the most significant impact. Smaller percentages (from 7 to almost 14 per cent) selected peer-to-peer collaboration, digital libraries, simulations and games, assistive technologies, and digital portfolios. In contrast, less than 5 per cent predicted that e-books, intelligent agents, Tablet PCs, virtual worlds, language support, and wearable technologies would have significant impact on the delivery of online learning.

These findings seem to reflect the perceived importance of online technologies for sharing and using pre-existing content. Additionally, respondents predicted that advances in Internet technology (for example, greatly extended bandwidth and wireless Internet connections) are likely to increase the use of multimedia and interactive simulations or games in online learning during the next five to 10 years. Only about one in 10, however, predicted that advances in Internet technology would enhance videoconferencing or international collaboration, and just one in 16 thought it might offer greater chances to interact with field experts or practitioners.

Again, the focus was on enhancing content and associated content delivery, not on the social interactions, cross-cultural exchanges, or new feedback channels that wider bandwidth could offer. Such responses indicate that respondents still see learning as content-driven, not based on social interactions and distributed intelligence. The emphasis remains on a knowledge-transmission approach to education, not one rich in peer feedback, online mentoring, or cognitive apprenticeship.

ENORMOUS LEARNER DEMANDS

Our study revealed a number of trends related to areas of growth in online education, future needs for online instructors, and the dominance of online versus face-to-face instruction.

Growth of Online Programmes/Degrees

Comparing current online offerings and projected future online offerings at respondents' institutions yields predictions about the areas of growth in online programmes and degrees. Most respondents expected considerable growth in online certification and recertification programmes in the next few years, as well as in associate's degrees. Yet, our survey respondents predicted little growth in the number of institutions that offer online master's or doctoral programmes in the future. Although more than half of the respondents (54 per cent) expected that their institutions would offer online master's or doctoral programmes in the coming years, almost the same number of respondents (53 per cent) reported that their institutions were presently offering online master's or doctoral programmes. In contrast, respondents predicted that certification and recertification programmes would see 10–20 per cent growth from present offerings. Such responses indicate that higher education institutions might be wise to explore certificate and shortprogram offerings rather than full degree programmes.

Online Instructors' Readiness

Will online instructors be ready to meet the challenges brought by the projected increases in learner demands for online education? About half of the respondents predicted that monetary support for and pedagogical competency of online instructors would most significantly affect the success of their online programmes. In addition, instructors' technical competency was the third most pressing factor. Nevertheless, pedagogical skill was deemed more important than technological skill for effective online teaching. With regard to the needs for pedagogical competency of online instructors, a majority of the respondents expected that online instructors would typically have received some sort of training in online teaching either internally or externally by the year 2010.

The Rise of Blended Learning

The survey asked respondents for their predictions related to the growth of online education in the next few years. Respondents indicated that more emphasis is expected on blended learning—instruction that combines face-to-face with online offerings—than on fully online courses. Those surveyed predicted a distinct shift from about onequarter of classes being blended today to perhaps the vast majority of courses having some Web component by the end of the decade.

ENHANCED PEDAGOGY

Although the use of CMSs in higher education has increased rapidly and is likely the foundation for the rapid increase in the number of online learners during the past decade, some researchers argue that CMSs are promoted as ways to manage learners rather than to promote rich, interactive experiences. As a result, enhancing pedagogy is perhaps the most important factor in navigating the perfect e-storm. In the present study, respondents made predictions about the quality of online education in the near future and about how online courses would be taught and evaluated.

GROUP WORK IN ONLINE EDUCATION

For the most part, he had favourable experiences with group work in an on-line environment, again a surprise to him. He assumed that the lack of face-to-face contact and non-verbal cues would limit the “human touch” of working cooperatively with people in groups.

My experience was just the opposite, an observation made by researchers in the computer-mediated communication (CMC) field. As pointed out by Rob Higgins (1991):

Without encompassing the full range of human sensory and expressive capabilities, text-based interaction is often thought to be an impersonal medium devoid of social context cues and non-verbal communication. Experience and research, however, are demonstrating that socio-emotional content can be communicated in text. Steinfeld

(1986) states that: "Evidence continues to mount showing that CMC will be used for emotional interaction. People seem to work around the non-verbal cue limitations and actively provide their own text-based translations of non-verbal cues" (p. 176). Tracz (1980) bears out this perception in a comment on his experience: "I was pleasantly surprised, nevertheless, that most users of electronic information exchange system (EIES) attempt to incorporate many little expressions to compensate for the lack of face-to-face contact, and on the whole, gentleness prevails." (p. 17). (p. 40-41)

Group methods or cooperative learning are widely written about in both adult and youth education. A common belief is that such approaches to learning are more human and productive than competitive approaches. Also, such approaches are held up as facilitative of the construction of knowledge (see for example Belenkey et. al.. (1988) and Vygotsky (1978)), a focus of many adult educators, myself included. Links have been made between cooperative learning and educational computing as pointed out by Higgins (1991).

Those involved with cooperative learning have not missed another growing innovation: educational computing. Johnson and Johnson (1986) discuss the complementary strengths of cooperative learning and computer-assisted instruction (not including Educational CMC). They cite their research involving cooperative, competitive, and individualistic learning in conjunction with the use of educational computer programmes featuring drill and practice, simulation and discovery, and word processing (pp. 16-17). Their data confirm the general effects of cooperative learning:

...computer-assisted cooperative learning promoted greater quantity and quality of daily achievement, more successful problem solving, and higher performance on factual recognition, application, and problem-solving test items than did computer-assisted competitive and individualistic learning (p. 15).

Another interesting finding was that the computer-assisted cooperative methods had an especially positive impact on female students in terms of their attitudes towards computers. Conversely, the competitive methods had an especially negative impact on their attitudes towards computers. Competitiveness also reduced the female students' confidence in their ability to work with computers (Johnson and Johnson, 1986, p. 15). (pp. 31- 32)

Concepts from cooperative learning, computer-assisted learning, and CMC are particularly relevant to educational classes being delivered using Internet/WWW technology. Again, Higgins (1991) clarifies the role of synchronous (simultaneous) and asynchronous (flex-time) in cooperative learning:

In the realm of educational computer-mediated communication, there are many studies that cover issues of social psychology and deal with socio-emotional factors, but nothing that addresses the cognitive foundations needed to help establish a theoretical and practical model for computer supported cooperative learning (CSCL). A variety of research efforts and numerous descriptive, or anecdotal reports appear in the literature.

Some address issues relating to the use of computer conferencing (CC) (Harsim, 1989; Hiltz et. al., 1990; Mason, 1990; Phillips et. al., 1988) in asynchronous mode. Others report on the application of synchronous communication via local area networks in the classroom (Foster, 1991; O'Kelly, 1991; Peyton, 1989; Wilton, 1988)....

Those involved with computer conferencing seem particularly resistant to the notion of an important role for synchronous CMC. This is not surprising in view of the fact that the features and capabilities provided by computer conferencing software have not changed significantly over the past 10 years, and appropriate synchronous capabilities have not been readily available. Another reason for lack of attention to the potential of synchronous CMC is that the asynchronous nature of computer conferencing is regarded as one of its most valued attributes in terms of intellectual activity. Levinson (1988) notes, "‘Asynchronous’ or non-immediate communication...may produce exchanges of richer intellectual quality than those resulting from immediate face-to-face dialogue" (p. 115).

In an on-line seminar, Turoff (1989c) took the extreme position and challenged the participants to produce examples to counter his proposition, "that there is no human group problem solving activity that would not be better served by asynchronous communications..." (conference note C1295 CC4). Further, he stated that, "...a pure synchronous system is worthless as far as I am concerned" (conference note C1295 CC16).

Obviously, then, an important debate exists. Synchronous interaction may be a critical feature of peer interaction and an important component in the developing theories of the social construction of knowledge as they pertain to cooperative learning. Asynchronous interaction, on the other hand, may improve group problem solving and lead to richer intellectual quality in the communications. (pp. 6-7)

Higgins research with nursing students working on a nursing case study demonstrated that...the synchronous mode of text-based CMC are more likely to include verbal elements reflecting important cognitive activities such as problem formulation, interactive arguing, and task management than similar discussions in asynchronous mode.

...greater focus on, and accuracy of outcomes are possible with synchronous text-based CMC than with asynchronous.

...greater mutual facilitation occurs in synchronous text-based CMC than in asynchronous mode. This facilitation is reflected in verbal elements demonstrating attempts to establish interpersonal ease, support, understanding, and encouragement.

... the novel and unique modes of interaction possible through CMC (synchronous and asynchronous) can have a motivating effect for learning activities undertaken in dyad or group situations. (p. 19)

His own experience as a learner in an on-line class and anecdotal reports from learners in previous classes I've taught confirm the importance of synchronous communication in collaborative learning. This course is designed around these considerations and,

thus, group work is considered essential to the overall success and enjoyment of participants, and synchronous chats are encouraged as a way to address the human element.

TECHNOLOGICAL DETERMINISM AND THE NETWORK OF LEARNING SOCIETY

The Internet is an evolving, growing entity aligned to continuous technological change. This constant change is instrumental in paradigm shifts in the development of learning technology. To some learners (and educators alike) it appears to have become too complex, too technical, and many make an assumption of technological determinism. There is no need to assume technological determinism, we can ensure that the development of technology will not be detrimental to learning and education.

Barry Jones (1990) uses the example of Los Angeles and the way the car and tyre industry was allowed to replace the public transport system with cars and freeways, as a warning that we should not accept the consequences of technological determinism, but that we have a choice in the way technology is used and developed. The development of the car was an uncontrolled, chaotic phenomenon with wide reaching “side-effects”, such as road deaths, urban sprawl and environmental pollution. This did not necessarily need to be the case, but was accepted as part of the development of technology.

The Alternative Network: In the educational field, as in the commercial, innovations may develop spontaneously. Some may be planned, such as in the commercial arena when a new calorie-controlled, portion-controlled, fixed weight chicken product is developed for sale to consumers at retail level because they are now cholesterol conscious and then diffuses to restaurants who in turn have expressed a need for such a product. Many developments have however occurred spontaneously from research with a different purpose, such as the many offshoots of the NASA space programmes and the Internet from its 1967 beginnings as ARPANET, initially commissioned by the US DoD and the Pentagon, for military purposes.

What is the Internet?: At a technological level, the Internet is millions of computers (no one is quite sure how many,) interconnected through the worldwide telecommunications systems. All these computers are able to share information with each other because they use common communications protocols.

At the human level, the Internet is the people who use those computers and the information they share. The people come from all walks of life, acting both as private individuals and representatives of organisations. Everyone on the Internet can publish information on any subject they wish, and almost everything published is available to everyone else. As a result, the content is staggeringly extensive and varied.

Finally, the Internet is a technological, social and cultural phenomenon, unlike anything the world has seen before. It has emerged as such not because of some ideology

or social manifesto, but simply because of its anarchic technological structure. It might be seen as historical irony the reason that this network was initially built during the period of the cold war, which was to ensure that there would always be some means of communication in case the USA was hit by enemy nuclear missiles.

Unlike other human-conceived networks which exist since the beginning of humanity (e.g. power and energy networks), the Internet is not owned by anyone. No one owns the Internet. It is shared, by the consensus of its users. It does not come from a place, or even a country. It recognises no political boundaries. It was not invented. It evolved, over three decades, from the desire of people worldwide to share knowledge and to communicate.

- “Why do people want to be “on the Internet?” One of the main reasons is simple freedom. The Internet is a rare example of a true, modern, functional anarchy. There is no “Internet Inc.” There are no official censors, no bosses, no board of directors, no stockholders. In principle, any node can speak as a peer to any other node, as long as it obeys the rules of the TCP/IP protocols, which are strictly technical, not social or political. (There has been some struggle over commercial use of the Internet, but that situation is changing as businesses supply their own links).” (Sterling, 1993)

Trends in Technological Change: Different technologies and methods for educational delivery. People retain about 20 per cent of what they see, 30 per cent of what they hear, 50 per cent of what they see and hear, and up to 80 per cent of what they see, hear, and do simultaneously; to the extent that computer-based learning systems integrate these techniques, they can be very effective.

Technological change, occurring at a faster rate now than ever before, is having incremental effects upon communication and social interaction. Increasing sophistication in the technologies of communication and computerisation are decreasing the cost and increasing the availability of instantaneous communication across the world. Electronic Mail, video-conferencing and multimedia applications are just a few examples of innovative technology now being used in education. The use of the Internet is growing exponentially, although it is not possible to measure accurately that growth rate. These figures are collated by one of the largest Internet hosts in the USA, using world wide surveys. They show that minimal figures of Internet hosts had reached approximately 13 million by mid 1996 and exceeded the 20 million mark by August 1997, of which approximately 5 million were European hosts representing a quarterly increase of nearly 15 per cent.. Of these, educational institutions as hosts (edu domains) had reached more than 2 million by June 1996 and neared 3 million in August 1997. One of the implications of this is that our learning institutions and practices will change. Moore (1995), refers to models explaining how educators will respond to new technologies

- The minimal change model-in which instructors make no fundamental changes but merely use technology as an instructional aid;

- The marginal change model-in which the pedagogy and organisation of education remain unchanged and students are added on to conventionally taught classes (the most common application of distance education in North America);
- Systemic change in which institutions change the fundamental organisation of teaching by reorganising it into a system driven by technology; and a virtual system in which universities and schools are “place-free, with little or no formal organisation”

Only the last model acknowledges the existence of a paradigm shift. It differs significantly from Scott’s notions of the British perspective (Scott, 1993), as it attempts to do away with the established institutional culture. This model should flow on to changes in the text-paper based emphasis on knowledge and content, and to the training of teachers, and perhaps a redefining of the position of teacher from “teacher as knowledge source” to “teacher as facilitator of the learning process” (D. Spender, 1995, pp114ff).

Other implications of this are that there will be fewer on-campus students, more education over the Internet, more universities online, and “virtual degrees” through virtual universities. The methodology of education will change, becoming more varied and flexible. Isolated and other marginalised students will benefit-assuming they have the technology of access. An example of this trend is Ken Eustace of Charles Sturt University, Australia, who has received accreditation for an MA from Paideia University. Eustace is the first Australian academic to be awarded an online degree from a “Virtual University”.

“The electronic transfer of a global Master’s degree over the Internet from Paideia University in Amsterdam to Perth in Australia six weeks ago signified the start of profound changes-and dilemmas-to the university system”, The ‘Australian’ reported on 6th September 1995. There is a wide spectrum of learning opportunities on and over the Internet, especially on the World Wide Web. The existence of Virtual Universities and Classrooms on the Net paves the way for wider access and participation for adult learners as it changes the philosophy and practice of ODL.

Hypertext, the non-linear medium, a term coined by computer utopian Theodor Nelson in his 1974 ‘Computer Lib/Dream Machines’ to describe electronic texts embedded with links to other texts is yet another tech-tool which enhances learning, breaks down the traditional linear narrative of the written word by encouraging readers/users/surfers to find their own paths through large amounts of information. His idea came to fruition with the advent of the World Wide Web, where “hypermedia” also includes sounds, pictures, and moving images. Hypertext was the first tool to enhance interactivity on the Net.

The capacity for learners to add to the dialogue through an interactive medium provides opportunity for development, application and linkage of new knowledge to the adults own learning context. The Internet recreates the ‘agora’ or meeting place in

which knowledge is not only shared but created and recreated. Learners engage in active learning within conferenced environments where they take responsibility for their own learning processes. Learners “are required to examine thinking and learning processes; collect, record, and analyse data; formulate and test hypothesis; reflect on previous understandings; and construct their own meaning” (Crotty 1994, as quoted in Jonassen et al., 1995, p. 11) within an environment that gives the opportunity for students to interact together to build a community of learners.

Computer conferencing (CC), now common on the Internet, is a technology that facilitates interaction between learner and instructor and among learners and, potentially, opens the door to active learning. It creates opportunities for students to engage in the kind of active learning activities that Meyers and Jones (1993) and others identify. For example, CC allows the formation of small groups, creates opportunities for collaborative learning, discussion using case studies, role playing, simulations, online journaling, and provides opportunities to discuss and make connections between content and their own lived experiences. One such example could be the online experience I have had in attending one of the newly established OU courses which are delivered over the Internet.

A Case Study within the Project: Because of its asynchronous nature, CC-based group discussion allows for more thoughtful, well-constructed responses than one might find in a face-to-face classroom. Also, within the group context, students have opportunity to interpret and transform content (*i.e.*, make content their own); they can integrate new material with what they already know about the subject. Both large and small group discussion facilitates the active exchange of ideas and opinions related to specific content. MZX205 is a computing OU course. It is delivered in the usual OU manner, plus the fact that OU offers learners access to its WWW and e-mail conferencing facilities over the Internet. Since the beginning of the term, I have accumulated 362 conferencing messages from the 21 fellow learners on the course. We have been discussing our common concerns on the course, facilitating each other and commenting on our work, flaming about and complaining, socialising in virtual reality, using our first names and showing intimacy rarely found in everyday acquaintance under similar circumstances.

CC mediated case studies and collaborative activities gave us opportunity to apply and test theory and knowledge to a “real-life” context. Specifically, collaborative activities encouraged mutual decision-making, and shared analysis amongst group members (skills that are valuable in the work world!). Generally, we were required to produce some product as evidence of our collaborative efforts, such as a final report or posting which was presented online for comments from other students and the instructor. Online journaling allows learners to reflect on content in a personal context, and to analyse and evaluate content in light of their experience. This reflection facilitates a personal level of integration and interpretation of content. Although an individual activity, journaling is, nonetheless, “active,” because it provides the opportunity for reflection on and “meaning-making” with regard to course content. (SCHANK, R. and CLEARY, C.1994)

ONLINE EDUCATIONAL COUNSELING

Well-prepared Web resources can be provided online, but asynchronous counseling assistance is required as well, particularly for learners who are experiencing difficulty. The online environment is one in which learners can “fall through the cracks” if assistance is not readily available. From time to time a learner may need someone to assist in keeping a positive outlook and determining if an intervention is needed. Learners need to know that help is there if they need it. The institution should provide this resource and all institutional staff should be trained to identify when a learner might benefit from a session with a counselor. It is important to remember, however, that while referrals can be made, the decision to pursue them belongs to the learner. The work of the counselor in an online learning environment has three aspects. The first is to work with Web designers to develop online resources to help learners to identify and address barriers to reaching their educational goals. The second is to interact with the learners when an intervention is required. The third is to work with other institutional staff to ensure that processes and procedures enhance learning. Educational and career counseling are well suited to the online environment, but personal counseling is less so. Personal counseling should be limited to immediate crisis resolution and referral. Counseling units need information about community resources to which they can refer clients.

ONGOING PROGRAMME ADVISING

Distance and online learners are often adults and they frequently spread their learning over a number of years. They report that they need programme planning that will help them achieve educational goals in an expeditious manner. Moreover, learners often transfer between institutions and jurisdictions, increasingly so in a global learning environment. Learners need to plan their coursework based on particular programmes, while remaining aware of course transferability. The role of advisors is to assist learners to understand programme requirements, match courses they may be transferring into a programme, and then plan the remainder of their programme accordingly. Since, the adult learner may change career and educational goals at any time during the process of completing a programme, advisors need to be readily available and to have access to all programme and transfer information. Software products are available to allow advisor and learner to navigate this process. Academics, counselors, and others within the institution need to know when to refer learners to an advisor.

THE DIGITAL LIBRARY

In the early years of distance education, providing library support to learners was a challenge. Courses were developed in print format, and comprehensive course packages were sent to each learner. The library typically developed a collection that was made available, by mail or fax, to the learner on request. Distance educators worried that

learners were not gaining the library search experience that would enhance their studies and their research skills.

Online sources of information have transformed libraries in distance education: where in the past libraries focused on holdings, they can now focus on access; where they used to be information repositories, they can now be gateways to information. This transformation has allowed the library experience to be more profound for the learner and more integrated within the learning process.

From a learner support perspective, a well designed online library

- Is easily found among other institutional Web pages;
- Provides an up-front tutorial for the new learner;
- Is integrated with the institution's online courses;
- Provides tools to assist with online searches; and
- Provides access to personal assistance, if needed.

The transformation of libraries in distance education has posed some interesting fiscal issues for institutions. In the past, acquisitions budgets provided for purchasing holdings that became part of the collection, part of the assets of the institution. Annual funding was provided for collection development and updating. Even subscriptions to periodicals, an annual expense, resulted in a "holding." Subscriptions to online resources are another matter entirely. As libraries move towards becoming gateways rather than repositories, a new way of viewing funding is needed.

ACCESS FOR STUDENTS WITH DISABILITIES

Online learning can enhance access for people with disabilities. New assistive technologies can allow access to learning opportunities previously denied to this population. Increasingly, the legal requirements to provide accommodations that allow access are being defined by the courts, and institutions are required to ensure that necessary accommodations are provided, without compromising academic rigour. In an online environment, the following services are expected:

- Administrative accommodation with respect to timed assignments and examinations;
- Alternative formats for learning materials;
- Advocacy within the institution;
- Advice about assistive technologies;
- Advice about funding sources;
- Referral for specific needs; and
- "Reasonable accommodation" as outlined in law.

STUDENT RIGHTS AND ACCESS TO OMBUDS SERVICES

Online learners have as much need of clearly articulated rights as do learners in traditional educational settings. An advocacy process designed for online learners is

one in which the learner is made aware of student rights and responsibilities. An institution can fulfill its basic legal responsibility by making a student code of conduct available on the Web and in print on request. However, a prudent institution will go well beyond this demonstration of due diligence. Web sites that clearly outline intellectual honesty expectations should be readily available on the Web and should be referred to frequently. Reminders just prior to assignment preparation reduce the chance that the learner will fail to provide proper scholarly acknowledgment.

Traditions regarding what constitutes intellectual property and what is generally accepted common knowledge are not universal concepts, nor are they always understood. Cultural views about ownership of knowledge vary, and if the rules that are to apply within the institution are to be understood, then the institution must ensure that they are properly explained. For example, although many educators do not accept the idea, the reality is that students do not always understand the concept of plagiarism, much less how to avoid it.

To address this problem, the University of Puget Sound has designed an excellent Web page that provides learners with exercises to enhance their understanding of the concept of plagiarism and to assist them in avoiding it. All efforts to provide smooth interactions between the learner and the institution notwithstanding, there will be situations in which the learner becomes ensnared and does not know where to turn. A highly visible ombuds office should be available. Moreover, from the institutional perspective, the ombuds office can assist in identifying policy and procedure problems that require attention within the institution.

THE ONLINE LEARNER'S ROLE IN GOVERNANCE

Online learners have valuable contributions to make to an educational institution. Institutions that specialise in distance and online learning can provide opportunities for students to participate in institutional governance. Student government is possible in such an institution as well, but administration must make special arrangements to facilitate the process. In many ways, the student union has the same issues in keeping in touch with its constituency as does the institution itself. Both are vying for the attention of a learner who may be juggling learning with other life demands. It is in the institution's best interests to have a healthy student union and to work with that union to meet the needs of the learners.

Some means by which to achieve these goals include

- Making networks available to the student union;
- Providing one main institutional contact with whom student union representatives can interact;
- Assisting in collecting student union fees;
- Making information available, within the confidentiality guidelines;
- Having student representatives on all decision-making bodies;

- Having decision-making bodies meet through electronic means (video-conference, online conference, etc.) to maximize participation;
- Keeping the student representatives and the student union apprised of significant events, initiatives, etc. (e.g., strategic planning, budgeting, tuition fee increases);
- Engaging in shared initiatives with the student union (include them in convocation, copublish newsletters, etc.);
- Seeking advice from the student union on important issues; and
- Demonstrating appreciation for the work of the student union.

LEARNER SATISFACTION MONITORING AND ENVIRONMENTAL SCANNING

Regular monitoring of the learners' satisfaction levels and scanning of the student services environment assist the institution to make continuous improvements. Year-over-year comparisons are possible if the survey instruments are carefully designed. In this way, trends and areas in need of attention will become apparent. Focus groups are being successfully conducted online by institutions that are at a distance from their constituents. Such groups can be used, for example, to test out the efficacy of a new course, programme, or service.

CURRENT ONLINE TEACHING SITUATION

ONLINE EXPERIENCE

While the vast majority of our survey respondents had been active in posting course resources, syllabi, or personal information on the Web, not all had previous experience in Web-based instruction. Nearly a quarter of the respondents had never taught even a portion of a course online. On the other hand, nearly 4 in 10 respondents had taught courses partially online; among this group, the average number of partially online courses taught was about four. Another 18 per cent had experience teaching fully online courses, with an average of five such courses. In addition, 19 per cent had done both—partial and completely online courses—with an average of 10 such online course experiences.

Calculations across these responses indicated that nearly 4 in 10 early Web adopters had taught completely online courses, while nearly 6 in 10 had taught at least part of a course online. Given these data, the respondents in this study certainly had extensive online teaching experiences on which to base their survey answers.

Respondent's Web-Related Skills

An instructor's degree of comfort in using different Web technologies has a direct bearing on classroom practices as well as the decision to teach even part of a course online. When instructors are hesitant or lacking in confidence, there is less likelihood for innovation and risk taking.

Therefore, we asked these early Web adopters about their degree of comfort with the following Web skills:

- Creating HTML pages,
- Hosting an online chat,
- Sending and receiving file attachments,
- Using Web-based courseware systems, and
- Moderating a Web-based asynchronous discussion.

The responses were interesting. For instance, over 90 per cent of these faculty members felt a high degree of comfort sending and receiving file attachments in e-mail. Fewer than one per cent of respondents were uncomfortable with this skill. Somewhat surprisingly, 62 per cent were highly comfortable with creating HTML pages and another 20 per cent had a medium level of comfort.

However, this acknowledged degree of comfort likely includes a range of skills from using standard software options such as “save as HTML” to actually being facile with HTML and other programming code. The degree of expertise with HTML remains a question for future surveys. These early adopting faculty were somewhat less comfortable moderating a Web-based asynchronous discussion forum or bulletin board. Still, nearly 50 per cent rated their degree of comfort as high, while another quarter of them reported a medium level of comfort. Similarly, 44 per cent were highly comfortable with Web-based courseware systems and another 34 per cent felt moderately comfortable.

On the low end was comfort with hosting an online chat session. Perceptions of online chat tools were roughly split across low, medium, and high comfort categories. These results indicate that these faculty members possessed at least some basic technology skills. Perhaps, as the NEA survey of traditional and distance learning higher education members revealed, workshops and training sessions on teaching via distance learning are now readily available.

While such a skill base and comfort level may be expected of these early Web adopters, many of these faculty members are either taking advantage of university training and support or are engaged in a heavy amount of self-teaching in regard to Web-based teaching tools. Or perhaps they are overstating their skills. In fact, latter parts of this report reveal a somewhat different picture.

Time Commitments

In terms of overall time investment, these college instructors almost unilaterally agreed that teaching online is more time-consuming than traditional classroom-based instruction. More than 4 in 5 faculty agreed that teaching online courses requires more time than traditional courses.

Fewer than 10 per cent disagreed with that statement. Once again, this is consistent with the NEA report finding that more than half of college faculty teaching via distance learning spent more time on their online courses than their traditional ones regardless

of the number of students or times they had previously taught the course. Such findings point to a need for greater course support and incentives that could ease time pressures felt by instructors involved in online teaching.

Attrition

Some reports and media releases contend that students are more likely to drop online courses than traditional ones. Those utilizing a mixed mode or blended approach—traditional and online in the same course—were less likely to experience significant student attrition than those teaching completely online courses.

In fact, only 29 per cent of those utilizing a blended approach experienced more than 10 per cent drop the their courses, whereas 44 per cent of those teaching completely online courses had more than 10 per cent drop their course. Perhaps more strikingly, only 2 per cent of blended courses experienced more than a 50 per cent attrition rate compared to 10 per cent of the completely online courses with such huge attrition rates.

Internet Access

Computer access does not appear to be a problem for these early adopters of Web technologies. Seventy-eight per cent of these college instructors had Internet access in their current or most recent classroom. Computer lab accessibility was even higher with 93 per cent indicating that they had access to an Internet-connected computer lab for class use. Even more, 97 per cent, had Web access from home.

This is more than double the 47 per cent of Americans who are users of the Internet at home as reported in a recent UCLA study. Such high level of technology access is not too surprising given that the majority of the respondents were early Web adopters who had a high level of education.² In effect, these findings indicate that access to computers and Internet resources is no longer an obstacle for many college faculty.

Platform Choices and Preferences

The delivery platform for online courses is a significant factor in faculty online teaching experiences. Eighty-three per cent of the respondents to this survey indicated that their institution provided a Web-based platform or courseware system for developing online courses or enhancing on-campus courses with online features. Our survey data also indicated that many institutions are utilizing more than one courseware package. In fact, 22 per cent of the respondents worked at institutions that provided access to more than one Web courseware or conferencing platform; when excluding those having yet to adopt a Web courseware system, this increases to 27 per cent. Moreover, 10 per cent provided access to three courseware systems or conferencing tools, and 5 per cent had four or more systems or tools available. When asked what is missing from the courseware tools that they use, slightly over half of the respondents at institutions supporting at least one courseware platform offered some ideas.

The specific features mentioned in their open-ended responses included:

- Ability to annotate documents and visuals in real-time,
- Better grade reporting systems,
- Collaborative white boards,
- Collaborative working tools,
- Drawing software,
- Easy ways to create animations,
- Effective drop box tools,
- Efficient ways to display mathematical notation,
- Electronic library resources,
- Good real-time chat tools,
- Improved quizzes,
- Options for chatting and using PowerPoint at the same time,
- Private asynchronous rooms for group work,
- Proctored testing,
- Streaming video,
- Three-dimensional concept visualization tools,
- Tools for tracking student statistics, and,
- 24/7 support.

Of course, many of the tools already exist in the common courseware platforms used in higher education. Other features, such as “options for chatting and using PowerPoint at the same time” are available in various synchronous presentation and collaboration tools often found in corporate training settings. Some general design features requested by these respondents included simplicity, ease of use, user friendliness, enhanced speed, less ugly designs, less cumbersome interfaces, customizability, integration across areas of campus, and flexibility to organize content. In general, there appeared to be a call for more professional appearance, easy to use features, and functional or usable tools. When asked what they liked about their present courseware tools or system, 56 per cent of the respondents offered ideas.

Instructors preferred:

- Ability to link in lectures with PowerPoint presentations,
- Assignment parts for students to pick up homework,
- Chatrooms,
- Comprehensive tools,
- Consistent course appearance,
- Customizability,
- Data and course security,
- Detailed statistics on bulletin board use,
- Ease of use,
- Flexibility,

- Good online help,
- Internal e-mail systems.
- Online discussion boards,
- Password access,
- Posting of assignments on the Web,
- Posting of deadlines and due dates,
- Randomized test banks,
- Reliability,
- Student drop boxes, and,
- Versatility in quiz types.

USEFULNESS OF WEB-BASED TOOLS FOR TEACHING AND LEARNING

We were also interested in the attitudes of these college instructors about Web-based instructional tools, resources, and activities.

As a result, the respondents were asked to rate the degree of usefulness for items categories:

- Online Class Tools.
- Collaboration and Sharing Tools.
- Instructional Activities.
- Web Resources.

After rating each item as low, medium, or high usefulness for online teaching and learning, the instructors were also asked whether they in fact used that item in their courses.

USEFUL ONLINE CLASS TOOLS

In general, these college educators perceived high utility for most of the online class tools considered in this part of the survey. Perhaps more importantly, at least one-third of the respondents actually used each of the items in this category. Not surprisingly, respondents tended to rate the tools that they actually used as more useful.

The highest rated tool was for posting syllabi online. Not only did 72 per cent of the faculty respondents report this feature as highly useful, 85 per cent actually used such a tool in their courses. These findings also match the Web-based Education Commission report, which documented the increased posting of course syllabi to the Web and incorporation of Web resources within college instructor syllabi. Of course, many of our survey respondents were selected for this survey because they had already posted their syllabus online.

The fact that more use this type of tool than rate it as highly useful indicates it is relatively easy to do. The large number of respondents using tools to post their syllabi

online reveals an initial area of penetration for the Web in college teaching and learning. For example, the University of Michigan School of Information has compiled a list of faculty course syllabi and placed it online. Similarly, the UCLA Humanities Department created the E-Campus for syllabi, assignment announcements, and other course related links. However, as indicated earlier, the most complete listing of college syllabi to date is located at the World Lecture Hall. This site hosts syllabi across disciplines for college instructors worldwide. A tool for posting cases, questions, or problems corresponding to course material on the Web was the next most valuable courseware feature of these early Web adopters.

Not only did 70 per cent rate this survey item as highly useful, but nearly 70 per cent also had engaged in such online activities. In fact, only 4 per cent rated this item as low in perceived usefulness. These college instructors also valued file uploading and downloading tools. Sixty-five per cent of the respondents felt they were highly useful, and 71 per cent had used such tools in their teaching. The next highest rated item in terms of usefulness was an online lecture notes utility, which was rated as highly useful by 57 per cent of the respondents and actually used by 69 per cent of them.

Once again, this indicates that while faculty members might view different tools as more useful, they generally rely on readily accessible tools that perform a useful function. Such findings also signify that online tools for posting lecture notes, cases, and syllabi are among the first wave of Web-based instruction courseware. In contrast, online databases received high ratings for usefulness from 51 per cent of the respondents but only 44 per cent were using such a tool. Perhaps such tools are not yet available to the degree that college instructors would like. Once a course is on the Web, there must be some student evaluation and assessment. Indeed, some scholars advocate the use of the Web for online testing and evaluation as a means for reducing costs and increasing speed.

In addition to quick and cost effective feedback, online evaluations provide more organized, individualized, and plentiful course feedback. Despite these benefits, Hmieski and Champagne report that 98 per cent of the most wired schools still use pen and paper course evaluations. Among the early Web-adopting faculty members of this study, however, 52 per cent rated student online evaluation tools as highly useful and 48 per cent were actually using such tools. Online quizzes or tests were deemed highly useful by 47 per cent of respondents and nearly the same per cent were actually using online exams in their teaching. One in five respondents gave a low usefulness rating to such tools, however.

Receiving even lower support was online student evaluations of course materials. Only 41 per cent rated these as highly useful, while just 36 per cent used such tools. Most of the findings are consistent with the research from Peffers and Bloom which found that online instructors tend to rely on common software such as e-mail, file uploading and downloading, and asynchronous conferencing as well as simple tools for

posting static or dynamic syllabi, Web links to course material, and lecture notes. Significantly fewer instructors used chatrooms, multimedia lectures, online examinations, animation, and video streaming.

However, this research also revealed that the instructional impact of Internet media tools in college settings is expected to dramatically increase in the next few years. Firdyiwek's review of courseware tools indicates that few such tools support pedagogy in an integrated fashion. As tool development proliferates, so, too, does resulting confusion about how to effectively use these online tools. Interestingly, in this study, only 49 per cent of respondents were highly supportive of tools to place their entire courses on the Web and 47 per cent were using such tools. Could such modest numbers among early Web adopters be due to the lack of pedagogical support in these tools? Or does it reflect a lack of time or training? Perhaps these early Web adopters simply do not want to give up traditional instruction. Or perhaps they rely on customised courseware tools. Whatever the answer, this seems a ripe area for additional research.

Useful Collaboration and Sharing Tools

There are decades of research studies detailing the clear advantages of cooperative and collaborative learning over more individual and competitive formats. Fortunately, many collaborative pedagogical strategies have relevance in Web-based instruction. In fact, a proliferation of collaborative learning technologies have recently emerged for both work and educational environments. In higher education, technologies are becoming more interactive and distributed, enabling learners and instructors to participate in an incredible array of information, resources, and instructional experiences.

The blending of technological and pedagogical advancements presents new opportunities for both research and teaching focused on online dialogue, information sharing, and facilitating learning. In part, such collaborative tools have come on the scene to meet the needs of an older and more diverse student population than in the past. Perhaps this survey will help educators design more powerful e-learning environments for Web-based collaboration and sharing. Collaborative Web-based learning tools offer unique ways for learners, instructors, and experts to interact.

There are now Web tools for student collaborative enquiry, problem-based learning, articulation and dialogue, debate, and personal reflection. Some research indicates that effective use of these new tools can actually foster communities of practice. To create a learning community, the tool or system must bring people together for some initial common interest or quest.

There not only is a need for a common reference point or issue for the online group, but members also need multiple ways to become informed about events of that community. Sharing information online often involves conferencing and computer-supported collaborative learning tools. Fortunately, such tools have begun to infiltrate online learning courseware. In addition, communities such as the World Lecture Hall,

MERLOT are now available for visitors to locate and share learning materials within specific discipline or interest areas. But what were the views about such resources and tools among the respondents to this particular survey who already had been involved in online information and resource sharing?

Surely, they would understand and promote collaboration and sharing tools more than the rest of the population. As research from Peffers and Bloom predicts, the respondents to this survey perceived less utility for collaborative and online sharing tools than for test, lecture note, and syllabus tools. For instance, when asked about the utility of tools to share success or failure stories with other instructors, only 27 per cent had done so and only 30 per cent listed this as a highly useful item. Another 51 per cent, however, rated the degree of usefulness as medium. Hence, more than 80 per cent would find some use for such tools; perhaps they simply are not yet available. Similarly, only 26 per cent used online tools to collaborate and form partnerships with other instructors. Still, 40 per cent saw this as a highly useful idea. Another 44 per cent saw it as of medium utility. Slightly more college instructors used Web-based tools in their courses for students to share success or failure stories with other students.

Forty-one per cent listed this as highly useful and another 45 per cent felt that it was of medium utility. Slightly higher, 46 per cent of the respondents used tools for students to collaborate and form partnerships with other students. In fact, 56 per cent felt that this was a highly useful endeavor and another 34 per cent found it of medium usefulness. The fact that 90 per cent perceived value in student online collaboration is of significance. Asynchronous discussion forums, synchronous chats, and annotation or feedback tools are common means for electronic collaboration. Sixty-one per cent of faculty members in this study utilized bulletin board or asynchronous types of discussion in their courses. While 60 per cent rated this type of tool as highly useful, another 31 per cent saw it as having medium utility. There was a significant drop in perceived utility and actual use in terms of synchronous collaborative environments compared to asynchronous environments.

Only 32 per cent of the instructors in this survey had used real-time chats, and only 37 per cent rated this item highly. In fact, 28 per cent of the respondents rated this item low in utility. In contrast, tools for interactive feedback, commenting, and annotations fared much better in terms of usefulness among these respondents. Forty-six per cent of the faculty respondents had used interactive feedback or annotation tools in their classes. Even more, 56 per cent perceived them as highly useful, while only 6 per cent rated this type of collaborative tool as low in utility. Perhaps software developers might want to target annotation and feedback tools; they are highly valued and yet not everyone is using them.

Personal profile tools are another means to share information online with peers and other instructors. Whereas 52 per cent claimed to use instructor profile tools in their courses, only 34 per cent utilized student profile tools. Such a finding seems odd. Perhaps

there was misinterpretation on this item or perhaps it is easier to reflect on tools one is personally using. Only 30 per cent considered instructor profiles important, indicating that they are using such a tool simply because it is there and it is easy to use. Even less, just 25 per cent, found student profile tools useful.

In fact, 35 per cent rated the degree of usefulness of student profile tools as low. Online guestbooks were even less appealing. Only 6 per cent used them and just 7 per cent rated them highly. In fact, 66 per cent of the respondents—the largest of any item—rated this type of tool as low in usefulness. Related to our findings about online evaluation and testing, only 7 per cent used the Internet for collaborating with other instructors for test-making. Still 22 per cent rated this as a highly useful item, while another 40 per cent felt it was of medium utility. Similarly, few instructors collaborated with other instructors on class tasks, activities, and discussion. Only 18 per cent had engaged in such collegial activities, while 34 per cent rated this as highly useful and another 41 per cent consider it of medium utility. Perhaps these are two immediate areas wherein universities and software development companies might partner together to develop and test new Web-based teaching and learning tools. Finally, online technology demonstrations received fairly favourable reactions from our respondents.

Thirty-one per cent of the faculty members had used this type of tool in their classes. In addition, 42 per cent rated this item as highly usable in their classes, while 38 per cent rated it of medium utility. Despite these findings, college instructors perceived a need for more collaborative tools. Tools with more than a 10 per cent gap between actual use and perceived high utility included tools for instructors to form collaborations with other instructors, tools for students to share stories with other students, tools for interactive feedback and annotations on student work, tools for instructor test-making collaboration, tools for instructor task collaboration, and tools for online technology demonstrations.

These large gaps between teaching practice and perceived utility indicate a need for more collaborative tools in e-learning environments. They may also point to the current direction of Web-based teaching and learning practices.

Useful Online Instructional Activities

Instructional activities that these instructors found useful were also of interest in this study.

The first four online activities asked about were:

1. Scientific simulations;
2. Data analysis;
3. Lab activities; and
4. Performance activities.

Examples of the latter activities might include band or music tasks as well as online decision making in any discipline including counseling, finance, or teaching. These

four activities were all infrequently used by the survey respondents. The actual use of these tools ranged from 23 to 26 per cent, with lab and performance being used slightly more often than scientific simulations and data analysis. All of these types of activities were deemed highly useful by approximately 45 per cent of the respondents. Such are interesting since the percentage of respondents who rated these items as highly useful was nearly double the percentage of who actually used them. When combining those who rated activities moderately or highly useful, more than 75 per cent of the respondents indicated utility for each of the four tools. Such data clearly indicate that there is a market for such tools, but college faculty members currently do not have access to them. A fifth and final instructional activity was online critical and creative thinking activities. This item was rated more favourably than the other four.

Forty-five per cent of these faculty members used such activities in their online teaching, and even more impressively, 62 per cent rated them as highly useful for their teaching discipline. An additional 28 per cent rated them of medium usefulness. Only 10 per cent considered their degree of usefulness low. Such results are further indication of the need for better pedagogical tools in online learning environments.

Useful Web Resources

The Web is highly touted as an online resource. Some suggest that it is a gigantic library sprawling in front of students and instructors alike. But in what ways do early Web adopters actually view it as a resource for teaching? Questions were asked about the utility of such Web resources as search engines, glossaries with links to examples, Web link suggestions, article and journal links, book recommendations, newsgroups, collegial Web sites, and general and discipline-specific online resources.

Given that research has revealed that college instructors tend to rely on easy to use tools, it is not surprising that search engines were the most commonly used Web resource with 83 per cent of these faculty members utilizing search engines in their teaching. Equally impressive, 70 per cent ranked search engines such as Yahoo or Lycos as highly useful and only 6 per cent ranked them low. The next most favourable ranking was for online article and journal links. Seventy-four per cent of the respondents used such tools and 70 per cent rated them as highly usable. Only 3 per cent rated this item low. Sixty-one per cent of these college educators used discipline specific resources in their teaching and 63 per cent found them highly useful. Along these same lines, 59 per cent had used Web sites created by colleagues in their teaching. Such collegial Web site use included syllabi and lecture notes. This is not surprising given where the sample was derived. In fact, only 8 per cent rated the utility of this item as low. Similarly, 58 per cent had used general teaching and learning resources or instructional strategies that had been posted online. Once again, only 8 per cent viewed this item as low in utility.

Online glossaries are another emerging Web-based teaching resource. In fact, 57 per cent of the survey respondents had used online glossaries with links to examples on the

Web in their teaching. Similarly, 55 per cent viewed this Web resource as highly usable, while another 35 per cent gave it a medium rating. In effect, the use of online glossaries, colleague Web sites, and general as well as discipline-specific online teaching and learning resources indicates that the Internet has spawned a new type of teaching—one that is reliant on the Web for a significant part of college instruction. Online teaching in an online world is different, and new faculty, as well as experienced ones, need to be prepared for it.

The three lowest rated areas, which were the only items used by less than 50 per cent of the respondents, were student Web link suggestions, online book reviews, and newsgroups. Slightly under half of the faculty members in this survey had used tools where students made Web link suggestions. Still, 45 per cent of the survey participants viewed this item as having high utility, while another 42 per cent rated it as medium in degree of usefulness. Book recommendations received roughly the same ratings; 47 per cent had used such a tool and 44 per cent deemed it as highly useful. In contrast, newsgroups were used by only 18 per cent of these faculty members, while just 17 per cent rated them as highly useful. Our findings suggest a relatively high and diverse use of Web resources in teaching. Web resources are highly valued by college educators since they can augment lecture notes with visual depictions of concepts, replace the need for textbooks with online articles and glossaries, and provide more current research and professional news. Tools to search, share, and evaluate online course materials are vital parts of one's Web-based teaching arsenal.

When asked to share URLs of Web resources they found particularly useful in their teaching, 15 per cent of these college instructors responded with extremely diverse suggestions. For instance, they listed course-sharing Web sites such as MERLOT, professional organization sites such as the American Psychological Association, textbook publisher Web sites, locations for instructional design models, and university teaching and learning centre resource listings. Only MERLOT was listed more than once.

The findings denote many areas wherein improvements in online teaching and learning could occur. The numbers reveal that tools for collaboration and resource sharing are highly valued by college faculty members but are not yet part of their typical online teaching life. Tools for annotation and feedback, article or journal linking, and online discussion were considered highly valuable. Additionally, activities for student labs, simulations, and critical and creative thinking have not been as prevalent as college faculty desire. Nevertheless, the number of tools and activities that were of substantial use already, as well as the high usefulness ratings that many additional tools received, was striking. Such ratings are signs that online teaching and learning is not going away in higher education settings, but, instead, is about to be enhanced, extended, and perhaps even transformed.

FUTURE ONLINE TEACHING SITUATION

PREDICTED INSTRUCTIONAL TIME ONLINE

Given that many of those surveyed were likely among the technology leaders at their respective institutions, it was important to ask about the per cent of time they anticipated teaching online in the next 1, 2, 5, and 10 years. Interestingly, while just under a third of these faculty members anticipated teaching more than one-fourth of their teaching load online one year from now, this increased to 43 per cent of the respondents in two years, 61 per cent in five years, and 59 per cent in 10 years.

The reason there was a drop-off in the 10 year data was due to a dramatic increase in those not anticipating to be teaching a decade from now. Once again, the age and experience level of these instructors would indicate that many of them plan to retire before the decade is out. When excluding the data related to those retiring or not teaching, the predictions regarding online teaching commitments were even more striking.

The per cent of respondents who anticipate devoting more than one-fourth of their teaching load to online activities increased as follows: 27 per cent in one year, 44 per cent in two years, 64 per cent in five years, and 73 per cent in ten years. Those predicting that at least half their teaching load would be online increased from 13 per cent in one year to nearly 50 per cent in ten years. And those expecting 75 to 100 per cent of their teaching to be online increased from 5 per cent a year in 2001 to 17 per cent at the end of the decade.

Hence, the college instructors responding to this survey expect the Web to become an even more vital instructional tool during the upcoming decade. Though most respondents do not view it as a replacement for all of their teaching activities and requirements, this finding indicates that Web-based teaching expectations will soon be common.

Freelance Instruction

In addition to predictions of increased online teaching loads within university settings, many college faculty members will likely encounter a myriad of new opportunities to teach for other institutions online. Whether “star” faculty members will be hired guns within the online teaching world is not yet known.

Instead, what is occurring already is the use of college instructors as freelance instructors in online institutions. These faculty members might work for multiple institutions, teach online during breaks or in the summer, or perhaps even take a leave from their institution to attempt to earn an income teaching online.

Other freelance instructors might include practitioners in the field wanting to keep one foot in academia, recently minted Ph.D.’s struggling to find tenure-track positions, and graduate students seeking relevant teaching experiences. Fueling such freelance

needs, many institutions are offering new online courses or programmes without expanding their faculty lines, thereby forcing them to find adjunct faculty or add to present faculty teaching loads.

The scenarios leading to freelance instruction are certainly complex. The 16 per cent of the faculty respondents in this study had experience as freelance or adjunct online instructors.

However, in the next five years, 75 per cent of these respondents indicated that they believed that they would be interested in teaching as freelance or adjunct online instructors. There definitely is potential here for someone to help coordinate and manage freelance instructor services.

Perhaps pending retirements of our respondents factor into these predictions, but other considerations may include additional online course opportunities and expected increases in Web tool availability and reliability

INSTITUTIONAL MOTIVES AND DECISION MAKING

Primary Institutional Motives for Online Education

As Ron Owston pointed out, during the past few years, perhaps nothing has captivated and excited the minds of administrators and educators more than the notion of teaching courses on the World Wide Web. He then argued “Before we introduce any new technology into our classrooms we must be able to justify its contribution”.

The three key areas wherein Owston suggested that Web-based learning might be evaluated were improved access to education, student learning, and cost efficiency. While he detailed many improvements to educational access as a result of online technologies, documenting learning outcomes and costs proved much more difficult.³ In order to establish the level of college instructor agreement with Owston’s key areas, our study participants were asked whether profit, improved learning, or access to education were among the primary motives behind the development of online education across institutions of higher education.

They could select all three. As Owston had documented, there appears to be more support among these early Web adopters for the use of Internet technology to increase access to education than for improving profit or learning. Of our respondents, 93 per cent agreed that access was a primary motive for developing online education. Only one person strongly disagreed with that statement. Additionally, 61 per cent agreed that improved learning was a primary motive. In contrast, only 41 per cent felt that profit was a primary motive.

Hence, those in the Web-based learning trenches put the emphasis on access and learning over profits. When asked the same questions about their own institutions, these general patterns hold. However, these instructors were slightly less likely to agree that each of the three motives were applicable to their particular institution; only 29 per

cent agreed or strongly agreed that profit was a motive while 53 per cent agreed that learning was a motive and 81 per cent felt that their own institution was concerned about access. The reasons for the lower agreement levels were unclear.

Reasons for Institutional Investment

These early Web adopters were asked to rate the level of importance of five key reasons why colleges and universities, in general, might be interested in investing in Web-based teaching and learning. Access to an external universe of libraries, information resources, and databases was the most important reason cited by respondents to explain university investment in Web-based teaching and learning.

The second most important reason, just as to these faculty members, was to support improved efficiency and effectiveness in teaching and research. Offering distance education to a potentially unlimited audience was rated third, while fostering closer inter-institutional cooperation, consortia relationships, and resource sharing within the higher education community was rated fourth. Finally, some respondents felt that building partnerships with private businesses and the government was a critical reason for investing in Web-based teaching and learning. Fewer than 10 per cent of the respondents offered additional reasons why higher education institutions should invest in Web-based teaching and learning.

Most of these reasons concerned student recruitment, student access to education, student skill development, contributing to the economy of the state, revenue enhancement, and staying up to date. Quotes from some respondents included, “to offer equal opportunity of high quality education to students in more rural areas,” “we are under a mandate to increase the number of students we serve. We cannot do it on campus, so we are trying distance learning...,” “to recruit and retain tech-savvy students,” “It’s a new revenue source, that’s #1,” and “because Web-based activities are becoming ubiquitous in ALL workplaces.” One person simply stated, “Students will demand Web-based courses or go somewhere else.”

Web-Based Teaching Technology Decision-Makers

We also asked about the organizational level in which decisions regarding Web-based teaching, including system purchases and policies, were made. Surprisingly, faculty governance also appears to play a key role in these institutions as 40 per cent of our respondents indicated that technology decisions regarding Web-based teaching were made at the faculty level. Similarly, 39 per cent indicated that it was a departmental responsibility or decision.

There were other key players here. For instance, 36 per cent of respondents thought that the technology support unit on campus made these decisions, while 27 per cent considered it a function of the Chief Technology Officer. Twenty-two per cent selected the teaching and learning centre director level as responsible for these decisions. Four

per cent listed others as responsible including the office of distance education, grant administrator, board of regents, or college provost.

Finally, only 5 per cent did not know who made these decisions. There were no statistically significant differences in instructional technology decisions across size and type of institution. However, there were some interesting trends. For instance, in institutions with fewer than 3,000 students, faculty-level decisions are made regarding courseware slightly over 50 per cent of the time, whereas this drops to 32 per cent in medium sized institutions and 41 per cent in large institutions. In comparisons of public and private institutions, we found that college instructors have a role in instructional technology decisions in nearly half of the 45 private institutions, whereas in this survey just 34 per cent of the faculty members in the 151 public institutions in this study helped formulate such decisions.

Besides asking for faculty input, smaller institutions also seem to rely on campus technology support units and the chief technology officer slightly more often than medium and large institutions. Larger institutions have a slight preference for learning centre and departmental-level decisions compared to smaller institutions. Still, most institutions appear to rely on high-level administrators to make the technology decisions that impact Web-based teaching and learning. Next, we looked at differences in the organizational level at which instructional technology decisions are made between institutions with fewer than 10,000 students and those with 10,000 or more students. At the larger institutions, the department or school is more involved in these instructional technology decisions than in smaller institutions.

Teaching and learning centre directors are also more involved in making these decisions in the larger colleges and universities than in smaller ones. This is not unexpected since larger institutions are more likely to have campus teaching and learning centres. Administrators are involved in Web-based teaching technology decisions at roughly the same rate at both types of institutions. None of these comparisons were statistically significant, however. That differences in the organizational level of technology decisions between public and private were minimal. Public institutions more often involved teaching and learning centre directors in their decision-making about the use and support of instructional technology than private institutions whereas private institutions more often involved faculty members in these decisions than public institutions. Administrative-level decisions were made at over 60 per cent of both public and private institutions. Once again, none of these differences were statistically significant.

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Index

A

Academic Achievement 181
Achievement 193, 209, 210, 211, 212, 214, 215, 216, 217
Acquisition 279
Administration 154, 160, 177, 178, 182, 183, 184
Ammons Quick Test 161
Analysis 11, 204, 211, 217
Approach 10, 193, 203, 204, 205, 206, 207, 208, 209, 213, 214, 215, 216
Appropriate 12, 123, 124, 125, 126, 193, 207
Archiving Knowledge 26
Assessment 121, 203, 207, 210, 211, 212, 213, 214, 215, 217, 218
Assurance 193, 194, 195
Asynchronous 281, 283, 287, 292, 295, 297
Attrition 282
Audio and Video 104
Autonomy 194

B

Based 9, 10, 11, 12, 13, 203, 204, 205, 207, 208, 210, 212, 213, 214, 217
Basic Education 1, 5, 7, 85, 86
Behaviorism 99
Belgrade Charter 51
Benefits 10, 14, 115

C

Care 11, 193, 195, 196, 209, 218
Caste System 243, 244
Characteristics 150, 153, 161, 170, 177, 192
Christian Feminism 227
Collaboration 196, 206, 294, 296, 298
Collaborative Learning 102
Combined Aptitude 153
Common 9, 10, 192, 205, 207, 212
Comprehensive 9, 10, 11, 12, 13, 14, 206
Computer-aided Assessment 109
Concerning 203, 207
Confidence-Based Learning 26
Constructivism 100
Consumer Education 58
Contemporary Theories 150, 174
Corporate and Professional 113
Curriculum 9, 10, 12, 210, 212, 213, 215, 216

D

Data Protection 120
Definition 46, 94
Depression 185
Design 11, 204, 208
Designing 195
Develop 12, 214
Development 9, 10, 12, 13, 194, 213
Difference 14, 205, 211, 215
Discussion 192, 205, 207, 208
Domestic Violence 224
Dramatic 215
Drive Theory 155

E

E-learning 298
Early 11, 206, 211
Ecology 207
Education 8, 9, 10, 11, 12, 13, 14, 192, 193, 194, 195, 196, 203, 204, 206, 209, 210, 211, 212, 213, 214, 215
Education and Civil Society 91

Effectiveness 10, 11, 13, 196, 210, 211, 212, 213, 214, 216
Eight Driving Forces 17
Emphasis 200, 288, 289, 302
Environmental Education 46
Evaluation 9, 11, 158, 162, 195, 206, 213, 214, 215, 216, 217, 218
Evidence 8, 9, 10, 11, 12, 13, 14, 206, 207, 208, 213, 217
Exponential Growth of Information 18

F

Family Organizations 245
Feminist Criticisms 226
Flipped Classroom 104
Forms of Pollution 64
Freelance Instruction 301, 302
Future 13, 14, 195, 212

G

Gender Equality 5, 227
Gender Relations 226, 227, 228
Group Similarities 187
Group Tests 154

H

Heredity 151, 152
Hierarchies 251
Higher Education 6, 8, 112
Highly 209, 212, 213
History 96
Human Health 67
Human Rights 1

I

Identify 193, 194, 208, 210, 213
Identity Options 115
Impact 12, 13, 211, 212, 217
Impediments 203, 204, 205, 207
Importance 195, 211, 212, 216
Improvement 193, 195, 204, 205, 206, 207, 210, 211, 213, 214, 215, 216, 217
Indian Society 243
Indicators 193, 195, 215

Individual 208, 211, 215, 216
Influence 138, 141, 145, 146
Integrity Inventory 157, 158
Islamic Feminism 227
Islamic Society 228, 232

J

Jewish Feminism 227
Judgment 167

K

Knowledge 10, 12, 13, 194, 195, 196, 203, 204, 205, 206, 208, 209, 210, 212, 213

L

Learning 193, 194, 195, 196, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214,
215, 216, 217, 218
Learning Camps 31
Learning Drivers 22
Learning Management System 108
Learning Objects 109
Learning Objects Standards 110
Lecturing 207, 208

M

Management 159, 160, 182
Materials 205, 207, 208
Measures of Intelligence 150, 168, 189
Method 150, 151, 161, 206
Methodological 203, 206
Motivation 150, 154, 155, 188
Motives as Inference 154

N

Neuroscientific Findings 186
Nonverbal Communication 135

O

Objective 216, 217
Observations 206, 213, 214

Online Instructors 296, 302
Online Teaching 280, 281, 282, 294, 299, 301, 302
Opportunities 8, 195, 203, 209, 212
Outcomes 11, 12, 13, 214
Over-stimulation 118

P

Parental 200
Parents 246
Participants 302
Participation 208
Pedagogical 288, 296, 299
Pedagogical Elements 110
Pedagogy 198, 200, 209, 296
Pedagogy and Teaching 197
Personality 157, 183, 189
Plastic Identification Code 58
Pollution Free Eco-system 62
Practice 14, 101, 205, 206, 209, 210, 213, 214, 217
Process 192, 193, 195, 196, 203, 204, 205, 206, 209, 214, 215, 216

Q

Qualities 205, 209, 210, 213
Quality 9, 192, 193, 194, 195, 196, 209, 212, 213, 214, 215, 216

R

Radical Feminism 225
Recycling Rates 58
Regulation and Monitoring 68
Relationships 12
Reproductive Rights 226
Requirements 192, 195, 205
Resources 14, 193, 195, 205, 207, 208, 214, 215

S

Sampling Theory 176, 177
Scientific 13, 14, 206
Scope 46, 214
Scoring 159, 162, 167, 168, 169, 170, 172, 173
Screencasting 107

Sex Discrimination 224
Situation 194, 204, 205, 206, 207, 208
Skills 10, 12, 13, 194, 196, 207, 211, 213, 214, 217
Synchronous and Asynchronous 101

T

Tbilisi Declaration 51
Teacher Enthusiasm 200
Teacher Training 120
Teaching 193, 195, 196, 203, 207, 208, 210, 212, 213, 214, 216, 217
Theories 149, 150, 155, 174, 176, 188, 204, 205, 206, 208
Theory 99, 203, 204, 205, 206, 207, 209
Training Management System 109

U

Underlying 12, 206
Understanding the System Entry-Point 30
Unique 205, 206, 208
University 202
Unlimited Shelf Space 24
Utilization 194

V

Values 194, 195
Variety 11, 203, 205, 215
Virtual Classroom 107
Vision 196

W

Webcams 107
Western Society 224
Whiteboards 107
Widow 228
Worker Productivity 68
Working Memory 188
World's Worst Polluted Places 71